Mathematics II (LA)

1 Inner Products (Olver & Shakiban)

Let $v, w \in \mathbb{R}^n$.

Definition : The Dot Product

The **dot product** of v and w is defined as:

$$v \cdot w = v_1 w_1 + \cdots + v_n w_n$$

Where the vectors are:

$$v = (v_1, \ldots, v_n)^t, \quad w = (w_1, \ldots, w_n)^t \in \mathbb{R}^n \text{ i.e., } v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, w = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$$

Matrix Notation

$v \cdot w = v^t w$ (Matrix multiplication of the row vector v^t and the column vector w).

$$v \cdot w = (v_1, \ldots, v_n) \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \sum_{i=1}^{n} v_i w_i$$

Euclidean Norm or Length

Definition : Norm

The norm of a vector is defined as:

$$\|v\| = \sqrt{v \cdot v} = \sqrt{v_1^2 + \cdots + v_n^2} = \sqrt{\sum_{i=1}^{n} v_i^2}$$

Note: The square root taken is the positive square root (non-negative).

Properties:

- $v \neq 0 \implies \|v\| > 0$
- $\|v\| = 0 \iff v = 0$

Definition : General Inner Product

An **inner product** on the real vector space V is a map $V \times V \xrightarrow{\langle \cdot, \cdot \rangle} \mathbb{R}$ which takes 2 vectors $v, w \in V$ and produces a real number $\langle v, w \rangle$ and satisfies the following three axioms:

1. **Bilinearity:** $\langle cu + dv, w \rangle = c\langle u, w \rangle + d\langle v, w \rangle$ for all $u, v, w \in V$ and for all $c, d \in \mathbb{R}$
2. **Symmetry:** $\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V$
3. **Positivity:** $\langle v, v \rangle \geq 0 \quad \forall v \in V$ and $\langle v, v \rangle = 0 \iff v = 0$

A (real) vector space V equipped with an inner product is called an **inner product space**.

Problem: Verify that the dot product defined above on $\mathbb{R}^n$ is an inner product on $\mathbb{R}^n$.

Associated Norm: Given an inner product on V, we can define the **associated norm** of any vector $v \in V$ by defining:

$$\|v\| = \sqrt{\langle v, v\rangle} \quad \text{(positive square root)}$$

By the positivity axiom: $\|v\| \geq 0$ & $\|v\| = 0 \iff v = 0$.

2 Examples of Inner Products

Example 3.2 (in the book)

Let $V = \mathbb{R}^2$ then verify that the following is an inner product on V:

$$\langle v, w\rangle = 2v_1w_1 + 5v_2w_2$$

where $v = (v_1, v_2)^t$ & $w = (w_1, w_2)^t$ in $\mathbb{R}^2$. Note this gives an example of an inner product which is different from the usual dot product on $\mathbb{R}^2$. It is an example of a **weighted inner product** & the associated norm is called a **weighted norm** (using the weights 2 & 5).

$$\|v\| = \sqrt{2v_1^2 + 5v_2^2} \quad \text{where } v = (v_1, v_2)^t$$

Example Alternative Inner Product

Check if the following defines an inner product on $\mathbb{R}^2$:

$$\langle v, w\rangle = v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2$$

Positivity follows from:

$$\langle v, v\rangle = v_1^2 - 2v_1v_2 + 4v_2^2 = (v_1 - v_2)^2 + 3v_2^2 \geq 0$$

(And $\langle v, v\rangle = 0 \iff v = 0$)

Example 3.3: Weighted Inner Product

Let $c_1, \ldots, c_n > 0$ be a set of true numbers. The corresponding **weighted inner product** and **weighted norm** on $\mathbb{R}^n$ are defined by:

$$\langle v, w\rangle = \sum_{i=1}^{n} c_i v_i w_i, \quad \|v\| = \sqrt{\sum_{i=1}^{n} c_i v_i^2}$$

The numbers c_i are called the **weights**. Check that this is an inner product on $\mathbb{R}^n$.

Exercises (page 132): 3.1.1, 3.1.3, 3.1.5, 3.1.6, 3.1.7, 3.1.9, 3.1.10, 3.1.11, 3.1.12, 3.1.13, 3.1.14, 3.1.15, 3.1.16, 3.1.17, 3.1.18, 3.1.19, 3.1.20

3 Inner Products on Function Spaces (Page 133)

We now extend the concept of inner products from finite-dimensional vector spaces (like $\mathbb{R}^n$) to infinite-dimensional vector spaces (function spaces).

3.1 The Vector Space $C^0[a, b]$

Let $[a, b] \subseteq \mathbb{R}$ be a bounded, closed interval. Let $C^0[a, b]$ denote the set of all **continuous functions** from $[a, b]$ to $\mathbb{R}$.

- $C^0[a, b]$ is a real vector space.
- **Question:** Can you prove that the dimension of this vector space is infinite?
- **Reasoning:** The set of monomials $\{1, x, x^2, \ldots, x^n, \ldots\}$ is linearly independent on $[a, b]$. Since this set is infinite, the basis of the space must be infinite.

3.2 Example 3.4: The Standard L^2 Inner Product

[Image of area under curve integration]

For any two functions $f, g \in C^0[a, b]$, we define their inner product by integrating their product over the interval:

$$\langle f, g \rangle = \int_a^b f(x)g(x)\, dx$$

Definition : L^2 Norm

The length (or magnitude) of the function f is defined as:

$$\|f\| = \sqrt{\langle f, f \rangle} = \sqrt{\int_a^b (f(x))^2\, dx}$$

This is known as the L^2 **norm** of f, and the operation is the L^2 **inner product** of f and g.

3.3 Example: Weighted Inner Product on Functions

Let $w : [a, b] \to \mathbb{R}$ be a continuous function that acts as a "weight." **Condition:** We must have $w(x) > 0$ for all $x \in [a, b]$ (strictly positive) to ensure the positivity axiom holds.

We define the **weighted inner product**:

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x)\, dx$$

And the **weighted norm**:

$$\|f\| = \sqrt{\int_a^b w(x)(f(x))^2\, dx}$$

Task: Show that this definition satisfies the axioms of an inner product on $C^0[a, b]$.

Proof Sketch of Positivity Proof

Consider $\langle f, f \rangle = \int_a^b w(x)(f(x))^2 dx$. Since $w(x) > 0$ and $(f(x))^2 \geq 0$, the integrand is always non-negative, so the integral is ≥ 0. If the integral is 0, since the function is continuous and non-negative, $f(x)$ must be 0 everywhere.

Exercises (Page 135): Practice problems for function spaces (do not need to submit):

3.1.21, 3.1.24, 3.1.25, 3.1.26, 3.1.29, 3.1.30

4 The Cauchy-Schwarz Inequality (Page 137)

This is a fundamental theorem in linear algebra that relates the inner product of two vectors to their norms.

Theorem Cauchy-Schwarz Inequality

Let V be a real vector space with an inner product $\langle \cdot, \cdot \rangle$. For all vectors $v, w \in V$, the following inequality holds:

$$|\langle v, w \rangle| \leq \|v\| \|w\|$$

Condition for Equality: Equality holds ($|\langle v, w \rangle| = \|v\| \|w\|$) if and only if v and w are **parallel** (linearly dependent). That is, $v = \lambda w$ or $w = \mu v$ for some scalars $\lambda, \mu \in \mathbb{R}$.

4.1 Proof of the Inequality (Summary)

Case 1: If $w = 0$, both sides are zero ($0 \leq 0$), so the inequality holds trivially.

Case 2: Assume $w \neq 0$. Let $t \in \mathbb{R}$ be any scalar. Consider the vector $v + tw$. By the positivity axiom, the square of its norm must be non-negative:

$$0 \leq \|v + tw\|^2 = \langle v + tw, v + tw \rangle$$

Expanding this using linearity and symmetry:

$$0 \leq \langle v, v \rangle + 2t\langle v, w \rangle + t^2 \langle w, w \rangle$$

$$0 \leq \|v\|^2 + 2t\langle v, w \rangle + t^2 \|w\|^2$$

Let $a = \|w\|^2$, $b = 2\langle v, w \rangle$, and $c = \|v\|^2$. Note that $a > 0$ since $w \neq 0$. The inequality becomes a quadratic polynomial in t:

$$p(t) = at^2 + bt + c \geq 0$$

Since the parabola $p(t)$ is always non-negative (it never crosses the horizontal axis), its minimum value occurs at the vertex $t = -b/2a$. Substituting this back in, or simply using the discriminant condition ($b^2 - 4ac \leq 0$), we get:

$$(2\langle v, w \rangle)^2 - 4(\|w\|^2)(\|v\|^2) \leq 0$$

$$4\langle v, w \rangle^2 \leq 4\|v\|^2 \|w\|^2$$

Taking the square root of both sides gives the Cauchy-Schwarz inequality:

$$|\langle v, w \rangle| \leq \|v\| \|w\|$$

5 The Cauchy-Schwarz Inequality (Detailed Proof)

Theorem

Let V be a real inner product space. For any vectors $v, w \in V$:

$$|\langle v, w \rangle| \leq \|v\| \|w\|$$

Equality Condition: Equality holds ($|\langle v, w \rangle| = \|v\| \|w\|$) if and only if v and w are linearly dependent (parallel), i.e., $v = \lambda w$ or $w = \mu v$ for some scalars λ, μ.

5.1 Step-by-Step Proof

Case 1: Trivial Case If $w = 0$, then:

- LHS: $|\langle v, 0 \rangle| = 0$
- RHS: $\|v\| \|0\| = 0$

Since $0 \leq 0$, the inequality holds.

Case 2: General Case ($w \neq 0$) Let $t \in \mathbb{R}$ be any scalar. Consider the vector $v + tw$. By the positivity axiom:

$$0 \leq \|v + tw\|^2 = \langle v + tw, v + tw \rangle$$

Expanding this using linearity:

$$0 \leq \langle v, v \rangle + \langle v, tw \rangle + \langle tw, v \rangle + \langle tw, tw \rangle$$

$$0 \leq \|v\|^2 + 2t\langle v, w \rangle + t^2\|w\|^2$$

This expression is a quadratic polynomial in t, of the form $P(t) = at^2 + bt + c$, where:

$$a = \|w\|^2, \quad b = 2\langle v, w \rangle, \quad c = \|v\|^2$$

Geometric Interpretation: Since $P(t) \geq 0$ for *all* real numbers t, the parabola representing this polynomial lies entirely above (or touches) the horizontal t-axis. This means it has a minimum value ≥ 0.

Finding the Minimum: To find the value of t that minimizes this quantity, we use calculus (derivative is zero) or simply complete the square. The minimum occurs at the vertex of the parabola:

$$P'(t) = 2at + b = 0 \implies t = -\frac{b}{2a} = -\frac{2\langle v, w \rangle}{2\|w\|^2} = -\frac{\langle v, w \rangle}{\|w\|^2}$$

Substitution: Substitute this optimal value of t back into the inequality $P(t) \geq 0$:

$$\|v\|^2 + 2\left(-\frac{\langle v, w \rangle}{\|w\|^2}\right)\langle v, w \rangle + \left(-\frac{\langle v, w \rangle}{\|w\|^2}\right)^2 \|w\|^2 \geq 0$$

Simplify the terms:

$$\|v\|^2 - \frac{2\langle v, w \rangle^2}{\|w\|^2} + \frac{\langle v, w \rangle^2}{\|w\|^4}\|w\|^2 \geq 0$$

$$\|v\|^2 - \frac{2\langle v, w \rangle^2}{\|w\|^2} + \frac{\langle v, w \rangle^2}{\|w\|^2} \geq 0$$

$$\|v\|^2 - \frac{\langle v, w \rangle^2}{\|w\|^2} \geq 0$$

Multiplying the entire inequality by $\|w\|^2$ (which is positive):

$$\|v\|^2\|w\|^2 - \langle v, w \rangle^2 \geq 0$$

$$\langle v, w \rangle^2 \leq \|v\|^2\|w\|^2$$

Taking the square root of both sides:

$$|\langle v, w \rangle| \leq \|v\|\|w\|$$

5.2 Application: Cauchy-Schwarz in $\mathbb{R}^n$ (Euclidean Space)

We now apply the general Cauchy-Schwarz inequality to the specific case of the standard dot product on $\mathbb{R}^n$.

The Setup: Suppose the inner product is the standard dot product $\langle v, w \rangle = v \cdot w$. Let the vectors be defined by their components:

$$v = (v_1, v_2, \ldots, v_n) \in \mathbb{R}^n, \quad w = (w_1, w_2, \ldots, w_n) \in \mathbb{R}^n$$

The Algebraic Inequality: Substituting the definitions of the dot product and the Euclidean norm into the general theorem $|\langle v, w \rangle| \leq \|v\|\|w\|$, we obtain:

$$|v_1w_1 + v_2w_2 + \cdots + v_nw_n| \leq \sqrt{\sum_{i=1}^{n} v_i^2} \cdot \sqrt{\sum_{i=1}^{n} w_i^2}$$

The Squared Form (Equivalent): Squaring both sides (which preserves the inequality since both sides are non-negative), we get the most common algebraic form of Cauchy-Schwarz:

$$\left(\sum_{i=1}^{n} v_i w_i\right)^2 \leq \left(\sum_{i=1}^{n} v_i^2\right)\left(\sum_{i=1}^{n} w_i^2\right)$$

Visual Representation:

$$(v \cdot w)^2 \leq \|v\|^2 \|w\|^2$$

This inequality guarantees that for any two sets of real numbers, the square of their "mixed sum" can never exceed the product of their individual "squared sums." This is the mathematical foundation for correlation coefficients in statistics.

6 Angles in Inner Product Spaces

In Euclidean space $\mathbb{R}^n$ with the standard dot product, we know:

$$v \cdot w = \|v\|\|w\| \cos(\theta)$$

Since Cauchy-Schwarz guarantees that $|\langle v, w\rangle| \leq \|v\|\|w\|$, we have:

$$-1 \leq \frac{\langle v, w\rangle}{\|v\|\|w\|} \leq 1$$

This allows us to define the "angle" between vectors in *any* inner product space (including spaces of functions).

Definition : Angle

The angle θ between two non-zero vectors $v, w \in V$ is defined by:

$$\cos(\theta) = \frac{\langle v, w\rangle}{\|v\|\|w\|}, \quad \text{where } 0 \leq \theta \leq \pi$$

6.1 Examples of Angles

Example 1: Standard Dot Product

Let $v = (1, 0, 1)^t$ and $w = (0, 1, 1)^t$ in $\mathbb{R}^3$ with the standard dot product.

- Inner Product: $\langle v, w\rangle = (1)(0) + (0)(1) + (1)(1) = 1$
- Norms: $\|v\| = \sqrt{1^2 + 0 + 1^2} = \sqrt{2}, \quad \|w\| = \sqrt{0 + 1^2 + 1^2} = \sqrt{2}$
- Angle: $\cos(\theta) = \frac{1}{\sqrt{2}\sqrt{2}} = \frac{1}{2} \implies \theta = \frac{\pi}{3}$ radians ($60°$)

Example 2: Weighted Inner Product

Using the same vectors v, w but with a weighted inner product:

$$\langle v, w\rangle = v_1 w_1 + 2v_2 w_2 + 3v_3 w_3$$

- New Inner Product: $0 + 0 + 3(1)(1) = 3$
- New Norms: $\|v\| = \sqrt{1(1)^2 + 3(1)^2} = \sqrt{4} = 2, \quad \|w\| = \sqrt{2(1)^2 + 3(1)^2} = \sqrt{5}$
- New Angle: $\cos(\theta) = \frac{3}{2\sqrt{5}} \approx 0.6708 \implies \theta \approx 0.835$ radians

Observation: The angle depends on the choice of inner product.

Example 3: Angle Between Functions

Find the angle between $p(x) = x$ and $q(x) = x^2$ in the space $C^0[0, 1]$ with the standard L^2 inner product.

- $\langle p, q\rangle = \int_0^1 x \cdot x^2\, dx = \int_0^1 x^3\, dx = \left[\frac{x^4}{4}\right]_0^1 = \frac{1}{4}$
- $\|p\|^2 = \int_0^1 x^2\, dx = \frac{1}{3} \implies \|p\| = \frac{1}{\sqrt{3}}$
- $\|q\|^2 = \int_0^1 (x^2)^2\, dx = \int_0^1 x^4\, dx = \frac{1}{5} \implies \|q\| = \frac{1}{\sqrt{5}}$
- $\cos(\theta) = \frac{1/4}{(1/\sqrt{3})(1/\sqrt{5})} = \frac{\sqrt{15}}{4} \approx 0.968$
- $\theta = \arccos(0.968) \approx 0.25$ radians

Exercises (Page 139): Practice problems for angles and inequalities:

3.2.1, 3.2.6, 3.2.7, 3.2.8, 3.2.9, 3.2.11

7 Orthogonality

We generalize the geometric concept of "perpendicularity" to general inner product spaces.

Definition : Orthogonality

Let V be an inner product space. Two vectors $v, w \in V$ are said to be **orthogonal** (with respect to the inner product $\langle \cdot, \cdot \rangle$) if:

$$\langle v, w\rangle = 0$$

Note on the Zero Vector: The zero vector 0 is orthogonal to *every* vector $v \in V$, because by linearity:

$$\langle 0, v\rangle = \langle 0 \cdot v, v\rangle = 0\langle v, v\rangle = 0$$

7.1 Examples of Orthogonality

7.1.1 Example 1: Standard Euclidean Space

Let $V = \mathbb{R}^2$ with the standard dot product. Consider the vectors:

$$v = (1, 2), \quad w = (6, -3)$$

Checking for orthogonality:

$$\langle v, w\rangle = v \cdot w = (1)(6) + (2)(-3) = 6 - 6 = 0$$

Conclusion: The vectors are orthogonal in the standard sense. (They form a 90° angle).

7.1.2 Example 2: Dependency on the Inner Product

Consider the *same vectors* $v = (1, 2)$ and $w = (6, -3)$ in $\mathbb{R}^2$, but equipped with a **weighted inner product**:

$$\langle v, w\rangle = 2v_1w_1 + 5v_2w_2$$

Checking for orthogonality:

$$\begin{aligned}\langle v, w\rangle &= 2(1)(6) + 5(2)(-3)\\ &= 12 - 30 = -18 \neq 0\end{aligned}$$

Conclusion: The vectors are **NOT** orthogonal with respect to this weighted inner product. *Key Takeaway:* Whether two vectors are "perpendicular" depends entirely on the definition of the inner product being used.

7.2 Orthogonal Vectors (Page 140)

Revisiting the definition with function space examples.

Definition

Two vectors $v, w \in V$ of an inner product space V are called **orthogonal** if:

$$\langle v, w\rangle = 0$$

Property: The zero vector $0 \in V$ is orthogonal to all vectors $v \in V$, since $\langle 0, v\rangle = 0$ for all $v \in V$.

Example Dependency on Inner Product (Summary)

Consider $v = (1, 2)^t$ and $w = (6, -3)^t$ in $\mathbb{R}^2$.

- **Standard Dot Product:**

$$\langle v, w\rangle = 1(6) + 2(-3) = 6 - 6 = 0$$

  The vectors **are** orthogonal.

- **Weighted Inner Product:** $\langle v, w\rangle = 2v_1w_1 + 5v_2w_2$.

$$\langle v, w\rangle = 2(1)(6) + 5(2)(-3) = 12 - 30 = -18 \neq 0$$

  The vectors are **not** orthogonal.

Conclusion: The property of orthogonality depends on the inner product being used.

Example Function Space

Let $V = C^0[0, 1]$ with $\langle p, q\rangle = \int_0^1 p(x)q(x)\,dx$. Consider $p(x) = x$ and $q(x) = x^2 - \frac{1}{2}$. Are they orthogonal?

$$\int_0^1 x(x^2 - 0.5)\,dx = \int_0^1 (x^3 - 0.5x)\,dx = \left[\frac{x^4}{4} - \frac{x^2}{4}\right]_0^1 = 0$$

Yes, they are orthogonal on $[0, 1]$. Note that if we changed the interval to $[0, 2]$, they might not be.

Exercises (Page 141): 3.2.15 – 3.2.25

8 The Triangle Inequality

Theorem Triangle Inequality

Let V be a real inner product space. For all vectors $v, w \in V$:

$$\|v + w\| \leq \|v\| + \|w\|$$

Geometric Interpretation: The length of any side of a triangle cannot exceed the sum of the lengths of the other two sides. Here, the vectors v, w, and $v + w$ form the sides of a triangle.

8.1 Proof

Proof

We begin by analyzing the square of the norm of the sum $v + w$:

$$\|v + w\|^2 = \langle v + w, v + w\rangle$$

Using the linearity and symmetry axioms of the inner product, we expand this term:

$$= \langle v, v\rangle + \langle v, w\rangle + \langle w, v\rangle + \langle w, w\rangle$$

$$= \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$$

Now, we apply the **Cauchy-Schwarz Inequality**, which states that $\langle v, w\rangle \leq |\langle v, w\rangle| \leq \|v\|\|w\|$. Substituting this inequality into our equation:

$$\|v + w\|^2 \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2$$

We recognize the right-hand side as a perfect square:

$$\|v + w\|^2 \leq (\|v\| + \|w\|)^2$$

Taking the non-negative square root of both sides gives the desired result:

$$\|v + w\| \leq \|v\| + \|w\|$$

8.2 Condition for Equality

Equality holds ($\|v + w\| = \|v\| + \|w\|$) if and only if the Cauchy-Schwarz inequality used in the proof is an equality **and** the inner product is positive ($\langle v, w\rangle = \|v\|\|w\|$).

This implies that v and w must be **parallel and point in the same direction**.

$$v = \lambda w \quad \text{for some scalar } \lambda \geq 0$$

(If $\lambda < 0$, the vectors point in opposite directions, and $\|v + w\| = |\|v\| - \|w\||$).

8.3 Example: Euclidean Space $\mathbb{R}^n$

In the standard coordinate space $\mathbb{R}^n$, the triangle inequality takes the algebraic form:

$$\sqrt{\sum_{i=1}^{n}(v_i + w_i)^2} \leq \sqrt{\sum_{i=1}^{n} v_i^2} + \sqrt{\sum_{i=1}^{n} w_i^2}$$

This confirms that the straight-line distance (Euclidean distance) is the shortest path between points.

9 Numerical Examples of the Triangle Inequality

We verify the inequality $\|v + w\| \leq \|v\| + \|w\|$ using concrete examples from different vector spaces.

9.1 Example 1: Euclidean Space $\mathbb{R}^3$

Let $V = \mathbb{R}^3$ with the standard dot product. Consider the vectors:

$$v = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad w = \begin{pmatrix} 2 \\ 0 \\ -3 \end{pmatrix}$$

Step 1: Calculate Individual Norms

$$\|v\| = \sqrt{1^2 + 2^2 + 1^2} = \sqrt{6}$$

$$\|w\| = \sqrt{2^2 + 0^2 + (-3)^2} = \sqrt{13}$$

Step 2: Calculate Norm of Sum

$$v + w = \begin{pmatrix} 1+2 \\ 2+0 \\ 1+(-3) \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -2 \end{pmatrix}$$

$$\|v + w\| = \sqrt{3^2 + 2^2 + (-2)^2} = \sqrt{9+4+4} = \sqrt{17}$$

Step 3: Verify Inequality We check if $\sqrt{17} \leq \sqrt{6} + \sqrt{13}$.

$$\text{LHS} \approx 4.123$$

$$\text{RHS} \approx 2.449 + 3.605 = 6.054$$

Since $4.123 < 6.054$, the inequality holds strictly.

9.2 Example 2: Function Space $C^0[0,1]$

Let $V = C^0[0,1]$ with the standard L^2 inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$. Consider the functions:

$$f(x) = x, \quad g(x) = x^2 + 1$$

Step 1: Calculate Individual Norms

$$\|f\| = \sqrt{\int_0^1 x^2\,dx} = \sqrt{\left[\frac{x^3}{3}\right]_0^1} = \sqrt{\frac{1}{3}}$$

$$\|g\| = \sqrt{\int_0^1 (x^2+1)^2\,dx} = \sqrt{\int_0^1 (x^4 + 2x^2 + 1)\,dx}$$
$$= \sqrt{\left[\frac{x^5}{5} + \frac{2x^3}{3} + x\right]_0^1} = \sqrt{\frac{1}{5} + \frac{2}{3} + 1} = \sqrt{\frac{28}{15}}$$

Step 2: Calculate Norm of Sum

$$(f+g)(x) = x^2 + x + 1$$

$$\|f + g\| = \sqrt{\int_0^1 (x^2 + x + 1)^2\,dx} = \sqrt{\frac{37}{10}}$$

Step 3: Verify Inequality We check if $\sqrt{\frac{37}{10}} \leq \sqrt{\frac{1}{3}} + \sqrt{\frac{28}{15}}$.

$$\text{LHS} \approx 1.923$$

$$\text{RHS} \approx 0.577 + 1.366 = 1.943$$

Since $1.923 < 1.943$, the inequality holds.

9.3 The Triangle Inequality (Page 142 Summary)

Theorem

Let V be an inner product space. Then:

$$\|v + w\| \leq \|v\| + \|w\| \quad \forall v, w \in V$$

where $\|\cdot\|$ is the associated norm.

Proof

Consider the square of the norm of the sum:

$$\|v+w\|^2 = \langle v+w, v+w\rangle = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$$

Using the Cauchy-Schwarz inequality ($\langle v, w\rangle \le \|v\|\|w\|$):

$$\le \|v\|^2 + 2\|v\|\|w\| + \|w\|^2$$

$$= (\|v\| + \|w\|)^2$$

Taking the positive square root of both sides gives:

$$\|v+w\| \le \|v\| + \|w\|$$

Equality Condition: Equality holds if and only if $\langle v, w\rangle = \|v\|\|w\|$. This implies that v and w are parallel vectors pointing in the **same direction** ($v = \lambda w$ with $\lambda \ge 0$).

Example Verification (Euclidean Space $\mathbb{R}^3$): For $v = (1, 2, 1)^t$ and $w = (2, 0, -3)^t$:

$$\|v\| = \sqrt{6}, \quad \|w\| = \sqrt{13}, \quad \|v+w\| = \sqrt{17}$$

$$\sqrt{17} \le \sqrt{6} + \sqrt{13} \quad \text{(True)}$$

Algebraic Forms: The Cauchy-Schwarz and Triangle inequalities for the Euclidean dot product in $\mathbb{R}^n$ are:

$$\left|\sum_{i=1}^{n} v_i w_i\right| \le \sqrt{\sum_{i=1}^{n} v_i^2}\sqrt{\sum_{i=1}^{n} w_i^2}$$

$$\sqrt{\sum_{i=1}^{n}(v_i + w_i)^2} \le \sqrt{\sum_{i=1}^{n} v_i^2} + \sqrt{\sum_{i=1}^{n} w_i^2}$$

10 Inequalities in Function Spaces (Page 143)

If $V = C^0[a, b]$ and $\langle\cdot,\cdot\rangle$ is the L^2 inner product, we have the integral forms of the inequalities.

10.1 Cauchy-Schwarz Integral Form

$$\left|\int_a^b f(x)g(x)\,dx\right| \le \sqrt{\int_a^b f(x)^2\,dx}\sqrt{\int_a^b g(x)^2\,dx}$$

10.2 Triangle Inequality Integral Form

$$\sqrt{\int_a^b (f(x) + g(x))^2\,dx} \le \sqrt{\int_a^b f(x)^2\,dx} + \sqrt{\int_a^b g(x)^2\,dx}$$

Exercises (Page 143): 3.2.31, 3.2.32, 3.2.39, 3.2.40

11 Norms (Lecture 2: Feb 2, 2026)

We have seen that an inner product on V gives rise to an associated norm. However, we can define a norm as a standalone concept. Note that **every norm may not come from an inner product**.

Definition : General Norm

Let V be a vector space over $\mathbb{R}$. A **norm** on V is a function $\|\cdot\| : V \to \mathbb{R}_{\geq 0}$ satisfying the following axioms for all $v, w \in V$ and $c \in \mathbb{R}$:

1. **Positivity:** $\|v\| \geq 0$ and $\|v\| = 0 \iff v = 0$.
2. **Homogeneity:** $\|cv\| = |c|\|v\|$.
3. **Triangle Inequality:** $\|v + w\| \leq \|v\| + \|w\|$.

11.1 Examples in $\mathbb{R}^n$

Example The 1-norm

Let $V = \mathbb{R}^n$. The **1-norm** is defined as the sum of the absolute values of the components:

$$\|v\|_1 = |v_1| + \cdots + |v_n| = \sum_{i=1}^{n} |v_i| \quad \text{where } v = (v_1, \ldots, v_n) \in \mathbb{R}^n$$

Example The Max-norm (∞-norm)

The **max-norm** (or infinity norm) picks the largest absolute component:

$$\|v\|_\infty = \max\{|v_1|, \ldots, |v_n|\}$$

Example The p-norm

For any real number $1 \leq p < \infty$, the p**-norm** on $\mathbb{R}^n$ is defined by:

$$\|v\|_p = \sqrt[p]{\sum_{i=1}^{n} |v_i|^p}$$

Note:

- If $p = 1$, this matches the 1-norm defined above.
- If $p = 2$, this is the standard **Euclidean norm**, which is the *only* norm in this list associated with an inner product (the dot product).
- All other p-norms do not come from an inner product.

Theorem Triangle Inequality for p-norms

For all $v, w \in \mathbb{R}^n$:

$$\sqrt[p]{\sum_{i=1}^{n} |v_i + w_i|^p} \leq \sqrt[p]{\sum_{i=1}^{n} |v_i|^p} + \sqrt[p]{\sum_{i=1}^{n} |w_i|^p}$$

11.2 Norms on Function Spaces $C^0[a,b]$

Example The L^p Norm

For $f \in C^0[a,b]$ and $1 \le p < \infty$:

$$\|f\|_p = \sqrt[p]{\int_a^b |f(x)|^p \, dx}$$

Special Cases:

- **L^1-norm:** $\|f\|_1 = \int_a^b |f(x)| \, dx$.
- **L^2-norm:** $\|f\|_2 = \sqrt{\int_a^b |f(x)|^2 \, dx}$. This is the only one arising from an inner product $\langle f, g \rangle = \int_a^b f(x)g(x)dx$.
- **L^∞-norm:** $\|f\|_\infty = \max\{|f(x)| : x \in [a,b]\}$.

12 Distance Between Vectors

Any norm $\|\cdot\|$ in a real vector space V defines a notion of **distance** between two elements.

Definition : Metric

The distance $d(v,w)$ between two vectors $v, w \in V$ is defined as:

$$d(v,w) = \|v - w\|$$

This distance function satisfies the following properties:

1. **Symmetry:** $d(v,w) = d(w,v)$ for all $v, w \in V$.
2. **Positivity:** $d(v,w) = 0 \iff v = w$ for all $v, w \in V$.
3. **Triangle Inequality:** $d(v,w) \le d(v,z) + d(z,w)$ for all $v, w, z \in V$.

Exercises (Page 147): 3.3.1, 3.3.3, 3.3.9, 3.3.12, 3.3.13, 3.3.14, 3.3.16, 3.3.19.

13 Unit Vectors and Unit Spheres

Definition : Unit Vector

Let V be a normed vector space. The elements $u \in V$ such that $\|u\| = 1$ are called **unit vectors**.

Theorem Normalization Lemma

Let V be a normed vector space and let $v \in V$ be any non-zero vector. Then the vector

$$u = \frac{v}{\|v\|}$$

is a unit vector which is parallel to v.

Example Normalization in $\mathbb{R}^2$

Let $v = (1, -2)^t \in \mathbb{R}^2$.

- **With Euclidean Norm ($\|\cdot\|_2$):** $\|v\|_2 = \sqrt{1^2 + (-2)^2} = \sqrt{5}$.

$$u = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{pmatrix}$$

- **With 1-Norm ($\|\cdot\|_1$):** $\|v\|_1 = |1| + |-2| = 3$.

$$\hat{u} = \frac{1}{3} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1/3 \\ -2/3 \end{pmatrix}$$

- **With Max-Norm ($\|\cdot\|_\infty$):** $\|v\|_\infty = \max\{|1|, |-2|\} = 2$.

$$\hat{u} = \frac{1}{2} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1 \end{pmatrix}$$

13.1 Unit Spheres and Balls

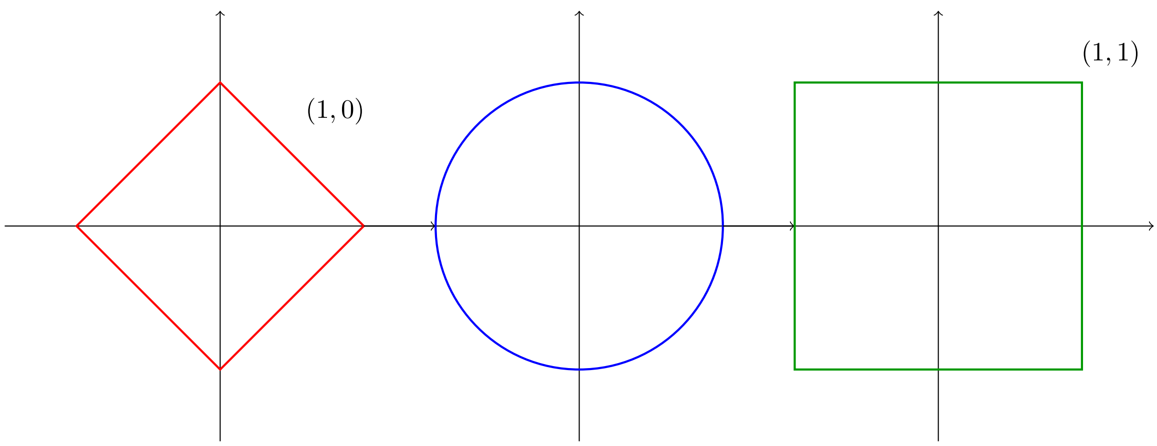
Definition : Spheres and Balls

In a real vector space V with norm $\|\cdot\|$:

- The **Unit Sphere** S_1 is the set of all unit vectors: $S_1 = \{u \in V : \|u\| = 1\}$.
- The **Sphere of radius** r is $S_r = \{u \in V : \|u\| = r\}$.
- The **Unit Ball** B_1 is the set: $B_1 = \{u \in V : \|u\| \leq 1\}$.

Visualization of Unit Spheres in $\mathbb{R}^2$

The geometric shape of the "sphere" depends heavily on the chosen norm.



L^1 **Norm**
Diamond

L^2 **Norm**
Circle

L^∞ **Norm**
Square

Exercises (Page 149): 3.3.20,3.3.22,3.3.23,3.3.27.

14 Equivalence of Norms and Convergence

Reference: Davidson & Donsig, Chapter 4, Section 2, Page 52.

14.1 Convergence in $\mathbb{R}^n$

Definition : Convergence

Let $(x_k)_{k=1}^{\infty}$ be a sequence of points in $\mathbb{R}^n$, where $x_k = (x_{k1}, \ldots, x_{kn})$. This sequence **converges** to a point $a \in \mathbb{R}^n$ if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall k \geq N, \|x_k - a\| < \epsilon.$$

We write $\lim_{k\to\infty} x_k = a$.

Theorem Lemma: Component-wise Convergence

A sequence (x_k) in $\mathbb{R}^n$ converges to a point $a = (a_1, \ldots, a_n)$ if and only if it converges component-wise:

$$\lim_{k\to\infty} x_k = a \iff \lim_{k\to\infty} x_{kj} = a_j \quad \forall 1 \leq j \leq n$$

Proof Sketch

($\Rightarrow$) Given $\lim x_k = a$, then for any $\epsilon > 0$, $\|x_k - a\| < \epsilon$ for large k. Since $|x_{ki} - a_i| \leq \|x_k - a\| < \epsilon$, each component converges.
($\Leftarrow$) Given each component converges, for any $\epsilon > 0$, there exists N_j for each component such that $|x_{kj} - a_j| < \epsilon/n$ (or similar bound). Let $N = \max\{N_1, \ldots, N_n\}$. Then for $k \geq N$:

$$\|x_k - a\| = \sqrt{\sum (x_{ki} - a_i)^2} < \epsilon$$

Thus the vector sequence converges.

14.2 Completeness

Definition : Cauchy Sequence

A sequence $(x_k)_{k\in\mathbb{N}}$ in $\mathbb{R}^n$ is **Cauchy** if:

$$\forall \epsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall k, l \geq N, \|x_k - x_l\| < \epsilon$$

Definition : Complete Set

A set $S \subseteq \mathbb{R}^n$ is **complete** if every Cauchy sequence of points in S converges to a point **in** S.

Theorem Completeness of $\mathbb{R}^n$

$\mathbb{R}^n$ is complete.

Proof

1. First, show that a convergent sequence is Cauchy (via Triangle Inequality). 2. Conversely, let (x_k) be a Cauchy sequence in $\mathbb{R}^n$. This implies each component sequence (x_{kj}) is Cauchy in $\mathbb{R}$. 3. By the completeness of $\mathbb{R}$, each component sequence converges to some a_j. 4. Let $a = (a_1, \ldots, a_n)$. By the previous Lemma, the vector sequence converges to a.

15 Closed and Open Subsets of $\mathbb{R}^n$

Definition : Limit Point

A point x is a **limit point** of a subset $A \subseteq \mathbb{R}^n$ if there exists a sequence $(a_n) \in A$ such that $x = \lim_{n\to\infty} a_n$.

Definition : Closed Set

A set $A \subseteq \mathbb{R}^n$ is **closed** if it contains all of its limit points.

Example Examples

1. $[a, b]$ is closed in $\mathbb{R}$.
2. $\emptyset$ and $\mathbb{R}^n$ are both closed.
3. $[0, +\infty)$ is closed in $\mathbb{R}$.
4. Intervals like $(0, 1)$, $(0, 1]$, and $[0, 1)$ are **not** closed.
5. The unit ball $\{x \in \mathbb{R}^n : \|x\| \leq 1\}$ is closed.
6. The set $\{x \in \mathbb{R}^n : \|x\| < 1\}$ is **not** closed.
7. Finite sets of real numbers are closed.

16 Properties of Closed Sets and Closure

Theorem Algebra of Closed Sets

Let A and B be subsets of $\mathbb{R}^n$.

1. If A and B are closed, then $A \cup B$ is closed.
2. If $\{A_i\}_{i\in I}$ is **any** family of closed subsets of $\mathbb{R}^n$, then $\bigcap_{i\in I} A_i$ is closed.

Note: Infinite unions of closed sets are not necessarily closed (e.g., $\bigcup_{n=1}^{\infty}[\frac{1}{n}, 1] = (0, 1]$).

Proof Sketch for Union

Let (x_k) be a sequence in $A \cup B$ with limit x. There are infinitely many terms in A or infinitely many terms in B (or both). WLOG, assume infinitely many x_k belong to A. This forms a subsequence in A. Since A is closed, the limit of this subsequence (which is x) must be in A. Thus $x \in A \subseteq A \cup B$.

16.1 The Closure of a Set

Definition : Closure

Let A be a subset of $\mathbb{R}^n$. The **closure** of A, denoted $\bar{A}$, is the set consisting of all limit points of A.

$$\bar{A} = \{x \in \mathbb{R}^n \mid \exists (a_n) \subseteq A \text{ such that } \lim_{n\to\infty} a_n = x\}$$

Theorem Properties of Closure

Let A be a subset of $\mathbb{R}^n$. Then:

1. $\bar{A}$ is a closed set.
2. $\bar{A}$ is the **smallest** closed set containing A.
3. $A \subseteq \bar{A}$.
4. A is closed $\iff$ $A = \bar{A}$.
5. If C is a closed set and $A \subseteq C$, then $\bar{A} \subseteq C$.

Proof Minimality of Closure

First, note $A \subseteq \bar{A}$ (using constant sequences). To show $\bar{A}$ is closed: Let (x_k) be a sequence in $\bar{A}$ converging to x. We must show $x \in \bar{A}$. Since $x_k \in \bar{A}$, for each k, there exists $a_k \in A$ such that $\|x_k - a_k\| < 1/k$. Consider the distance from a_k to x:

$$\|a_k - x\| = \|(a_k - x_k) + (x_k - x)\| \le \|a_k - x_k\| + \|x_k - x\|$$

As $k \to \infty$, both terms on the RHS go to 0. Thus $a_k \to x$. Since (a_k) is in A, x is a limit point of A, so $x \in \bar{A}$.

17 Open Sets

Definition : Open Ball

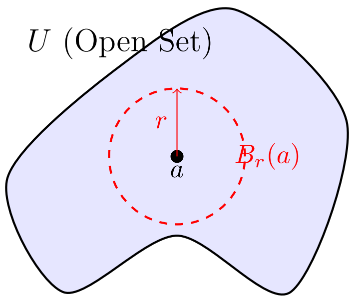
The **open ball** about $a \in \mathbb{R}^n$ of radius $r > 0$ is the set:

$$B_r(a) = \{x \in \mathbb{R}^n \mid \|x - a\| < r\}$$

Definition : Open Set

A subset $U \subseteq \mathbb{R}^n$ is **open** in $\mathbb{R}^n$ if for every $a \in U$, there exists some radius $r = r(a) > 0$ such that the ball is fully contained in U:

$$B_r(a) \subseteq U$$



For any point a in an open set U, we can find a small "buffer" radius r such that the entire ball stays inside U.

Example Examples of Open/Not Open Sets

1. Any open interval (a, b) is open in $\mathbb{R}$.
2. $\emptyset$ and $\mathbb{R}^n$ are open in $\mathbb{R}^n$.
3. Half-open intervals like $(0, 1]$ and closed intervals $[0, 1]$ are **not** open.
4. The open ball $B_r(a)$ is open in $\mathbb{R}^n$.

5. The closed ball $\overline{B_r(a)} = \{x \mid \|x-a\| \le r\}$ is **not** open.
6. The set $\{(x,y) \in \mathbb{R}^2 \mid xy < 1\}$ is open in $\mathbb{R}^2$.

18 Relationship Between Open and Closed Sets

There is a duality between open and closed sets via the complement operation.

Theorem Complement Theorem

A set $A \subseteq \mathbb{R}^n$ is **open** in $\mathbb{R}^n$ if and only if its complement $A^c = \mathbb{R}^n \setminus A$ is **closed** in $\mathbb{R}^n$.

Proof

($\Rightarrow$) Suppose A is open. Let (x_k) be a sequence in A^c with limit x. We show $x \in A^c$. Suppose for contradiction $x \notin A^c$, i.e., $x \in A$. Since A is open, $\exists r > 0$ s.t. $B_r(x) \subseteq A$. Since $x_k \to x$, for large k, $\|x_k - x\| < r$, meaning $x_k \in B_r(x) \subseteq A$. But $x_k \in A^c$, a contradiction. Thus $x \in A^c$, so A^c is closed.
($\Leftarrow$) Suppose A^c is closed. Assume A is **not** open. Then $\exists a \in A$ such that for every $r > 0$, $B_r(a)$ is not contained in A. In particular, for $r = 1/k$, choose $x_k \in B_{1/k}(a)$ such that $x_k \notin A$ (so $x_k \in A^c$). Then $\|x_k - a\| < 1/k \implies x_k \to a$. Since $x_k \in A^c$ and A^c is closed, the limit a must be in A^c. But we started with $a \in A$. Contradiction. Thus A is open.

Theorem Algebra of Open Sets

Using De Morgan's Laws and the properties of closed sets:

1. If U and V are open subsets of $\mathbb{R}^n$, then $U \cap V$ is open.
2. If $\{U_i\}_{i \in I}$ is **any** family of open subsets, then $\bigcup_{i \in I} U_i$ is open.

Proof Sketch

Using the previous theorem:

$$\left(\bigcup_{i \in I} U_i \right)^c = \bigcap_{i \in I} (U_i^c)$$

Since each U_i is open, U_i^c is closed. The intersection of closed sets is closed. Thus the complement of the union is closed, making the union open.

19 Compact Sets (Lecture 3)

Reference: Davidson-Donsig, Chapter 4, Section 4, Page 61.

We now introduce the concept of compactness, which generalizes the properties of closed and bounded intervals in $\mathbb{R}$ to higher dimensions.

Definition : Compactness (Sequential Definition)

A subset $A \subseteq \mathbb{R}^n$ is called **compact** if every sequence $(x_k)_{k \in \mathbb{N}}$ contained in A has a convergent subsequence $(x_{k_j})_{j \in \mathbb{N}}$ whose limit x lies in A.

$$x = \lim_{j \to \infty} x_{k_j} \in A$$

Recall: Bolzano-Weierstrass Theorem

In Math I, we learned the Bolzano-Weierstrass Theorem: Every **bounded** sequence in $\mathbb{R}$ has a convergent subsequence. This implies that any closed and bounded subset of $\mathbb{R}$ (like $[a, b]$) is compact. We will generalize this to $\mathbb{R}^n$.

19.1 Examples of Non-Compact Sets

To understand compactness, it is helpful to visualize sets that fail to be compact.

Example Failure by Missing Limit Points

Consider the interval $S = (0, 1] \subseteq \mathbb{R}$. Let the sequence be $x_k = \frac{1}{k}$.

- The limit is $\lim_{k\to\infty} \frac{1}{k} = 0$.
- However, $0 \notin (0, 1]$.
- Thus, the subsequence converges to a point **outside** the set.

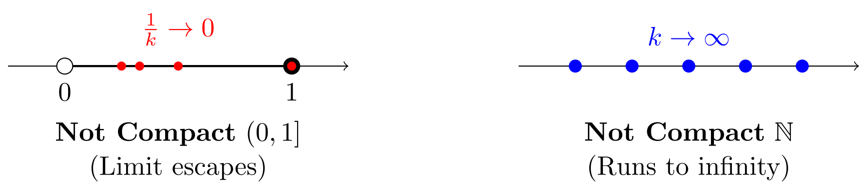
Conclusion: $(0, 1]$ is **not** compact.

Example Failure by Unboundedness

Consider the set of natural numbers $\mathbb{N} \subseteq \mathbb{R}$. Let the sequence be $x_k = k$.

- This sequence is unbounded.
- No subsequence can converge to a real number.

Conclusion: $\mathbb{N}$ is **not** compact.



20 Boundedness and Properties

Definition : Bounded Set

A subset $S \subseteq \mathbb{R}^n$ is **bounded** if there exists a real number $R > 0$ such that S is contained entirely within the ball of radius R centered at the origin:

$$S \subseteq B_R(0) = \{x \in \mathbb{R}^n : \|x\| < R\}$$

Theorem Lemma 1: Compact implies Closed and Bounded

If $C \subseteq \mathbb{R}^n$ is a compact set, then:

1. C is closed.
2. C is bounded.

Proof

1. C is closed: Let x be a limit point of C. By definition, there exists a sequence $(c_k) \subseteq C$ such that $\lim c_k = x$. Since C is compact, (c_k) has a subsequence converging to a point in C. Since the original sequence converges to x, every subsequence must converge to x. Thus $x \in C$. Since C contains its limit points, it is closed.
2. C is bounded: Suppose C is not bounded. Then for every $k \in \mathbb{N}$, there exists $c_k \in C$ such that $\|c_k\| > k$. If we take any subsequence (c_{k_j}), we have $\|c_{k_j}\| > k_j \to \infty$. Thus, no subsequence can converge to a point $c \in C$ (since convergent sequences are bounded). This contradicts compactness.

Theorem Lemma 2: Closed Subset of Compact is Compact

Let $K \subseteq \mathbb{R}^n$ be compact, and let $C \subseteq K$ be a closed set. Then C is compact.

Proof

Let (x_n) be a sequence in C. Since $C \subseteq K$, this is also a sequence in K. Because K is compact, there exists a subsequence (x_{n_i}) converging to some point $x \in K$. Since the sequence lies entirely in C and C is closed, the limit x must belong to C. Thus, every sequence in C has a subsequence converging to a point in C.

21 The Heine-Borel Theorem

Before stating the theorem, we need one crucial lemma regarding hypercubes.

Theorem Lemma 3: Compactness of Hypercubes

The hypercube $[a, b]^n = [a, b] \times \cdots \times [a, b]$ is a compact subset of $\mathbb{R}^n$.

Proof Idea: Coordinate-wise Bolzano-Weierstrass

Let (x_k) be a sequence in $[a, b]^n$.

- Consider the first coordinate sequence $(x_{k,1})$. It is bounded in $[a, b]$. By Bolzano-Weierstrass, there is a subsequence converging to some $z_1 \in [a, b]$.
- Look at the second coordinates of *that* subsequence. They are bounded, so refine the subsequence again to make the second coordinates converge to z_2.
- Repeat this process n times.
- The final "diagonal" subsequence converges in every coordinate component.

Thus, the vector sequence converges to $z = (z_1, \ldots, z_n)$, which lies in $[a, b]^n$.

Theorem The Heine-Borel Theorem

A subset of $\mathbb{R}^n$ is compact if and only if it is **closed** and **bounded**.

Proof

($\Rightarrow$) Proven in Lemma 1.
($\Leftarrow$) Let C be closed and bounded. Since C is bounded, there exists $M > 0$ such that $\|x\| \leq M$ for all $x \in C$. This implies $C \subseteq [-M, M]^n$. By Lemma 3, the box $[-M, M]^n$ is compact. Since C is a closed subset of a compact set, by Lemma 2, C is compact.

22 Cantor Intersection Theorem

This theorem describes the behavior of "nested" compact sets.

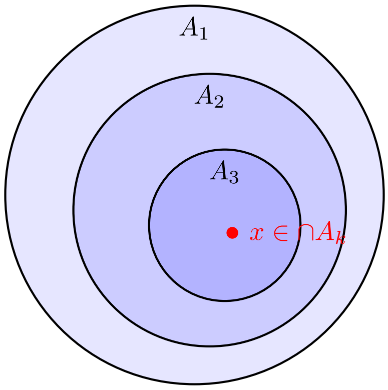
Theorem Cantor Intersection Theorem

If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$ is a decreasing sequence of **non-empty compact** subsets of $\mathbb{R}^n$, then their intersection is non-empty:

$$\bigcap_{k \in \mathbb{N}} A_k \neq \emptyset$$

Proof

For each k, the set A_k is non-empty, so choose a point $a_k \in A_k$. Consider the sequence $(a_k)_{k \in \mathbb{N}}$. Since $A_1 \supseteq A_k$ for all k, the entire sequence lies inside A_1. Since A_1 is compact, there exists a subsequence (a_{k_j}) converging to some limit $x \in A_1$.
For any fixed integer m, consider the tail of the subsequence where $k_j \geq m$. Since the sets are nested, $a_{k_j} \in A_{k_j} \subseteq A_m$. Since A_m is closed (being compact) and the subsequence lies in A_m, the limit x must be in A_m. Since this holds for all m, $x \in \bigcap A_m$. Thus the intersection is non-empty.



Nested compact sets trap at least one point in the limit. If these were open sets, the center could be "missing."

23 Limits and Continuity (Chapter 5)

We now shift our focus to functions mapping subsets of $\mathbb{R}^n$ to $\mathbb{R}^m$.

23.1 Functions on $\mathbb{R}^n$

Let $S \subseteq \mathbb{R}^n$. Consider a function $f : S \to \mathbb{R}^m$. Note that for any $x = (x_1, \dots, x_n) \in S$, the output is a vector $f(x) \in \mathbb{R}^m$. We can write f in terms of its component functions:

$$f(x) = (f_1(x), f_2(x), \dots, f_m(x))$$

where each $f_i : S \to \mathbb{R}$ is a real-valued scalar function. **Key Idea:** Analyzing f is often equivalent to analyzing the m component functions f_i individually.

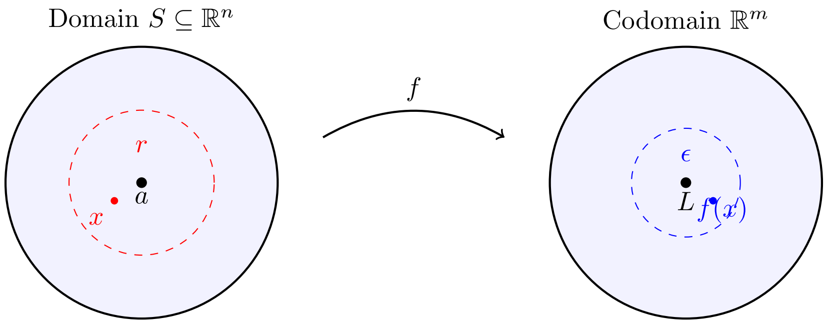
23.2 The Limit of a Function

Definition : Limit

Let $S \subseteq \mathbb{R}^n$ and $f : S \to \mathbb{R}^m$. Suppose $a \in \mathbb{R}^n$ is a **limit point** of $S \setminus \{a\}$ (i.e., we can approach a using points in S). A point $L \in \mathbb{R}^m$ is said to be the **limit** of f at a if:

$$\forall \epsilon > 0, \exists r > 0 \text{ such that } \forall x \in S, \text{ if } 0 < \|x - a\| < r \text{ then } \|f(x) - L\| < \epsilon$$

We write: $\lim_{x \to a} f(x) = L$.



Visualizing Limits: If inputs x are trapped within radius r of a, outputs $f(x)$ must land within radius ϵ of L.

- $f(a)$ need not be defined.
- Even if $f(a)$ is defined, it need not equal the limit L.

24 Continuity

Definition : Continuity

Let $S \subseteq \mathbb{R}^n$, $a \in S$, and $f : S \to \mathbb{R}^m$. f is said to be **continuous at** a if for any $\epsilon > 0$, there exists $r > 0$ such that:

$$\text{whenever } x \in S \text{ and } \|x - a\| < r, \text{ then } \|f(x) - f(a)\| < \epsilon.$$

f is **continuous on** S if it is continuous at every point $a \in S$.

Relation to Limits

If a is a limit point of $S \setminus \{a\}$, then:

$$f \text{ is continuous at } a \iff \lim_{x \to a} f(x) = f(a).$$

If a is an **isolated point** of S, f is automatically continuous at a by definition.

Example Continuity of the Norm

Let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = \|x\|$ (the Euclidean norm). We prove f is continuous at any $a \in \mathbb{R}^n$.
Proof: Recall the **Reverse Triangle Inequality**:

$$|\|x\| - \|a\|| \leq \|x - a\|$$

Given any $\epsilon > 0$, choose $r = \epsilon$. If $\|x - a\| < r$, then:

$$|f(x) - f(a)| = |\|x\| - \|a\|| \leq \|x - a\| < r = \epsilon.$$

Thus, the norm function is continuous everywhere.

25 Lipschitz Functions

Definition : Lipschitz Continuity

A function $f : S \to \mathbb{R}^m$ is called **Lipschitz** if there exists a constant $C \geq 0$ such that:

$$\|f(x) - f(y)\| \leq C\|x - y\| \quad \forall x, y \in S.$$

Theorem Proposition: Lipschitz implies Continuous

Every Lipschitz function is continuous.

Proof

Given $\epsilon > 0$, let $r = \epsilon/C$ (assuming $C > 0$). If $\|x - y\| < r$, then:

$$\|f(x) - f(y)\| \leq C\|x - y\| < C\left(\frac{\epsilon}{C}\right) = \epsilon.$$

Theorem Linear Maps are Lipschitz

Any linear map $A : \mathbb{R}^n \to \mathbb{R}^m$ is Lipschitz, and therefore continuous.

Proof

Let the matrix representation of A be $(a_{ij}) \in M_{m\times n}(\mathbb{R})$. Let $x, y \in \mathbb{R}^n$. By linearity, $\|Ax - Ay\| = \|A(x - y)\|$. Let $z = x - y$. The i-th component of Az is $\sum_{j=1}^n a_{ij} z_j$.

$$\|Az\|^2 = \sum_{i=1}^m \left(\sum_{j=1}^n a_{ij} z_j \right)^2$$

By Cauchy-Schwarz on the inner sum:

$$\left(\sum_{j=1}^{n} a_{ij} z_j\right)^2 \leq \left(\sum_{j=1}^{n} a_{ij}^2\right)\left(\sum_{j=1}^{n} z_j^2\right) = \left(\sum_{j=1}^{n} a_{ij}^2\right) \|z\|^2$$

Summing over i:

$$\|Az\|^2 \leq \left(\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2\right) \|z\|^2$$

Let $C = \sqrt{\sum_{i,j} a_{ij}^2}$ (The Frobenius norm of the matrix). Then $\|Ax - Ay\| \leq C\|x - y\|$.

26 Characterizations of Continuity

Definition : Relative Open Set

Let $S \subseteq \mathbb{R}^n$. A subset $V \subseteq S$ is **open in** S if there exists an open set $U \subseteq \mathbb{R}^n$ such that $V = U \cap S$.

Theorem The Big Theorem on Continuity

Let $S \subseteq \mathbb{R}^n$ and $f : S \to \mathbb{R}^m$. The following are equivalent (TFAE):

1. f is continuous on S ($\epsilon - r$ definition).
2. **Sequential Continuity:** For every sequence (x_k) in S with limit $a \in S$, we have $\lim_{k\to\infty} f(x_k) = f(a)$.
3. **Topological Continuity:** If U is open in $\mathbb{R}^m$, then the pre-image $f^{-1}(U) = \{x \in S \mid f(x) \in U\}$ is open in S.

Proof Sketch: (1) $\iff$ (2)

(1) $\implies$ (2): Given f is continuous. If $x_k \to a$, then for large k, $\|x_k - a\| < r$. By continuity, $\|f(x_k) - f(a)\| < \epsilon$. Thus $f(x_k) \to f(a)$.
(2) $\implies$ (1): We prove by contrapositive. Suppose f is *not* continuous at a. Then $\exists \epsilon > 0$ such that for all $r = 1/k$, there exists a point x_k with $\|x_k - a\| < 1/k$ but $\|f(x_k) - f(a)\| \geq \epsilon$. The sequence $x_k \to a$, but $f(x_k)$ does *not* converge to $f(a)$ (it stays ϵ-far away). This contradicts sequential continuity.

27 Compactness and The Extreme Value Theorem

Continuity preserves the topological property of compactness.

Theorem Theorem: Continuous Image of Compact is Compact

Let $C \subseteq \mathbb{R}^n$ be a compact set and $f : C \to \mathbb{R}^m$ be a continuous function. Then the image set $f(C)$ is compact in $\mathbb{R}^m$.

Proof

Let (y_k) be a sequence in the image $f(C)$. We need to find a convergent subsequence in $f(C)$. For each y_k, choose $x_k \in C$ such that $f(x_k) = y_k$. Since C is compact, the sequence (x_k) has a subsequence (x_{k_i}) converging to some $x \in C$. Since f is continuous, $f(x_{k_i}) \to f(x)$. Let $y = f(x)$.

Since $x \in C$, $y \in f(C)$. Thus, $y_{k_i} \to y$, so $f(C)$ is compact.

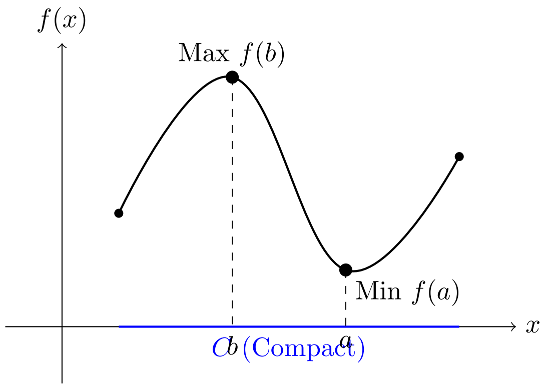
Theorem Extreme Value Theorem

Let C be a compact subset of $\mathbb{R}^n$ and $f : C \to \mathbb{R}$ be a continuous real-valued function. Then f attains its maximum and minimum values on C. That is, there exist $a, b \in C$ such that:

$$f(a) \leq f(x) \leq f(b) \quad \forall x \in C$$

Proof

1. Since C is compact, the image set $f(C) \subseteq \mathbb{R}$ is compact. 2. By Heine-Borel, $f(C)$ is **closed** and **bounded**. 3. Since $f(C)$ is bounded, its supremum $M = \sup f(C)$ and infimum $m = \inf f(C)$ exist (are finite). 4. Since $f(C)$ is closed, it contains its limit points, so $m \in f(C)$ and $M \in f(C)$. 5. Therefore, there exist points $a, b \in C$ such that $f(a) = m$ and $f(b) = M$.



Because the domain C is compact (closed & bounded),
the continuous function cannot "escape" to infinity
and must actually hit its highest and lowest points.

28 Equivalence of Norms (Lecture 4)

Reference: Olver & Shakiban, Page 150.

We investigate whether different norms on $\mathbb{R}^n$ lead to different concepts of convergence. The answer is no: all norms on finite-dimensional spaces are "structurally the same."

28.1 Continuity of Norms

Theorem Proposition: Norms are Continuous

Let $\|\cdot\|$ be *any* norm on $\mathbb{R}^n$. Then the function $f : \mathbb{R}^n \to \mathbb{R}$ given by $f(x) = \|x\|$ is continuous (with respect to the standard Euclidean metric).

Proof

Let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{R}^n$. Any x can be written as $x = \sum x_i e_i$. Using the triangle inequality and homogeneity:

$$\|x\| = \left\| \sum_{i=1}^{n} x_i e_i \right\| \leq \sum_{i=1}^{n} |x_i| \|e_i\|$$

Apply Cauchy-Schwarz to the sum on the right (viewing it as dot product of vector $(|x_i|)$ and

($\|e_i\|$)):

$$\sum_{i=1}^{n} |x_i|\|e_i\| \leq \left(\sum |x_i|^2\right)^{1/2} \left(\sum \|e_i\|^2\right)^{1/2} = \|x\|_2 \cdot C$$

where $C = \sqrt{\sum \|e_i\|^2}$ is a constant. Thus, $\|x\| \leq C\|x\|_2$. Using the reverse triangle inequality: $|\|x\| - \|y\|| \leq \|x - y\| \leq C\|x - y\|_2$. This implies f is Lipschitz, and therefore continuous.

28.2 The Equivalence Theorem

Definition : Equivalent Norms

Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ on a vector space V are called **equivalent** if there exist constants $m, M > 0$ such that for all $x \in V$:

$$m\|x\|_a \leq \|x\|_b \leq M\|x\|_a$$

Theorem Theorem: All Norms on $\mathbb{R}^n$ are Equivalent

Let $\|\cdot\|$ be any arbitrary norm on $\mathbb{R}^n$, and let $\|\cdot\|_2$ be the standard Euclidean norm. Then $\|\cdot\|$ and $\|\cdot\|_2$ are equivalent. Consequently, any two norms on $\mathbb{R}^n$ are equivalent.

Proof

1. The Unit Sphere is Compact: Consider the standard unit sphere $S_1 = \{x \in \mathbb{R}^n : \|x\|_2 = 1\}$. Since S_1 is closed and bounded in $\mathbb{R}^n$, it is compact.
2. Extreme Value Theorem: The function $f(x) = \|x\|$ (the arbitrary norm) is continuous. By the Extreme Value Theorem, f attains a maximum M and a minimum m on the compact set S_1.

$$m = \min_{x \in S_1} \|x\|, \quad M = \max_{x \in S_1} \|x\|$$

3. Positivity: Since $x \in S_1$ implies $x \neq 0$, and f is a norm, we must have $f(x) > 0$. Thus the minimum m is strictly positive: $m > 0$.
4. Scaling: For any non-zero vector $y \in \mathbb{R}^n$, let $x = \frac{y}{\|y\|_2}$. Then $x \in S_1$. The inequality $m \leq \|x\| \leq M$ becomes:

$$m \leq \left\| \frac{y}{\|y\|_2} \right\| \leq M \implies m \leq \frac{\|y\|}{\|y\|_2} \leq M$$

Multiplying by $\|y\|_2$ gives:

$$m\|y\|_2 \leq \|y\| \leq M\|y\|_2$$

28.3 Example: Euclidean vs. Max Norm

Let's find the specific constants m and M relating $\|\cdot\|_2$ and $\|\cdot\|_\infty$.

Example Comparing L^2 and L^∞

We want to satisfy $m\|x\|_2 \leq \|x\|_\infty \leq M\|x\|_2$.
1. Finding bounds: We know that for any x, $\|x\|_\infty = \max |x_i|$. Also, $\|x\|_2 = \sqrt{x_1^2 + \cdots + x_n^2}$. Clearly, $\max |x_i|^2 \leq \sum x_i^2 \leq n(\max |x_i|)^2$. Taking square roots:

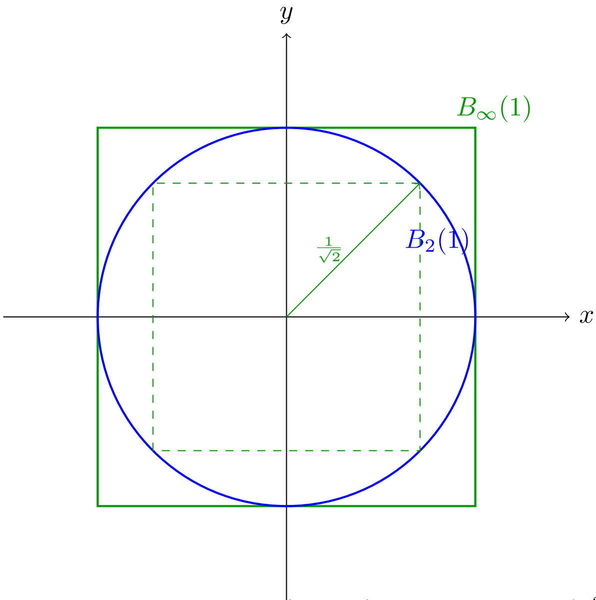
$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n}\|x\|_\infty$$

2. Rearranging for Equivalence Form: From $\|x\|_\infty \leq \|x\|_2$, we get the upper bound constant (relative to infinity norm) is 1? No, we need $\|x\|_\infty \leq \dots$ From $\|x\|_2 \leq \sqrt{n}\|x\|_\infty$, we divide by

$\sqrt{n}$ to get: $\frac{1}{\sqrt{n}}\|x\|_2 \le \|x\|_\infty$. From $\|x\|_\infty \le \|x\|_2$, we have the other side. Combined:

$$\frac{1}{\sqrt{n}}\|x\|_2 \le \|x\|_\infty \le 1 \cdot \|x\|_2$$

Thus, $m = \frac{1}{\sqrt{n}}$ and $M = 1$.



Geometric Interpretation ($n = 2$): The unit circle (L^2) is "sandwiched" between the unit square (L^∞) and a scaled down square of radius $1/\sqrt{2}$. This proves that convergence to the origin in one shape implies convergence in the others.

29 Consequences of Norm Equivalence

Since all norms on $\mathbb{R}^n$ are equivalent, the fundamental concepts of analysis (convergence and topology) do not depend on the choice of norm.

29.1 Independence of Convergence

Theorem Theorem: Convergence is Independent of Norm

Let $(a_k)_{k\in\mathbb{N}}$ be a sequence in $\mathbb{R}^n$ and $a \in \mathbb{R}^n$. The sequence converges to a with respect to the Euclidean norm $\|\cdot\|_2$ if and only if it converges to a with respect to **any** arbitrary norm $\|\cdot\|$ on $\mathbb{R}^n$.

Proof

Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$. By the Equivalence Theorem, there exist constants $m, M > 0$ such that:

$$m\|x\|_2 \le \|x\| \le M\|x\|_2 \quad \forall x \in \mathbb{R}^n$$

($\Rightarrow$) Suppose $\lim_{k\to\infty} \|a_k - a\|_2 = 0$. Using the right-hand inequality:

$$\|a_k - a\| \le M\|a_k - a\|_2$$

Since $\|a_k - a\|_2 \to 0$, it follows that $\|a_k - a\| \to 0$.

(⇐) Suppose $\lim_{k\to\infty} \|a_k - a\| = 0$. Using the left-hand inequality:

$$m\|a_k - a\|_2 \leq \|a_k - a\| \implies \|a_k - a\|_2 \leq \frac{1}{m}\|a_k - a\|$$

Since $\|a_k - a\| \to 0$, it follows that $\|a_k - a\|_2 \to 0$.

29.2 Independence of Topology

Since the definitions of Open and Closed sets rely on convergence and distances, these concepts are also norm-independent.

Theorem Theorem: Topology is Independent of Norm

Let $S \subseteq \mathbb{R}^n$.

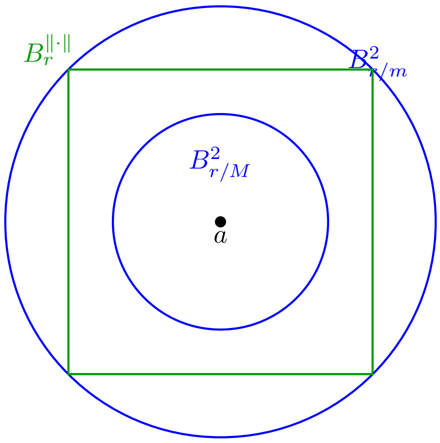
1. S is a **Closed Set** with respect to $\|\cdot\|_2 \iff S$ is closed with respect to any norm $\|\cdot\|$.
2. S is an **Open Set** with respect to $\|\cdot\|_2 \iff S$ is open with respect to any norm $\|\cdot\|$.

Proof Sketch for Open Sets

Recall that a set U is open if for every $a \in U$, there exists a ball $B_r(a)$ contained in U. Let $B_r^{\|\cdot\|}(a)$ denote the ball of radius r in the arbitrary norm, and $B_r^2(a)$ be the Euclidean ball. From the equivalence $m\|x\|_2 \leq \|x\| \leq M\|x\|_2$, we have the nesting of balls:

$$B^2_{r/M}(a) \subseteq B_r^{\|\cdot\|}(a) \subseteq B^2_{r/m}(a)$$

This "nesting" ensures that if you can fit a ball of one type inside U, you can fit a ball of the other type inside as well (possibly with a smaller radius).



Topology Independence: Because we can always fit a Euclidean ball inside any other norm's ball (and vice versa), a set that is "open" in one metric is automatically "open" in the other.

30 Infinite Dimensional Spaces (Counter-Example)

Warning: The statement "all norms are equivalent" is **FALSE** for infinite-dimensional vector spaces. We demonstrate this with a counter-example.

Example Counter-Example in $C^0[0,1]$

Let $V = C^0[0,1]$ be the vector space of continuous real-valued functions on $[0,1]$. Consider two norms on V:

1. L^2 **Norm:** $\|f\|_2 = \sqrt{\int_0^1 (f(x))^2 \, dx}$
2. L^∞ **Norm:** $\|f\|_\infty = \sup\{|f(x)| : x \in [0,1]\}$

Consider the sequence of "spike" functions $f_n : [0,1] \to \mathbb{R}$ defined by:

$$f_n(x) = \begin{cases} 1 - nx & \text{if } 0 \leq x \leq \frac{1}{n} \\ 0 & \text{if } \frac{1}{n} \leq x \leq 1 \end{cases}$$

1. Calculation in L^∞ Norm: The maximum height of the triangle is always 1 (at $x = 0$).

$$\|f_n\|_\infty = 1 \quad \forall n \in \mathbb{N}$$

Thus, f_n does **not** converge to the zero function in the L^∞ norm.
2. Calculation in L^2 Norm: We integrate the square of the function:

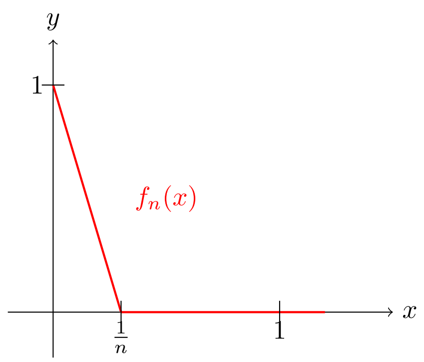
$$\|f_n\|_2 = \sqrt{\int_0^{1/n} (1 - nx)^2 \, dx}$$

Let $u = 1 - nx$, then $du = -ndx$. Limits: $1 \to 0$.

$$\int (1 - nx)^2 dx = \int_1^0 u^2 \frac{du}{-n} = \frac{1}{n} \int_0^1 u^2 du = \frac{1}{n} \left[\frac{u^3}{3} \right]_0^1 = \frac{1}{3n}$$

$$\implies \|f_n\|_2 = \sqrt{\frac{1}{3n}} = \frac{1}{\sqrt{3n}}$$

As $n \to \infty$, $\|f_n\|_2 \to 0$.
Conclusion: The sequence converges to 0 in $\|\cdot\|_2$ but does not converge to 0 in $\|\cdot\|_\infty$. Therefore, these two norms are **not equivalent**.



The "Spike" Function: As n increases, the base $\frac{1}{n}$ shrinks. The area (and thus L^2 norm) goes to 0. However, the height stays fixed at 1, so the L^∞ norm never decreases.

30.1 Proof of Non-Equivalence

If the norms were equivalent, there would exist a constant M such that $\|f\|_\infty \leq M\|f\|_2$ for all f. Applying this to our sequence:

$$1 \leq M\left(\frac{1}{\sqrt{3n}}\right) \implies \sqrt{3n} \leq M$$

This must hold for all $n \in \mathbb{N}$. But as $n \to \infty$, $\sqrt{3n} \to \infty$, which contradicts the existence of a finite constant M.

31 Transitivity of Norm Equivalence

Equivalence of norms is indeed an "equivalence relation" in the mathematical sense.

Theorem Proposition

The relation "$\|\cdot\|_a$ is equivalent to $\|\cdot\|_b$" is:

1. **Reflexive:** $\|\cdot\| \sim \|\cdot\|$ (Trivial with $m = M = 1$).
2. **Symmetric:** $\|\cdot\|_a \sim \|\cdot\|_b \implies \|\cdot\|_b \sim \|\cdot\|_a$.
3. **Transitive:** $\|\cdot\|_1 \sim \|\cdot\|_2$ and $\|\cdot\|_2 \sim \|\cdot\|_3 \implies \|\cdot\|_1 \sim \|\cdot\|_3$.

Proof Proof of Transitivity

Given: 1. $m_1\|x\|_1 \leq \|x\|_2 \leq M_1\|x\|_1$ 2. $m_2\|x\|_2 \leq \|x\|_3 \leq M_2\|x\|_2$
Substitute (1) into (2):

$$\|x\|_3 \leq M_2\|x\|_2 \leq M_2(M_1\|x\|_1) = (M_1M_2)\|x\|_1$$

$$\|x\|_3 \geq m_2\|x\|_2 \geq m_2(m_1\|x\|_1) = (m_1m_2)\|x\|_1$$

Thus, $\|\cdot\|_1 \sim \|\cdot\|_3$ with constants $m = m_1m_2$ and $M = M_1M_2$.

Exercises (Page 152): Practice problems: 3.3.31, 3.3.32, 3.3.33, 3.3.34, 3.3.35, 3.3.36, 3.3.37, 3.3.38.

32 Matrix Norms

Reference: Olver & Shakiban, Page 153.

Just as we assign a "size" to vectors using norms, we can assign a "size" to linear transformations (matrices).

32.1 The Operator Norm

Definition : Associated Matrix Norm

Let $\|\cdot\|$ be a norm on $\mathbb{R}^n$. For any $n \times n$ matrix $A \in M_n(\mathbb{R})$, we define its **operator norm** (or associated matrix norm) as:

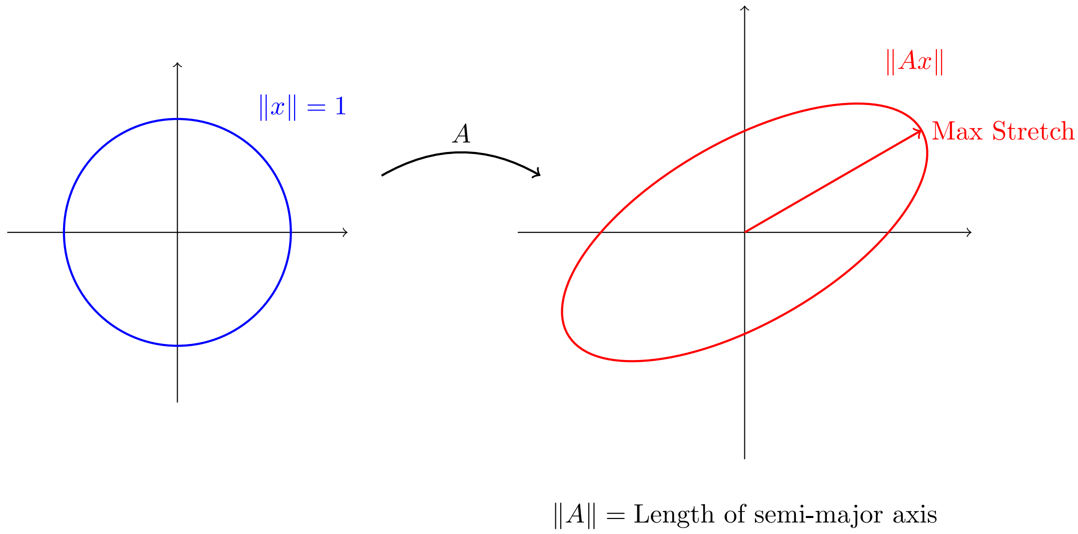
$$\|A\| = \max\{\|Ax\| : \|x\| = 1\}$$

Geometrically, this measures the maximum "stretch" the matrix A applies to the unit sphere.

Proof Proof that $\|A\|$ is well-defined

Consider the function $f : \mathbb{R}^n \to \mathbb{R}$ defined by $f(x) = \|Ax\|$. Since matrix multiplication is continuous and vector norms are continuous, their composition f is continuous. The set $S = \{x \in \mathbb{R}^n : \|x\| = 1\}$ (the unit sphere) is closed and bounded, hence **compact**. By the Extreme

Value Theorem, the continuous function f attains a maximum on the compact set S. Thus, $\|A\|$ exists and is a finite non-negative number.



32.2 Properties of Matrix Norms

We verify that the operator norm satisfies the three vector norm axioms, plus a special fourth property.

Theorem Theorem: Norm Properties

For any $A, B \in M_n(\mathbb{R})$ and $c \in \mathbb{R}$:

1. **Positivity:** $\|A\| \geq 0$, and $\|A\| = 0 \iff A = 0$.
2. **Homogeneity:** $\|cA\| = |c|\|A\|$.
3. **Triangle Inequality:** $\|A + B\| \leq \|A\| + \|B\|$.
4. **Sub-multiplicativity:** $\|AB\| \leq \|A\|\|B\|$.

Also, for any vector v, the following compatibility holds:

$$\|Av\| \leq \|A\|\|v\|$$

Proof Proof of Sub-multiplicativity

Let u be a unit vector ($\|u\| = 1$). We know $\|ABu\| = \|A(Bu)\|$. Using the compatibility property ($\|Ay\| \leq \|A\|\|y\|$):

$$\|A(Bu)\| \leq \|A\|\|Bu\|$$

Applying it again to B:

$$\|Bu\| \leq \|B\|\|u\| = \|B\|$$

Combining these:

$$\|ABu\| \leq \|A\|\|B\|$$

Taking the maximum over all unit vectors u:

$$\|AB\| = \max_{\|u\|=1} \|ABu\| \leq \|A\|\|B\|$$

32.3 Calculating Matrix Norms

The value of $\|A\|$ depends on which vector norm we use on $\mathbb{R}^n$.

Example Matrix Norm Induced by the ∞-Norm

If we use the vector norm $\|x\|_\infty = \max_i |x_i|$, the associated matrix norm is the **Maximum Absolute Row Sum**:

$$\|A\|_\infty = \max_{1\le i\le n} \sum_{j=1}^{n} |a_{ij}|$$

Proof Proof Sketch

Let $R_i = \sum_{j=1}^{n} |a_{ij}|$ be the absolute sum of row i, and let $S = \max_i R_i$.
Step 1: $\|A\|_\infty \le S$ (Upper Bound). Let v be a unit vector, so $\|v\|_\infty = 1 \implies |v_j| \le 1$. The i-th entry of Av is $\sum_{j=1}^{n} a_{ij}v_j$.

$$|(Av)_i| = \left|\sum_{j=1}^{n} a_{ij}v_j\right| \le \sum_{j=1}^{n} |a_{ij}||v_j| \le \sum_{j=1}^{n} |a_{ij}| = R_i \le S$$

Since every component is bounded by S, the max component $\|Av\|_\infty \le S$. Thus $\|A\|_\infty \le S$.
Step 2: $\|A\|_\infty \ge S$ (Attainability). Let k be the index of the row with the maximum sum ($S = R_k$). Define a specific vector u such that $u_j = \text{sign}(a_{kj})$. Then $|u_j| = 1$, so $\|u\|_\infty = 1$. The k-th entry of Au is:

$$(Au)_k = \sum_{j=1}^{n} a_{kj}u_j = \sum_{j=1}^{n} a_{kj}\text{sign}(a_{kj}) = \sum_{j=1}^{n} |a_{kj}| = S$$

Thus $\|Au\|_\infty \ge S$. Combining both steps, $\|A\|_\infty = S$.

33 Generalized Inner Products

Example Example Computation

Let $A = \begin{pmatrix} 1/2 & -1/3 \\ -1/3 & 1/4 \end{pmatrix}$. Then $\|A\|_\infty = \max\{\frac{1}{2} + \frac{1}{3}, \frac{1}{3} + \frac{1}{4}\} = \frac{5}{6}$.

We generalize the standard Euclidean dot product on $\mathbb{R}^n$. Let $K \in M_n(\mathbb{R})$. We define a pairing $\langle x, y\rangle_K$ by:

$$\langle x, y\rangle = x^t K y$$

where $x, y \in \mathbb{R}^n$ (column vectors). In coordinates:

$$\langle x, y\rangle = \left\langle \sum x_i e_i, \sum y_j e_j \right\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i y_j \langle e_i, e_j\rangle$$

If we define $k_{ij} = \langle e_i, e_j\rangle$, then:

$$\langle x, y\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i y_j k_{ij} = x^t K y$$

Requirement 1 (Symmetry): For this to be a valid inner product, we need $\langle x, y\rangle = \langle y, x\rangle$.

$$x^t K y = (x^t K y)^t = y^t K^t x$$

This implies we must have $K = K^t$. Thus, K must be a **symmetric matrix**.
Requirement 2 (Positivity): We require $\langle x, x\rangle > 0$ for all $x \neq 0$.

$$\langle x, x\rangle = x^t K x = \sum_{i,j} k_{ij} x_i x_j > 0 \quad \forall x \in \mathbb{R}^n \setminus \{0\}$$

Definition : Positive Definite Matrix

A symmetric matrix $K \in M_n(\mathbb{R})$ is called **Positive Definite** if:

$$x^t K x > 0 \quad \text{for all } x \neq 0 \text{ in } \mathbb{R}^n.$$

Notation: We write $K > 0$.

34 Quadratic Forms

Definition : Quadratic Form

Given a symmetric matrix $K \in M_n(\mathbb{R})$, the homogeneous quadratic polynomial defined by:

$$q(x) = x^t K x = \sum_{i=1}^{n} \sum_{j=1}^{n} k_{ij} x_i x_j$$

is known as a **Quadratic Form** on $\mathbb{R}^n$.

The quadratic form is called **positive definite** if $q(x) > 0$ for all $x \neq 0$. This is equivalent to saying the matrix K is positive definite.

Example Testing 2×2 Matrices

Let $K = \begin{pmatrix} 4 & -2 \\ -2 & 3 \end{pmatrix}$. The quadratic form is:

$$q(x_1, x_2) = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 4 & -2 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 4x_1^2 - 4x_1x_2 + 3x_2^2$$

By completing the square:

$$q(x) = (2x_1 - x_2)^2 - x_2^2 + 3x_2^2 = (2x_1 - x_2)^2 + 2x_2^2$$

Since this is a sum of squares, $q(x) \geq 0$. $q(x) = 0 \iff 2x_1 - x_2 = 0$ and $x_2 = 0 \iff x = (0, 0)$. Thus, K is Positive Definite.

Example Positive Entries do not guarantee Positive Definiteness

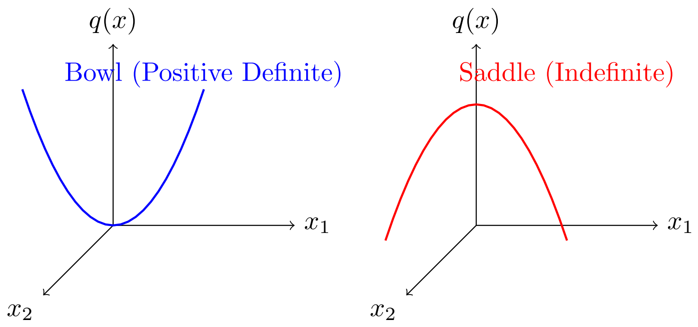
Let $K = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

$$q(x) = x_1^2 + 4x_1x_2 + x_2^2$$

This matrix has all positive entries, but consider the vector $x = (1, -1)$:

$$q(1, -1) = 1^2 - 4(1)(1) + 1^2 = -2 < 0$$

Thus, K is **not** positive definite.



35 Testing Definiteness

Theorem 2×2 Condition

Let $K = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. K is positive definite if and only if:

$$a > 0 \quad \text{and} \quad \det(K) = ac - b^2 > 0$$

Example 3×3 Completing the Square

Let $K = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 6 & 0 \\ -1 & 0 & 9 \end{pmatrix}$. The quadratic form is:

$$\begin{aligned} q(x) &= x^2 + 4xy + 6y^2 - 2xz + 9z^2 \\ &= (x^2 + 4xy - 2xz - 4yz + 4y^2 + z^2) + 4yz + 8z^2 + 2y^2 \\ &= (x + 2y - z)^2 + 2(y^2 + 2yz + z^2) + 6z^2 \\ &= (x + 2y - z)^2 + 2(y + z)^2 + 6z^2 \end{aligned}$$

Since this is a sum of squares with positive coefficients:

- $q(x) \geq 0$ for all x.
- $q(x) = 0 \iff y + z = 0, z = 0 \implies y = 0, z = 0$. Then $x + 0 - 0 = 0 \implies x = 0$.

Thus $q(x) > 0$ for $x \neq 0$, so K is **Positive Definite**.

Definition : Other Forms of Definiteness

A symmetric matrix K is called:

- **Positive Semidefinite (PSD):** If $q(x) = x^t K x \geq 0$ for all x. (Written $K \geq 0$).
- **Negative Definite:** If $q(x) < 0$ for all $x \neq 0$.
- **Negative Semidefinite:** If $q(x) \leq 0$ for all x.
- **Indefinite:** If $q(x)$ takes both positive and negative values.

A positive definite matrix is always **non-singular** (invertible). If K is singular, there exists $x \neq 0$ such that $Kx = 0$, which implies $x^t K x = 0$, so it cannot be strictly positive definite.

36 Gram Matrices

We define a special class of matrices derived from inner products.

Definition : Gram Matrix

Let V be a real inner product space. Let $v_1, \ldots, v_n \in V$. The $n \times n$ matrix defined by:

$$K_{ij} = \langle v_i, v_j \rangle$$

is called the **Associated Gram Matrix** of the vectors $v_1, \ldots, v_n$.

In the specific case where $V = \mathbb{R}^m$ and the inner product is the dot product: Let A be the $m \times n$

matrix with columns $v_1, \dots, v_n$:

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

Then the Gram matrix is given by:

$$K = A^t A$$

Proof: The (i, j)-entry of $A^t A$ is (row i of A^t) $\cdot$ (col j of A) $= v_i^t v_j = \langle v_i, v_j \rangle$.

Theorem Properties of Gram Matrices

Let $K = A^t A$ be a Gram matrix.

1. K is always **Positive Semidefinite**.

$$q(x) = x^t K x = x^t A^t A x = (Ax)^t (Ax) = \|Ax\|^2 \geq 0$$

2. K is **Positive Definite** if and only if the vectors $v_1, \dots, v_n$ are **Linearly Independent**.

Proof Proof of (2)

K is positive definite $\iff$ $x^t K x = 0$ implies $x = 0$. From above, $x^t K x = \|Ax\|^2$. So K is PD $\iff$ $\|Ax\|^2 = 0 \implies x = 0$. $\iff$ $Ax = 0 \implies x = 0$. $\iff$ The columns of A (which are $v_1, \dots, v_n$) are linearly independent.

Example Example in $\mathbb{R}^3$

Let $v_1 = (1, 2, -1)^t$ and $v_2 = (3, 0, 6)^t$. Inner products:

- $v_1 \cdot v_1 = 1 + 4 + 1 = 6$
- $v_1 \cdot v_2 = 3 + 0 - 6 = -3$
- $v_2 \cdot v_2 = 9 + 0 + 36 = 45$

Gram Matrix $K = \begin{pmatrix} 6 & -3 \\ -3 & 45 \end{pmatrix}$. Since v_1, v_2 are linearly independent, K is positive definite.

Theorem Theorem: The Big Equivalence

Given a real matrix A of size $m \times n$: The following are equivalent (TFAE):

1. $K = A^t A$ is Positive Definite.
2. The columns of A are linearly independent.
3. $\text{Rank}(A) = n$ (which implies $n \leq m$).
4. $\ker(A) = \{0\}$.

37 Matrix Norms

Reference: Olver & Shakiban, Page 153.

We now extend the concept of norms to matrices. We want to measure the "size" or the maximum stretching power of a linear transformation $A \in M_n(\mathbb{R})$.

37.1 The Associated Matrix Norm (Operator Norm)

Theorem Theorem: Associated Matrix Norm

Let $\|\cdot\|$ be any vector norm on $\mathbb{R}^n$. For any matrix $A \in M_n(\mathbb{R})$, we define:

$$\|A\| = \max\{\|Au\| \mid \|u\| = 1\}$$

This definition gives a valid norm on the vector space of matrices $M_n(\mathbb{R}) = \mathbb{R}^{n^2}$, known as the **associated matrix norm** (or operator norm).

Proof Proof of Existence/Well-definedness

Consider the function from $\mathbb{R}^n \to \mathbb{R}$ given by the composition:

$$u \mapsto Au \mapsto \|Au\|$$

Because matrix multiplication is a continuous operation and the vector norm is a continuous function, their composite function is continuous (this can be proven using sequential or $\epsilon - \delta$ definitions of continuity).
Let $S = \{u \in \mathbb{R}^n \mid \|u\| = 1\}$ be the unit sphere. S is a closed and bounded set in $\mathbb{R}^n$, which means it is a **compact set**. By the Extreme Value Theorem, a continuous map $S \to \mathbb{R}$ attains its maximum value. Hence, $\|A\| = \max_{\|u\|=1} \|Au\|$ is attained and is a finite real number ≥ 0.

37.2 Proving the Norm Axioms

We verify that $\|A\|$ satisfies the fundamental properties of a norm.

Proof Proof of Axiom 1: Positivity

We already noted $\|A\| \geq 0$. We must show $\|A\| = 0 \iff A = 0$.

- ($\Leftarrow$) Clearly, if $A = 0$, then $Au = 0$ for all u, so $\|A\| = 0$.
- ($\Rightarrow$) Suppose $\|A\| = 0$. This implies $\max_{\|u\|=1} \|Au\| = 0$, so $\|Au\| = 0$ for every unit vector u. Thus $Au = 0$ for all u with $\|u\| = 1$. Let $v \in \mathbb{R}^n$ be any non-zero vector. Let $u = \frac{v}{\|v\|}$. Then u has norm 1.

$$A\left(\frac{v}{\|v\|}\right) = 0 \implies \frac{1}{\|v\|}Av = 0 \implies Av = 0$$

Also, $A(0) = 0$. Thus $Av = 0$ for all $v \in \mathbb{R}^n$, which implies the matrix A itself is the zero matrix ($A = 0$).

Proof Proof of Axiom 2: Homogeneity

Let $c \in \mathbb{R}$ and $A \in M_n(\mathbb{R})$.

$$\begin{aligned}
\|cA\| &= \max\{\|(cA)u\| \mid \|u\| = 1\} \\
&= \max\{|c|\|Au\| \mid \|u\| = 1\} \\
&= |c| \max\{\|Au\| \mid \|u\| = 1\} \\
&= |c|\|A\|
\end{aligned}$$

Proof Proof of Axiom 3: Triangle Inequality

Let $A, B \in M_n(\mathbb{R})$.

$$\begin{aligned} \|A+B\| &= \max\{\|(A+B)u\| \mid \|u\|=1\} \\ &= \max\{\|Au+Bu\| \mid \|u\|=1\} \\ &\leq \max\{\|Au\|+\|Bu\| \mid \|u\|=1\} \quad \text{(by vector triangle inequality)} \\ &\leq \max\{\|Au\| \mid \|u\|=1\} + \max\{\|Bu\| \mid \|u\|=1\} \\ &= \|A\|+\|B\| \end{aligned}$$

38 Sub-multiplicativity and Compatibility

Matrix norms possess additional properties related to multiplication that standard vector spaces do not have.

Theorem Theorem: Compatibility and Sub-multiplicativity

An associated matrix norm satisfies the following inequalities:

1. **Compatibility:** $\|Av\| \leq \|A\|\|v\|$ for all $A \in M_n(\mathbb{R})$ and for all $v \in \mathbb{R}^n$.
2. **Sub-multiplicativity:** $\|AB\| \leq \|A\|\|B\|$ for all $A, B \in M_n(\mathbb{R})$.

Proof Proof of Compatibility

Let $v \in \mathbb{R}^n$. If $v = 0$, then $\|A(0)\| = 0$ and $\|A\|\|0\| = 0$, so the inequality $0 \leq 0$ holds trivially. Assume $v \neq 0$. Let $u = \frac{v}{\|v\|}$, which is a unit vector ($\|u\| = 1$). By definition of the matrix norm as a maximum over all unit vectors:

$$\|Au\| \leq \|A\|$$

Substituting u:

$$\left\| A\left(\frac{v}{\|v\|}\right) \right\| \leq \|A\| \implies \frac{1}{\|v\|}\|Av\| \leq \|A\| \implies \|Av\| \leq \|A\|\|v\|$$

Proof Proof of Sub-multiplicativity

By definition:

$$\|AB\| = \max\{\|ABu\| \mid \|u\| = 1\}$$

Apply the compatibility property to the matrix A and vector (Bu):

$$\|A(Bu)\| \leq \|A\|\|Bu\|$$

Now take the maximum over all unit vectors:

$$\begin{aligned} \|AB\| &\leq \max\{\|A\|\|Bu\| \mid \|u\|=1\} \\ &= \|A\| \max\{\|Bu\| \mid \|u\|=1\} \\ &= \|A\|\|B\| \end{aligned}$$

Definition : Matrix Norm

A norm on the vector space of matrices $M_n(\mathbb{R}) = \mathbb{R}^{n^2}$ is strictly called a **matrix norm** if, in addition to the 3 standard properties of a norm, it also satisfies the sub-multiplicative property:

$$\|AB\| \leq \|A\|\|B\| \quad \forall A, B \in M_n(\mathbb{R})$$

Corollary

By applying the sub-multiplicative property iteratively to a matrix multiplied by itself, we get:

$$A \in M_n(\mathbb{R}) \implies \|A^k\| \leq \|A\|^k \quad \text{for any integer } k > 0.$$

39 Computing the Infinity Matrix Norm

How do we actually calculate $\|A\|$? The result depends entirely on the vector norm used to define it. We examine the matrix norm associated with the L^∞ vector norm.

Theorem Theorem: The Infinity Matrix Norm

Consider the ∞-norm on $\mathbb{R}^n$, defined as $\|v\|_\infty = \max\{|v_1|, \dots, |v_n|\}$. The associated matrix norm on $M_n(\mathbb{R})$ is given by the **Maximum Absolute Row Sum**:

$$\|A\|_\infty = \max_{1 \leq i \leq n} \left\{ \sum_{j=1}^{n} |a_{ij}| \right\}$$

Proof Proof

Let $s_i = \sum_{j=1}^{n} |a_{ij}|$ be the i-th absolute row sum of matrix A. Let $S = \max\{s_1, \dots, s_n\}$. We claim that $\|A\|_\infty = S$.

Part 1: Upper Bound ($\|A\|_\infty \leq S$)

Let $v \in \mathbb{R}^n$ be any unit vector, so $\|v\|_\infty = 1$. This implies $|v_j| \leq 1$ for all j. The i-th component of the vector Av is given by $\sum_{j=1}^{n} a_{ij} v_j$. Taking the absolute value:

$$|(Av)_i| = \left| \sum_{j=1}^{n} a_{ij} v_j \right| \leq \sum_{j=1}^{n} |a_{ij} v_j| = \sum_{j=1}^{n} |a_{ij}||v_j|$$

Since $|v_j| \leq 1$, we have:

$$\sum_{j=1}^{n} |a_{ij}||v_j| \leq \sum_{j=1}^{n} |a_{ij}| = s_i$$

Thus, the magnitude of every component i is bounded by s_i, which in turn is bounded by the maximum S.

$$\|Av\|_\infty = \max_{1 \leq i \leq n} |(Av)_i| \leq \max_{1 \leq i \leq n} s_i = S$$

Since $\|Av\|_\infty \leq S$ for all unit vectors v, taking the maximum over all v gives $\|A\|_\infty \leq S$.

Part 2: Attainability ($\|A\|_\infty \geq S$)

We must show that the bound S can actually be reached. Suppose the maximum row sum occurs at row k, so $S = s_k = \sum_{j=1}^{n} |a_{kj}|$. Construct a specific test vector $u = (u_1, \dots, u_n) \in \mathbb{R}^n$ by choosing:

$$u_j = \begin{cases} 1 & \text{if } a_{kj} \geq 0 \\ -1 & \text{if } a_{kj} < 0 \end{cases}$$

By this construction, every component of u has absolute value 1, so $\|u\|_\infty = 1$. Crucially, $a_{kj} u_j = |a_{kj}|$. Now calculate the k-th entry of the output vector Au:

$$(Au)_k = \sum_{j=1}^{n} a_{kj} u_j = \sum_{j=1}^{n} |a_{kj}| = s_k = S$$

Because at least one component of Au equals S, the maximum component norm $\|Au\|_\infty \geq S$. Therefore, by definition of the operator norm, $\|A\|_\infty \geq S$.

Conclusion: Since $\|A\|_\infty \leq S$ and $\|A\|_\infty \geq S$, we conclude $\|A\|_\infty = S$. Claim proved.

$$A = \begin{pmatrix} -1 & 2 & -3 \\ 4 & 0 & 1 \\ -2 & -2 & -2 \end{pmatrix} \begin{array}{l} \longrightarrow |-1| + |2| + |-3| = \mathbf{6} \quad \text{(Maximum)} \\ \longrightarrow |4| + |0| + |1| = 5 \\ \longrightarrow |-2| + |-2| + |-2| = 6 \end{array}$$

Example: To find $\|A\|_\infty$, calculate the sum of the absolute values across each row. The largest sum is the norm. Here, $\|A\|_\infty = 6$.

40 Generalized Inner Products

Example Example Computation

Let $A = \begin{pmatrix} 1/2 & -1/3 \\ -1/3 & 1/4 \end{pmatrix}$. Then $\|A\|_\infty = \max\left\{\left|\frac{1}{2}\right| + \left|-\frac{1}{3}\right|, \left|-\frac{1}{3}\right| + \left|\frac{1}{4}\right|\right\} = \max\left\{\frac{5}{6}, \frac{7}{12}\right\} = \frac{5}{6}$.

We generalize the standard Euclidean dot product on $\mathbb{R}^n$. Let $K \in M_n(\mathbb{R})$. We define a pairing $\langle x, y\rangle_K$ by:

$$\langle x, y\rangle = x^T K y$$

where $x, y \in \mathbb{R}^n$ (column vectors). In coordinates, let $\{e_1, \ldots, e_n\}$ be the standard basis of $\mathbb{R}^n$. Let $x = \sum x_i e_i$ and $y = \sum y_i e_i$.

$$\langle x, y\rangle = \left\langle \sum x_i e_i, \sum y_j e_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i y_j \langle e_i, e_j\rangle$$

If we define $k_{ij} = \langle e_i, e_j\rangle$, then:

$$K = (k_{ij}) \in M_n(\mathbb{R}) \implies \langle x, y\rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i y_j k_{ij} = x^T K y$$

Requirement 1 (Symmetry): For this to be a valid inner product, we need $\langle x, y\rangle = \langle y, x\rangle$.

$$x^T K y = (x^T K y)^T = y^T K^T x$$

This implies we must have $K = K^T$. Thus, K must be a **symmetric matrix**.

Requirement 2 (Positivity): We require $\langle x, x\rangle > 0$ for all $x \neq 0$.

$$\langle x, x\rangle = x^T K x = \sum_{i=1}^{n} \sum_{j=1}^{n} k_{ij} x_i x_j > 0 \quad \forall x \in \mathbb{R}^n \setminus \{0\}$$

Definition : Positive Definite Matrix

A symmetric matrix $K \in M_n(\mathbb{R})$ is called **Positive Definite** if:

$$x^T K x > 0 \quad \text{for all } x \neq 0 \text{ in } \mathbb{R}^n.$$

Notation: If K is a true positive definite matrix, we write $K > 0$.

41 Quadratic Forms

Definition : Quadratic Form

Given a symmetric matrix $K \in M_n(\mathbb{R})$, the homogeneous quadratic polynomial defined by:

$$q(x) = x^T K x = \sum_{i=1}^{n} \sum_{j=1}^{n} k_{ij} x_i x_j$$

is known as a **Quadratic Form** on $\mathbb{R}^n$.

The quadratic form is called **positive definite** if $q(x) > 0$ for all $x \neq 0$. This is logically equivalent to saying the matrix K is positive definite.

Example Testing 2×2 Matrices via Completing the Square

Let $K = \begin{pmatrix} 4 & -2 \\ -2 & 3 \end{pmatrix}$. The associated quadratic form is:

$$q(x_1, x_2) = x^T K x = 4x_1^2 - 4x_1 x_2 + 3x_2^2$$

We can complete the square:

$$q(x) = (2x_1 - x_2)^2 - x_2^2 + 3x_2^2 = (2x_1 - x_2)^2 + 2x_2^2 \geq 0$$

Since this is a sum of squares, $q(x) \geq 0$. Furthermore, $q(x) = 0 \iff (2x_1 - x_2) = 0$ and $x_2 = 0 \iff (x_1, x_2) = (0, 0)$. Thus, K is **Positive Definite**.

Example Positive Entries do not guarantee Positive Definiteness

Let $K = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$.

$$q(x) = x_1^2 + 4x_1 x_2 + x_2^2$$

This matrix has all positive entries. However, consider the vector $x = (1, -1)^T$:

$$q(1, -1) = 1^2 - 4(1)(1) + 1^2 = -2 < 0$$

Thus, K is **not** positive definite.

42 Testing Definiteness

Theorem 2×2 Condition

Let $K = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ be a symmetric real 2×2 matrix. K is positive definite if and only if:

$$q(x) = ax_1^2 + 2bx_1 x_2 + cx_2^2 > 0 \quad \forall (x_1, x_2) \neq (0, 0)$$

$$\iff \quad a > 0 \quad \text{and} \quad \det(K) = ac - b^2 > 0$$

Example 3×3 Completing the Square

Consider the 3×3 symmetric real matrix $K = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 6 & 0 \\ -1 & 0 & 9 \end{pmatrix}$. The quadratic form is:

$$\begin{aligned} q(x) &= x^2 + 4xy + 6y^2 - 2xz + 9z^2 \\ &= (x^2 + 4xy - 2xz) + 6y^2 + 9z^2 \\ &= (x^2 + 4xy - 2xz + 4y^2 + z^2 - 4yz) - 4y^2 - z^2 + 4yz + 6y^2 + 9z^2 \\ &= (x + 2y - z)^2 + 2y^2 + 4yz + 8z^2 \\ &= (x + 2y - z)^2 + 2(y^2 + 2yz + z^2) - 2z^2 + 8z^2 \\ &= (x + 2y - z)^2 + 2(y + z)^2 + 6z^2 \end{aligned}$$

Since this is a sum of squares with positive coefficients, $q(x, y, z) \geq 0$ for all $(x, y, z) \in \mathbb{R}^3$. Also, $q(x, y, z) = 0 \iff x + 2y - z = 0$, $y + z = 0$, and $z = 0 \implies (x, y, z) = (0, 0, 0)$. Thus K is a true **Positive Definite** matrix.

Definition : Other Forms of Definiteness

A quadratic form and its associated symmetric matrix K are called:

- **Positive Semidefinite (PSD):** If $q(x) = x^T K x \geq 0$ for all $x \in \mathbb{R}^n$.
- **Negative Definite:** If $q(x) = x^T K x < 0$ for all $x \neq 0$.
- **Negative Semidefinite:** If $q(x) = x^T K x \leq 0$ for all $x \in \mathbb{R}^n$.

43 Singularity and Gram Matrices

Theorem Corollary: Positive Definite Matrices are Non-Singular

A positive definite matrix is always non-singular (invertible).

Proof

If a matrix K is singular, then there exists a non-zero vector $x \in \mathbb{R}^n$ such that $Kx = 0$. This implies $x^T K x = x^T(0) = 0$. Since there exists $x \neq 0$ where $q(x) = 0$, K cannot be strictly positive definite. Thus, true positive definite implies non-singular.

Example Positive Semidefinite vs Positive Definite

Let $K = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$.

$$q(x) = x_1^2 - 2x_1x_2 + x_2^2 = (x_1 - x_2)^2 \geq 0$$

This is Positive Semidefinite. However, if $x = (1, 1)^T$, then $q(1, 1) = 0$. Since $x \neq 0$ but $q(x) = 0$, K is **not** positive definite.

43.1 Gram Matrices

Definition : Associated Gram Matrix

Let V be an inner product space. Let $v_1, \ldots, v_n \in V$. The $n \times n$ real matrix defined by:

$$K = (k_{ij}) = \begin{pmatrix} \langle v_1, v_1 \rangle & \ldots & \langle v_1, v_n \rangle \\ \vdots & \ddots & \vdots \\ \langle v_n, v_1 \rangle & \ldots & \langle v_n, v_n \rangle \end{pmatrix}$$

is called the **Associated Gram Matrix** of the elements $v_1, \ldots, v_n$.

K is strictly symmetric because $\langle v_i, v_j \rangle = \langle v_j, v_i \rangle$.

Theorem Theorem: Properties of Gram Matrices

All Gram matrices are Positive Semidefinite. A Gram matrix is strictly Positive Definite if and only if $v_1, \ldots, v_n$ are linearly independent elements of V.

Proof

First, we show $K \geq 0$.

$$q(x) = x^T K x = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i x_j \langle v_i, v_j \rangle = \left\langle \sum_{i=1}^{n} x_i v_i, \sum_{j=1}^{n} x_j v_j \right\rangle$$

Let $v = \sum_{i=1}^{n} x_i v_i$. Then $q(x) = \langle v, v \rangle = \|v\|^2 \geq 0$. Thus, K is Positive Semidefinite. Now, suppose $x^T K x = \|v\|^2 = 0$. This means $v = \sum x_i v_i = 0$. If $v_1, \ldots, v_n$ are linearly independent, then $\sum x_i v_i = 0$ implies $x_i = 0$ for all i (so $x = 0$). Thus K is Positive Definite. Conversely, if they are linearly dependent, there exist $x_1, \ldots, x_n$ not all zero such that $\sum x_i v_i = 0$. Then $q(x) = 0$ for $x \neq 0$, so K is not positive definite.

Example Example in $\mathbb{R}^3$

Let $V = \mathbb{R}^3$ with the standard dot product. Let $v_1 = (1, 2, -1)^T$ and $v_2 = (3, 0, 6)^T$.

- $v_1 \cdot v_1 = 1^2 + 2^2 + (-1)^2 = 6$
- $v_1 \cdot v_2 = 1(3) + 2(0) + (-1)(6) = -3$
- $v_2 \cdot v_2 = 3^2 + 0^2 + 6^2 = 45$

The Gram Matrix is $K = \begin{pmatrix} 6 & -3 \\ -3 & 45 \end{pmatrix}$. Now v_1, v_2 are linearly independent, hence K is strictly positive definite.

43.2 Matrix Formulation

If $V = \mathbb{R}^m$, we can arrange the vectors $v_1, \ldots, v_n \in V$ as columns of an $m \times n$ matrix:

$$A = \begin{pmatrix} v_1 & v_2 & \ldots & v_n \end{pmatrix}$$

Then the (i, j) entry of the Gram Matrix is $v_i \cdot v_j = v_i^T v_j$. This equals the i-th row of A^T multiplied by the j-th column of A. Hence:

$$K = A^T A$$

And $q(x) = x^T K x = x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 \geq 0$.

Theorem Theorem: The Big Equivalence

Given an $m \times n$ real matrix A, The Following Are Equivalent (TFAE):

1. $K = A^T A$ is strictly positive definite.
2. A has linearly independent columns.
3. $\text{rank}(A) = n \le m$.
4. $\ker(A) = \{0\}$.

44 Exercise Solutions: Inner Products

Reference: Olver & Shakiban, Chapter 3, Section 1, Pages 132-133.

The following problems test the axioms of inner products, specifically focusing on bilinear forms, symmetric matrices, and positive definiteness.

44.1 Testing Inner Product Formulas

Example Problem 3.1.1

Prove that $\langle v, w \rangle = v_1 w_1 - v_1 w_2 - v_2 w_1 + b w_2 w_2$ defines an inner product on $\mathbb{R}^2$ if and only if $b > 1$.

Proof

The associated real matrix for this bilinear form is $K = \begin{pmatrix} 1 & -1 \\ -1 & b \end{pmatrix}$. This matrix is symmetric. To define an inner product, it must be **Positive Definite** ($q(v) > 0$ for $v \neq 0$). The quadratic form is:

$$q(v) = x^T K x = v_1^2 - 2v_1 v_2 + b v_2^2$$

By completing the square, we rewrite this as:

$$q(v) = (v_1 - v_2)^2 - v_2^2 + b v_2^2 = (v_1 - v_2)^2 + (b-1)v_2^2$$

For this sum to be strictly greater than 0 for all $(v_1, v_2) \neq (0,0)$, the coefficient of v_2^2 must be strictly positive. Thus, $b - 1 > 0 \implies b > 1$.

Example Problem 3.1.2 & 3.1.3: Which formulas define inner products on $\mathbb{R}^2$?

Let $v = (v_1, v_2)$ and $w = (w_1, w_2)$.

(a) $2v_1 w_1 + 3v_2 w_2$: **Yes.** The associated matrix is $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, which is strictly positive definite.

(b) $v_1 w_2 + v_2 w_1$: **No.** The matrix is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. This is indefinite. Counter-example: Let $v = (1, -1)$. Then $\langle v, v \rangle = (1)(-1) + (-1)(1) = -2 \not> 0$.

(c) $(v_1 + v_2)(w_1 + w_2)$: **No.** It expands to $v_1 w_1 + v_1 w_2 + v_2 w_1 + v_2 w_2$. The matrix is $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. The determinant is 0. Counter-example: Let $v = (1, -1)$. Then $\langle v, v \rangle = (1-1)(1-1) = 0$, but $v \neq 0$. Fails strict positivity.

(d) $v_1^2 w_1^2 + v_2^2 w_2^2$: **No.** It is not bilinear (contains squared terms of components).

(e) $\sqrt{v_1^2 + v_2^2}\sqrt{w_1^2 + w_2^2}$: **No.** It is not bilinear.

(f) $2v_1w_1 + (v_1 - v_2)(w_1 - w_2)$**: Yes.** Expanding yields $3v_1w_1 - v_1w_2 - v_2w_1 + v_2w_2$. The matrix is $\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$. It is positive definite since $q(v) = 2v_1^2 + (v_1 - v_2)^2 > 0$ for $v \neq 0$.

(g) $4v_1w_1 - 2v_1w_2 - 2v_2w_1 + 4v_2w_2$**: Yes.** The matrix $\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$ has $a = 4 > 0$ and $\det = 16 - 4 = 12 > 0$, so it is positive definite.

Example Problem 3.1.4: Inner Products on $\mathbb{R}^3$

Verify that the following define inner products by checking the quadratic form $q(v)$:

(a) $\langle v, w \rangle = v_1w_1 + 2v_2w_2 + 3v_3w_3$. Matrix is diagonal with positive entries. $q(v) = v_1^2 + 2v_2^2 + 3v_3^2 > 0$. **Yes.**

(b) $v_1w_1 + v_1w_2 + v_2w_1 + 2v_2w_2 + v_2w_3 + v_3w_2 + 3v_3w_3$. Matrix $K = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 3 \end{pmatrix}$. $q(v) = v_1^2 + 2v_1v_2 + 2v_2^2 + 2v_2v_3 + 3v_3^2 = (v_1 + v_2)^2 + (v_2 + v_3)^2 + 2v_3^2 > 0$. **Yes.**

(c) $2v_1w_1 - 2v_1w_2 - 2v_2w_1 + 3v_2w_2 - v_2w_3 - v_3w_2 + 2v_3w_3$. Matrix $K = \begin{pmatrix} 2 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. $q(v) = 2v_1^2 - 4v_1v_2 + 3v_2^2 - 2v_2v_3 + 2v_3^2 = 2(v_1 - v_2)^2 + (v_2 - v_3)^2 + v_3^2 > 0$. **Yes.**

45 The Geometry of Inner Products

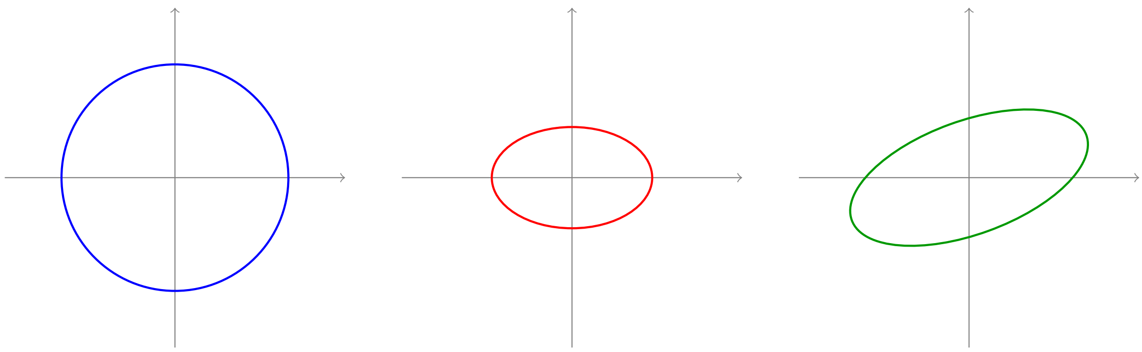
Example Problem 3.1.5: Unit Circles

The "unit circle" for an inner product on $\mathbb{R}^2$ is the set of vectors satisfying $\|v\| = 1 \implies \langle v, v \rangle = 1$.

(a) Euclidean: $v_1^2 + v_2^2 = 1$ (Standard Circle).

(b) $\langle v, w \rangle = 2v_1w_1 + 5v_2w_2 \implies 2v_1^2 + 5v_2^2 = 1$ (Axis-aligned Ellipse).

(c) $\langle v, w \rangle = v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2 \implies v_1^2 - 2v_1v_2 + 4v_2^2 = 1$ (Tilted Ellipse).



(a) Euclidean Circle (b) Weighted Ellipse (c) Tilted Ellipse

Proof Problem 3.1.7: Homogeneity of the Induced Norm

Prove that $\|cv\| = |c|\|v\|$ for all $c \in \mathbb{R}$ and $v \in V$.

$$\|cv\|^2 = \langle cv, cv \rangle = c\langle v, cv \rangle = c^2\langle v, v \rangle = c^2\|v\|^2$$

Taking the positive square root on both sides:

$$\|cv\| = \sqrt{c^2\|v\|^2} = |c|\|v\|$$

Proof Problems 3.1.8 & 3.1.9: Linearity Expansions

By applying the bilinearity axioms (additivity and scalar multiplication):

$$\begin{aligned}\langle au + bw, cv + dw\rangle &= a\langle u, cv + dw\rangle + b\langle w, cv + dw\rangle \\ &= a(c\langle u, v\rangle + d\langle u, w\rangle) + b(c\langle w, v\rangle + d\langle w, w\rangle) \\ &= ac\langle u, v\rangle + ad\langle u, w\rangle + bc\langle w, v\rangle + bd\langle w, w\rangle\end{aligned}$$

46 Identities and Transpose Properties

Theorem Problem 3.1.10 & 3.1.11: The Zero Vector Property

Let V be an inner product space.

1. If $\langle x, v\rangle = 0$ for all $v \in V$, then $x = 0$. (Proof: Choose $v = x$, then $\langle x, x\rangle = 0 \implies x = 0$).
2. If $\langle x, v\rangle = \langle y, v\rangle$ for all $v \in V$, then $x = y$. (Proof: $\langle x - y, v\rangle = 0 \implies x - y = 0$).
3. Application: If $x^T A^T v = b^T v$ for all $v \in \mathbb{R}^n$, then $(Ax)^T v = b^T v \implies \langle Ax, v\rangle = \langle b, v\rangle \implies Ax = b$.

Proof Problem 3.1.12: The Polarization Identity

We express the inner product purely in terms of norms:

$$\begin{aligned}\frac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2\right) &= \frac{1}{4}\left(\langle u+v, u+v\rangle - \langle u-v, u-v\rangle\right) \\ &= \frac{1}{4}\left((\|u\|^2 + 2\langle u, v\rangle + \|v\|^2) - (\|u\|^2 - 2\langle u, v\rangle + \|v\|^2)\right) \\ &= \frac{1}{4}\left(4\langle u, v\rangle\right) = \langle u, v\rangle\end{aligned}$$

Proof Problem 3.1.13: The Parallelogram Law

For any inner product space:

$$\begin{aligned}\|x+y\|^2 + \|x-y\|^2 &= (\langle x+y, x+y\rangle) + (\langle x-y, x-y\rangle) \\ &= (\|x\|^2 + 2\langle x, y\rangle + \|y\|^2) + (\|x\|^2 - 2\langle x, y\rangle + \|y\|^2) \\ &= 2\|x\|^2 + 2\|y\|^2 = 2(\|x\|^2 + \|y\|^2)\end{aligned}$$

Note: The max norm $\|\cdot\|_\infty$ on $\mathbb{R}^2$ does **not** satisfy this law, meaning the max norm *cannot* be derived from any inner product.

Example Problem 3.1.14: Applying the Parallelogram Law

Suppose $\|u\| = 3$, $\|u+v\| = 4$, and $\|u-v\| = 6$. Assuming this is an inner product norm, find $\|v\|$. By the Parallelogram Law:

$$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2)$$

$$4^2 + 6^2 = 2(3^2 + \|v\|^2) \implies 16 + 36 = 18 + 2\|v\|^2$$

$$52 - 18 = 2\|v\|^2 \implies 34 = 2\|v\|^2 \implies \|v\|^2 = 17 \implies \|v\| = \sqrt{17}$$

Proof Problem 3.1.15 & 3.1.16: Matrix Transpose over Dot Product

Prove that $v \cdot (Aw) = (A^T v) \cdot w$. Using matrix multiplication mechanics ($v \cdot u = v^T u$):

$$v \cdot (Aw) = v^T (Aw) = (v^T A)w = (A^T v)^T w = (A^T v) \cdot w$$

Corollary: A matrix A is symmetric ($A = A^T$) if and only if $(Av) \cdot w = v \cdot (Aw)$ for all $v, w \in \mathbb{R}^n$.

46.1 Abstract Inner Products (Page 8)

Proof Problem 3.1.17: Inner Product on Matrix Space

Let $M_n(\mathbb{R})$ be the vector space of $n \times n$ matrices. The operation $\langle A, B\rangle = \operatorname{tr}(A^T B)$ defines a valid inner product. This is because $\operatorname{tr}(A^T B) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ij}$, which is mathematically equivalent to flattening the matrices into n^2-dimensional vectors and taking their standard Euclidean dot product. Thus, it inherits all inner product axioms.

Problems 3.1.18 - 3.1.20: Creating New Inner Products

We can construct new inner products using existing ones:

1. **Restriction:** If $\langle \cdot, \cdot \rangle$ is an inner product on V, and $W \subseteq V$ is a subspace, the same formula defines an inner product strictly on W.
2. **Addition:** If $\ll \cdot, \cdot \gg$ and $\ll \cdot, \cdot \ggg$ are two valid inner products on V, their sum $\langle u, w\rangle = \ll u, w \gg + \ll u, w \ggg$ is also a valid inner product on V.
3. **Direct Sum:** If V and W are inner product spaces, we can define an inner product on the Cartesian product space $V \times W$ by adding their respective inner products: $\langle (v, w), (\tilde{v}, \tilde{w})\rangle = \langle v, \tilde{v}\rangle_V + \langle w, \tilde{w}\rangle_W$.

47 Exercise Solutions: Function Spaces

Reference: Olver & Shakiban, Chapter 3, Pages 135-136.

We now explore inner products defined via integrals on the continuous function space $C^0[a, b]$.

47.1 Computing L^2 Norms and Cauchy-Schwarz

Example Problem 3.1.21

For each pair of functions in $C^0[0, 1]$, find their L^2 inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)dx$ and their L^2 norms $\|f\|, \|g\|$. Verify the Cauchy-Schwarz inequality.

(a) $f(x) = 1, \quad g(x) = x$

- $\langle f, g\rangle = \int_0^1 (1)(x)\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$
- $\|f\|^2 = \int_0^1 1^2\,dx = 1 \implies \|f\| = 1$
- $\|g\|^2 = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3} \implies \|g\| = \frac{1}{\sqrt{3}}$
- **C-S Check:** $|\langle f, g\rangle| = \frac{1}{2} \le \|f\|\|g\| = \frac{1}{\sqrt{3}}$. Since $0.5 \le 0.577$, it holds.

(b) $f(x) = \cos(2\pi x), \quad g(x) = \sin(2\pi x)$

- Using the identity $\sin(2\theta) = 2\sin\theta\cos\theta$:

$$\langle f, g\rangle = \int_0^1 \cos(2\pi x)\sin(2\pi x)\,dx = \int_0^1 \frac{\sin(4\pi x)}{2}\,dx = \left[-\frac{\cos(4\pi x)}{8\pi}\right]_0^1 = -\frac{1}{8\pi} - \left(-\frac{1}{8\pi}\right) = 0$$

- Using $\cos^2(2\theta) = \frac{1+\cos(4\theta)}{2}$:

$$\begin{aligned}
\|f\|^2 &= \int_0^1 \cos^2(2\pi x)\,dx \\
&= \int_0^1 \left(\frac{1}{2} + \frac{\cos(4\pi x)}{2}\right) dx \\
&= \left[\frac{x}{2} + \frac{\sin(4\pi x)}{8\pi}\right]_0^1 = \frac{1}{2}
\end{aligned}$$

  So $\|f\| = \frac{1}{\sqrt{2}}$.

- Using $\sin^2(2\theta) = \frac{1-\cos(4\theta)}{2}$, we get:

$$\begin{aligned}
\|g\|^2 &= \int_0^1 \sin^2(2\pi x)\,dx \\
&= \int_0^1 \left(\frac{1}{2} - \frac{\cos(4\pi x)}{2}\right) dx \\
&= \frac{1}{2}
\end{aligned}$$

  So $\|g\| = \frac{1}{\sqrt{2}}$.

- **C-S Check:** $0 \leq \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{2}$. It holds.

(c) $f(x) = x, \quad g(x) = e^x$

- Integration by parts $(u = x, dv = e^x dx)$:

$$\langle f, g\rangle = \int_0^1 xe^x\,dx = [xe^x]_0^1 - \int_0^1 e^x\,dx = e - (e-1) = 1$$

- $\|f\|^2 = \int_0^1 x^2\,dx = \frac{1}{3} \implies \|f\| = \frac{1}{\sqrt{3}}$
- $\|g\|^2 = \int_0^1 e^{2x}\,dx = \left[\frac{e^{2x}}{2}\right]_0^1 = \frac{e^2-1}{2} \implies \|g\| = \sqrt{\frac{e^2-1}{2}}$
- **C-S Check:** $1 \leq \frac{1}{\sqrt{3}}\sqrt{\frac{e^2-1}{2}} = \sqrt{\frac{e^2-1}{6}}$. It holds.

(d) $f(x) = (x+1)^2, \quad g(x) = \frac{1}{x+1}$

- $\langle f, g\rangle = \int_0^1 (x+1)^2\left(\frac{1}{x+1}\right)dx = \int_0^1 (x+1)\,dx = \left[\frac{x^2}{2} + x\right]_0^1 = \frac{3}{2}$
- $\|f\|^2 = \int_0^1 (x+1)^4\,dx = \left[\frac{(x+1)^5}{5}\right]_0^1 = \frac{32-1}{5} = \frac{31}{5} \implies \|f\| = \sqrt{\frac{31}{5}}$
- $\|g\|^2 = \int_0^1 \frac{1}{(x+1)^2}\,dx = \left[-\frac{1}{x+1}\right]_0^1 = -\frac{1}{2} - (-1) = \frac{1}{2} \implies \|g\| = \frac{1}{\sqrt{2}}$
- **C-S Check:** $\frac{3}{2} \leq \sqrt{\frac{31}{5}}\frac{1}{\sqrt{2}} = \sqrt{\frac{31}{10}}$. Since $1.5 \leq 1.76$, it holds.

Example Problem 3.1.22: Varying Intervals and Weights

Let $f(x) = x$ and $g(x) = 1 + x^2$. Compute inner products and norms: **(a) Standard L^2 on** $[0,1]$

$$\langle f, g\rangle = \int_0^1 x(1+x^2)dx = \left[\frac{x^2}{2} + \frac{x^4}{4}\right]_0^1 = \frac{3}{4}$$

$$\|f\| = \frac{1}{\sqrt{3}}, \quad \|g\|^2 = \int_0^1 (1+2x^2+x^4)dx = 1 + \frac{2}{3} + \frac{1}{5} = \frac{28}{15} \implies \|g\| = \sqrt{\frac{28}{15}}$$

(b) Standard L^2 on $[-1,1]$

$$\langle f, g\rangle = \int_{-1}^1 (x+x^3)dx = 0 \quad \text{(Odd function over symmetric interval)}$$

$$\|f\|^2 = \int_{-1}^1 x^2 dx = \frac{2}{3} \implies \|f\| = \sqrt{\frac{2}{3}}$$

$$\|g\|^2 = \int_{-1}^1 (1+2x^2+x^4)dx = 2 + \frac{4}{3} + \frac{2}{5} = \frac{56}{15} \implies \|g\| = \sqrt{\frac{56}{15}}$$

(c) Weighted inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)x\,dx$

$$\langle f, g\rangle = \int_0^1 x(1+x^2)x\,dx = \int_0^1 (x^2+x^4)dx = \frac{1}{3} + \frac{1}{5} = \frac{8}{15}$$

$$\|f\|^2 = \int_0^1 x^2 \cdot x\,dx = \frac{1}{4} \implies \|f\| = \frac{1}{2}$$

$$\|g\|^2 = \int_0^1 (1+x^2)^2 x\,dx = \int_0^1 (x+2x^3+x^5)dx = \frac{1}{2} + \frac{1}{2} + \frac{1}{6} = \frac{7}{6} \implies \|g\| = \sqrt{\frac{7}{6}}$$

47.2 Validating Inner Products on Function Spaces

Theorem Problem 3.1.23

Which of the following define inner products on the space $C^0[-1,1]$?

(a) $\int_{-1}^1 f(x)g(x)e^{-x}\,dx$: **Yes.** $e^{-x} > 0$ on $[-1,1]$, ensuring positivity.

(b) $\int_{-1}^1 f(x)g(x)x\,dx$: **No.** Fails positivity. Let $f(x) = 1$. Then $\int_{-1}^1 (1)(1)x\,dx = 0$, but $f \neq 0$.

(c) $\int_{-1}^1 f(x)g(x)(x+2)\,dx$: **Yes.** $x + 2 \geq 1 > 0$ on $[-1,1]$.

(d) $\int_{-1}^1 f(x)g(x)x^2\,dx$: **Yes.** Clearly bilinear and symmetric. For positivity: $\int_{-1}^1 f(x)^2x^2dx \geq 0$. If the integral is 0, since $f^2x^2 \geq 0$ and continuous, $f(x)^2x^2 = 0$ everywhere. This implies $f(x) = 0$ for all $x \neq 0$. By continuity, $f(0) = 0$ as well, so $f \equiv 0$.

Proof Problem 3.1.24: The Danger of Subdomains

Prove that $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$ does NOT define an inner product on $V = C^0[-1,1]$.
Proof: Consider the continuous function $f \in C^0[-1,1]$ defined by:

$$f(x) = \begin{cases} x & \text{on } [-1,0] \\ 0 & \text{on } [0,1] \end{cases}$$

Calculate the norm: $\langle f, f\rangle = \int_0^1 f(x)^2\,dx = \int_0^1 0\,dx = 0$. However, f is not the zero function on

$[-1,1]$ (e.g., $f(-1)=-1$). This violates the positivity axiom.
Note: This does define a valid inner product on the subspace $P^{(n)} \subseteq C^0[-1,1]$ consisting of all polynomials. If $\int_0^1 p(x)^2 dx = 0$, then $p(x)=0$ for all $x \in [0,1]$. A non-zero polynomial has only finitely many roots, so $p(x)$ must be the identically zero polynomial on the whole domain $[-1,1]$.

Example Problem 3.1.25: Point Evaluations

Does either of the following define an inner product on $C^0[0,1]$?

(a) $\langle f,g\rangle = f(0)g(0)+f(1)g(1)$. **No.** Choose a continuous function that is zero at the endpoints but non-zero elsewhere, e.g., $f(x)=x(x-1)$. Then $\langle f,f\rangle = 0^2+0^2=0$, but $f \neq 0$.

(b) $\langle f,g\rangle = f(0)g(0)+f(1)g(1)+\int_0^1 f(x)g(x)\,dx$. **Yes.** It is bilinear and symmetric. For positivity: $\langle f,f\rangle = f(0)^2+f(1)^2+\int_0^1 f(x)^2dx \geq 0$. If $\langle f,f\rangle = 0$, then $\int_0^1 f(x)^2dx = 0 \implies f(x)=0$ everywhere on $[0,1]$.

Example Problem 3.1.26: Norm of a Square

Is $\|f^2\| = \|f\|^2$? **No.** Let $f(x)=x$ on $[0,1]$. $\|f\|^2 = \int_0^1 x^2\,dx = \frac{1}{3}$. $\|f^2\| = \sqrt{\int_0^1 (x^2)^2\,dx} = \sqrt{\int_0^1 x^4\,dx} = \sqrt{\frac{1}{5}} = \frac{1}{\sqrt{5}}$. Clearly, $\frac{1}{\sqrt{5}} \neq \frac{1}{3}$.

48 Derivatives in Inner Products

Proof Problem 3.1.27: Sobolev H^1-Norm

Prove that $\langle f,g\rangle = \int_a^b [f(x)g(x)+f'(x)g'(x)]\,dx$ defines an inner product on $C^1[a,b]$.
Proof: Bilinearity and symmetry are straightforward. For positivity:

$$\langle f,f\rangle = \int_a^b [f(x)^2+f'(x)^2]\,dx \geq 0$$

If $\langle f,f\rangle = 0$, since the integrand is a sum of non-negative continuous functions, we must have $f(x)^2+f'(x)^2=0$ everywhere. This immediately forces $f(x)=0$ for all x. The corresponding norm is the Sobolev H^1-norm: $\|f\|_{H^1} = \sqrt{\int_a^b (f^2+(f')^2)dx}$.

Proof Problem 3.1.28: Pure Derivative Inner Product

Let $V = C^1[-1,1]$. **(a)** Does $\langle f,g\rangle = \int_{-1}^1 f'(x)g'(x)\,dx$ define an inner product on V? **No.** Suppose $\langle f,f\rangle = 0 \implies \int_{-1}^1 f'(x)^2dx = 0 \implies f'(x)=0 \implies f(x)=c$ (a constant). Let $f(x)=1$. Then $f\neq 0$, but $\langle f,f\rangle = 0$. Positivity fails.
(b) Answer the same question for the subspace $W = \{f \in V \mid f(0)=0\}$. **Yes.** If we restrict to W, then $\langle f,f\rangle = 0 \implies f(x)=c$. But since $f\in W$, we know $f(0)=0$, which forces $c=0$. Thus, f must be the zero function.

48.1 Rigorous Analysis of Weight Functions

Theorem Problem 3.1.29

Let $h(x)\geq 0$ be a continuous non-negative function on $[a,b]$. Prove that $\int_a^b h(x)\,dx = 0 \iff h(x)\equiv 0$.
Proof: ($\Leftarrow$) If $h(x)=0$, the integral is trivially 0. ($\Rightarrow$) Suppose $h(x)\not\equiv 0$. Since $h\geq 0$,

there exists some point $x_0 \in [a, b]$ where $h(x_0) > 0$. By continuity, there exists a small interval $[c, d] \subseteq [a, b]$ around x_0 where $h(x) > 0$ strictly. Then $\int_a^b h(x)\,dx \geq \int_c^d h(x)\,dx > 0$. This contradicts the assumption that the integral is 0.
Counter-example for discontinuous functions: Let $h(a) = 1$ and $h(x) = 0$ for all $x \in (a, b]$. Then $\int_a^b h(x)\,dx = 0$, but $h \not\equiv 0$.

Theorem Problem 3.1.30: Conditions for Weighted Inner Products

Analyze the weighted inner product $\langle f, g\rangle = \int_a^b f(x)g(x)w(x)\,dx$.

- **(a) If** $w(x) > 0$**:** It is a valid inner product. By Problem 3.1.29, if $\int f^2 w dx = 0$, since $f^2 w$ is continuous and non-negative, $f^2 w \equiv 0 \implies f \equiv 0$.
- **(b) If** $w(x_0) < 0$ **for some** x_0**:** It is **NOT** a valid inner product. By continuity, $w(x) < 0$ on some interval $[c, d]$. We can choose a non-zero function $f(x)$ that is zero everywhere except on $[c, d]$. Then $\langle f, f\rangle = \int_c^d f^2 w dx < 0$, violating positivity.
- **(c) If** $w(x) \geq 0$ **but** $w(x) = 0$ **on a subinterval** $[c, d]$**:** It is **NOT** a valid inner product. We can choose a non-zero function $f(x)$ that lives only inside $[c, d]$. Then $f(x)^2 w(x) = 0$ everywhere, so $\langle f, f\rangle = 0$ even though $f \neq 0$.

49 Vector-Valued Functions

Proof Problem 3.1.33: Inner Products on Vector Fields

Let V be the vector space of all continuous, vector-valued functions $f(x) = (f_1(x), f_2(x))^T$ for $x \in [0, 1]$.
(a) Standard Dot Product Extension: Prove $\ll f, g \gg = \int_0^1 (f_1 g_1 + f_2 g_2)\,dx$ is an inner product. *Positivity:* $\ll f, f \gg = \int_0^1 (f_1^2 + f_2^2)\,dx \geq 0$. If $= 0$, then $f_1^2 + f_2^2 \equiv 0 \implies f_1 \equiv 0$ and $f_2 \equiv 0 \implies f \equiv 0$.
(b) General Extension Theorem: More generally, let $\langle\cdot,\cdot\rangle$ be *any* inner product on $\mathbb{R}^2$. Prove that $\ll f, g \gg = \int_0^1 \langle f(x), g(x)\rangle\,dx$ defines an inner product on V. *Positivity:* Since the $\mathbb{R}^2$ inner product guarantees $\langle f(x), f(x)\rangle \geq 0$ for all x, the integral of this non-negative function must be ≥ 0. If $\int_0^1 \langle f(x), f(x)\rangle\,dx = 0$, the continuous integrand must be 0 everywhere. $\langle f(x), f(x)\rangle = 0 \implies f(x) = (0, 0)^T$ for all x.
(c) Application: Use part (b) to show that $\ll f, g \gg = \int_0^1 [f_1 g_1 - f_1 g_2 - f_2 g_1 + 3 f_2 g_2]\,dx$ is an inner product. *Proof:* The integrand is the quadratic form associated with the matrix $K = \begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}$. Since $\det(K) = 3 - 1 = 2 > 0$ and $a = 1 > 0$, K is positive definite, so it defines a valid inner product on $\mathbb{R}^2$. By Theorem (b), its integral defines a valid inner product on V.

50 Exercise Solutions: Angles and Inequalities

Reference: Olver & Shakiban, Chapter 3, Pages 139-140.
These exercises explore the geometric implications of inner products, specifically focusing on the Cauchy-Schwarz inequality and the definition of angles in abstract spaces.

50.1 Angles in Euclidean Space

Example Problem 3.2.1: Verifying Cauchy-Schwarz and Finding Angles

Verify the Cauchy-Schwarz inequality ($|\langle v, w\rangle| \leq \|v\|\|w\|$) using the standard dot product, and determine the angle between the vectors.

(a) $v = (2, 1)^T, \quad w = (-1, 2)^T$

- $\langle v, w\rangle = 2(-1) + 1(2) = 0$.
- $\|v\| = \sqrt{4+1} = \sqrt{5}, \quad \|w\| = \sqrt{1+4} = \sqrt{5}$.
- C-S Check: $0 \leq \sqrt{5}\sqrt{5} \implies 0 \leq 5$ (True).
- Angle: $\cos\theta = \frac{0}{5} = 0 \implies \theta = \frac{\pi}{2}$ (90°).

(b) $v = (-1, 0)^T, \quad w = (1, 1)^T$

- $\langle v, w\rangle = -1(1) + 0(1) = -1$.
- $\|v\| = 1, \quad \|w\| = \sqrt{1+1} = \sqrt{2}$.
- C-S Check: $|-1| \leq 1 \cdot \sqrt{2} \implies 1 \leq \sqrt{2}$ (True).
- Angle: $\cos\theta = -\frac{1}{\sqrt{2}} \implies \theta = \frac{3\pi}{4}$ (135°).

(c) $v = (-1, 1)^T, \quad w = (2, 2)^T$

- $\langle v, w\rangle = -2 + 2 = 0$.
- $\|v\| = \sqrt{2}, \quad \|w\| = \sqrt{8} = 2\sqrt{2}$.
- C-S Check: $0 \leq 4$ (True).
- Angle: $\cos\theta = 0 \implies \theta = \frac{\pi}{2}$ (90°).

(d) $v = (1, -1, 1)^T, \quad w = (-1, 1, -1)^T$

- $\langle v, w\rangle = -1 - 1 - 1 = -3$.
- $\|v\| = \sqrt{1+1+1} = \sqrt{3}, \quad \|w\| = \sqrt{3}$.
- C-S Check: $|-3| \leq \sqrt{3}\sqrt{3} \implies 3 \leq 3$ (True).
- Angle: $\cos\theta = \frac{-3}{3} = -1 \implies \theta = \pi$ (180°).

(e) $v = (3, -1, 0)^T, \quad w = (-1, 1, 2)^T$

- $\langle v, w\rangle = -3 - 1 + 0 = -4$.
- $\|v\| = \sqrt{9+1} = \sqrt{10}, \quad \|w\| = \sqrt{1+1+4} = \sqrt{6}$.
- C-S Check: $|-4| \leq \sqrt{10}\sqrt{6} \implies 4 \leq \sqrt{60} = 2\sqrt{15}$ (True).
- Angle: $\cos\theta = \frac{-4}{\sqrt{60}} = \frac{-2}{\sqrt{15}} \approx 121°$.

Example Problem 3.2.2: Angles in $\mathbb{R}^4$

(a) Find the Euclidean angle between $v = (1, 1, 1, 1)^T$ and $w = (1, 1, 1, -1)^T$.

$$\|v\| = \sqrt{4} = 2, \quad \|w\| = \sqrt{4} = 2$$

$$\langle v, w\rangle = 1 + 1 + 1 - 1 = 2$$

$$\cos\theta = \frac{2}{2 \cdot 2} = \frac{1}{2} \implies \theta = 60°$$

(b) List all possible angles between $v = (1,1,1,1)^T$ and vectors of the form $w = (a_1, a_2, a_3, a_4)^T$ where each $a_i \in \{1, -1\}$. The norm is always $\|v\| = 2$ and $\|w\| = \sqrt{a_1^2 + \cdots + a_4^2} = \sqrt{4} = 2$. The dot product is $\langle v, w\rangle = \sum_{i=1}^4 a_i$. Since $a_i \in \{1, -1\}$, the possible sums are $-4, -2, 0, 2, 4$. Thus, $\cos\theta \in \{-1, -0.5, 0, 0.5, 1\}$. The possible angles are: $180°, 120°, 90°, 60°, 0°$.

Proof Problem 3.2.3: Geometry of a Tetrahedron

Prove that the points $(0,0,0), (1,1,0), (1,0,1), (0,1,1)$ form the vertices of a regular tetrahedron, and find the angle between rays from the center to the vertices.
1. Side Lengths: The vectors representing the 6 edges between the vertices are:

$$\begin{aligned}
(1,1,0) - (0,0,0) &= (1,1,0)\\
(1,0,1) - (0,0,0) &= (1,0,1)\\
(0,1,1) - (0,0,0) &= (0,1,1)\\
(1,1,0) - (1,0,1) &= (0,1,-1)\\
(1,1,0) - (0,1,1) &= (1,0,-1)\\
(1,0,1) - (0,1,1) &= (1,-1,0)
\end{aligned}$$

The Euclidean norm of *every single edge vector* is $\sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$. Since all 6 sides have equal length, it is a regular tetrahedron.
2. Central Angle: The center of the tetrahedron is $P = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})^T$. The vectors from the center to the vertices $O(0,0,0)$ and $A(1,1,0)$ are:

$$v_1 = \left(-\frac{1}{2}, -\frac{1}{2}, -\frac{1}{2}\right)^T, \quad v_2 = \left(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)^T$$

Their norms are $\|v_1\| = \|v_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{3}{4}}$. Their dot product is $\langle v_1, v_2\rangle = (-\frac{1}{4}) + (-\frac{1}{4}) + (\frac{1}{4}) = -\frac{1}{4}$.

$$\cos\theta = \frac{\langle v_1, v_2\rangle}{\|v_1\|\|v_2\|} = \frac{-1/4}{3/4} = -\frac{1}{3}$$

Thus, $\theta = \arccos(-1/3) \approx 109.5°$.

51 Cauchy-Schwarz with Custom Matrices

Example Problem 3.2.4: $\mathbb{R}^2$ Custom Inner Products

Verify the Cauchy-Schwarz inequality for $v = (1,2)^T$ and $w = (-1,3)^T$ using different inner products.
(a) Standard Dot Product: $\|v\|^2 = 5 \implies \|v\| = \sqrt{5}$. $\|w\|^2 = 10 \implies \|w\| = \sqrt{10}$. $\langle v, w\rangle = -1 + 6 = 5$. C-S Check: $5 \le \sqrt{5}\sqrt{10} = \sqrt{50} = 5\sqrt{2}$. (True).
(b) Weighted Product: $\langle v, w\rangle = v_1w_1 + 2v_2w_2$ $\|v\|^2 = 1^2 + 2(2^2) = 9 \implies \|v\| = 3$. $\|w\|^2 = (-1)^2 + 2(3^2) = 19 \implies \|w\| = \sqrt{19}$. $\langle v, w\rangle = 1(-1) + 2(2)(3) = 11$. C-S Check: $11 \le 3\sqrt{19} = \sqrt{171}$. Since $11^2 = 121 \le 171$, it holds.
(c) Matrix Product: $\langle v, w\rangle = v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2$ $\|v\|^2 = 1^2 - 2(1)(2) + 4(2^2) = 1 - 4 + 16 = 13 \implies \|v\| = \sqrt{13}$. $\|w\|^2 = (-1)^2 - 2(-1)(3) + 4(3^2) = 1 + 6 + 36 = 43 \implies \|w\| = \sqrt{43}$. $\langle v, w\rangle = 1(-1) - 1(3) - 2(-1) + 4(2)(3) = -1 - 3 + 2 + 24 = 22$. C-S Check: $22 \le \sqrt{13}\sqrt{43} = \sqrt{559}$. Since $22^2 = 484 \le 559$, it holds.

Example Problem 3.2.5: $\mathbb{R}^3$ Custom Inner Products

Verify Cauchy-Schwarz for $v = (3, -1, 2)^T$ and $w = (1, -1, 1)^T$.
(a) Standard Dot Product: $\|v\| = \sqrt{9 + 1 + 4} = \sqrt{14}$. $\|w\| = \sqrt{1 + 1 + 1} = \sqrt{3}$. $\langle v, w\rangle =$

$3+1+2=6$. C-S Check: $6 \le \sqrt{14}\sqrt{3} = \sqrt{42}$. (Since $36 \le 42$, True).
(b) Weighted Product: $\langle v, w\rangle = v_1w_1 + 2v_2w_2 + 3v_3w_3$ $\|v\|^2 = 9+2(1)+3(4) = 23 \implies \|v\| = \sqrt{23}$. $\|w\|^2 = 1+2(1)+3(1) = 6 \implies \|w\| = \sqrt{6}$. $\langle v, w\rangle = 3(1)+2(-1)(-1)+3(2)(1) = 3+2+6 = 11$. C-S Check: $11 \le \sqrt{23}\sqrt{6} = \sqrt{138}$. (Since $121 \le 138$, True).
(c) General Matrix Product: Associated with $K = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. $\|v\|^2 = v^T K v = 38 \implies \|v\| = \sqrt{38}$. $\|w\|^2 = w^T K w = 10 \implies \|w\| = \sqrt{10}$. $\langle v, w\rangle = v^T K w = v^T \begin{pmatrix} 3 \\ -4 \\ 3 \end{pmatrix} = 9+4+6 = 19$. C-S Check: $19 \le \sqrt{38}\sqrt{10} = \sqrt{380}$. (Since $19^2 = 361 \le 380$, True).

52 Recovering Euclidean Geometry

Theorem Problem 3.2.6: Calculating Angles from Norms

Show that one can determine the angle θ between v and w purely from lengths via the formula:

$$\cos\theta = \frac{\|v+w\|^2 - \|v-w\|^2}{4\|v\|\|w\|}$$

Proof

Expand the squared norms using inner product properties:

$$\|v+w\|^2 = \langle v+w, v+w\rangle = \|v\|^2 + 2\langle v, w\rangle + \|w\|^2$$

$$\|v-w\|^2 = \langle v-w, v-w\rangle = \|v\|^2 - 2\langle v, w\rangle + \|w\|^2$$

Subtract the second equation from the first:

$$\|v+w\|^2 - \|v-w\|^2 = 4\langle v, w\rangle$$

Divide both sides by $4\|v\|\|w\|$:

$$\frac{\|v+w\|^2 - \|v-w\|^2}{4\|v\|\|w\|} = \frac{4\langle v, w\rangle}{4\|v\|\|w\|} = \frac{\langle v, w\rangle}{\|v\|\|w\|}$$

By definition, $\frac{\langle v, w\rangle}{\|v\|\|w\|} = \cos\theta$.

Theorem Problem 3.2.7: The Law of Cosines

Prove that the formula $\|v-w\|^2 = \|v\|^2 + \|w\|^2 - 2\|v\|\|w\|\cos\theta$ is valid in every inner product space.

Proof

By definition of the angle θ in an inner product space, we have:

$$\cos\theta = \frac{\langle v, w\rangle}{\|v\|\|w\|} \implies \langle v, w\rangle = \|v\|\|w\|\cos\theta$$

Expanding the norm squared on the left-hand side:

$$\begin{aligned} \|v-w\|^2 &= \langle v-w, v-w\rangle \\ &= \langle v, v\rangle - 2\langle v, w\rangle + \langle w, w\rangle \\ &= \|v\|^2 - 2\langle v, w\rangle + \|w\|^2 \end{aligned}$$

Substitute the angle definition for the inner product term:

$$\|v - w\|^2 = \|v\|^2 + \|w\|^2 - 2\|v\|\|w\| \cos\theta$$

Example Problem 3.2.8: Algebraic Inequality via Cauchy-Schwarz

Use the Cauchy-Schwarz inequality to prove that for any real numbers θ, a, b:

$$(a\cos\theta + b\sin\theta)^2 \leq a^2 + b^2$$

Proof: Consider the space $\mathbb{R}^2$ with the standard dot product. Define two vectors:

$$v = \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{and} \quad w = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

Compute their inner product and norms:

- $\langle v, w\rangle = a\cos\theta + b\sin\theta$
- $\|v\|^2 = a^2 + b^2$
- $\|w\|^2 = \cos^2\theta + \sin^2\theta = 1$

By the Cauchy-Schwarz inequality, $\langle v, w\rangle^2 \leq \|v\|^2\|w\|^2$. Substituting our values:

$$(a\cos\theta + b\sin\theta)^2 \leq (a^2 + b^2)(1) = a^2 + b^2$$

Note: Equality holds if and only if (a, b) is a scalar multiple of $(\cos\theta, \sin\theta)$.

53 Algebraic and Geometric Inequalities

Example Problem 3.2.9: Algebraic Inequality via Vectors

Prove that $(a_1 + a_2 + \cdots + a_n)^2 \leq n(a_1^2 + a_2^2 + \cdots + a_n^2)$ for any real numbers $a_1, \ldots, a_n$. When does equality hold?

Proof: Consider the space $\mathbb{R}^n$ equipped with the standard Euclidean dot product. Define two vectors:

$$v = (a_1, a_2, \ldots, a_n)^T \quad \text{and} \quad w = (1, 1, \ldots, 1)^T$$

Compute their dot product and norms:

- $\langle v, w\rangle = a_1(1) + a_2(1) + \cdots + a_n(1) = \sum_{i=1}^{n} a_i$
- $\|v\| = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$
- $\|w\| = \sqrt{1^2 + 1^2 + \cdots + 1^2} = \sqrt{n}$

By the Cauchy-Schwarz inequality, $\langle v, w\rangle^2 \leq \|v\|^2\|w\|^2$. Substituting our expressions:

$$(a_1 + \cdots + a_n)^2 \leq \left(a_1^2 + \cdots + a_n^2\right)(n)$$

Equality Condition: Equality holds if and only if v and w are parallel, meaning $v = \lambda w$ for some scalar $\lambda \in \mathbb{R}$. This implies $(a_1, \ldots, a_n) = (\lambda, \ldots, \lambda)$, so equality holds if and only if $a_1 = a_2 = \cdots = a_n$.

53.1 The 2D Cross Product

Theorem Problem 3.2.10: The 2D Cross Product

The cross product of two vectors in $\mathbb{R}^2$ is defined as the scalar $v \times w = v_1 w_2 - v_2 w_1$, where $v = (v_1, v_2)^T$ and $w = (w_1, w_2)^T$.
(a) Does it define an inner product? No.

- Bilinearity holds.
- Symmetry **does not hold**: $v \times w = v_1 w_2 - v_2 w_1 = -(w_1 v_2 - w_2 v_1) = -(w \times v)$.
- Positivity **does not hold**: $v \times v = v_1 v_2 - v_2 v_1 = 0$ for all vectors $v \neq 0$.

(b) Prove that $v \times w = \|v\|\|w\| \sin\theta$ We use the fact that $\cos\theta = \frac{v \cdot w}{\|v\|\|w\|}$. We know $\cos^2\theta + \sin^2\theta = 1$. Let's evaluate the sum of the squares of our proposed formulas:

$$\left(\frac{v \cdot w}{\|v\|\|w\|}\right)^2 + \left(\frac{v \times w}{\|v\|\|w\|}\right)^2 = \frac{(v_1 w_1 + v_2 w_2)^2 + (v_1 w_2 - v_2 w_1)^2}{\|v\|^2 \|w\|^2}$$

Expanding the numerators:

$$= \frac{v_1^2 w_1^2 + 2 v_1 w_1 v_2 w_2 + v_2^2 w_2^2 + v_1^2 w_2^2 - 2 v_1 w_2 v_2 w_1 + v_2^2 w_1^2}{(v_1^2 + v_2^2)(w_1^2 + w_2^2)}$$

The cross terms cancel out. Factoring the remaining terms:

$$= \frac{v_1^2 (w_1^2 + w_2^2) + v_2^2 (w_1^2 + w_2^2)}{(v_1^2 + v_2^2)(w_1^2 + w_2^2)} = \frac{(v_1^2 + v_2^2)(w_1^2 + w_2^2)}{(v_1^2 + v_2^2)(w_1^2 + w_2^2)} = 1$$

Since $\cos^2\theta + \left(\frac{v \times w}{\|v\|\|w\|}\right)^2 = 1$, we conclude $\frac{v \times w}{\|v\|\|w\|} = \pm \sin\theta$. By convention of orientation in $\mathbb{R}^2$, it evaluates to exactly $\sin\theta$.
(c) Parallel Vectors: $v \times w = 0 \iff \sin\theta = 0 \iff \theta = 0$ or π. This exactly means v and w are parallel.

Proof Part (d): Area of a Parallelogram

Show that $|v \times w|$ is the area of the parallelogram defined by v and w. Without loss of generality, let v lie on the x-axis, so $v = (v_1, 0)^T$. The area of a parallelogram is Base $\times$ Height.

- Base $= \|v\|$
- Height $= \|w\| |\sin\theta|$ (the perpendicular length from the tip of w to v).

Area $= \|v\|\|w\||\sin\theta| = |\|v\|\|w\| \sin\theta| = |v \times w|$.

Problem 3.2.11: Inequality Notation

Explain why the inequality $\langle u, w\rangle \leq \|u\|\|w\|$ (obtained by omitting the absolute value signs from Cauchy-Schwarz) is valid. **Answer:** For any real number x, we know $x \leq |x|$. Therefore:

$$\langle u, w\rangle \leq |\langle u, w\rangle| \leq \|u\|\|w\|$$

The first step is algebraic definition, the second is Cauchy-Schwarz.

Example Problem 3.2.12: Verifying Cauchy-Schwarz Integrals

Verify Cauchy-Schwarz for $f(x) = x$ and $g(x) = e^x$ under various inner products.
(a) Standard L^2 on $[0, 1]$:

- $\|f\|^2 = \int_0^1 x^2\,dx = \frac{1}{3} \implies \|f\| = \frac{1}{\sqrt{3}}$
- $\|g\|^2 = \int_0^1 e^{2x}\,dx = \left[\frac{e^{2x}}{2}\right]_0^1 = \frac{e^2-1}{2} \implies \|g\| = \sqrt{\frac{e^2-1}{2}}$
- $\langle f, g\rangle = \int_0^1 xe^x\,dx = [xe^x - e^x]_0^1 = 1$

C-S requires $1 \le \frac{1}{\sqrt{3}}\sqrt{\frac{e^2-1}{2}} = \sqrt{\frac{e^2-1}{6}}$. Squaring both sides gives $1 \le \frac{e^2-1}{6} \implies 7 \le e^2$. Since $e \approx 2.718$, $e^2 \approx 7.389 \ge 7$. True.
(b) Standard L^2 on $[-1, 1]$:

- $\|f\|^2 = \int_{-1}^1 x^2\,dx = \frac{2}{3} \implies \|f\| = \sqrt{\frac{2}{3}}$
- $\|g\|^2 = \int_{-1}^1 e^{2x}\,dx = \frac{e^2-e^{-2}}{2} \implies \|g\| = \sqrt{\frac{e^2-e^{-2}}{2}}$
- $\langle f, g\rangle = \int_{-1}^1 xe^x\,dx = [xe^x - e^x]_{-1}^1 = (e-e) - (-e^{-1} - e^{-1}) = 2e^{-1} = \frac{2}{e}$

C-S requires $\frac{2}{e} \le \sqrt{\frac{2}{3}}\sqrt{\frac{e^2-e^{-2}}{2}} = \sqrt{\frac{e^2-e^{-2}}{3}}$. Squaring gives $\frac{4}{e^2} \le \frac{e^2-e^{-2}}{3} \implies 12 \le e^4 - 1 \implies 13 \le e^4$. (Since $e^2 > 7$, $e^4 > 49$, True).
(c) Weighted Inner Product on $[0, 1]$ with $w(x) = e^{-x}$:

- $\|f\|^2 = \int_0^1 x^2 e^{-x}\,dx = \left[-x^2e^{-x} - 2xe^{-x} - 2e^{-x}\right]_0^1 = (-e^{-1} - 2e^{-1} - 2e^{-1}) - (-2) = 2 - 5e^{-1}$
- $\|g\|^2 = \int_0^1 (e^x)^2 e^{-x}\,dx = \int_0^1 e^x\,dx = e - 1$
- $\langle f, g\rangle = \int_0^1 (x)(e^x)(e^{-x})\,dx = \int_0^1 x\,dx = \frac{1}{2}$

C-S requires $\frac{1}{2} \le \sqrt{(2-5e^{-1})(e-1)}$. Squaring gives $\frac{1}{4} \le (2-\frac{5}{e})(e-1) \implies e \le 4(e-1)(2e-5)$. This algebraic bound holds true.

Example Problem 3.2.13: Angles between Functions

Using the standard L^2 inner product on $[0, \pi]$, find the angle between the following functions. First, compute the squared norms:

- $\|1\|^2 = \int_0^\pi 1\,dx = \pi \implies \|1\| = \sqrt{\pi}$
- $\|\cos x\|^2 = \int_0^\pi \cos^2 x\,dx = \int_0^\pi \left(\frac{1}{2} + \frac{\cos 2x}{2}\right)dx = \frac{\pi}{2} \implies \|\cos x\| = \sqrt{\frac{\pi}{2}}$
- $\|\sin x\|^2 = \int_0^\pi \sin^2 x\,dx = \int_0^\pi \left(\frac{1}{2} - \frac{\cos 2x}{2}\right)dx = \frac{\pi}{2} \implies \|\sin x\| = \sqrt{\frac{\pi}{2}}$

(a) Angle between 1 and $\cos(x)$:

$$\langle 1, \cos x\rangle = \int_0^\pi \cos x\,dx = [\sin x]_0^\pi = 0$$

Since the inner product is 0, $\cos\theta = 0 \implies \theta = \frac{\pi}{2}$ (90°).
(b) Angle between 1 and $\sin(x)$:

$$\langle 1, \sin x\rangle = \int_0^\pi \sin x\,dx = [-\cos x]_0^\pi = -(-1) - (-1) = 2$$

$$\cos\theta = \frac{\langle 1, \sin x\rangle}{\|1\|\|\sin x\|} = \frac{2}{\sqrt{\pi}\sqrt{\pi/2}} = \frac{2}{\pi/\sqrt{2}} = \frac{2\sqrt{2}}{\pi}$$

Thus, $\theta = \arccos\left(\frac{2\sqrt{2}}{\pi}\right)$.

(c) Angle between $\cos(x)$ and $\sin(x)$:

$$\langle \cos x, \sin x\rangle = \int_0^\pi \cos x \sin x \, dx = \int_0^\pi \frac{\sin 2x}{2}\, dx = \left[-\frac{\cos 2x}{4}\right]_0^\pi = -\frac{1}{4} - \left(-\frac{1}{4}\right) = 0$$

Since the inner product is 0, $\cos\theta = 0 \implies \theta = \frac{\pi}{2}$ $(90°)$.

55 Exercise Solutions: Orthogonal Vectors

Reference: Olver & Shakiban, Chapter 3, Pages 141-142.

Unless otherwise mentioned, the inner product is the standard dot product on $\mathbb{R}^n$.

55.1 Solving for Orthogonal Vectors

Example Problem 3.2.15

(a) Find $a \in \mathbb{R}$ such that $v = (2, a, -3)^T$ is orthogonal to $w = (-1, 3, -2)^T$.

$$\langle v, w\rangle = 0 \implies 2(-1) + a(3) + (-3)(-2) = 0$$

$$-2 + 3a + 6 = 0 \implies 3a = -4 \implies a = -4/3$$

(b) Is there any value of a such that v is parallel to w? For $v \parallel w$, there must exist a scalar λ such that $v = \lambda w$.

$$(2, a, -3)^T = \lambda(-1, 3, -2)^T$$

Comparing the first components: $2 = -\lambda \implies \lambda = -2$. Comparing the third components: $-3 = -2\lambda \implies \lambda = 3/2$. Since λ must be uniquely defined, no such a exists.

Example Problem 3.2.16

Find all vectors in $\mathbb{R}^3$ orthogonal to both $(1, 2, 3)^T$ and $(-2, 0, 1)^T$.

Let the vector be $v = (x, y, z)^T$. We establish a system of equations: 1. $x + 2y + 3z = 0$ 2. $-2x + z = 0 \implies z = 2x$

Substitute $z = 2x$ into the first equation:

$$x + 2y + 3(2x) = 0 \implies 7x + 2y = 0 \implies y = -\frac{7}{2}x$$

All such vectors are of the form $(x, -\frac{7}{2}x, 2x)^T$ for $x \in \mathbb{R}$. Equivalently, this is the 1-dimensional subspace spanned by all scalar multiples of $(2, -7, 4)^T$.

Example Problem 3.2.17: Changing the Inner Product

Answer 3.2.15 and 3.2.16 using the weighted inner product: $\langle v, w\rangle = 3v_1w_1 + 2v_2w_2 + v_3w_3$.

Redoing 3.2.15(a):

$$\langle (2, a, -3)^T, (-1, 3, -2)^T\rangle = 0$$

$$3(2)(-1) + 2(a)(3) + 1(-3)(-2) = 0 \implies -6 + 6a + 6 = 0 \implies a = 0$$

Redoing 3.2.15(b): Parallelism does not depend on the inner product. The answer is still "Not possible".

Redoing 3.2.16: 1. $3(x)(1) + 2(y)(2) + 1(z)(3) = 0 \implies 3x + 4y + 3z = 0$ 2. $3(x)(-2) + 2(y)(0) + 1(z)(1) = 0 \implies -6x + z = 0 \implies z = 6x$

Substitute $z = 6x$ into the first equation:

$$3x + 4y + 3(6x) = 0 \implies 21x + 4y = 0 \implies y = -\frac{21}{4}x$$

All such vectors are of the form $(x, -\frac{21}{4}x, 6x)^T$, or all real multiples of $(4, -21, 24)^T$.

Example Problem 3.2.18 & 3.2.19: Orthogonal Subspaces in $\mathbb{R}^4$

3.2.18: Find all vectors in $\mathbb{R}^4$ orthogonal to $(1, 2, 3, 4)^T$ and $(5, 6, 7, 8)^T$. Let $v = (x, y, z, w)^T$. 1. $x + 2y + 3z + 4w = 0$ 2. $5x + 6y + 7z + 8w = 0$
Multiply (1) by 5 and subtract from (2):

$$(5x + 6y + 7z + 8w) - (5x + 10y + 15z + 20w) = 0$$

$$-4y - 8z - 12w = 0 \implies y + 2z + 3w = 0 \implies y = -2z - 3w$$

Substitute y back into (1):

$$x + 2(-2z - 3w) + 3z + 4w = 0 \implies x - z - 2w = 0 \implies x = z + 2w$$

Choosing free variables $(z, w) = (1, 0) \implies v_1 = (1, -2, 1, 0)^T$. Choosing free variables $(z, w) = (0, 1) \implies v_2 = (2, -3, 0, 1)^T$. The vectors form a 2-dimensional subspace spanned by $\{(1, -2, 1, 0)^T, (2, -3, 0, 1)^T\}$.
3.2.19: Determine a basis for subspace $W \subseteq \mathbb{R}^4$ orthogonal to $(1, 2, -1, 3)^T$.

$$x + 2y - z + 3w = 0 \implies x = -2y + z - 3w$$

We have three free variables (y, z, w). Set $y = 1, z = 0, w = 0 \implies (-2, 1, 0, 0)^T$. Set $y = 0, z = 1, w = 0 \implies (1, 0, 1, 0)^T$. Set $y = 0, z = 0, w = 1 \implies (-3, 0, 0, 1)^T$. Basis of $W = \{(-2, 1, 0, 0)^T, (1, 0, 1, 0)^T, (-3, 0, 0, 1)^T\}$.

56 Properties of Orthogonality

Example Problem 3.2.20: Dependencies among Orthogonal Vectors

Find 3 vectors $u, v, w \in \mathbb{R}^3$ such that $u \perp v$ and $u \perp w$, but $v \not\perp w$.
Case 1: Linearly Independent Vectors Let $u = (0, 0, 1)^T$, $v = (1, 0, 0)^T$, and $w = (1, 1, 0)^T$. $u \cdot v = 0$ and $u \cdot w = 0$. However, $v \cdot w = 1 \neq 0$. These vectors are linearly independent.
Case 2: Linearly Dependent Vectors Let $u = (0, 0, 1)^T$, $v = (1, 0, 0)^T$, and $w = (-1, 0, 0)^T$. $u \cdot v = 0$ and $u \cdot w = 0$. However, $v \cdot w = -1 \neq 0$. Here, $w = -v$, so the vectors are linearly dependent.

Theorem Problem 3.2.22 & 3.2.23: The Zero Vector

1. When is a vector orthogonal to itself?

$$v \perp v \iff \langle v, v \rangle = 0 \iff \|v\|^2 = 0 \iff v = 0$$

2. Prove that the only vector w in an inner product space orthogonal to every vector is the zero vector. Suppose $\langle v, w \rangle = 0$ for all $v \in V$. Since it holds for all vectors, we can choose $v = w$. Then $\langle w, w \rangle = 0 \implies \|w\|^2 = 0 \implies w = 0$.

Proof Problem 3.2.24: Diagonals of a Rhombus

A vector with $\|v\| = 1$ is a unit vector. Prove that if v, w are unit vectors, then $v + w$ and $v - w$ are orthogonal. Are they also unit vectors?
Orthogonality:

$$\begin{aligned}\langle v + w, v - w\rangle &= \langle v, v\rangle - \langle v, w\rangle + \langle w, v\rangle - \langle w, w\rangle \\ &= \|v\|^2 - \|w\|^2 = 1 - 1 = 0\end{aligned}$$

Yes, they are orthogonal. Geometrically, the diagonals of a rhombus are always perpendicular.
Unit Vectors?

$$\|v + w\|^2 = \langle v + w, v + w\rangle = \|v\|^2 + \|w\|^2 + 2\langle v, w\rangle = 2 + 2\cos\theta$$

This equals 1 if and only if $2\cos\theta = -1 \implies \cos\theta = -1/2 \implies \theta = 120°$. So, they are **not** unit vectors in general.

Proof Problem 3.2.25: Orthogonal Complement is a Subspace

Let V be an inner product space and $v \in V$. Prove that the set $W = \{w \in V \mid \langle w, v\rangle = 0\}$ is a subspace of V.
To show W is a subspace, we check closure under scalar multiplication and addition.

1. **Scalar Multiplication:** Let $w_1 \in W$ and $\lambda \in \mathbb{R}$.

$$\langle \lambda w_1, v\rangle = \lambda\langle w_1, v\rangle = \lambda(0) = 0 \implies \lambda w_1 \in W$$

2. **Addition:** Let $w_1, w_2 \in W$.

$$\langle w_1 + w_2, v\rangle = \langle w_1, v\rangle + \langle w_2, v\rangle = 0 + 0 = 0 \implies w_1 + w_2 \in W$$

Thus, W is a valid subspace.

57 Orthogonal Functions

Proof Problem 3.2.26: Orthogonality in L^2

(a) Orthogonal Polynomials: Show that $p_1(x) = 1$, $p_2(x) = x - 1/2$, and $p_3(x) = x^2 - x + 1/6$ are mutually orthogonal with respect to the standard L^2 inner product on $[0, 1]$.

- $\langle p_1, p_2\rangle = \int_0^1 1(x - 1/2)\, dx = \left[\frac{x^2}{2} - \frac{x}{2}\right]_0^1 = \frac{1}{2} - \frac{1}{2} = 0.$
- $\langle p_1, p_3\rangle = \int_0^1 1(x^2 - x + 1/6)\, dx = \left[\frac{x^3}{3} - \frac{x^2}{2} + \frac{x}{6}\right]_0^1 = \frac{1}{3} - \frac{1}{2} + \frac{1}{6} = 0.$
- $\langle p_2, p_3\rangle = \int_0^1 (x - 1/2)(x^2 - x + 1/6)\, dx = \int_0^1 (x^3 - \frac{3}{2}x^2 + \frac{2}{3}x - \frac{1}{12})\, dx$

$$= \left[\frac{x^4}{4} - \frac{3x^3}{6} + \frac{2x^2}{6} - \frac{x}{12}\right]_0^1 = \frac{1}{4} - \frac{1}{2} + \frac{1}{3} - \frac{1}{12} = \frac{3 - 6 + 4 - 1}{12} = 0.$$

(b) Orthogonal Sines (Fourier Basis): Show that functions $\sin(n\pi x)$ for $n = 1, 2, 3\ldots$ are mutually orthogonal in $L^2[0, 1]$. Let $n \neq m$. We evaluate $\int_0^1 \sin(n\pi x)\sin(m\pi x)\, dx$. Using the trigonometric identity $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$:

$$\frac{1}{2}\int_0^1 [\cos((n - m)\pi x) - \cos((n + m)\pi x)]\, dx$$

$$= \frac{1}{2}\left[\frac{\sin((n-m)\pi x)}{(n-m)\pi} - \frac{\sin((n+m)\pi x)}{(n+m)\pi}\right]_0^1$$

Since n and m are integers, $(n-m)$ and $(n+m)$ are integers. The sine of any integer multiple of π is exactly 0. Thus, the integral evaluates to $0 - 0 = 0$.

58 Orthogonal Polynomials and Functions

Example Problem 3.2.27: Finding Orthogonal Polynomials

Find a non-zero quadratic polynomial $p(x) = ax^2 + bx + c$ that is orthogonal to both $p_1(x) = 1$ and $p_2(x) = x$ under the standard L^2 inner product on $[-1, 1]$.

Solution: We require two conditions to be met: 1. $\langle p, p_1 \rangle = 0 \implies \int_{-1}^{1}(ax^2 + bx + c)(1)\,dx = 0$
2. $\langle p, p_2 \rangle = 0 \implies \int_{-1}^{1}(ax^2 + bx + c)(x)\,dx = 0$
Evaluating the first integral:

$$\int_{-1}^{1}(ax^2 + bx + c)\,dx = \left[\frac{a}{3}x^3 + \frac{b}{2}x^2 + cx\right]_{-1}^{1} = \left(\frac{a}{3} + \frac{b}{2} + c\right) - \left(-\frac{a}{3} + \frac{b}{2} - c\right) = \frac{2a}{3} + 2c = 0$$

This simplifies to $a + 3c = 0$.
Evaluating the second integral:

$$\int_{-1}^{1}(ax^3 + bx^2 + cx)\,dx = \left[\frac{a}{4}x^4 + \frac{b}{3}x^3 + \frac{c}{2}x^2\right]_{-1}^{1} = \left(\frac{a}{4} + \frac{b}{3} + \frac{c}{2}\right) - \left(\frac{a}{4} - \frac{b}{3} + \frac{c}{2}\right) = \frac{2b}{3} = 0$$

This implies $b = 0$.
We have the system $b = 0$ and $a = -3c$. We can choose any non-zero value for c. Let $c = -1$, which gives $a = 3$. Thus, a valid polynomial is:

$$p(x) = 3x^2 - 1$$

Note: This is a multiple of the Legendre polynomial $P_2(x) = \frac{1}{2}(3x^2 - 1)$.

Example Problem 3.2.28: Orthogonality to Exponentials

Find all quadratic polynomials $p(x) = ax^2 + bx + c$ that are orthogonal to e^x under the L^2 inner product on $[0, 1]$.

Solution: We require $\int_0^1 (ax^2 + bx + c)e^x\,dx = 0$. We evaluate the integral of each term using integration by parts:

- $\int_0^1 ce^x\,dx = c[e^x]_0^1 = c(e - 1)$
- $\int_0^1 bxe^x\,dx = b[xe^x - e^x]_0^1 = b(0 - (-1)) = b(1) = b$
- $\int_0^1 ax^2e^x\,dx = a[x^2e^x - 2xe^x + 2e^x]_0^1 = a((e - 2e + 2e) - (2)) = a(e - 2)$

Summing these results gives the linear relation that the coefficients must satisfy:

$$a(e - 2) + b + c(e - 1) = 0$$

Any choice of a, b, c satisfying this equation yields a valid orthogonal polynomial.

59 Pairwise Orthogonality in $L^2[-1,1]$

A Note on Symmetry

When integrating over a symmetric interval $[-a,a]$, the integral of any **odd** function ($f(-x) = -f(x)$) is exactly 0. Recall that:

- x and $\sin(\pi x)$ are odd.
- 1 and $\cos(\pi x)$ are even.
- Odd $\times$ Even = Odd.
- Odd $\times$ Odd = Even.

Theorem Problem 3.2.29

Determine all pairs among the functions $\{1, x, \cos(\pi x), \sin(\pi x), e^x\}$ that are orthogonal with respect to the L^2 inner product on $[-1,1]$.

1. Pairs involving 1:

- $\{1, x\}$: $\int_{-1}^{1} x\,dx = 0$. **(Orthogonal)**
- $\{1, \cos(\pi x)\}$: $\int_{-1}^{1} \cos(\pi x)\,dx = \left[\frac{\sin(\pi x)}{\pi}\right]_{-1}^{1} = 0$. **(Orthogonal)**
- $\{1, \sin(\pi x)\}$: $\int_{-1}^{1} \sin(\pi x)\,dx = 0$ (Odd function). **(Orthogonal)**
- $\{1, e^x\}$: $\int_{-1}^{1} e^x\,dx = e - e^{-1} \neq 0$. (Not orthogonal)

2. Pairs involving x:

- $\{x, \cos(\pi x)\}$: $x\cos(\pi x)$ is an odd function, so $\int_{-1}^{1} x\cos(\pi x)\,dx = 0$. **(Orthogonal)**
- $\{x, \sin(\pi x)\}$: $x\sin(\pi x)$ is even, integral is > 0. (Not orthogonal)
- $\{x, e^x\}$: $\int_{-1}^{1} xe^x\,dx = 2e^{-1} \neq 0$. (Not orthogonal)

3. Pairs involving Trig Functions and Exponentials:

- $\{\cos(\pi x), \sin(\pi x)\}$: Product is odd, so $\int_{-1}^{1} \cos(\pi x)\sin(\pi x)\,dx = 0$. **(Orthogonal)**
- $\{\cos(\pi x), e^x\}$: $\int_{-1}^{1} e^x \cos(\pi x)\,dx = \left[\frac{e^x(\cos(\pi x)+\pi\sin(\pi x))}{1+\pi^2}\right]_{-1}^{1} = -\frac{e+e^{-1}}{1+\pi^2} \neq 0$. (Not orthogonal)
- $\{\sin(\pi x), e^x\}$: $\int_{-1}^{1} e^x \sin(\pi x)\,dx = \left[\frac{e^x(\sin(\pi x)-\pi\cos(\pi x))}{1+\pi^2}\right]_{-1}^{1} = \frac{\pi(e+e^{-1})}{1+\pi^2} \neq 0$. (Not orthogonal)

Correction to Handwritten Notes

In the original notes, the integral $\int_{-1}^{1}\sin(\pi x)dx$ was evaluated as $2/\pi$, leading to the conclusion that 1 and $\sin(\pi x)$ are not orthogonal. This calculation mistakenly used 0 as the lower limit instead of -1. Because $\sin(\pi x)$ is odd, its integral over $[-1,1]$ is precisely 0, making them orthogonal.

60 Weighted Orthogonality

Example Problem 3.2.30

Find two non-zero functions that are orthogonal with respect to the weighted inner product:

$$\langle f, g\rangle = \int_0^1 f(x)g(x)x\,dx$$

Solution: Let $f(x) = 1$. Let us construct a linear function $g(x) = x - a$ and find the value of a that forces orthogonality.

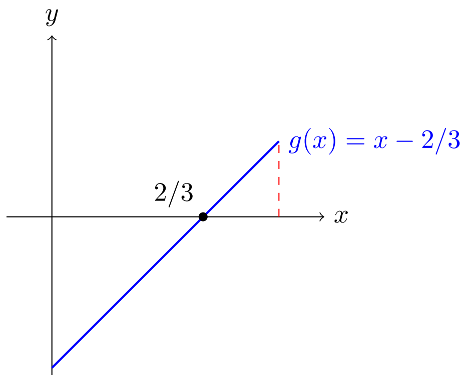
$$\langle 1, x-a\rangle = \int_0^1 (1)(x-a)x\,dx = \int_0^1 (x^2 - ax)\,dx = 0$$

Evaluate the integral:

$$\left[\frac{x^3}{3} - \frac{ax^2}{2}\right]_0^1 = \frac{1}{3} - \frac{a}{2} = 0$$

$$\frac{a}{2} = \frac{1}{3} \implies a = \frac{2}{3}$$

Thus, the pair of functions $f(x) = 1$ and $g(x) = x - \frac{2}{3}$ are orthogonal under this weight.



Geometric Intuition: Under standard L^2, a line crossing at $x = 0.5$ is orthogonal to 1. Because our weight $w(x) = x$ "values" the right side of the graph more heavily, the root must shift to the right (to $x = 2/3$) to balance the positive and negative areas.

61 Computational Verification of Inequalities

Reference: Olver & Shakiban, Chapter 3, Pages 143-144.

These exercises focus on computationally verifying the Cauchy-Schwarz and Triangle inequalities across different spaces and inner products.

61.1 Inequalities in $\mathbb{R}^3$

Example Problem 3.2.31

Use the dot product in $\mathbb{R}^3$ to answer the following for the vectors $v = (1, 2, 3)^T$ and $w = (1, -1, 2)^T$.
(a) Find the angle between v and w.

- $\|v\|^2 = 1^2 + 2^2 + 3^2 = 14 \implies \|v\| = \sqrt{14}$
- $\|w\|^2 = 1^2 + (-1)^2 + 2^2 = 6 \implies \|w\| = \sqrt{6}$
- $\langle v, w\rangle = (1)(1) + (2)(-1) + (3)(2) = 1 - 2 + 6 = 5$

$$\cos\theta = \frac{\langle v, w\rangle}{\|v\|\|w\|} = \frac{5}{\sqrt{14}\sqrt{6}} = \frac{5}{\sqrt{84}}$$

(b) Verify the Cauchy-Schwarz and Triangle Inequalities.

- **Cauchy-Schwarz:** $|\langle v, w\rangle| \le \|v\|\|w\| \implies 5 \le \sqrt{84}$. Squaring both sides yields $25 \le 84$, which is strictly true.
- **Triangle Inequality:** First, compute the sum vector: $v + w = (2, 1, 5)^T$.

$$\|v + w\| = \sqrt{2^2 + 1^2 + 5^2} = \sqrt{4 + 1 + 25} = \sqrt{30}$$

We must verify that $\sqrt{30} \le \sqrt{14} + \sqrt{6}$. Squaring both sides:

$$30 \le 14 + 6 + 2\sqrt{(14)(6)} \implies 30 \le 20 + 2\sqrt{84} \implies 10 \le 2\sqrt{84} \implies 5 \le \sqrt{84}$$

Squaring again yields $25 \le 84$, which is true. Thus, the inequality holds.

(c) Find all vectors orthogonal to both v and w. Let $u = (x, y, z)^T$. We require: 1. $\langle u, v\rangle = 0 \implies x + 2y + 3z = 0$ 2. $\langle u, w\rangle = 0 \implies x - y + 2z = 0$
Subtracting the second equation from the first:

$$(x + 2y + 3z) - (x - y + 2z) = 0 \implies 3y + z = 0 \implies z = -3y$$

Substitute z back into the second equation:

$$x - y + 2(-3y) = 0 \implies x - 7y = 0 \implies x = 7y$$

Thus, all orthogonal vectors take the form $(7y, y, -3y)^T$. The set of all such vectors is the 1-dimensional subspace spanned by $(7, 1, -3)^T$.

61.2 Triangle Inequality with Custom Norms

Example Problem 3.2.32 & 3.2.33: Verification across metrics

Verify the triangle inequality $\|v + w\| \le \|v\| + \|w\|$ for various vectors and inner products.
1. Standard Dot Product (Ex 3.2.1 pairs):

(a) $v = (1, 2)^T, w = (-1, 2)^T \implies v + w = (0, 4)^T$. Check: $4 \le \sqrt{5} + \sqrt{5} = 2\sqrt{5} \approx 4.47$ (True).

(b) $v = (-1, 0)^T, w = (1, 1)^T \implies v + w = (0, 1)^T$. Check: $1 \le 1 + \sqrt{2}$ (True).

(c) $v = (-1, 1)^T, w = (2, 2)^T \implies v + w = (1, 3)^T$. Check: $\sqrt{10} \le \sqrt{2} + \sqrt{8} = 3\sqrt{2} = \sqrt{18}$

(True).

2. Custom Inner Products for $v = (1,2)^T, w = (1,-3)^T$:

- **(a) Dot Product:** $v + w = (2,-1)^T$. Check: $\sqrt{5} \le \sqrt{5} + \sqrt{10}$ (True).
- **(b) Weighted ($\langle u, z\rangle = u_1 z_1 + 2u_2 z_2$):** $\|v\| = \sqrt{1+8} = 3$. $\|w\| = \sqrt{1+18} = \sqrt{19}$. $\|v+w\| = \|(2,-1)^T\| = \sqrt{4+2(1)} = \sqrt{6}$. Check: $\sqrt{6} \le 3 + \sqrt{19}$ (True).
- **(c) Matrix Norm ($\langle u, z\rangle = u_1 z_1 - u_1 z_2 - u_2 z_1 + 4u_2 z_2$):** $\|v\| = \sqrt{13}$, $\|w\| = \sqrt{43}$. $\|v+w\| = \|(2,-1)^T\| = \sqrt{4-(-2)-(-2)+4(1)} = \sqrt{12}$. Check: $\sqrt{12} \le \sqrt{13} + \sqrt{43}$ (True).

62 Triangle Inequality for Functions

Example Problem 3.2.34: Verifying the Triangle Inequality for Integrals

Verify the triangle inequality $\|f+g\| \le \|f\| + \|g\|$ for the functions $f(x) = x$ and $g(x) = e^x$ under the indicated inner products.
Strategy: We will compute $\|f+g\|^2$ and compare it to $(\|f\| + \|g\|)^2 = \|f\|^2 + \|g\|^2 + 2\|f\|\|g\|$.
(a) Standard L^2 inner product on $[0,1]$: Recall from previous exercises: $\|f\| = \frac{1}{\sqrt{3}}$ and $\|g\| = \sqrt{\frac{e^2-1}{2}}$.

$$\|f+g\|^2 = \int_0^1 (x+e^x)^2\,dx = \int_0^1 (x^2 + 2xe^x + e^{2x})\,dx$$

Using the fact that $\int_0^1 2xe^x dx = 2[xe^x - e^x]_0^1 = 2(1) = 2$:

$$= \frac{1}{3} + 2 + \frac{e^2-1}{2} = \frac{e^2}{2} + \frac{11}{6}$$

We must check if:

$$\frac{e^2}{2} + \frac{11}{6} \le \left(\frac{1}{\sqrt{3}} + \sqrt{\frac{e^2-1}{2}}\right)^2 = \frac{1}{3} + \frac{e^2-1}{2} + \frac{2}{\sqrt{3}}\sqrt{\frac{e^2-1}{2}}$$

Simplifying the inequality:

$$\frac{e^2}{2} + \frac{11}{6} \le \frac{e^2}{2} - \frac{1}{6} + 2\sqrt{\frac{e^2-1}{6}} \implies \frac{12}{6} \le 2\sqrt{\frac{e^2-1}{6}} \implies 1 \le \sqrt{\frac{e^2-1}{6}}$$

Squaring yields $1 \le \frac{e^2-1}{6} \implies 7 \le e^2$. Since $e^2 \approx 7.389$, the inequality holds.
(b) Standard L^2 inner product on $[-1,1]$: Recall $\|f\| = \sqrt{\frac{2}{3}}$ and $\|g\| = \sqrt{\frac{e^2-e^{-2}}{2}}$.

$$\|f+g\|^2 = \int_{-1}^1 (x^2 + 2xe^x + e^{2x})\,dx = \frac{2}{3} + 2(2e^{-1}) + \frac{e^2-e^{-2}}{2}$$

We must check if:

$$\frac{2}{3} + \frac{4}{e} + \frac{e^2-e^{-2}}{2} \le \left(\sqrt{\frac{2}{3}} + \sqrt{\frac{e^2-e^{-2}}{2}}\right)^2 = \frac{2}{3} + \frac{e^2-e^{-2}}{2} + 2\sqrt{\frac{2(e^2-e^{-2})}{6}}$$

Subtracting common terms leaves:

$$\frac{4}{e} \le 2\sqrt{\frac{e^2-e^{-2}}{3}} \implies \frac{2}{e} \le \sqrt{\frac{e^2-e^{-2}}{3}}$$

Squaring both sides:

$$\frac{4}{e^2} \le \frac{e^2-e^{-2}}{3} \implies 12 \le e^4 - 1 \implies 13 \le e^4$$

Since $e^4 \approx 54.6$, this is clearly true.
(c) Weighted inner product $\langle u, v\rangle = \int_0^1 u(x)v(x)e^{-x}\,dx$: Recall $\|f\| = \sqrt{2-5e^{-1}}$ and $\|g\| = \sqrt{e-1}$.

$$\|f+g\|^2 = \int_0^1 (x+e^x)^2 e^{-x}\,dx = \int_0^1 (x^2e^{-x} + 2x + e^x)\,dx$$

We calculate the three integrals:

- $\int_0^1 x^2e^{-x}\,dx = 2-5e^{-1}$
- $\int_0^1 2x\,dx = 1$
- $\int_0^1 e^x\,dx = e-1$

Summing these gives $\|f+g\|^2 = (2-5e^{-1}) + 1 + (e-1) = 2-5e^{-1}+e$. We must check if:

$$2-5e^{-1}+e \le \left(\sqrt{2-5e^{-1}}+\sqrt{e-1}\right)^2 = (2-5e^{-1}) + (e-1) + 2\sqrt{(2-5e^{-1})(e-1)}$$

Subtracting common terms:

$$1 \le 2\sqrt{(2-5e^{-1})(e-1)}$$

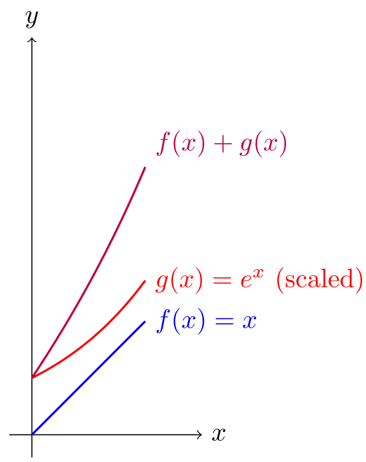
Squaring both sides:

$$1 \le 4(2-5e^{-1})(e-1) = 4(2e-2-5+5e^{-1}) = 8e-28+\frac{20}{e}$$

Evaluating the right side for $e \approx 2.718$ gives approximately 29.1, which is strictly greater than 1. Thus, the inequality holds.

Omitted Exercises

- **Problem 3.2.35:** Omitted temporarily as it requires techniques from multivariable integration.
- **Problem 3.2.36:** Will proceed in the next section.



Triangle Inequality in Function Space: Just as the length of a vector sum is bounded by the individual lengths, the "area" or energy under the squared sum curve $f(x)+g(x)$ is strictly bounded by the combinations of the individual curves.

63 Orthogonality and Weighted Spaces

Example Problem: Angles and Orthogonal Polynomials

Consider the functions 1 and x in the space $C^0[-1,1]$ equipped with the standard L^2 inner product.

(a) Find the angle between 1 and x. Are they orthogonal?

$$\langle 1, x\rangle = \int_{-1}^{1} (1)(x)\, dx = \left[\frac{x^2}{2}\right]_{-1}^{1} = \frac{1}{2} - \frac{1}{2} = 0$$

Yes, they are perfectly orthogonal. The angle between them is $\theta = \frac{\pi}{2}$ (90°).

(b) Verify the Cauchy-Schwarz and Triangle Inequalities.

- $\|1\|^2 = \int_{-1}^{1} 1\, dx = 2 \implies \|1\| = \sqrt{2}$
- $\|x\|^2 = \int_{-1}^{1} x^2\, dx = \frac{2}{3} \implies \|x\| = \sqrt{\frac{2}{3}}$
- $\|1+x\|^2 = \int_{-1}^{1} (1+x)^2\, dx = \int_{-1}^{1} (1+2x+x^2)\, dx = 2 + 0 + \frac{2}{3} = \frac{8}{3}$

Cauchy-Schwarz: $0 \le \sqrt{2}\sqrt{\frac{2}{3}} = \sqrt{\frac{4}{3}}$. (Strictly True).

Triangle Inequality: $\sqrt{\frac{8}{3}} \le \sqrt{2} + \sqrt{\frac{2}{3}}$. Since $\sqrt{2.66} \approx 1.63$ and $1.41 + 0.81 = 2.22$, the inequality holds ($1.63 \le 2.22$).

(c) Find all quadratic polynomials orthogonal to both 1 and x. Let $p(x) = ax^2 + bx + c$. 1. $\langle p(x), 1\rangle = 0 \implies \int_{-1}^{1} (ax^2 + bx + c)\, dx = \frac{2a}{3} + 2c = 0 \implies a + 3c = 0$ 2. $\langle p(x), x\rangle = 0 \implies \int_{-1}^{1} (ax^3 + bx^2 + cx)\, dx = \frac{2b}{3} = 0 \implies b = 0$ Thus, any polynomial of the form $p(x) = c(3x^2 - 1)$ for $c \in \mathbb{R}$ is orthogonal to both.

Example Problem 3.2.37: Weighted Inequalities

Write down the explicit formulas for the Cauchy-Schwarz and Triangle inequalities based on the weighted inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)e^x\, dx$.

Cauchy-Schwarz:

$$\left|\int_0^1 f(x)g(x)e^x\, dx\right| \le \sqrt{\int_0^1 f(x)^2 e^x\, dx}\sqrt{\int_0^1 g(x)^2 e^x\, dx}$$

Triangle Inequality:

$$\sqrt{\int_0^1 (f(x)+g(x))^2 e^x\, dx} \le \sqrt{\int_0^1 f(x)^2 e^x\, dx} + \sqrt{\int_0^1 g(x)^2 e^x\, dx}$$

Verification for $f(x) = 1$ and $g(x) = e^x$:

- $\|f\|^2 = \int_0^1 e^x\, dx = e - 1 \implies \|f\| = \sqrt{e-1}$
- $\|g\|^2 = \int_0^1 e^{3x}\, dx = \frac{e^3-1}{3} \implies \|g\| = \sqrt{\frac{e^3-1}{3}}$
- $\langle f, g\rangle = \int_0^1 e^{2x}\, dx = \frac{e^2-1}{2}$

Cauchy-Schwarz yields: $\frac{e^2-1}{2} \le \sqrt{(e-1)\frac{e^3-1}{3}}$. Using a calculator ($e \approx 2.718$): $3.194 \le \sqrt{1.718 \times 6.361} = \sqrt{10.93} \approx 3.30$. (True).

64 The Reverse Triangle Inequality

Theorem Problem 3.2.39: The Reverse Triangle Inequality

Prove that for any vectors v, w in an inner product space:

$$\|v - w\| \geq |\|v\| - \|w\||$$

Proof

By the standard triangle inequality, the length of a sum of vectors is less than or equal to the sum of their lengths. We can write $v = (v - w) + w$. Applying the triangle inequality to this sum:

$$\|v\| = \|(v - w) + w\| \leq \|v - w\| + \|w\|$$

Subtracting $\|w\|$ from both sides gives:

$$\|v\| - \|w\| \leq \|v - w\| \quad \text{— (Equation 1)}$$

By symmetry, we can write $w = (w - v) + v$. Applying the triangle inequality:

$$\|w\| = \|(w - v) + v\| \leq \|w - v\| + \|v\|$$

Since $\|w - v\| = \| - (v - w)\| = \|v - w\|$, we have:

$$\|w\| - \|v\| \leq \|v - w\| \implies -(\|v\| - \|w\|) \leq \|v - w\| \quad \text{— (Equation 2)}$$

Combining Equation 1 and Equation 2 implies that both the positive and negative versions of $(\|v\| - \|w\|)$ are bounded by $\|v - w\|$. Therefore:

$$|\|v\| - \|w\|| \leq \|v - w\|$$

Pictorial Interpretation: In any triangle, the length of one side ($\|v - w\|$) must be strictly greater than or equal to the absolute difference of the lengths of the other two sides.

Example Problem 3.2.40: Vector Sum Bounds

True or False: $\|w\| \leq \|v\| + \|v + w\|$ for all $v, w \in V$. **Answer: True.** We use the algebraic trick of adding and subtracting the same vector:

$$w = (v + w) - v$$

Apply the standard triangle inequality:

$$\|w\| = \|(v + w) + (-v)\| \leq \|v + w\| + \| - v\|$$

Since $\| - v\| = \|v\|$, we get:

$$\|w\| \leq \|v + w\| + \|v\|$$

65 The Infinite Sequence Space ℓ^2

We now move from finite-dimensional vectors to infinite-dimensional sequences.

Definition : The Space $\mathbb{R}^\infty$

Let $\mathbb{R}^\infty$ denote the space of all infinite sequences of real numbers:

$$X = (x_1, x_2, x_3, \dots) \text{ where } x_i \in \mathbb{R}$$

This forms a vector space under component-wise addition and scalar multiplication:

- $X + Y = (x_1 + y_1, x_2 + y_2, \dots)$
- $\lambda X = (\lambda x_1, \lambda x_2, \dots)$

Definition : The ℓ^2 Subspace

The space $\ell^2 \subset \mathbb{R}^\infty$ is the set of all square-summable sequences. That is, a sequence X belongs to ℓ^2 if and only if:

$$\sum_{k=1}^{\infty} x_k^2 < \infty$$

Theorem Problem 3.2.41 (b): ℓ^2 is a Subspace

Prove that ℓ^2 is a valid vector subspace of $\mathbb{R}^\infty$.

Proof

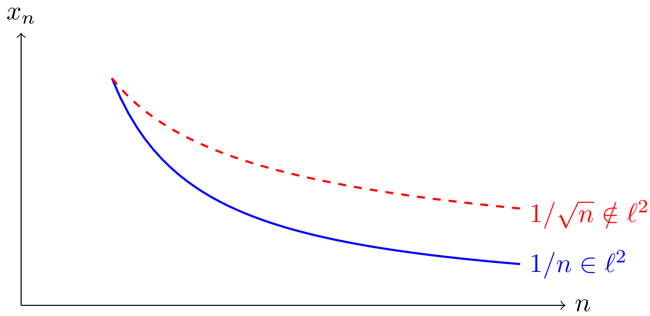
We must show closure under scalar multiplication and vector addition. **1. Scalar Multiplication:** Let $X \in \ell^2$ and $\lambda \in \mathbb{R}$.

$$\sum_{i=1}^{\infty} (\lambda x_i)^2 = \lambda^2 \sum_{i=1}^{\infty} x_i^2$$

Since $\sum x_i^2 < \infty$ and λ^2 is finite, the new sum is finite. Thus $\lambda X \in \ell^2$.
2. Addition: Let $X, Y \in \ell^2$. We must show $\sum (x_i + y_i)^2 < \infty$. Note that for any real numbers a, b, we have $(a-b)^2 \geq 0 \implies 2ab \leq a^2 + b^2$. Thus $2|x_i y_i| \leq x_i^2 + y_i^2$.

$$\sum_{i=1}^{n} (x_i + y_i)^2 = \sum_{i=1}^{n} (x_i^2 + 2x_i y_i + y_i^2) \leq \sum_{i=1}^{n} (x_i^2 + x_i^2 + y_i^2 + y_i^2) = 2\sum_{i=1}^{n} x_i^2 + 2\sum_{i=1}^{n} y_i^2$$

Since both sequences independently sum to a finite value, the sum of their combination is bounded by $2(\Sigma x_i^2) + 2(\Sigma y_i^2) < \infty$. Thus $X + Y \in \ell^2$.



Both sequences go to zero, but $1/\sqrt{n}$ decays too slowly. When squared, it becomes the Harmonic Series $\sum 1/n$, which diverges to infinity.

65.1 Properties of Sequences in ℓ^2

Example Examples of Sequences

Sequences in ℓ^2:

- $x_n = \frac{1}{n}$. The sum is $\sum \frac{1}{n^2}$, which converges (p-series with $p = 2$).
- $x_n = a^n$ for $0 < a < 1$. The sum is $\sum (a^2)^n$, which is a convergent geometric series.

Sequences NOT in ℓ^2:

- $x_n = 1$. The sum is $1 + 1 + 1 + \cdots = \infty$.
- $x_n = \frac{1}{\sqrt{n}}$. The squared sum is $\sum \frac{1}{n}$, the harmonic series, which diverges.

Theorem Convergence to Zero

- **True:** If $X \in \ell^2$, then $\lim_{k\to\infty} x_k = 0$. *Proof:* If the terms did not approach 0, there would be some $\epsilon > 0$ such that infinitely many terms are larger than ϵ. The sum of infinitely many ϵ^2 terms is infinite, contradicting $X \in \ell^2$.
- **False:** If $\lim_{k\to\infty} x_k = 0$, then $X \in \ell^2$. *Counter-example:* $x_n = \frac{1}{\sqrt{n}}$ approaches 0, but its squared sum diverges.

Mathematical Clarification on Parametric Sequences

Part (f): For $x_k = \alpha^k$, $\sum x_k^2 = \sum (\alpha^2)^k$. This is a geometric series, which converges if and only if $|\alpha^2| < 1 \implies |\alpha| < 1$.
Part (g): For $x_k = k^\alpha$ (or $\frac{1}{k^\alpha}$), we examine the p-series. The sum $\sum (k^\alpha)^2 = \sum k^{2\alpha}$ converges if and only if $2\alpha < -1 \implies \alpha < -1/2$. *(Note: If the sequence was defined as $x_k = 1/k^\alpha$, it would converge for $2\alpha > 1 \implies \alpha > 1/2$.)*

65.2 Inner Product on ℓ^2

Theorem The ℓ^2 Inner Product

The operation $\langle X, Y \rangle = \sum_{k=1}^{\infty} x_k y_k$ defines a valid inner product on ℓ^2.

Proof

First, we ensure the infinite sum actually converges to a finite real number. From our earlier algebraic inequality, we know $x_k y_k \le |x_k y_k| \le \frac{1}{2}(x_k^2 + y_k^2)$. Since $\sum x_k^2$ and $\sum y_k^2$ are both finite (because $X, Y \in \ell^2$), the series $\sum x_k y_k$ is absolutely convergent. Thus, $\langle X, Y\rangle$ is well-defined. It clearly satisfies symmetry ($\sum x_k y_k = \sum y_k x_k$), bilinearity, and positivity ($\sum x_k^2 \ge 0$). The corresponding norm is the fundamental ℓ^2 norm:

$$\|X\|_{\ell^2} = \sqrt{\sum_{k=1}^{\infty} x_k^2}$$

Inequalities in ℓ^2: With the inner product verified, the standard abstract inequalities automatically apply to infinite series:

- **Cauchy-Schwarz:** $\left|\sum_{k=1}^{\infty} x_k y_k\right| \le \sqrt{\sum_{k=1}^{\infty} x_k^2}\sqrt{\sum_{k=1}^{\infty} y_k^2}$
- **Triangle Inequality:** $\sqrt{\sum_{k=1}^{\infty}(x_k + y_k)^2} \le \sqrt{\sum_{k=1}^{\infty} x_k^2} + \sqrt{\sum_{k=1}^{\infty} y_k^2}$

66 Pivots and Determinants (Problem 1.9.17)

Reference: Olver & Shakiban, Page 74.

We establish a profound connection between the pivots obtained during standard Gaussian elimination (without row swaps) and the determinants of the matrix sub-blocks. A matrix is called **regular** if Gaussian elimination can be performed without any row interchanges.

Theorem Problem 1.9.17: Pivots of Regular Matrices

Show that:

(a) If $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is regular, then its pivots are a and $\frac{\det A}{a}$.

(b) If $A = \begin{pmatrix} a & b & e \\ c & d & f \\ g & h & j \end{pmatrix}$ is regular, then its pivots are a, $\frac{ad-bc}{a}$, and $\frac{\det A}{ad-bc}$.

(c) Generalize this observation to regular $n \times n$ matrices.

Proof Proof of (a)

Since $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is regular, the first pivot is a, and $a \ne 0$. We use this fact to make the entry below a (which is c) zero. We perform the row operation $R_2 \leftarrow R_2 - \frac{c}{a}R_1$:

$$\begin{pmatrix} 1 & 0 \\ -\frac{c}{a} & 1 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ 0 & d - \frac{cb}{a} \end{pmatrix}$$

Hence, the second pivot of A is:

$$d - \frac{cb}{a} = \frac{ad-bc}{a} = \frac{\det A}{a}$$

Proof Proof of (b)

Because A is regular, its 1st pivot is $a \neq 0$. We use a to eliminate the entries below it in the first column (c and g):

$$\begin{pmatrix} 1 & 0 & 0 \\ -\frac{c}{a} & 1 & 0 \\ -\frac{g}{a} & 0 & 1 \end{pmatrix} \begin{pmatrix} a & b & e \\ c & d & f \\ g & h & j \end{pmatrix} = \begin{pmatrix} a & b & e \\ 0 & d - \frac{cb}{a} & f - \frac{ce}{a} \\ 0 & h - \frac{gb}{a} & j - \frac{ge}{a} \end{pmatrix}$$

Now, the second pivot is exactly the $(2, 2)$ entry:

$$d - \frac{cb}{a} = \frac{ad - bc}{a}$$

Since the matrix is regular, this pivot is non-zero. We use it to make the entry in the $(3, 2)$ position (which is $h - \frac{gb}{a}$) zero. The row operation is $R_3 \leftarrow R_3 - \frac{h - gb/a}{d - cb/a} R_2$. The new $(3, 3)$ entry (the 3rd pivot) becomes:

$$\text{3rd Pivot} = \left(j - \frac{ge}{a}\right) - \left(\frac{h - \frac{gb}{a}}{d - \frac{bc}{a}}\right)\left(f - \frac{ce}{a}\right)$$

Finding a common denominator and expanding the massive algebra:

$$= \frac{(aj - ge)(ad - bc) - (ah - bg)(af - ce)}{a(ad - bc)}$$

$$= \frac{a(ajad - adge - ajbc + gbc) - a(ahaf - ahce - bgaf + bgce)}{a(ad - bc)}$$

Canceling the common factor of a in the numerator and denominator:

$$= \frac{adj - deg - bcj - afh + ceh + bfg}{ad - bc} = \frac{\det A}{ad - bc}$$

66.1 The General Case and Principal Minors

Let A_i denote the $i \times i$ top-left submatrix of A (known as the i-th **leading principal submatrix**).

$$\begin{array}{c} \begin{array}{cccc} A_1 & A_2 & A_3 & \end{array} \\ \begin{pmatrix} a_{11} & a_{12} & a_{13} & \cdots \\ a_{21} & a_{22} & a_{23} & \cdots \\ a_{31} & a_{32} & a_{33} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{array}$$

Theorem (c) General Case

For an $n \times n$ regular matrix A, the i-th pivot is given by the ratio of successive principal minors:

$$p_i = \frac{\det(A_i)}{\det(A_{i-1})} \quad \text{for } 1 \leq i \leq n$$

(where we formally define $\det(A_0) = 1$).

67 Sylvester's Criterion for Positive Definiteness

We connect the algebraic properties of pivots to the geometric property of positive definiteness.

Theorem Theorem: Pivots and Positive Definiteness

An $n \times n$ real symmetric matrix A is strictly Positive Definite if and only if A is regular and all the pivots of A are strictly positive.

Using the pivot formula $p_i = \frac{\det(A_i)}{\det(A_{i-1})}$ from the previous section, we can reframe this theorem purely in terms of determinants.

Proof Deriving Sylvester's Criterion

Case 1: $n = 2$ Suppose $n = 2$. The pivots of $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ are a and $\frac{ac-b^2}{a}$. A is positive definite $\iff$ the pivots are positive:

$$a > 0 \quad \text{and} \quad \frac{ac - b^2}{a} > 0$$

Since $a > 0$, the second condition implies $ac - b^2 > 0$. Thus, both $\det(A_1) > 0$ and $\det(A_2) > 0$. We have already seen this result geometrically by completing the square.

Case 2: $n = 3$ Suppose $n = 3$. The pivots of $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$ are:

$$p_1 = a_{11}, \quad p_2 = \frac{\det(A_2)}{a_{11}}, \quad p_3 = \frac{\det(A)}{\det(A_2)}$$

Assuming A is symmetric, it is positive definite $\iff$ all pivots are positive:

- $p_1 > 0 \implies a_{11} > 0 \implies \det(A_1) > 0$
- $p_2 > 0 \implies \frac{\det(A_2)}{a_{11}} > 0$. Since $a_{11} > 0$, we must have $\det(A_2) > 0$.
- $p_3 > 0 \implies \frac{\det(A)}{\det(A_2)} > 0$. Since $\det(A_2) > 0$, we must have $\det(A) > 0$.

(Note: A is automatically regular because the strictly positive principal minors guarantee that no pivot is ever zero, so row swaps are never needed).

Definition : Sylvester's Criterion

We have a similar statement for the general case. An $n \times n$ symmetric matrix A is strictly Positive Definite if and only if all of its leading principal minors are strictly positive:

$$\det(A_i) > 0 \quad \forall\, 1 \leq i \leq n$$

68 Exercise Solutions: Norms and Distances

Reference: Olver & Shakiban, Chapter 3, Pages 147-148.

68.1 Computations in $\mathbb{R}^n$

Example Problems 3.3.1 & 3.3.2: Computing Vector Norms

Compute the 1-norm, 2-norm, and ∞-norm of the given vectors and verify the triangle inequality $\|v + w\| \leq \|v\| + \|w\|$.
(1) $v = (1, 0)^T, \quad w = (0, 1)^T \; v + w = (1, 1)^T$.

- **L^1:** $\|v+w\|_1 = 1+1 = 2$. Check: $2 \le 1+1 = 2$.
- **L^2:** $\|v+w\|_2 = \sqrt{1^2+1^2} = \sqrt{2}$. Check: $\sqrt{2} \le 1+1 = 2$.
- **L^∞:** $\|v+w\|_\infty = \max\{1,1\} = 1$. Check: $1 \le 1+1 = 2$.

(2) $v = (-2,-1)^T, \quad w = (1,-3)^T \; v+w = (-1,-4)^T$.

- **L^1:** $\|v\|_1 = 3, \|w\|_1 = 4, \|v+w\|_1 = 5$. Check: $5 \le 3+4 = 7$.
- **L^2:** $\|v\|_2 = \sqrt{5}, \|w\|_2 = \sqrt{10}, \|v+w\|_2 = \sqrt{17}$. Check: $\sqrt{17} \le \sqrt{5}+\sqrt{10}$.
- **L^∞:** $\|v\|_\infty = 2, \|w\|_\infty = 3, \|v+w\|_\infty = 4$. Check: $4 \le 2+3 = 5$.

(3) $v = (-1,-1)^T, \quad w = (2,-3)^T \; v+w = (1,-4)^T$.

- **L^1:** $\|v\|_1 = 2, \|w\|_1 = 5, \|v+w\|_1 = 5$. Check: $5 \le 2+5 = 7$.
- **L^2:** $\|v\|_2 = \sqrt{2}, \|w\|_2 = \sqrt{13}, \|v+w\|_2 = \sqrt{17}$. Check: $\sqrt{17} \le \sqrt{2}+\sqrt{13}$.
- **L^∞:** $\|v\|_\infty = 1, \|w\|_\infty = 3, \|v+w\|_\infty = 4$. Check: $4 \le 1+3 = 4$.

Example Problem 3.3.3: Distances between Vectors

Which two of the vectors $u = (-3,0)^T$, $v = (1,4)^T$, $w = (0,-1)^T$ are closest to each other in distance for the (a) L^2 norm, (b) L^∞ norm, and (c) L^1 norm?
First, compute the difference vectors:

$$u - v = (-4,-4)^T, \quad u - w = (-3,1)^T, \quad v - w = (1,5)^T$$

(a) Euclidean Norm (L^2):

- $\|u-v\|_2 = \sqrt{16+16} = \sqrt{32}$
- $\|u-w\|_2 = \sqrt{9+1} = \sqrt{10}$ **(Closest: u and w)**
- $\|v-w\|_2 = \sqrt{1+25} = \sqrt{26}$

(b) Infinity Norm (L^∞):

- $\|u-v\|_\infty = 4$
- $\|u-w\|_\infty = 3$ **(Closest: u and w)**
- $\|v-w\|_\infty = 5$

(c) 1-Norm (L^1):

- $\|u-v\|_1 = 4+4 = 8$
- $\|u-w\|_1 = 3+1 = 4$ **(Closest: u and w)**
- $\|v-w\|_1 = 1+5 = 6$

69 Norms on Function Spaces

Example Problems 3.3.4 & 3.3.5: Function Norms

Compute the L^∞ and L^1 norms on $[0,1]$ for $f(x) = \frac{1}{3} - x$ and $g(x) = x - x^2$. Verify the triangle inequality.

1. The L^∞ Norm ($\max |h(x)|$):

- $\|f\|_\infty = \max_{x\in[0,1]} |1/3 - x| = |1/3 - 1| = \frac{2}{3}$.
- $\|g\|_\infty = \max_{x\in[0,1]} |x - x^2|$. The vertex is at $x = 1/2$, yielding $\frac{1}{4}$.
- $f + g = \frac{1}{3} - x^2$. Maximum magnitude occurs at $x = 1$: $|1/3 - 1| = \frac{2}{3}$. Thus $\|f + g\|_\infty = \frac{2}{3}$.
- **Check:** $\frac{2}{3} \le \frac{2}{3} + \frac{1}{4} = \frac{11}{12}$. (True).

2. The L^1 Norm ($\int |h(x)|dx$): We must split the integrals at the roots.

- $\|f\|_1 = \int_0^{1/3}(1/3 - x)dx + \int_{1/3}^1 (x - 1/3)dx = \left[\frac{x}{3} - \frac{x^2}{2}\right]_0^{1/3} + \left[\frac{x^2}{2} - \frac{x}{3}\right]_{1/3}^1 = \frac{1}{18} + \frac{4}{18} = \frac{5}{18}$.
- $\|g\|_1 = \int_0^1 (x - x^2)dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{6}$.
- $\|f + g\|_1 = \int_0^1 |1/3 - x^2|dx$. Root is at $x = 1/\sqrt{3}$.

$$= \int_0^{1/\sqrt{3}} (1/3 - x^2)dx + \int_{1/\sqrt{3}}^1 (x^2 - 1/3)dx$$

$$= \left[\frac{x}{3} - \frac{x^3}{3}\right]_0^{1/\sqrt{3}} + \left[\frac{x^3}{3} - \frac{x}{3}\right]_{1/\sqrt{3}}^1 = \frac{2}{3\sqrt{3}} + 0 - \left(-\frac{2}{3\sqrt{3}}\right) = \frac{4}{3\sqrt{3}} \approx 0.77$$

- **Check:** $\frac{4}{3\sqrt{3}} \le \frac{5}{18} + \frac{1}{6} = \frac{8}{18} = \frac{4}{9}$. Since $\frac{4}{5.19} \le \frac{4}{4.5}$, the inequality holds.

Example Problem 3.3.6: Distances in Function Space

Which two of $f(x) = 1$, $g(x) = x$, and $h(x) = \sin(\pi x)$ are closest in $[0,1]$ under the L^∞ and L^2 norms?

1. The L^∞ Norm:

- $\|f - g\|_\infty = \max |1 - x| = 1$ (at $x = 0$).
- $\|f - h\|_\infty = \max |1 - \sin(\pi x)| = 1$ (at $x = 0$ or 1).
- $\|g - h\|_\infty = \max |x - \sin(\pi x)| = 1$ (at $x = 1$).
- **Conclusion:** All three pairs are equidistant in L^∞.

2. The L^2 Norm:

- $\|f - g\|_2 = \sqrt{\int_0^1 (1 - x)^2 dx} = \sqrt{\frac{1}{3}} \approx 0.577$
- $\|f - h\|_2 = \sqrt{\int_0^1 (1 - \sin(\pi x))^2 dx} = \sqrt{1 + \frac{1}{2} - \frac{4}{\pi}} = \sqrt{\frac{3}{2} - \frac{4}{\pi}} \approx 0.476$
- $\|g - h\|_2 = \sqrt{\int_0^1 (x - \sin(\pi x))^2 dx} = \sqrt{\frac{1}{3} + \frac{1}{2} - \frac{2}{\pi}} = \sqrt{\frac{5}{6} - \frac{2}{\pi}} \approx 0.443$
- **Conclusion:** $g(x)$ and $h(x)$ are the closest pair in L^2.

title

3.3.6(a) - L^1 Distances:

- $\|f-g\|_1 = \int_0^1 |1-x|dx = 0.5$.
- $\|f-h\|_1 = \int_0^1 (1-\sin(\pi x))dx = 1 - \frac{2}{\pi} \approx 0.363$.
- $\|g-h\|_1 = \int_0^1 (\sin(\pi x) - x)dx = \frac{2}{\pi} - \frac{1}{2} \approx 0.136$. **(Closest in L^1)**

3.3.7 - $f = 1, g = x - 3/4$ on $[0,1]$:

- L^1: $\|f\|_1 = 1$, $\|g\|_1 = \frac{5}{16}$, $\|f+g\|_1 = \frac{3}{4}$. $(0.75 \le 1 + 0.3125)$.
- L^2: $\|f\|_2 = 1$, $\|g\|_2 = \sqrt{\frac{7}{48}}$, $\|f+g\|_2 = \sqrt{\frac{13}{48}}$. $(0.52 \le 1 + 0.38)$.

3.3.8 - Exponentials $f = e^x, g = e^{-x}$ for L^∞: $\|f\|_\infty = e$, $\|g\|_\infty = 1$, $\|f+g\|_\infty = e + e^{-1}$. Triangle inequality: $e + 1/e \le e + 1$. This is logically sound since $1/e < 1$.

70 Proving Custom Norms

To prove a function is a norm, it must satisfy Positivity, Homogeneity, and the Triangle Inequality.

Proof Problem 3.3.9: A Custom Norm on $\mathbb{R}^2$

Prove that $\|(x_1, x_2)^T\| = |x_1| + 2|x_1 - x_2|$ defines a valid norm. Let $x = (x_1, x_2)^T$ and $y = (y_1, y_2)^T$.

1. **Positivity:** Since it is a sum of absolute values, $\|x\| \ge 0$. If $\|x\| = 0$, then $|x_1| = 0 \implies x_1 = 0$, and $2|x_1 - x_2| = 0 \implies x_1 = x_2 \implies x_2 = 0$. Thus $x = 0$.
2. **Homogeneity:** $\|\lambda x\| = |\lambda x_1| + 2|\lambda x_1 - \lambda x_2| = |\lambda|(|x_1| + 2|x_1 - x_2|) = |\lambda|\|x\|$.
3. **Triangle Inequality:**

$$\begin{aligned}
\|x+y\| &= |x_1 + y_1| + 2|(x_1 + y_1) - (x_2 + y_2)| \\
&= |x_1 + y_1| + 2|(x_1 - x_2) + (y_1 - y_2)| \\
&\le (|x_1| + |y_1|) + 2(|x_1 - x_2| + |y_1 - y_2|) \\
&= (|x_1| + 2|x_1 - x_2|) + (|y_1| + 2|y_1 - y_2|) = \|x\| + \|y\|
\end{aligned}$$

Theorem Problem 3.3.10: Identifying Norms on $\mathbb{R}^2$

Determine which of the following formulas define norms. Let $v = (v_1, v_2)^T$.

- **(a) $\|v\| = \sqrt{2v_1^2 + 3v_2^2}$: Yes.** This is the norm associated with the weighted inner product defined by the strictly positive definite matrix $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.
- **(b) $\|v\| = \sqrt{2v_1^2 - v_1v_2 + 2v_2^2}$: Yes.** This is the norm associated with the inner product matrix $K = \begin{pmatrix} 2 & -1/2 \\ -1/2 & 2 \end{pmatrix}$. Since $\det(K) = 4 - 1/4 = 15/4 > 0$ and $2 > 0$, K is positive definite.
- **(c) $\|v\| = 2|v_1| + |v_2|$: Yes.** It inherits positivity and homogeneity from absolute values. Triangle inequality: $2|u_1 + w_1| + |u_2 + w_2| \le 2|u_1| + 2|w_1| + |u_2| + |w_2| = \|u\| + \|w\|$.
- **(d) $\|v\| = \max\{2|v_1|, |v_2|\}$: Yes.** Positivity and homogeneity hold. Triangle inequality holds because $\max\{A+B, C+D\} \le \max\{A, C\} + \max\{B, D\}$.
- **(e) $\|v\| = \max\{|v_1 - v_2|, |v_1 + v_2|\}$: Yes.** *Positivity:* If max is 0, both $|v_1 - v_2| = 0$ and

$|v_1 + v_2| = 0$. Adding these gives $2v_1 = 0 \implies v_1 = 0$, thus $v_2 = 0$. *Triangle Ineq:* $\max\{|(u_1 + w_1) - (u_2 + w_2)|, \dots\} \leq \max\{|u_1 - u_2| + |w_1 - w_2|, \dots\} \leq \|u\| + \|w\|$.

- **(f)** $\|v\| = |v_1 - v_2| + |v_1 + v_2|$**: Yes.** Proved via the exact same absolute value bounding mechanics as (e) and 3.3.9.

71 Advanced Norm Properties and Constructions

Reference: Olver & Shakiban, Chapter 3, Pages 148-149.

71.1 Identifying Valid Norms in $\mathbb{R}^n$

Example Problem 3.3.11: Which formulas define norms on $\mathbb{R}^3$?

Let $v = (v_1, v_2, v_3)^T$. We test the following formulas:
(b) $\|v\| = \max\{|v_1|, |v_2|, |v_3|\}$. **Answer: Yes.** This is the standard ∞-norm on $\mathbb{R}^3$.
(c) $\|v\| = |v_1 - v_2| + |v_2 - v_3| + |v_3 - v_1|$. **Answer: No.** It is not a norm because it fails the positivity axiom. Let $v = (1, 1, 1)^T$. Then $v \neq 0$, but:

$$\|v\| = |1 - 1| + |1 - 1| + |1 - 1| = 0$$

(Note: It does satisfy homogeneity and the triangle inequality).
(d) $\|v\| = \sqrt{v_1^2 + 2v_1v_2 + v_2^2 + v_3^2}$. **Answer: No.** We can rewrite the term under the square root as $(v_1 + v_2)^2 + v_3^2$. This also fails positivity. Let $v = (1, -1, 0)^T$. Then $v \neq 0$, but:

$$\|v\| = \sqrt{(1 - 1)^2 + 0^2} = 0$$

(It is associated with a positive *semidefinite* matrix, not a positive definite one).
(e) $\|v\| = |v_1| + \max\{|v_2|, |v_3|\}$. **Answer: Yes.**

- **Positivity:** $\|v\| \geq 0$. If $\|v\| = 0$, then $|v_1| = 0 \implies v_1 = 0$, and $\max\{|v_2|, |v_3|\} = 0 \implies v_2 = v_3 = 0$. Thus $v = 0$.
- **Homogeneity:** $\|\lambda v\| = |\lambda v_1| + \max\{|\lambda v_2|, |\lambda v_3|\} = |\lambda|(|v_1| + \max\{|v_2|, |v_3|\}) = |\lambda|\|v\|$.
- **Triangle Inequality:**

$$\begin{aligned}
\|u + w\| &= |u_1 + w_1| + \max\{|u_2 + w_2|, |u_3 + w_3|\} \\
&\leq |u_1| + |w_1| + \max\{|u_2| + |w_2|, |u_3| + |w_3|\} \\
&\leq |u_1| + |w_1| + \max\{|u_2|, |u_3|\} + \max\{|w_2|, |w_3|\} \\
&= (|u_1| + \max\{|u_2|, |u_3|\}) + (|w_1| + \max\{|w_2|, |w_3|\}) = \|u\| + \|w\|
\end{aligned}$$

71.2 Parallelism and the Parallelogram Law

Theorem Problem 3.3.12: Parallel Vectors with Equal Norms

Prove that two parallel vectors v and w have the same norm if and only if $v = \pm w$.

Proof

Since v and w are parallel, $v = \lambda w$ for some scalar $\lambda \in \mathbb{R}$. We are given $\|v\| = \|w\|$. Substituting v:

$$\|\lambda w\| = \|w\| \implies |\lambda|\|w\| = \|w\|$$

Case 1: If $w = 0$, then $v = \lambda(0) = 0$, so $v = \pm w$ holds trivially.
Case 2: If $w \neq 0$, we can divide by $\|w\|$ to get $|\lambda| = 1$. This implies $\lambda = 1$ or $\lambda = -1$. Therefore,

$v = w$ or $v = -w$.

Example Problem 3.3.13: Equality in Triangle Inequality

True or False: If $\|v + w\| = \|v\| + \|w\|$, then v and w are parallel vectors. **Answer: False.** While this statement is TRUE for norms associated with inner products (where equality holds $\iff v = \lambda w, \lambda \geq 0$), it is FALSE in general. *Counter-example:* Consider the L^1 norm on $\mathbb{R}^2$, $\|x\|_1 = |x_1| + |x_2|$. Let $v = (1, 0)^T$ and $w = (0, 1)^T$.

- $\|v\|_1 = 1$, $\|w\|_1 = 1 \implies \|v\|_1 + \|w\|_1 = 2$
- $v + w = (1, 1)^T \implies \|v + w\|_1 = |1| + |1| = 2$

Hence, $\|v+w\|_1 = \|v\|_1 + \|w\|_1$, but $(1, 0)^T$ and $(0, 1)^T$ are clearly not parallel. *Conclusion:* This proves that the L^1 norm cannot be associated with any inner product.

Proof Problem 3.3.14 & 3.3.15: The Parallelogram Law

Prove that the ∞-norm on $\mathbb{R}^2$ does not come from an inner product.
Proof: If a norm comes from an inner product, it must satisfy the **Parallelogram Law** (Exercise 3.1.13):

$$\|x + y\|^2 + \|x - y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

Let $x = (1, 0)^T$ and $y = (0, 1)^T$. We use the ∞-norm: $\|v\|_\infty = \max\{|v_1|, |v_2|\}$.

- $\|x\|_\infty = 1$ and $\|y\|_\infty = 1$.
- $x + y = (1, 1)^T \implies \|x + y\|_\infty = 1$.
- $x - y = (1, -1)^T \implies \|x - y\|_\infty = 1$.

Substitute these into the law:

$$\text{LHS} = 1^2 + 1^2 = 2$$

$$\text{RHS} = 2(1^2) + 2(1^2) = 4$$

Since $2 \neq 4$, the ∞-norm fails the Parallelogram Law and cannot be derived from an inner product. Consequently, the Polarization Identity formula (3.11) cannot be used to define an inner product for the L^1 or L^∞ norms.

72 Limits of p-Norms

Theorem Problem 3.3.16

Prove that $\lim_{p\to\infty} \|v\|_p = \|v\|_\infty$ for all $v \in \mathbb{R}^2$.

Proof

Let $v = (v_1, v_2)^T$. The p-norm is $\|v\|_p = (|v_1|^p + |v_2|^p)^{1/p}$. Without loss of generality, assume $|v_1| \geq |v_2|$. Let $a = |v_1|$ and $b = |v_2|$. By definition, $\|v\|_\infty = \max\{a, b\} = a$.
If $v = 0$, then $a = b = 0$, and $\lim_{p\to\infty}(0)^{1/p} = 0 = \|v\|_\infty$. Suppose $v \neq 0$, so $a > 0$. We factor a^p out of the expression:

$$\|v\|_p = (a^p + b^p)^{1/p} = \left(a^p \left(1 + \left(\frac{b}{a}\right)^p\right)\right)^{1/p} = a\left(1 + \left(\frac{b}{a}\right)^p\right)^{1/p}$$

Since $b \leq a$, the ratio $\frac{b}{a} \leq 1$.

- **Case 1 ($b < a$):** $\frac{b}{a} < 1$. As $p \to \infty$, $\left(\frac{b}{a}\right)^p \to 0$. The limit becomes $a(1 + 0)^0 = a \cdot 1 = a$.

- **Case 2** $(b = a)$**:** $\frac{b}{a} = 1$. The expression is $a(1 + 1^p)^{1/p} = a(2)^{1/p}$. As $p \to \infty$, $1/p \to 0$, so $2^{1/p} \to 1$. The limit is $a \cdot 1 = a$.

In all cases, $\lim_{p\to\infty} \|v\|_p = a = \|v\|_\infty$.

73 Constructing New Norms

Proof Problem 3.3.17 & 3.3.18: Weighted Function Norms

Let $w(x) > 0$ for $x \in [a, b]$ be a continuous weight function. We verify the triangle inequality for weighted L^1 and L^∞ norms on $C^0[a, b]$.
(a) Weighted L^1 norm: $\|f\|_{1,w} = \int_a^b |f(x)|w(x)\,dx$. Since $|f(x) + g(x)| \le |f(x)| + |g(x)|$ for all real numbers, and $w(x) > 0$:

$$|f(x) + g(x)|w(x) \le |f(x)|w(x) + |g(x)|w(x)$$

Integrating both sides preserves the inequality:

$$\int_a^b |f(x) + g(x)|w(x)\,dx \le \int_a^b |f(x)|w(x)\,dx + \int_a^b |g(x)|w(x)\,dx$$

Thus, $\|f + g\|_{1,w} \le \|f\|_{1,w} + \|g\|_{1,w}$.
(b) Weighted L^∞ norm: $\|f\|_{\infty,w} = \max_{x\in[a,b]}\{|f(x)|w(x)\}$. Using the same scalar inequality:

$$|f(x) + g(x)|w(x) \le |f(x)|w(x) + |g(x)|w(x)$$

Since $|f(x)|w(x) \le \max(|f|w)$ and $|g(x)|w(x) \le \max(|g|w)$:

$$|f(x) + g(x)|w(x) \le \max_{x\in[a,b]}\{|f(x)|w(x)\} + \max_{x\in[a,b]}\{|g(x)|w(x)\}$$

Because the RHS is a constant upper bound that holds for all x, the maximum of the LHS must also be bounded by it:

$$\max\{|f(x) + g(x)|w(x)\} \le \max\{|f(x)|w(x)\} + \max\{|g(x)|w(x)\}$$

Theorem Problem 3.3.19: Combining Norms

Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two different valid norms on a vector space V.
(a) Prove that $\|v\| = \max\{\|v\|_1, \|v\|_2\}$ is a norm.

- *Positivity:* $\max\{\|v\|_1, \|v\|_2\} \ge 0$. If the max is 0, both are 0, meaning $v = 0$.
- *Homogeneity:* $\|\lambda v\| = \max\{|\lambda|\|v\|_1, |\lambda|\|v\|_2\} = |\lambda| \max\{\|v\|_1, \|v\|_2\} = |\lambda|\|v\|$.
- *Triangle Inequality:*

$$\|u + w\| = \max\{\|u + w\|_1, \|u + w\|_2\} \le \max\{\|u\|_1 + \|w\|_1, \|u\|_2 + \|w\|_2\}$$

  Since the maximum of sums is less than or equal to the sum of maximums:

$$\le \max\{\|u\|_1, \|u\|_2\} + \max\{\|w\|_1, \|w\|_2\} = \|u\| + \|w\|$$

(b) Does $\|v\| = \min\{\|v\|_1, \|v\|_2\}$ define a norm? No. It fails the triangle inequality. *Counter-example:* Let $V = \mathbb{R}^2$. Let the two norms be heavily weighted L^1 norms:

$$\|x\|_1 = |x_1| + 2|x_2| \quad \text{and} \quad \|x\|_2 = 2|x_1| + |x_2|$$

Consider the vectors $x = (2, 1)^T$ and $y = (1, 2)^T$.

- $\|x\|_1 = 2 + 2(1) = 4$, and $\|x\|_2 = 2(2) + 1 = 5 \implies \|x\| = \min(4, 5) = \mathbf{4}$.
- $\|y\|_1 = 1 + 2(2) = 5$, and $\|y\|_2 = 2(1) + 2 = 4 \implies \|y\| = \min(5, 4) = \mathbf{4}$.

Now consider their sum $x + y = (3, 3)^T$:

- $\|x + y\|_1 = 3 + 2(3) = 9$.
- $\|x + y\|_2 = 2(3) + 3 = 9$.
- $\|x + y\| = \min(9, 9) = \mathbf{9}$.

Checking the triangle inequality: $\|x+y\| \leq \|x\| + \|y\| \implies 9 \leq 4+4 \implies 9 \leq 8$. This is strictly false.
(c) Does $\|v\| = \frac{1}{2}(\|v\|_1 + \|v\|_2)$ define a norm? Yes. This is a linear combination of norms with positive coefficients. It trivially inherits positivity and homogeneity. The triangle inequality holds because:

$$\frac{1}{2}(\|u + w\|_1 + \|u + w\|_2) \leq \frac{1}{2}(\|u\|_1 + \|w\|_1 + \|u\|_2 + \|w\|_2) = \frac{1}{2}(\|u\|_1 + \|u\|_2) + \frac{1}{2}(\|w\|_1 + \|w\|_2)$$

74 Cholesky Decomposition (Lecture 6)

Let A be an $n \times n$ real symmetric positive definite matrix. We have proved that A is regular and all its pivots are strictly positive. Since A is regular, it has an LU decomposition:

$$A = LU$$

where L is a lower triangular matrix with 1's on the diagonal, and U is an upper triangular matrix. Note that the diagonal elements of U are precisely the pivots of A.

Because A is symmetric, we can further decompose U to isolate the pivots. We can write:

$$A = LDL^T$$

where D is a diagonal matrix whose entries are the pivots of A, and L^T is a unit upper triangular matrix.

Since A is positive definite, all the entries of the diagonal matrix D are strictly positive. Hence, we can take the (positive) square root of each of them to form a new diagonal matrix S which satisfies $S^2 = D$.

$$A = LDL^T = LSSL^T$$

Since S is a diagonal matrix, $S = S^T$, so we can rewrite this as:

$$A = LSS^TL^T = (LS)(LS)^T$$

Let $M = LS$. Because the product of two lower triangular matrices is lower triangular, M is a lower triangular matrix. Therefore:

$$A = MM^T$$

This factorization is called the **Cholesky Decomposition** of A.

Example Calculating the Cholesky Decomposition

Let $A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 6 & 0 \\ -1 & 0 & 9 \end{pmatrix}$. We have previously seen (Quiz 1) that A is a positive definite matrix.
First, we find the LU decomposition using Gaussian elimination.
Step 1: Eliminate entries below a_{11}

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 2 & 6 & 0 \\ -1 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 2 & 2 \\ 0 & 2 & 8 \end{pmatrix}$$

Step 2: Eliminate entry below a_{22}

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & -1 \\ 0 & 2 & 2 \\ 0 & 2 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 2 & 2 \\ 0 & 0 & 6 \end{pmatrix} = U$$

The pivots of A are $1, 2, 6$ (the diagonal of U).
Step 3: Construct L and S The matrix L contains the inverse of the elimination multipliers:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$$

The diagonal matrix of pivots is $D = \operatorname{diag}(1, 2, 6)$. The square root matrix is $S = \operatorname{diag}(1, \sqrt{2}, \sqrt{6}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{6} \end{pmatrix}$.
Step 4: Compute $M = LS$

$$M = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{6} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & \sqrt{2} & 0 \\ -1 & \sqrt{2} & \sqrt{6} \end{pmatrix}$$

Thus, the Cholesky decomposition is $A = MM^T$.

Application: Diagonalizing the Quadratic Form

The inner product on $\mathbb{R}^n$ given by A is $\langle x, y \rangle = x^T A y$, and the norm squared is $\|x\|^2 = x^T A x$. Using the Cholesky decomposition $A = MM^T$:

$$\|x\|^2 = x^T (MM^T) x = (M^T x)^T (M^T x)$$

If we define a new coordinate vector $z = M^T x$, then:

$$\|x\|^2 = z^T z = z_1^2 + z_2^2 + \cdots + z_n^2$$

Alternatively, using $A = LDL^T$, if we let $y = L^T x$:

$$\|x\|^2 = x^T LDL^T x = (L^T x)^T D (L^T x) = y^T D y = d_1 y_1^2 + d_2 y_2^2 + \cdots + d_n y_n^2$$

Both substitutions transform the complex quadratic form into a simple sum of squares.

75 Exercise Solutions: Unit Vectors and Spheres

Reference: Olver & Shakiban, Chapter 3, Pages 149-150.

Example Problem 3.3.20: Normalizing a Vector

Find a unit vector in the same direction as $v = (1, 2, -3)^T$ for the following norms. *Strategy:* We compute the norm $\|v\|$, then scale the vector by its reciprocal $u = \frac{v}{\|v\|}$.
(a) The Euclidean norm (L^2):

$$\|v\|_2 = \sqrt{1^2 + 2^2 + (-3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14} \implies u = \left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, -\frac{3}{\sqrt{14}} \right)^T$$

(b) The norm $\|v\| = \sqrt{2v_1^2 + v_2^2 + \frac{1}{3}v_3^2}$:

$$\|v\| = \sqrt{2(1) + 4 + \frac{1}{3}(9)} = \sqrt{2+4+3} = \sqrt{9} = 3 \implies u = \left(\frac{1}{3}, \frac{2}{3}, -1\right)^T$$

(c) The L^1-norm:

$$\|v\|_1 = |1| + |2| + |-3| = 6 \implies u = \left(\frac{1}{6}, \frac{2}{6}, -\frac{3}{6}\right)^T = \left(\frac{1}{6}, \frac{1}{3}, -\frac{1}{2}\right)^T$$

(d) The ∞-norm:

$$\|v\|_\infty = \max\{|1|, |2|, |-3|\} = 3 \implies u = \left(\frac{1}{3}, \frac{2}{3}, -1\right)^T$$

(e) The norm associated with $q(v) = 2v_1^2 - 2v_1v_2 + 2v_2^2 - 2v_2v_3 + 2v_3^2$:

$$\|v\|^2 = 2(1)^2 - 2(1)(2) + 2(2)^2 - 2(2)(-3) + 2(-3)^2 = 2 - 4 + 8 + 12 + 18 = 36$$

$$\|v\| = \sqrt{36} = 6 \implies u = \left(\frac{1}{6}, \frac{2}{6}, -\frac{3}{6}\right)^T = \left(\frac{1}{6}, \frac{1}{3}, -\frac{1}{2}\right)^T$$

Proof Problem 3.3.21: Spherical Coordinates

Show that for every choice of angles θ, φ, ψ, the following parameterizations yield unit vectors in the Euclidean norm.

(a) $v = (\cos\theta\cos\varphi, \cos\theta\sin\varphi, \sin\theta)^T$:

$$\begin{aligned}\|v\|^2 &= \cos^2\theta\cos^2\varphi + \cos^2\theta\sin^2\varphi + \sin^2\theta \\ &= \cos^2\theta(\cos^2\varphi + \sin^2\varphi) + \sin^2\theta \\ &= \cos^2\theta(1) + \sin^2\theta = 1\end{aligned}$$

(b) $v = \frac{1}{\sqrt{2}}(\cos\theta, \sin\theta, \cos\varphi, \sin\varphi)^T$:

$$\|v\|^2 = \frac{1}{2}(\cos^2\theta + \sin^2\theta + \cos^2\varphi + \sin^2\varphi) = \frac{1}{2}(1+1) = 1$$

(c) Hyperspherical coordinates in $\mathbb{R}^4$:

$$v = (\cos\theta\cos\varphi\cos\psi, \cos\theta\cos\varphi\sin\psi, \cos\theta\sin\varphi, \sin\theta)^T$$

$$\begin{aligned}\|v\|^2 &= \cos^2\theta\cos^2\varphi\cos^2\psi + \cos^2\theta\cos^2\varphi\sin^2\psi + \cos^2\theta\sin^2\varphi + \sin^2\theta \\ &= \cos^2\theta\cos^2\varphi(\cos^2\psi + \sin^2\psi) + \cos^2\theta\sin^2\varphi + \sin^2\theta \\ &= \cos^2\theta(\cos^2\varphi + \sin^2\varphi) + \sin^2\theta \\ &= \cos^2\theta + \sin^2\theta = 1\end{aligned}$$

Theorem Problem 3.3.22: Parallel Unit Vectors

How many unit vectors are parallel to a given nonzero vector v? **Answer: 2.** A vector w parallel to v must satisfy $w = \lambda v$ for some scalar $\lambda \in \mathbb{R}$. If w is a unit vector, then $\|w\| = 1$.

$$\|\lambda v\| = 1 \implies |\lambda|\|v\| = 1 \implies |\lambda| = \frac{1}{\|v\|}$$

Thus, $\lambda = \frac{1}{\|v\|}$ or $\lambda = -\frac{1}{\|v\|}$. The vectors are $\frac{v}{\|v\|}$ and $-\frac{v}{\|v\|}$.

Example Problem 3.3.23: Plotting 2D Unit Circles

Identify the shape of the unit circle $\|v\| = 1$ for the following norms:

- **(a)** $\|v\| = \sqrt{v_1^2 + 4v_2^2}$**:** The equation is $x^2 + 4y^2 = 1$. This forms an axis-aligned **ellipse** with an x-intercept at ± 1 and y-intercept at ± 0.5.
- **(b) The norm associated with** $\langle v, w\rangle = v_1w_1 - v_1w_2 - v_2w_1 + 4v_2w_2$**:** The equation is $x^2 - 2xy + 4y^2 = 1$. The cross-term $-2xy$ indicates this is a **tilted ellipse**.
- **(c)** $\|(x, y)\| = |x| + 2|x - y|$**:** The equation is $|x| + 2|x - y| = 1$. This forms a **skewed parallelogram** centered at the origin.

Solutions to Skipped Problems 3.3.24 & 3.3.25

3.3.24: Unit circles for Exercise 3.3.10 norms:

- *(a)* $\sqrt{2x^2 + 3y^2}$*:* Ellipse.
- *(b)* $\sqrt{2x^2 - xy + 2y^2}$*:* Tilted ellipse.
- *(c)* $2|x| + |y|$*:* Diamond (stretched L^1 circle).
- *(d)* $\max\{2|x|, |y|\}$*:* Rectangle (stretched L^∞ circle).

3.3.25: Shape of the unit sphere $S_1 \subset \mathbb{R}^3$:

- **(a)** L^1 **norm:** A regular **Octahedron**.
- **(b)** L^∞ **norm:** A **Cube**.
- **(c)** $\|v\| = \sqrt{v_1^2 + v_2^2 + 3v_3^2}$**:** An **Ellipsoid** (squashed along the z-axis).
- **(d)** $\max\{|v_1 + v_2|, |v_2 + v_3|, |v_1 + v_3|\}$**:** A convex polyhedron known as a **Rhombic Dodecahedron**.

Theorem Problem 3.3.26: Customizing the Unit Sphere

Let $v \neq 0$ be any nonzero vector in a normed vector space V. Construct a new norm on V that makes v a unit vector.

Proof

Let $\|\cdot\|$ be the original norm on V. We define a new norm by scaling the space:

$$\|w\|_{new} = \frac{\|w\|}{\|v\|} \quad \text{for all } w \in V$$

Since $\frac{1}{\|v\|}$ is a positive real constant, multiplying a valid norm by a positive constant strictly preserves Positivity, Homogeneity, and the Triangle Inequality. Thus, $\|\cdot\|_{new}$ is a valid norm. By definition, evaluating this norm on v gives:

$$\|v\|_{new} = \frac{\|v\|}{\|v\|} = 1$$

77 Uniqueness and Normalization in Function Spaces

Theorem Problem 3.3.27: Uniqueness of the Unit Sphere

True or False: Two norms on a vector space have the same unit sphere if and only if they are the same norm. **Answer: True.**

Proof

Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be two norms on V. Let S_1 and S_2 be their respective unit spheres. ($\Leftarrow$) If $\|\cdot\|_1 = \|\cdot\|_2$, then trivially $S_1 = S_2$. ($\Rightarrow$) Suppose $S_1 = S_2$. Let $y \in V$ be any non-zero vector. We can normalize y using the first norm: let $x = \frac{y}{\|y\|_1}$. By definition, $\|x\|_1 = 1$, which means $x \in S_1$. Since the spheres are identical, $x \in S_2$. This implies that $\|x\|_2 = 1$. Substitute x back into the equation:

$$\left\| \frac{y}{\|y\|_1} \right\|_2 = 1 \implies \frac{1}{\|y\|_1}\|y\|_2 = 1 \implies \|y\|_2 = \|y\|_1$$

Since this holds for all $y \in V$, the two norms are identical.

Example Problem 3.3.28: Normalizing a Function

Find the unit function (a constant multiple of $f(x)$) for $f(x) = x - 1/3$ on $[0, 1]$ with respect to various norms.

(a) The L^1 Norm:

$$\|f\|_1 = \int_0^1 \left| x - \frac{1}{3} \right| dx = \int_0^{1/3} \left(\frac{1}{3} - x \right) dx + \int_{1/3}^1 \left(x - \frac{1}{3} \right) dx$$

$$= \left[\frac{x}{3} - \frac{x^2}{2} \right]_0^{1/3} + \left[\frac{x^2}{2} - \frac{x}{3} \right]_{1/3}^1 = \left(\frac{1}{9} - \frac{1}{18} \right) + \left(\left(\frac{1}{2} - \frac{1}{3} \right) - \left(\frac{1}{18} - \frac{1}{9} \right) \right)$$

$$= \frac{1}{18} + \left(\frac{1}{6} + \frac{1}{18} \right) = \frac{1}{18} + \frac{4}{18} = \frac{5}{18}$$

The unit function is $g(x) = \frac{f(x)}{\|f\|_1} = \frac{18}{5}\left(x - \frac{1}{3}\right)$.

(b) The L^2 Norm:

$$\|f\|_2^2 = \int_0^1 \left(x - \frac{1}{3} \right)^2 dx = \left[\frac{(x - 1/3)^3}{3} \right]_0^1 = \frac{(2/3)^3}{3} - \frac{(-1/3)^3}{3} = \frac{8/27}{3} + \frac{1/27}{3} = \frac{9}{81} = \frac{1}{9}$$

Thus, $\|f\|_2 = 1/3$. The unit function is $g(x) = 3\left(x - \frac{1}{3}\right) = 3x - 1$.

(c) The L^∞ Norm:

$$\|f\|_\infty = \max_{x \in [0,1]} \left| x - \frac{1}{3} \right|$$

Evaluating at the endpoints: $|0 - 1/3| = 1/3$, and $|1 - 1/3| = 2/3$. The maximum is $2/3$. The unit function is $g(x) = \frac{3}{2}\left(x - \frac{1}{3}\right)$.

Example Problem 3.3.29: The Constant Function

For which standard norms on $[0,1]$ is the constant function $f(x)=1$ a unit function?

- **(a)** L^1 **norm:** $\|1\|_1 = \int_0^1 1\,dx = [x]_0^1 = 1$. **(Yes)**
- **(b)** L^2 **norm:** $\|1\|_2 = \sqrt{\int_0^1 1^2\,dx} = \sqrt{1} = 1$. **(Yes)**
- **(c)** L^∞ **norm:** $\|1\|_\infty = \max_{x\in[0,1]} |1| = 1$. **(Yes)**

Because the area of a 1×1 rectangle is 1, $f(x)=1$ serves as a natural unit vector for all unweighted standard norms on the unit interval.

78 Equivalence Inequalities in $\mathbb{R}^n$

Reference: Olver & Shakiban, Chapter 3, Pages 152-153.

Example Problem 3.3.31: Checking Equivalence Bounds

Check the validity of the inequalities $\frac{1}{\sqrt{n}}\|v\|_2 \le \|v\|_\infty \le \|v\|_2$ for the following vectors in $\mathbb{R}^n$:
(a) $v=(-1)$ **in** $\mathbb{R}^1$ $(n=1)$**:** $\|v\|_2 = 1$, $\|v\|_\infty = 1$. $\frac{1}{\sqrt{1}}(1) \le 1 \le 1 \implies 1 \le 1 \le 1$. (True)
(b) $v=(1,2,3)^T$ **in** $\mathbb{R}^3$ $(n=3)$**:** $\|v\|_2 = \sqrt{1^2+2^2+3^2} = \sqrt{14}$. $\|v\|_\infty = \max\{|1|,|2|,|3|\} = 3$. $\frac{1}{\sqrt{3}}\sqrt{14} \le 3 \le \sqrt{14} \implies 2.16 \le 3 \le 3.74$. (True)
(c) $v=(1,-1)^T$ **in** $\mathbb{R}^2$ $(n=2)$**:** $\|v\|_2 = \sqrt{2}$, $\|v\|_\infty = 1$. $\frac{1}{\sqrt{2}}\sqrt{2} \le 1 \le \sqrt{2} \implies 1 \le 1 \le 1.414$. (True)
(d) $v=(2,-2,0)^T$ **in** $\mathbb{R}^3$ $(n=3)$**:** $\|v\|_2 = \sqrt{4+4+0} = \sqrt{8}$. $\|v\|_\infty = 2$. $\frac{1}{\sqrt{3}}\sqrt{8} \le 2 \le \sqrt{8} \implies 1.63 \le 2 \le 2.82$. (True)

Theorem Problem 3.3.32: Conditions for Equality

Find all vectors $v=(v_1,v_2)^T \in \mathbb{R}^2$ such that the following equalities hold:
(a) $\|v\|_1 = \|v\|_\infty$

$$|v_1|+|v_2| = \max\{|v_1|,|v_2|\}$$

This holds if and only if one of the terms is zero. Thus, $v_1=0$ or $v_2=0$. (The vector lies on the x or y axis).
(b) $\|v\|_1 = \|v\|_2$ Squaring both sides:

$$(|v_1|+|v_2|)^2 = v_1^2+v_2^2 \implies v_1^2+v_2^2+2|v_1v_2| = v_1^2+v_2^2 \implies 2|v_1v_2| = 0$$

This implies $v_1=0$ or $v_2=0$.
(c) $\|v\|_2 = \|v\|_\infty$ Squaring both sides:

$$v_1^2+v_2^2 = \max\{v_1^2,v_2^2\}$$

If $|v_1|\ge|v_2|$, this becomes $v_1^2+v_2^2 = v_1^2 \implies v_2^2 = 0 \implies v_2 = 0$. Thus, $v_1=0$ or $v_2=0$.
(d) $\|v\|_\infty = \frac{1}{\sqrt{2}}\|v\|_2$ Squaring both sides:

$$\max\{v_1^2,v_2^2\} = \frac{1}{2}(v_1^2+v_2^2)$$

Without loss of generality, assume $v_1^2 \ge v_2^2$. Then $v_1^2 = \frac{1}{2}v_1^2+\frac{1}{2}v_2^2 \implies \frac{1}{2}v_1^2 = \frac{1}{2}v_2^2 \implies |v_1|=|v_2|$. (The vector lies on the diagonals $y=x$ or $y=-x$).

Problem 3.3.33: Quantifying "Small" Norms

Statement: The norm of a vector is small if and only if all its entries are small. **Proof:** By the equivalence of norms, there exist constants $m, M > 0$ such that for any valid norm $\|\cdot\|$ on $\mathbb{R}^n$:

$$m\|v\|_\infty \leq \|v\| \leq M\|v\|_\infty$$

Recall that $\|v\|_\infty = \max\{|v_1|, \dots, |v_n|\}$. If all entries are small, $\|v\|_\infty$ is small, and thus $\|v\| \leq M\|v\|_\infty$ is small. Conversely, if $\|v\|$ is small, the maximum entry is bounded by $\|v\|_\infty \leq \frac{1}{m}\|v\|$, forcing all individual components to be small.

79 Proving Equivalence Bounds

Proof Problem 3.3.34: L^2 and L^∞

Find an elementary proof of the inequalities $\|v\|_\infty \leq \|v\|_2 \leq \sqrt{n}\|v\|_\infty$.
Let $v = (v_1, \dots, v_n)^T \in \mathbb{R}^n$. Let $\|v\|_\infty = \max|v_i| = |v_k|$ for some specific index k. **Lower Bound:**

$$\|v\|_2^2 = v_1^2 + \cdots + v_k^2 + \cdots + v_n^2$$

Since all squared terms are non-negative, the sum is clearly greater than or equal to any single term:

$$\|v\|_2^2 \geq v_k^2 = \|v\|_\infty^2 \implies \|v\|_2 \geq \|v\|_\infty$$

Upper Bound: Since $|v_k|$ is the maximum absolute entry, $v_i^2 \leq v_k^2$ for all $i = 1, \dots, n$.

$$\|v\|_2^2 = \sum_{i=1}^n v_i^2 \leq \sum_{i=1}^n v_k^2 = nv_k^2 = n\|v\|_\infty^2$$

Taking the square root yields $\|v\|_2 \leq \sqrt{n}\|v\|_\infty$.

Proof Problem 3.3.35: L^1 and L^2

Show the equivalence of the Euclidean norm and 1-norm by proving $\|v\|_2 \leq \|v\|_1 \leq \sqrt{n}\|v\|_2$.
Lower Bound: Square both sides of $\|v\|_2 \leq \|v\|_1$:

$$v_1^2 + \cdots + v_n^2 \leq (|v_1| + \cdots + |v_n|)^2$$

Expanding the right side gives $\sum_{i=1}^n v_i^2 + 2\sum_{i<j}|v_i v_j|$. Since absolute values are non-negative, adding them to the sum of squares strictly increases (or maintains) the value. Thus, the inequality holds.
Upper Bound: We apply the Cauchy-Schwarz inequality to the standard dot product of two vectors. Let $x = (|v_1|, \dots, |v_n|)^T$ and let $y = (1, 1, \dots, 1)^T$.

$$\langle x, y \rangle \leq \|x\|_2 \|y\|_2$$

$$\sum_{i=1}^n |v_i|(1) \leq \sqrt{\sum_{i=1}^n |v_i|^2}\sqrt{\sum_{i=1}^n 1^2}$$

$$\|v\|_1 \leq \|v\|_2\sqrt{n}$$

Example Problem 3.3.36 & 3.3.37: Custom Equivalence Constants

1. Bounds for L^1 and L^∞:

$$\|v\|_\infty \leq \|v\|_1 \leq n\|v\|_\infty$$

Proof: $\max|v_i| \leq \sum_{i=1}^n |v_i| \leq \sum_{i=1}^n \max|v_i| = n\max|v_i|$.

2. Custom Norms vs L^2: Let $m\|v\| \le \|v\|_2 \le M\|v\|$.

- **(a)** $\|v\| = \sqrt{2v_1^2 + 3v_2^2}$. Observe that $2(v_1^2 + v_2^2) \le 2v_1^2 + 3v_2^2 \le 3(v_1^2 + v_2^2)$. Thus, $2\|v\|_2^2 \le \|v\|^2 \le 3\|v\|_2^2$. Rearranging gives: $\frac{1}{3}\|v\|^2 \le \|v\|_2^2 \le \frac{1}{2}\|v\|^2 \implies \frac{1}{\sqrt{3}}\|v\| \le \|v\|_2 \le \frac{1}{\sqrt{2}}\|v\|$.

- **(b)** $\|v\| = \max\{|v_1+v_2|, |v_1-v_2|\}$. Squaring the norm: $\|v\|^2 = \max\{(v_1+v_2)^2, (v_1-v_2)^2\} = v_1^2 + v_2^2 + 2|v_1v_2| = \|v\|_2^2 + 2|v_1v_2|$. Clearly, $\|v\|_2^2 \le \|v\|^2 \implies \|v\|_2 \le 1\|v\|$. (So $M = 1$). Also, $2|v_1v_2| \le v_1^2 + v_2^2 \implies \|v\|^2 \le 2(v_1^2 + v_2^2) = 2\|v\|_2^2 \implies \frac{1}{\sqrt{2}}\|v\| \le \|v\|_2$. (So $m = 1/\sqrt{2}$).

80 Equivalence Failure in Function Spaces

While all norms are equivalent in $\mathbb{R}^n$, this property completely fails in infinite-dimensional spaces.

Theorem Problem 3.3.39: Bounding L^2 by L^∞ on a Finite Interval

Prove that if $f \in C^0[a,b]$, then $\|f\|_2 \le \sqrt{b-a}\|f\|_\infty$.

Proof

By definition, $|f(x)| \le \max_{x\in[a,b]} |f(x)| = \|f\|_\infty$ for all $x \in [a,b]$. Squaring both sides: $f(x)^2 \le \|f\|_\infty^2$. Integrate both sides over $[a,b]$:

$$\|f\|_2^2 = \int_a^b f(x)^2\,dx \le \int_a^b \|f\|_\infty^2\,dx = \|f\|_\infty^2 \int_a^b 1\,dx = \|f\|_\infty^2(b-a)$$

Taking the square root yields $\|f\|_2 \le \sqrt{b-a}\|f\|_\infty$. *Note: This works because the interval length $(b-a)$ is finite. If the interval was $(-\infty,\infty)$, this bound would not exist.*

Example Problem 3.3.40: Counter-examples to Equivalence

We show that no constants exist to mutually bound the L^1 and L^2 norms for all continuous functions.

Part (a) & (b): Proving $\|f\|_2 \le C\|f\|_1$ is impossible. Let $f_n(x)$ be defined as:

$$f_n(x) = \begin{cases} 1 & \text{for } -\frac{1}{n} \le x \le \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

(Assume we smooth the edges slightly so it remains in C^0).

- $\|f_n\|_1 = \int_{-1/n}^{1/n} 1\,dx = \frac{2}{n}$. As $n\to\infty$, $\|f_n\|_1 \to 0$.
- $\|f_n\|_2 = \sqrt{\int_{-1/n}^{1/n} 1^2\,dx} = \sqrt{\frac{2}{n}}$. As $n\to\infty$, $\|f_n\|_2 \to 0$.

Suppose there exists a constant C such that $\|f_n\|_2 \le C\|f_n\|_1$ for all n.

$$\sqrt{\frac{2}{n}} \le C\left(\frac{2}{n}\right) \implies C \ge \frac{\sqrt{2/n}}{2/n} = \sqrt{\frac{n}{2}}$$

As $n\to\infty$, the right side goes to ∞. No finite constant C can satisfy this for all functions.

Part (c): Proving $\|f\|_1 \le C\|f\|_2$ is impossible. Let $f_n(x)$ be defined as:

$$f_n(x) = \begin{cases} \frac{1}{\sqrt{2n}} & \text{for } -n \le x \le n \\ 0 & \text{otherwise} \end{cases}$$

- $\|f_n\|_1 = \int_{-n}^{n} \frac{1}{\sqrt{2n}}\,dx = \frac{1}{\sqrt{2n}}(2n) = \sqrt{2n}$.

- $\|f_n\|_2 = \sqrt{\int_{-n}^{n} \left(\frac{1}{\sqrt{2n}}\right)^2 dx} = \sqrt{\int_{-n}^{n} \frac{1}{2n} dx} = \sqrt{\frac{2n}{2n}} = 1.$

Suppose there exists a constant C such that $\|f_n\|_1 \leq C\|f_n\|_2$ for all n.

$$\sqrt{2n} \leq C(1) \implies C \geq \sqrt{2n}$$

As $n \to \infty$, C must be infinite. This is a contradiction. *Conclusion:* In infinite-dimensional spaces, norms are strictly **not equivalent**.

81 Equivalence Failure in Infinite Dimensions

Reference: Olver & Shakiban, Chapter 3.

We have seen that all norms are equivalent in finite-dimensional spaces like $\mathbb{R}^n$. However, this property breaks down in infinite-dimensional function spaces.

Example Problem 3.3.41: L^∞ vs. Integral Norms

Prove that the L^∞ norm is not equivalent to the L^2 or L^1 norms on the space $C^0[-1,1]$.
The Counter-Example: Let us construct a sequence of "tent" functions $f_n(x)$ defined by:

$$f_n(x) = \begin{cases} n - n^2|x| & \text{for } -\frac{1}{n} \leq x \leq \frac{1}{n} \\ 0 & \text{otherwise} \end{cases}$$

This represents a triangular spike centered at $x = 0$ with a height of n and a base width of $2/n$. Let us calculate its norms:

- **L^∞ Norm:** The maximum value is exactly the peak height: $\|f_n\|_\infty = n$.
- **L^1 Norm:** This is the area of the triangle: $\|f_n\|_1 = \frac{1}{2}(\text{base})(\text{height}) = \frac{1}{2}\left(\frac{2}{n}\right)(n) = 1$.
- **L^2 Norm:** We integrate the square of the function:

$$\|f_n\|_2^2 = \int_{-1/n}^{1/n} (n - n^2|x|)^2\, dx = 2\int_0^{1/n} (n - n^2 x)^2\, dx$$

$$= 2\left[\frac{(n-n^2x)^3}{-3n^2}\right]_0^{1/n} = 2\left(0 - \frac{n^3}{-3n^2}\right) = \frac{2n}{3} \implies \|f_n\|_2 = \sqrt{\frac{2n}{3}}$$

(a) Why L^∞ and L^2 are not equivalent: If they were equivalent, there would exist a constant $C > 0$ such that $\|f_n\|_\infty \leq C\|f_n\|_2$ for all n. Substituting our values:

$$n \leq C\sqrt{\frac{2n}{3}} \implies \sqrt{n} \leq C\sqrt{\frac{2}{3}}$$

As $n \to \infty$, the left side grows without bound, making it impossible for any finite constant C to satisfy the inequality.
(c) Why L^∞ and L^1 are not equivalent: Similarly, assume $\|f_n\|_\infty \leq C\|f_n\|_1$.

$$n \leq C(1)$$

Again, as $n \to \infty$, no constant C can bound n. Thus, equivalence fails.
(b) Bounding Integrals by the Maximum: While L^∞ cannot be bounded by the integral norms, the integral norms *can* be bounded by L^∞ on a finite interval $[a, b]$.

$$\|f\|_2^2 = \int_{-1}^{1} f(x)^2\, dx \leq \int_{-1}^{1} \|f\|_\infty^2\, dx = 2\|f\|_\infty^2 \implies \|f\|_2 \leq \sqrt{2}\|f\|_\infty$$

$$\|f\|_1 = \int_{-1}^{1} |f(x)|\, dx \leq \int_{-1}^{1} \|f\|_\infty\, dx = 2\|f\|_\infty$$

82 Linear Combinations of Inner Products

Problem 3.3.42: Equal Equivalence Constants

Suppose two norms satisfy $m\|v\|_1 \leq \|v\|_2 \leq M\|v\|_1$ for all vectors v. What does it mean if $m = M$? **Answer:** If the constants are equal, the inequality collapses to an equality: $M\|v\|_1 \leq \|v\|_2 \leq M\|v\|_1$, which strictly implies $\|v\|_2 = M\|v\|_1$. This means the two norms are simply positive scalar multiples of one another.

Theorem Problem 3.3.43: Mixing Inner Products

Suppose $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ are two legitimate inner products on a finite-dimensional space V. For which $\alpha, \beta \in \mathbb{R}$ is the linear combination:

$$\langle v, w \rangle = \alpha \langle v, w \rangle_1 + \beta \langle v, w \rangle_2$$

also a legitimate inner product?

Proof

Bilinearity and symmetry are linear properties and are trivially preserved for any $\alpha, \beta \in \mathbb{R}$. We must check **Positivity**: We require $\langle v, v \rangle > 0$ for all $v \neq 0$.

$$\alpha \|v\|_1^2 + \beta \|v\|_2^2 > 0$$

Since V is finite-dimensional, the two induced norms are equivalent. There exist strict, optimal bounds $m^2, M^2 > 0$ such that:

$$m^2 \|v\|_1^2 \leq \|v\|_2^2 \leq M^2 \|v\|_1^2$$

Since $v \neq 0$, we can divide the positivity requirement by $\|v\|_1^2$ to get:

$$\alpha + \beta \frac{\|v\|_2^2}{\|v\|_1^2} > 0$$

Let $Q(v) = \frac{\|v\|_2^2}{\|v\|_1^2}$ (this is akin to a Rayleigh quotient). We know $Q(v) \in [m^2, M^2]$. Thus, we require the linear function $L(Q) = \alpha + \beta Q > 0$ for all $Q \in [m^2, M^2]$. A linear function is strictly positive on an interval if and only if it is positive at both endpoints. Therefore, the conditions on α and β are:

$$\alpha + \beta m^2 > 0 \quad \text{and} \quad \alpha + \beta M^2 > 0$$

(Note: If $\alpha > 0, \beta > 0$, this is trivially satisfied. If signs are mixed, it dictates how "strong" the negative coefficient can be before breaking positivity.)

82.1 Equivalence of Matrix Norms

Proof Problem 3.3.44: Equivalence on $M_n(\mathbb{R})$

Suppose $\|\cdot\|_1$ and $\|\cdot\|_2$ are two valid norms on $\mathbb{R}^n$. Prove that the corresponding induced matrix norms satisfy $m\|A\|_1 \leq \|A\|_2 \leq M\|A\|_1$ for some positive constants $m \leq M$, for all $n \times n$ matrices A.
Proof: The space of all real $n \times n$ matrices, denoted $M_n(\mathbb{R})$, is a vector space of dimension n^2 under standard matrix addition and scalar multiplication. Because it has a finite dimension, it is isomorphic to $\mathbb{R}^{n^2}$. A fundamental theorem of linear algebra states that **all valid norms on a finite-dimensional vector space are equivalent**. Since the induced matrix norms $\|\cdot\|_1$ and $\|\cdot\|_2$ both satisfy the axioms of a norm on the space $M_n(\mathbb{R})$, they must be equivalent. Therefore, there strictly exist constants $0 < m \leq M$ such that:

$$m\|A\|_1 \leq \|A\|_2 \leq M\|A\|_1 \quad \forall A \in M_n(\mathbb{R})$$

83 Exercise Solutions: Matrix Norms

Reference: Olver & Shakiban, Chapter 3, Pages 155-156.

We will utilize the fact that the matrix norm induced by the ∞-norm on $\mathbb{R}^n$ (where $\|v\|_\infty = \max |v_i|$) is given by the **Maximum Absolute Row Sum**:

$$\|A\|_\infty = \max_{1\le i\le n} \left\{ \sum_{j=1}^{n} |a_{ij}| \right\}$$

83.1 Computing Infinity Norms

Example Problem 3.3.45: Computing $\|A\|_\infty$

Compute the ∞-matrix norm for the following matrices:

- **(a)** $A = \begin{pmatrix} 1/2 & 1/4 \\ 1/3 & 1/6 \end{pmatrix}$

$$\|A\|_\infty = \max\left\{ \left|\frac{1}{2}\right| + \left|\frac{1}{4}\right|, \left|\frac{1}{3}\right| + \left|\frac{1}{6}\right| \right\} = \max\left\{\frac{3}{4}, \frac{1}{2}\right\} = \frac{3}{4}$$

- **(b)** $A = \begin{pmatrix} 5/3 & 4/3 \\ -7/6 & -5/6 \end{pmatrix}$

$$\|A\|_\infty = \max\left\{ \frac{5}{3} + \frac{4}{3}, \frac{7}{6} + \frac{5}{6} \right\} = \max\{3, 2\} = 3$$

- **(c)** $A = \begin{pmatrix} 0 & .1 & .8 \\ -0.1 & 0 & .1 \\ -.8 & -.1 & 0 \end{pmatrix}$ By inspection, the absolute row sums are $0.9, 0.2$, and 0.9. Thus, $\|A\|_\infty = 0.9$.

- **(d)** $A = \begin{pmatrix} 1/3 & 0 & 0 \\ -1/3 & 0 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix}$ The absolute row sums are $1/3, 2/3$, and 1. Thus, $\|A\|_\infty = 1$.

Example Problem 3.3.46 & 3.3.47: Counter-examples in Matrix Norms

3.3.46: Find a matrix A such that $\|A^2\|_\infty \neq \|A\|_\infty^2$. We know from sub-multiplicativity that $\|A^2\|_\infty \le \|A\|_\infty^2$. We seek a strict inequality. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. The row sums are 3 and 7, so $\|A\|_\infty = 7$. Thus, $\|A\|_\infty^2 = 49$. Now compute A^2:

$$A^2 = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$$

The row sums of A^2 are 17 and 37. Thus, $\|A^2\|_\infty = 37$. Since $37 \neq 49$, we have found our matrix.
3.3.47: True or False: If $B = S^{-1}AS$, then $\|B\|_\infty = \|A\|_\infty$. Answer: False. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ as above ($\|A\|_\infty = 7$). Choose $S = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$, so $S^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix}$.

$$B = S^{-1}AS = \begin{pmatrix} 1 & 0 \\ 0 & 1/2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 3/2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 3/2 & 4 \end{pmatrix}$$

The row sums of B are 5 and 5.5. Thus $\|B\|_\infty = 5.5$. Clearly, $7 \neq 5.5$.

84 The 1-Matrix Norm

Theorem Problem 3.3.48 (i): Formula for the 1-Matrix Norm

Find an explicit formula for the matrix norm induced by the L^1 vector norm. **Claim:** $\|A\|_1$ is the **Maximum Absolute Column Sum**:

$$\|A\|_1 = \max_{1\le j\le n}\left\{\sum_{i=1}^{n}|a_{ij}|\right\}$$

Proof

Let $A \in M_n(\mathbb{R})$. Let A_j denote the j-th column vector of A. For any vector $x = (x_1, \dots, x_n)^T$, the matrix-vector product can be written as a linear combination of the columns of A:

$$Ax = x_1A_1 + x_2A_2 + \cdots + x_nA_n$$

By definition, $\|A\|_1 = \max_{\|x\|_1=1}\|Ax\|_1$. Taking the 1-norm of Ax:

$$\|Ax\|_1 = \|x_1A_1 + \cdots + x_nA_n\|_1$$

By the triangle inequality and homogeneity of the vector norm:

$$\|Ax\|_1 \le |x_1|\|A_1\|_1 + \cdots + |x_n|\|A_n\|_1$$

Let $M = \max_{1\le j\le n}\|A_j\|_1$. We can bound the expression by replacing each column norm with M:

$$\|Ax\|_1 \le (|x_1| + \cdots + |x_n|)M = \|x\|_1 M$$

Since we are constrained to unit vectors ($\|x\|_1 = 1$), we have $\|Ax\|_1 \le M$. Thus, $\|A\|_1 \le M$.
Attainability: To show the bound is achieved, suppose the maximum column sum occurs at column k (so $M = \|A_k\|_1$). Choose the specific test vector $x = e_k$ (the standard basis vector with a 1 in the k-th position and 0 elsewhere). Clearly $\|e_k\|_1 = 1$. The product Ae_k simply extracts the k-th column:

$$\|Ae_k\|_1 = \|A_k\|_1 = M$$

Thus, the maximum value is precisely M, proving $\|A\|_1 = \max_{1\le j\le n}\sum_{i=1}^{n}|a_{ij}|$.

Example Problem 3.3.48 (ii): Computing $\|A\|_1$

Compute the 1-matrix norm for the matrices from Exercise 3.3.45 by finding the maximum column sum.

- **(a)** $A = \begin{pmatrix} 1/2 & 1/4 \\ 1/3 & 1/6 \end{pmatrix} \implies$ Col sums: $\left\{\frac{5}{6}, \frac{5}{12}\right\} \implies \|A\|_1 = \frac{5}{6}$
- **(b)** $A = \begin{pmatrix} 5/3 & 4/3 \\ -7/6 & -5/6 \end{pmatrix} \implies$ Col sums: $\left\{\frac{10}{6} + \frac{7}{6}, \frac{8}{6} + \frac{5}{6}\right\} = \left\{\frac{17}{6}, \frac{13}{6}\right\} \implies \|A\|_1 = \frac{17}{6}$
- **(c)** $A = \begin{pmatrix} 0 & .1 & .8 \\ -0.1 & 0 & .1 \\ -.8 & -.1 & 0 \end{pmatrix} \implies$ Col sums: $\{0.9, 0.2, 0.9\} \implies \|A\|_1 = 0.9$
- **(d)** $A = \begin{pmatrix} 1/3 & 0 & 0 \\ -1/3 & 0 & 1/3 \\ 0 & 2/3 & 1/3 \end{pmatrix} \implies$ Col sums: $\left\{\frac{2}{3}, \frac{2}{3}, \frac{2}{3}\right\} \implies \|A\|_1 = \frac{2}{3}$

84.1 Direct Proofs of Norm Axioms

Proof Problem 3.3.49: Proving Axioms for Maximum Row Sum

Prove directly from the axioms that $\|A\|_\infty = \max_{1 \le i \le n} \left\{ \sum_{j=1}^n |a_{ij}| \right\}$ defines a valid norm on the space of $n \times n$ matrices.

1. Positivity: Since absolute values are non-negative, the sum of absolute values is non-negative, and the maximum of such sums is non-negative. Hence $\|A\|_\infty \ge 0$. Suppose $\|A\|_\infty = 0$.

$$\max_{1 \le i \le n} \sum_{j=1}^n |a_{ij}| = 0$$

Since it's a sum of non-negative terms, this forces $\sum_{j=1}^n |a_{ij}| = 0$ for all rows i. This in turn forces $|a_{ij}| = 0$ for all i and j, meaning $a_{ij} = 0$. Thus, A is the zero matrix.

2. Homogeneity:

$$\|cA\|_\infty = \max_{1 \le i \le n} \sum_{j=1}^n |ca_{ij}| = \max_{1 \le i \le n} \sum_{j=1}^n |c||a_{ij}|$$

Factoring out the constant $|c|$ from the sum and then the max function:

$$= |c| \max_{1 \le i \le n} \sum_{j=1}^n |a_{ij}| = |c|\|A\|_\infty$$

3. Triangle Inequality:

$$\|A + B\|_\infty = \max_{1 \le i \le n} \sum_{j=1}^n |a_{ij} + b_{ij}|$$

By the scalar triangle inequality $|a + b| \le |a| + |b|$:

$$\le \max_{1 \le i \le n} \sum_{j=1}^n (|a_{ij}| + |b_{ij}|) = \max_{1 \le i \le n} \left(\sum_{j=1}^n |a_{ij}| + \sum_{j=1}^n |b_{ij}| \right)$$

Since the maximum of a sum is less than or equal to the sum of the maximums:

$$\le \max_{1 \le i \le n} \sum_{j=1}^n |a_{ij}| + \max_{1 \le i \le n} \sum_{j=1}^n |b_{ij}| = \|A\|_\infty + \|B\|_\infty$$

85 Custom Matrix Norms

Example Problem 3.3.50: Computing Custom Induced Norms

Let $A = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$. Compute the natural matrix norm induced by custom vector norms.

(a) Induced by $\|v\| = \max\{2|v_1|, 3|v_2|\}$: We seek to maximize $\|Ax\|$ subject to $\|x\| = 1$. Let $x = (x_1, x_2)^T$.

$$Ax = \begin{pmatrix} x_1 - x_2 \\ x_1 + 2x_2 \end{pmatrix}$$

$$\|Ax\| = \max\{2|x_1 - x_2|, 3|x_1 + 2x_2|\}$$

We are constrained by $\|x\| = \max\{2|x_1|, 3|x_2|\} = 1$, which implies $2|x_1| \le 1 \implies |x_1| \le 1/2$, and $3|x_2| \le 1 \implies |x_2| \le 1/3$. Let's bound the terms of $\|Ax\|$:

- $2|x_1 - x_2| \le 2(|x_1| + |x_2|) \le 2(1/2 + 1/3) = 2(5/6) = 5/3$.
- $3|x_1 + 2x_2| \le 3(|x_1| + 2|x_2|) \le 3(1/2 + 2/3) = 3(7/6) = 7/2$.

Since $7/2 > 5/3$, the maximum possible value is $\mathbf{7/2}$. This is attained when $x_1 = 1/2$ and $x_2 = 1/3$ (which satisfies $\|x\| = 1$).
(b) Induced by $\|v\| = 2|v_1| + 3|v_2|$:

$$\|Ax\| = 2|x_1 - x_2| + 3|x_1 + 2x_2|$$

Subject to $2|x_1| + 3|x_2| = 1$. Let us substitute $u = 2|x_1|$ and $v = 3|x_2|$, so $u + v = 1$. We rewrite $\|Ax\|$ to bound it:

$$\|Ax\| \le 2|x_1| + 2|x_2| + 3|x_1| + 6|x_2| = 5|x_1| + 8|x_2|$$

Expressing this in terms of u and v:

$$= \frac{5}{2}u + \frac{8}{3}v$$

Since $u + v = 1$, this is a convex combination of $5/2$ and $8/3$. The maximum value occurs at the boundary where the coefficient is largest. Since $8/3 > 5/2$, the maximum is $\mathbf{8/3}$, attained when $u = 0, v = 1$ (i.e., $x_1 = 0, x_2 = 1/3$).

86 The Frobenius Norm and Sub-Multiplicativity

Theorem Problem 3.3.51: The Frobenius Norm

The Frobenius norm of an $n \times n$ matrix A is defined as:

$$\|A\|_F = \sqrt{\sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}^2}$$

Prove that this defines a valid matrix norm.

Proof

The Frobenius norm is mathematically equivalent to flattening the matrix into an n^2-dimensional vector and taking the standard Euclidean L^2 norm. Therefore, the properties of **Positivity**, **Homogeneity**, and the **Triangle Inequality** automatically follow from the standard vector space $\mathbb{R}^{n^2}$.
To be a true matrix norm, it must additionally satisfy the **Multiplicative Identity (Sub-multiplicativity)**:

$$\|AB\|_F \le \|A\|_F \|B\|_F$$

Let $C = AB$. The entry c_{ij} is the dot product of the i-th row of A (let's call it R_i) and the j-th column of B (C_j).

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

By the Cauchy-Schwarz inequality for standard vectors:

$$c_{ij}^2 = \left(\sum_{k=1}^{n} a_{ik} b_{kj}\right)^2 \le \left(\sum_{k=1}^{n} a_{ik}^2\right)\left(\sum_{k=1}^{n} b_{kj}^2\right)$$

Now, compute $\|AB\|_F^2$ by summing over all entries i and j:

$$\|AB\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} c_{ij}^2 \le \sum_{i=1}^{n}\sum_{j=1}^{n}\left(\sum_{k=1}^{n} a_{ik}^2\right)\left(\sum_{m=1}^{n} b_{mj}^2\right)$$

Notice that the summations decouple perfectly:

$$= \left(\sum_{i=1}^{n}\sum_{k=1}^{n} a_{ik}^2\right)\left(\sum_{j=1}^{n}\sum_{m=1}^{n} b_{mj}^2\right) = \|A\|_F^2 \|B\|_F^2$$

Taking the square root completes the proof.

Example Problem 3.3.52: A Vector Norm that is NOT a Matrix Norm

Explain why $\|A\| = \max|a_{ij}|$ defines a norm on the space of matrices, but show by counter-example that it is *not* a matrix norm.
Explanation: The formula $\|A\| = \max|a_{ij}|$ is perfectly valid as a norm because it is simply the L^∞ vector norm applied to the flattened n^2-dimensional space. It satisfies Positivity, Homogeneity, and the Triangle Inequality.
Counter-example: To be a *matrix norm*, it must satisfy sub-multiplicativity ($\|AB\| \leq \|A\|\|B\|$).
Let $A = B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$. The maximum entry in A is 4, so $\|A\| = 4$.

$$AB = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix}$$

The maximum entry in AB is 22, so $\|AB\| = 22$. Checking the sub-multiplicative condition:

$$\|AB\| \leq \|A\|\|B\| \implies 22 \leq (4)(4) \implies 22 \leq 16$$

This is clearly **false**. Therefore, the maximum entry formula is not a valid matrix norm.

87 Exercise Solutions: Positive Definite Matrices

Reference: Olver & Shakiban, Chapter 3, Pages 159-160.

These exercises explore the algebraic and geometric properties of positive definite matrices. Recall that a matrix K is strictly positive definite if it is symmetric ($K = K^T$) and its quadratic form $q(x) = x^T K x > 0$ for all $x \neq 0$.

87.1 Identifying Positive Definiteness

Example Problem 3.4.1: Checking 2×2 Matrices

Which of the following 2×2 matrices are positive definite?

- **(a)** $\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$: **True definite.** It is symmetric, and $q(x) = x_1^2 + 2x_2^2 > 0$.
- **(b)** $\begin{pmatrix} 0 & 1 \\ 2 & 0 \end{pmatrix}$: **Not true definite.** It is not symmetric ($1 \neq 2$).
- **(c)** $\begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$: **Not true definite.** Symmetric, but determinant is $1 - 4 = -3 < 0$.
- **(d)** $\begin{pmatrix} 5 & 3 \\ 3 & -2 \end{pmatrix}$: **Not true definite.** Diagonal contains a negative number, and det is $-10 - 9 = -19 < 0$.
- **(e)** $\begin{pmatrix} 1 & -1 \\ -1 & 3 \end{pmatrix}$: **True definite.** Symmetric, $a_{11} = 1 > 0$, and det is $3 - 1 = 2 > 0$. The inner product is $\langle x, y \rangle = x_1y_1 + 3x_2y_2 - x_1y_2 - x_2y_1$.
- **(f)** $\begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}$: **Not true definite.** It is not symmetric ($1 \neq -1$).

Proof Problem 3.4.2: Proving Indefiniteness

Let $K = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. Prove the associated quadratic form is indefinite by finding points x^+ and x^- that yield positive and negative values, respectively.
The quadratic form is:
$$q(x) = x^T K x = x_1^2 + 4x_1x_2 + 3x_2^2$$
We can complete the square to understand its behavior:
$$q(x) = (x_1^2 + 4x_1x_2 + 4x_2^2) - x_2^2 = (x_1 + 2x_2)^2 - x_2^2$$
To find a strictly positive result, let $x_2 = 0$ to eliminate the negative term. Choose $x^+ = e_1 = (1, 0)^T \implies q(x^+) = (1+0)^2 - 0 = 1 > 0$.
To find a strictly negative result, we want the first term to be zero. Set $x_1 + 2x_2 = 0 \implies x_1 = -2x_2$. Choose $x^- = (-2, 1)^T \implies q(x^-) = 0^2 - (1)^2 = -1 < 0$. Since the form takes both positive and negative values, it is indefinite.

Theorem Problem 3.4.3: Diagonal Matrices

Prove that a diagonal matrix $D = \operatorname{diag}(c_1, \ldots, c_n)$ is positive definite if and only if $c_i > 0$ for all $1 \le i \le n$.

Proof

The quadratic form associated with a diagonal matrix is:
$$q(x) = x^T D x = c_1x_1^2 + c_2x_2^2 + \cdots + c_nx_n^2$$
($\Leftarrow$) If all $c_i > 0$, then for any $x \neq 0$, at least one $x_i^2 > 0$. Thus, the sum of positive terms is strictly positive, $q(x) > 0$. ($\Rightarrow$) Assume D is positive definite. Then $q(x) > 0$ for all $x \neq 0$. If we test the standard basis vector $x = e_i$ (where the i-th entry is 1 and all others are 0), we get:
$$q(e_i) = c_i(1)^2 = c_i$$
Since $e_i \neq 0$, the definition of positive definiteness forces $q(e_i) > 0$, hence $c_i > 0$ for all i. The associated inner product is the weighted Euclidean dot product: $\langle x, y\rangle = \sum_{i=1}^n c_ix_iy_i$.

Example Problem 3.4.4: Abstract vs. Explicit Inequalities

Write out the explicit Cauchy-Schwarz and Triangle inequalities for the inner product defined by $K = \begin{pmatrix} 4 & -2 \\ -2 & 3 \end{pmatrix}$.
The inner product is $\langle x, y\rangle = 4x_1y_1 + 3x_2y_2 - 2x_1y_2 - 2x_2y_1$. The norm is $\|x\| = \sqrt{4x_1^2 + 3x_2^2 - 4x_1x_2}$.
Cauchy-Schwarz ($|\langle x, y\rangle| \le \|x\|\|y\|$):
$$|4x_1y_1 + 3x_2y_2 - 2x_1y_2 - 2x_2y_1| \le \sqrt{4x_1^2 + 3x_2^2 - 4x_1x_2}\sqrt{4y_1^2 + 3y_2^2 - 4y_1y_2}$$
Triangle Inequality ($\|x + y\| \le \|x\| + \|y\|$):
$$\sqrt{4(x_1 + y_1)^2 + 3(x_2 + y_2)^2 - 4(x_1 + y_1)(x_2 + y_2)} \le \sqrt{4x_1^2 + 3x_2^2 - 4x_1x_2} + \sqrt{4y_1^2 + 3y_2^2 - 4y_1y_2}$$

88 Traps and Operations

Theorem Problem 3.4.5: The Diagonal and Positive Entries

1. **Prove that every diagonal entry of a true definite matrix must be positive.** Let a_{ii} be the i-th diagonal entry of A. Consider the standard basis vector e_i. The quadratic form yields $e_i^T A e_i = a_{ii}$. Since $e_i \neq 0$, positive definiteness guarantees $e_i^T A e_i > 0$, thus $a_{ii} > 0$.

2. **Give an example of a symmetric matrix with all positive entries that is NOT positive definite.** Consider $A = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. The quadratic form is $q(x) = x_1^2 + 4x_1x_2 + x_2^2$. If we choose $x = (1, -1)^T$, then $q(1, -1) = 1 - 4 + 1 = -2 < 0$. It is indefinite.

3. **Find a nonzero matrix with one or more zero diagonal entries that is positive semidefinite.** Consider $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$. Then $q(x) = x_1^2$. Clearly $x_1^2 \geq 0$ for all x, making it positive semidefinite. However, if $x = (0, 1)^T \neq 0$, then $q(x) = 0$, meaning it is not strictly positive definite.

Proof Problem 3.4.6 & 3.4.7: Matrix Scaling and Addition

1. Scalar Multiplication: If K is positive definite and $c > 0$, then cK is positive definite. *Proof:* $x^T(cK)x = c(x^T Kx)$. Since $c > 0$ and $x^T Kx > 0$ for all $x \neq 0$, their product is strictly > 0.
2. Matrix Addition: If K and L are positive definite, then $K + L$ is positive definite. *Proof:* $x^T(K + L)x = x^T Kx + x^T Lx$. Since both individual forms are strictly positive for $x \neq 0$, their sum is strictly positive.
3. Sum of Indefinite Matrices: Can the sum of two non-positive definite matrices be positive definite? *Yes.* Consider the indefinite matrices $\begin{pmatrix} -2 & 0 \\ 0 & 4 \end{pmatrix}$ and $\begin{pmatrix} 4 & 0 \\ 0 & -2 \end{pmatrix}$. Their sum is $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, which is strictly positive definite.

Example Problem 3.4.8: Product of Positive Definite Matrices

Find two positive definite matrices A and B whose product AB is **not** positive definite.
Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Both are symmetric and positive definite (by Sylvester's Criterion). Compute their product:

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 4 \end{pmatrix}$$

Notice that the resulting matrix is **not symmetric** (the $(1, 2)$ entry is 1, but the $(2, 1)$ entry is 2). Since symmetry is a strict prerequisite for positive definiteness, AB cannot be positive definite.

89 Inverses and Geometric Interpretation

Theorem Problem 3.4.10: The Inverse of a Positive Definite Matrix

Let K be a non-singular symmetric matrix. If K is positive definite, prove that its inverse K^{-1} is also positive definite.

Proof

First, we confirm K^{-1} is symmetric. Since $K = K^T$, taking the inverse of both sides gives $K^{-1} = (K^T)^{-1} = (K^{-1})^T$. Now we evaluate the quadratic form for the inverse: $x^T K^{-1} x$. Let $x \in \mathbb{R}^n$ be any non-zero vector ($x \neq 0$). Because K is non-singular (invertible), we can define a unique vector y such that $y = K^{-1}x$. Crucially, since $x \neq 0$ and the matrix is invertible, $y \neq 0$. Now, substitute $x = Ky$ into the quadratic form:

$$\begin{aligned} x^T K^{-1} x &= (Ky)^T K^{-1}(Ky) \\ &= y^T K^T K^{-1} K y \\ \text{Since } K \text{ is symmetric } (K^T = K): & \\ &= y^T K K^{-1} K y \\ &= y^T I K y = y^T K y \end{aligned}$$

Since K is known to be positive definite and $y \neq 0$, we know $y^T K y > 0$. Therefore, $x^T K^{-1} x > 0$ for all $x \neq 0$, proving K^{-1} is positive definite.

Theorem Problem 3.4.11: The Geometric Angle of Positive Definiteness

Prove that an $n \times n$ symmetric matrix K is positive definite if and only if for every non-zero vector $v \in \mathbb{R}^n$, the input vector v and the output vector Kv meet at an acute Euclidean angle θ (meaning $|\theta| < \pi/2$).

Proof

Let θ be the standard Euclidean angle between v and Kv. By the geometric definition of the standard dot product:

$$\cos(\theta) = \frac{v \cdot (Kv)}{\|v\| \|Kv\|}$$

Notice that the numerator is exactly the quadratic form: $v \cdot (Kv) = v^T K v$.

$$\cos(\theta) = \frac{v^T K v}{\|v\| \|Kv\|}$$

Since the norms in the denominator are strictly positive, the sign of $\cos(\theta)$ is entirely determined by the numerator $v^T K v$.

- ($\Rightarrow$) If K is positive definite, $v^T K v > 0$. Thus $\cos(\theta) > 0$. The only angles in $[0, \pi]$ with a positive cosine are in the range $[0, \pi/2)$, meaning the angle is strictly acute.
- ($\Leftarrow$) If the angle is always acute, $\cos(\theta) > 0$, forcing the numerator $v^T K v > 0$ for all $v \neq 0$, which is the exact definition of positive definiteness.

90 Polarization and Symmetrization

Reference: Olver & Shakiban, Chapter 3.

Theorem Problem 3.4.12: The Polarization Formula

Prove that the inner product associated with a positive definite quadratic form $q(x)$ is given by the polarization formula:

$$\langle x, y \rangle = \frac{1}{2} \left[q(x + y) - q(x) - q(y) \right]$$

Proof

By definition, the quadratic form is the inner product of a vector with itself: $q(v) = \langle v, v \rangle$. Expand the term $q(x+y)$ using the bilinearity and symmetry of the inner product:

$$\begin{aligned} q(x+y) &= \langle x+y, x+y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= q(x) + 2\langle x, y \rangle + q(y) \end{aligned}$$

Rearranging the terms to isolate $\langle x, y \rangle$:

$$2\langle x, y \rangle = q(x+y) - q(x) - q(y) \implies \langle x, y \rangle = \frac{1}{2}\left[q(x+y) - q(x) - q(y)\right]$$

Example Problem 3.4.13: Unique Roots and Positivity

(a) Is it possible for a quadratic form to be strictly positive ($q(x_+) > 0$) at only one point $x_+ \in \mathbb{R}^n$?
Answer: No. If $q(x_+) > 0$, consider the ray defined by all scalar multiples λx_+ for $\lambda \in \mathbb{R}$. By homogeneity, $q(\lambda x_+) = \lambda^2 q(x_+)$. For any $\lambda \neq 0$, $\lambda^2 > 0$, meaning $q(\lambda x_+) > 0$. Thus, it must be positive on an entire line (infinitely many points).
(b) Under what conditions is $q(x_0) = 0$ at only one point?
Answer: If the quadratic form is either **strictly positive definite** ($q(x) > 0$ for all $x \neq 0$) or **strictly negative definite** ($q(x) < 0$ for all $x \neq 0$), then the only point where it equals zero is the origin ($x_0 = 0$).

Theorem Problem 3.4.14 & 3.4.15: The Necessity of Symmetry

1. Uniqueness of Symmetric Matrices: Let K and L be symmetric $n \times n$ matrices. Prove that $x^T K x = x^T L x$ for all $x \in \mathbb{R}^n \iff K = L$.
Proof: ($\Leftarrow$) Trivial. ($\Rightarrow$) If $x^T K x = x^T L x$, then $x^T (K - L) x = 0$. Let $M = K - L$. Since K and L are symmetric, M is symmetric. We are given $x^T M x = 0$ for all x. The expansion is $\sum_{i,j} m_{ij} x_i x_j = 0$. Let $x = e_i$ (standard basis vector). Then $e_i^T M e_i = m_{ii} = 0$. So all diagonal entries are 0. Let $x = e_i + e_j$ (where $i \neq j$). Then $x^T M x = m_{ii} + m_{jj} + 2m_{ij} = 0$. Since $m_{ii} = 0$ and $m_{jj} = 0$, we have $2m_{ij} = 0 \implies m_{ij} = 0$. Thus all entries of M are 0, meaning $M = 0 \implies K = L$.
2. Counter-example for Non-Symmetric Matrices: If we drop the symmetry requirement, uniqueness fails. Let $K = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $L = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$. For K: $x^T K x = x_1(x_1 + x_2) = x_1^2 + x_1 x_2$. For L: $x^T L x = x_1^2 + x_1 x_2$. $K \neq L$, but their quadratic forms are identical.
3. Symmetrization: Suppose $q(x) = x^T A x$ is a general quadratic form where A is not symmetric. Show we can always rewrite it as $x^T K x$ where $K = \frac{1}{2}(A + A^T)$. *Proof:* K is guaranteed symmetric because $K^T = \frac{1}{2}(A^T + A) = K$.

$$x^T K x = x^T \left[\frac{1}{2}(A + A^T)\right] x = \frac{1}{2} x^T A x + \frac{1}{2} x^T A^T x$$

Since $x^T A x$ is a 1×1 scalar, it equals its own transpose: $(x^T A x)^T = x^T A^T x$.

$$= \frac{1}{2} x^T A x + \frac{1}{2} x^T A x = x^T A x = q(x)$$

Thus, any quadratic form can be perfectly represented by a symmetric matrix.

91 Negative Definiteness

Definition : Negative Definite

A symmetric matrix $N \in M_n(\mathbb{R})$ is said to be **negative definite** if its quadratic form is strictly negative:

$$q(x) = x^T N x < 0 \quad \text{for all } x \neq 0 \text{ in } \mathbb{R}^n$$

Theorem Problem 3.4.16: Testing Negative Definiteness

(a) Equivalence to Positive Definiteness: Show that N is negative definite $\iff$ $K = -N$ is positive definite. *Proof:* $x^T N x < 0 \iff -(x^T N x) > 0 \iff x^T(-N)x > 0 \iff K$ is positive definite.

(b) 2×2 Explicit Criteria: Let $N = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$. N is negative definite if $K = \begin{pmatrix} -a & -b \\ -b & -c \end{pmatrix}$ is positive definite. By Sylvester's criterion on K, we require the top-left entry to be positive ($-a > 0 \implies a < 0$) and the determinant to be positive ($(-a)(-c) - (-b)^2 > 0 \implies ac - b^2 > 0$). Thus, N is negative definite $\iff$ $a < 0$ and $\det(N) > 0$.

Example Testing specific matrices

Use the criteria $a < 0, \det > 0$ to test if the following matrices are negative definite.

- $\begin{pmatrix} -1 & -2 \\ -2 & -5 \end{pmatrix}$: $a = -1 < 0$. Det $= 5 - 4 = 1 > 0$. **(Yes, Negative Definite)**
- $\begin{pmatrix} -4 & -5 \\ -5 & -6 \end{pmatrix}$: $a = -4 < 0$. Det $= 24 - 25 = -1 < 0$. **(No, Indefinite)**
- $\begin{pmatrix} -3 & -1 \\ -1 & -2 \end{pmatrix}$: $a = -3 < 0$. Det $= 6 - 1 = 5 > 0$. **(Yes, Negative Definite)**

92 Null Directions and Light Cones

Definition : Null Direction

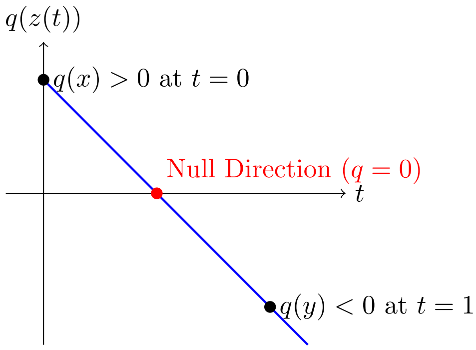
A nonzero vector $x \in \mathbb{R}^n$ is called a **null direction** for a symmetric matrix K if $x^T K x = 0$.

Example Problem 3.4.17 & 3.4.19(a): Null Directions vs Kernels

Show that $x = (1, 1)^T$ is a null direction for $K = \begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix}$, but $x \notin \ker(K)$.

$$x^T K x = (1, 1) \begin{pmatrix} 1 & -2 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = (1, 1) \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 + 1 = 0$$

Since $x^T K x = 0$ and $x \neq 0$, it is a null direction. However, $Kx = \begin{pmatrix} -1 \\ 1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. Thus, x is not in the null space (kernel) of K. This proves that a matrix can have a null direction even if its kernel is $\{0\}$ (which is true here since $\det(K) = -1 \neq 0$).



Proof Problem 3.4.18 & 3.4.19(b): Indefiniteness guarantees Null Directions

Explain why an indefinite quadratic form necessarily has a nonzero null direction.
Proof: Let $q(z) = z^T K z$ be an indefinite quadratic form. By definition, there exist nonzero vectors $x, y \in \mathbb{R}^n$ such that $q(x) > 0$ and $q(y) < 0$. Consider the straight line connecting x and y parameterized by $t \in [0, 1]$:

$$z(t) = (1 - t)x + ty$$

We must ensure this line does not pass through the origin. If $z(t) = 0$ for some t, then x and y are linearly dependent ($x = \lambda y$). By homogeneity, $q(x) = \lambda^2 q(y)$. Since $\lambda^2 > 0$ and $q(y) < 0$, this would force $q(x) < 0$, contradicting our premise. Thus, $z(t) \neq 0$ for all t.
Now consider the function $f(t) = q(z(t))$. This is a continuous polynomial function of t. We know $f(0) = q(x) > 0$ and $f(1) = q(y) < 0$. By the **Intermediate Value Theorem**, there must exist some $t_0 \in (0, 1)$ where $f(t_0) = 0$. Thus, the vector $z_0 = z(t_0)$ is non-zero and satisfies $q(z_0) = 0$, making it a null direction. *Corollary:* If a symmetric matrix has no null directions, it must be either strictly positive definite or strictly negative definite.

92.1 The Light Cone in Minkowski Space

In Special Relativity, light rays in Minkowski spacetime travel along the "light cone", which consists of all null directions associated with a specific indefinite quadratic form.

Example Problem 3.4.20: Finding Light Cones

Find and sketch the light cone for the following matrices by setting $q(x) = 0$.
(a) $K = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ **in** $\mathbb{R}^2$**:**

$$q(x) = x_1^2 - x_2^2 = 0 \iff x_1^2 = x_2^2 \iff x_1 = \pm x_2$$

This forms two intersecting lines through the origin: $y = x$ and $y = -x$.
(b) $K = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ **in** $\mathbb{R}^2$**:**

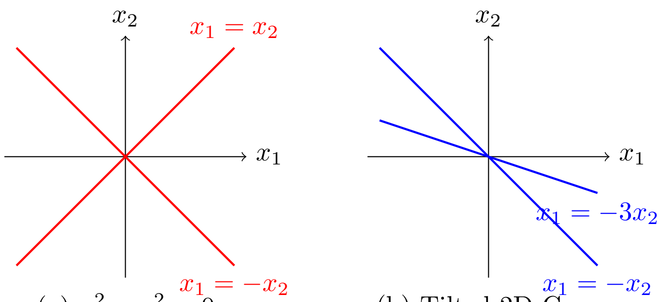
$$q(x) = x_1^2 + 4x_1x_2 + 3x_2^2 = (x_1 + 2x_2)^2 - x_2^2 = 0$$

$$(x_1 + 2x_2)^2 = x_2^2 \implies x_1 + 2x_2 = \pm x_2$$

This yields two intersecting lines: $x_1 = -x_2$ (or $y = -x$) and $x_1 = -3x_2$ (or $y = -x/3$).
(c) $K = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ **in** $\mathbb{R}^3$**:**

$$q(x) = x_1^2 - x_2^2 - x_3^2 = 0 \iff x_1^2 = x_2^2 + x_3^2$$

This forms a standard 3-dimensional double cone extending along the x_1-axis.



(a) $x_1^2 - x_2^2 = 0$ (b) Tilted 2D Cone

92.2 Homogeneous Functions

Definition : Homogeneous Function

A function $f(x)$ on $\mathbb{R}^n$ is called **homogeneous of degree** k if for every scalar $c \in \mathbb{R}$:

$$f(cx) = c^k f(x)$$

Example Problem 3.4.21: Degrees of Homogeneity

- **(a) Linear Forms:** $f(x) = a \cdot x$ is degree 1.

$$f(cx) = a \cdot (cx) = c(a \cdot x) = c^1 f(x)$$

- **(b) Quadratic Forms:** $q(x) = x^T K x$ is degree 2.

$$q(cx) = (cx)^T K(cx) = cx^T K cx = c^2 (x^T K x) = c^2 q(x)$$

- **(c) Non-Polynomials:** Is there a degree 2 homogeneous function that is not a quadratic form? Yes. Consider $f(x_1, x_2) = \sqrt{x_1^4 + x_2^4}$.

$$f(cx_1, cx_2) = \sqrt{(cx_1)^4 + (cx_2)^4} = \sqrt{c^4(x_1^4 + x_2^4)} = c^2 \sqrt{x_1^4 + x_2^4} = c^2 f(x_1, x_2)$$

This function scales exactly like a quadratic form, but it cannot be generated by matrix multiplication.

93 Testing for Positive Definiteness

Reference: Olver & Shakiban, Chapter 3, Pages 170-171.

We utilize two primary tools to test symmetric matrices for positive definiteness without having to calculate eigenvalues:

1. **Sylvester's Criterion:** All leading principal minors (upper-left sub-determinants) must be strictly positive.
2. LDL^T **Factorization:** The diagonal matrix D (which contains the pivots) must have strictly positive entries.

93.1 Sylvester's Criterion Exercises

Example Problem 3.5.1: Are the following matrices positive definite?

(a) $\begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$ *Solution:* $D_1 = 4 > 0$. $D_2 = \det(A) = 16 - 4 = 12 > 0$. **Yes.**

(b) $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ *Solution:* $D_1 = 1 > 0$, but $D_2 = \det(A) = 1 - 1 = 0$. **No.**

(c) $\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 1 & 1 \end{pmatrix}$ *Solution:* We check the 3×3 determinant using cofactor expansion along the first row:

$$\det(A) = 1(2-1) - 1(1-2) + 2(1-4) = 1(1) - 1(-1) + 2(-3) = 1 + 1 - 6 = -4$$

Since $\det(A) < 0$, it is **No.**

(d) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & -2 \\ 1 & -2 & 4 \end{pmatrix}$ *Solution:* Expand the determinant along the first row:

$$\det(A) = 1(8-4) - 1(4-(-2)) + 1(-2-2) = 4 - 6 - 4 = -6$$

Since $\det(A) < 0$, it is **No.**

(e) $\begin{pmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 1 & 1 & 2 \end{pmatrix}$ *Solution:* We check the principal minors. $D_1 = 2 > 0$. $D_2 = \det \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 4 - 1 = 3 > 0$. $D_3 = \det \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} = 2(3) - 1(1) + 1(-1) = 4 > 0$. By pattern (for $I + J_n$ matrices), $D_4 = 5 > 0$. All principal minors are positive. **Yes.**

(f) $\begin{pmatrix} -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \end{pmatrix}$ *Solution:* $D_1 = a_{11} = -1 < 0$. It fails immediately. **No.**

93.2 The LDL^T Factorization

Example Problem 3.5.2: Finding LDL^T

Find the LDL^T factorization of the following symmetric matrices and determine if they are positive definite.

(a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ Apply row operation $R_2 \leftarrow R_2 - 2R_1$:

$$U = \begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix} \implies L = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Since D contains a negative pivot (-1), it is **not positive definite**.

(b) $A = \begin{pmatrix} 5 & -1 \\ -1 & 3 \end{pmatrix}$ Apply row operation $R_2 \leftarrow R_2 + \frac{1}{5}R_1$:

$$U = \begin{pmatrix} 5 & -1 \\ 0 & 3 - 1/5 \end{pmatrix} = \begin{pmatrix} 5 & -1 \\ 0 & 14/5 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 \\ -1/5 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 5 & 0 \\ 0 & 14/5 \end{pmatrix}, \quad L^T = \begin{pmatrix} 1 & -1/5 \\ 0 & 1 \end{pmatrix}$$

Since all diagonal entries of D are strictly positive, **Yes, it is positive definite.**

(c) $A = \begin{pmatrix} 3 & -1 & 3 \\ -1 & 5 & 1 \\ 3 & 1 & 5 \end{pmatrix}$ Step 1 ($R_2 \leftarrow R_2 + \frac{1}{3}R_1$ and $R_3 \leftarrow R_3 - R_1$):

$$\begin{pmatrix} 3 & -1 & 3 \\ 0 & 14/3 & 2 \\ 0 & 2 & 2 \end{pmatrix}$$

Step 2 ($R_3 \leftarrow R_3 - \frac{6}{14}R_2$):

$$\begin{pmatrix} 3 & -1 & 3 \\ 0 & 14/3 & 2 \\ 0 & 0 & 16/14 \end{pmatrix}$$

Extracting the components:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -1/3 & 1 & 0 \\ 1 & 6/14 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 14/3 & 0 \\ 0 & 0 & 16/14 \end{pmatrix}$$

Since $3 > 0, 14/3 > 0$, and $16/14 > 0$, **Yes, it is positive definite.**

Example Problem 3.5.3: Parameterized Definiteness

Consider $A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & c & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

(a) For which values of c is A positive definite? By Sylvester's criterion:

- $D_1 = 1 > 0$ (Always true).
- $D_2 = \det \begin{pmatrix} 1 & 1 \\ 1 & c \end{pmatrix} = c - 1 > 0 \implies c > 1$.
- $D_3 = \det(A) = 1(c-1) - 1(1) = c - 2 > 0 \implies c > 2$.

The most restrictive condition is $c > 2$.

(b) For $c = 3$, write $A = LDL^T$.

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_2 - R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 - \frac{1}{2}R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 1/2 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1/2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}, \quad L^T = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{pmatrix}$$

94 Quadratic Forms and Completing the Square

Example Problem 3.5.3 (c) & (d): Diagonalizing the Quadratic Form

(c) Use the previous LDL^T factorization to rewrite the quadratic form $q(x, y, z) = x^2 + 2xy + 3y^2 + 2yz + z^2$ as a sum of squares. Notice that $q(x, y, z) = \mathbf{x}^T A\mathbf{x}$, where $\mathbf{x} = (x, y, z)^T$. Substitute $A = LDL^T$:

$$\mathbf{x}^T(LDL^T)\mathbf{x} = (L^T\mathbf{x})^T D(L^T\mathbf{x})$$

Let us define a new coordinate vector $\mathbf{u} = L^T\mathbf{x}$:

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1/2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + y \\ y + z/2 \\ z \end{pmatrix}$$

In these new coordinates, the form is purely diagonalized by the pivots in D:

$$q(\mathbf{x}) = \mathbf{u}^T D\mathbf{u} = 1u^2 + 2v^2 + \frac{1}{2}w^2$$

Substituting back yields the perfect sum of squares:

$$q(x, y, z) = (x + y)^2 + 2(y + z/2)^2 + \frac{1}{2}z^2$$

(d) Explain how this relates to positive definiteness. Since $q(x, y, z)$ is a sum of squared terms multiplied by strictly positive coefficients $(1, 2, 1/2)$, it is guaranteed that $q(x, y, z) \geq 0$ for all real inputs. Furthermore, $q(x, y, z) = 0$ if and only if $u = 0, v = 0, w = 0$, which directly forces $z = 0, y = 0, x = 0$. This fulfills the exact definition of strict positive definiteness.

Example Problem 3.5.4: Extracting the Matrix from the Polynomial

Write $q(x) = x_1^2 + 2x_2^2 - x_1x_3 + 3x_3^2$ in the form x^TKx for a symmetric matrix K, and check if it is positive definite.
To build K, the squared terms go on the diagonal, and cross-terms are divided equally across the symmetry line:

$$K = \begin{pmatrix} 1 & 0 & -1/2 \\ 0 & 2 & 0 \\ -1/2 & 0 & 3 \end{pmatrix}$$

Apply Sylvester's Criterion:

- $D_1 = 1 > 0$
- $D_2 = \det \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = 2 > 0$
- $D_3 = \det(K) = 1(6) - 0 + (-1/2)(-(-1/2)(2)) = 6 - 1/2 = 5.5 > 0$

Since all leading principal minors are positive, $q(x)$ is **positive definite**.

Example Problem 3.5.5: Manual Sum of Squares on $\mathbb{R}^2$

Write the following forms as a sum of squares and classify them.

- **(a)** $x^2 + 8xy + y^2 = (x^2 + 8xy + 16y^2) - 15y^2 = (x + 4y)^2 - 15y^2$. (Difference of squares $\implies$ **Not** positive def).
- **(b)** $x^2 - 4xy + 7y^2 = (x^2 - 4xy + 4y^2) + 3y^2 = (x - 2y)^2 + 3y^2$. (Sum of positive squares $\implies$ **Positive def**).
- **(c)** $x^2 - 2xy - y^2 = (x^2 - 2xy + y^2) - 2y^2 = (x - y)^2 - 2y^2$. (**Not** positive def).
- **(d)** $x^2 + 6xy = (x^2 + 6xy + 9y^2) - 9y^2 = (x + 3y)^2 - 9y^2$. (**Not** positive def).

95 Theoretical Properties of Definite Matrices

Reference: Olver & Shakiban, Chapter 3.

Example Problem 3.5.8: Parameterized Positive Definiteness

For what values of $a, b, c \in \mathbb{R}$ is the quadratic form $q(x, y, z) = x^2 + axy + y^2 + bxz + cyz + z^2$ positive definite?
Solution: We extract the symmetric matrix K corresponding to this quadratic form:

$$K = \begin{pmatrix} 1 & a/2 & b/2 \\ a/2 & 1 & c/2 \\ b/2 & c/2 & 1 \end{pmatrix}$$

We apply Sylvester's Criterion, requiring all leading principal minors to be strictly positive:

1. $D_1 = 1 > 0$ (Trivially satisfied).

2. $D_2 = \det \begin{pmatrix} 1 & a/2 \\ a/2 & 1 \end{pmatrix} = 1 - \frac{a^2}{4} > 0 \implies a^2 < 4 \implies -2 < a < 2.$

3. $D_3 = \det(K) > 0$. Expanding the 3×3 determinant:

$$D_3 = 1\left(1 - \frac{c^2}{4}\right) - \frac{a}{2}\left(\frac{a}{2} - \frac{bc}{4}\right) + \frac{b}{2}\left(\frac{ac}{4} - \frac{b}{2}\right)$$

$$= 1 - \frac{c^2}{4} - \frac{a^2}{4} + \frac{abc}{8} + \frac{abc}{8} - \frac{b^2}{4} = 1 - \frac{a^2+b^2+c^2}{4} + \frac{abc}{4}$$

For $D_3 > 0$, we require:

$$1 - \frac{a^2+b^2+c^2-abc}{4} > 0 \implies a^2+b^2+c^2-abc < 4$$

Thus, the form is positive definite if and only if $a^2 < 4$ and $a^2+b^2+c^2-abc < 4$.

Theorem Problem 3.5.9 & 3.5.10: Trace and Determinant

3.5.9: True or False: Every quadratic form $q(x, y) = ax^2 + 2bxy + cy^2$ can be written as a sum of squares. **Answer: False.** It can only be written as a sum of *positive* squares if the matrix is positive definite ($a > 0$ and $ac - b^2 > 0$). Indefinite forms result in differences of squares.
3.5.10 (a): A positive definite matrix has a positive determinant. (**True** - It is the product of positive pivots).
3.5.10 (b): Show a positive definite matrix must have a positive trace. *Proof:* The trace is the sum of the main diagonal entries. In Problem 3.4.5, we proved that every diagonal entry of a positive definite matrix must be strictly positive. The sum of strictly positive numbers is strictly positive.
3.5.10 (c): Prove that a 2×2 symmetric matrix is positive definite if $a + c > 0$ and $ac > b^2$. *Proof:* We are given $ac > b^2$. Since $b^2 \geq 0$, it follows that $ac > 0$. This implies a and c must have the same sign (both positive or both negative). We are also given $a + c > 0$. If they were both negative, their sum would be negative. Therefore, both $a > 0$ and $c > 0$. Since $a > 0$ (first principal minor) and $ac - b^2 > 0$ (second principal minor), Sylvester's criterion guarantees the matrix is positive definite.

Problem 3.5.11 (d): Determinant Trap

Find a symmetric 3×3 matrix with a strictly positive determinant which is **not** positive definite. Let $K = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$. The determinant is $(-1)(-2)(4) = 8 > 0$. However, $D_1 = -1 < 0$. Therefore, K is indefinite. A positive determinant is necessary, but not sufficient, for positive definiteness.

95.1 Shifting the Spectrum

Theorem Problem 3.5.12 & 3.5.13: Shifting to Positive Definiteness

(a) Show that $q(x)$ is a positive definite quadratic form $\iff$ $q(u) > 0$ for all unit vectors $\|u\| = 1$. *Proof:* ($\Rightarrow$) Obvious, since $u \neq 0$. ($\Leftarrow$) Let x be any nonzero vector. Then $u = x/\|x\|$ is a unit vector. Since $q(u) > 0$, by homogeneity $q(x) = \|x\|^2 q(u) > 0$.
(b) Prove that if $S = S^T$ is any symmetric matrix, then $K = S + cI$ is positive definite if c is sufficiently large. *Proof:* Consider the quadratic form $q_S(x) = x^T S x$. We evaluate this function exclusively on the unit sphere $\{u \in \mathbb{R}^n \mid \|u\| = 1\}$. Because the unit sphere is a **compact set**

(closed and bounded) in $\mathbb{R}^n$, and q_S is continuous, the Extreme Value Theorem guarantees that q_S achieves an absolute minimum value on this sphere. Let this minimum value be m. Now, choose a constant c such that $c > -m$ (i.e., $c + m > 0$). Evaluate the quadratic form of $K = S + cI$ for any unit vector u:

$$q_K(u) = u^T(S + cI)u = u^T Su + c(u^T u) = q_S(u) + c\|u\|^2$$

Since $\|u\| = 1$ and $q_S(u) \geq m$:

$$q_K(u) \geq m + c > 0$$

Since $q_K(u) > 0$ for all unit vectors, K is positive definite.
Corollary (3.5.13): Every symmetric matrix S can be written as the sum of a positive definite matrix K and a negative definite matrix N. *Proof:* From (b), choose $c > 0$ large enough so that $K = S+cI$ is positive definite. Let $N = -cI$. Since $c > 0$, the quadratic form of N is $-c\|x\|^2 < 0$, making N negative definite. Thus, $S = (S + cI) - cI = K + N$.

Theorem Problem 3.5.14: Rank-1 Decomposition

Prove that every symmetric regular matrix K can be decomposed as a linear combination of symmetric rank 1 matrices: $K = d_1 L_1 L_1^T + \cdots + d_n L_n L_n^T$.

Proof

Since K is symmetric and regular, it possesses an LDL^T factorization, where L is a unit lower triangular matrix and D is a diagonal matrix of pivots. Let $L_1, L_2, \ldots, L_n$ be the column vectors of L. By the definition of block matrix multiplication, LD scales each column j of L by the scalar d_j. Multiplying this result by L^T (which has rows $L_1^T, L_2^T, \ldots, L_n^T$) directly yields the sum of the outer products of the columns and rows:

$$K = LDL^T = \sum_{j=1}^{n} d_j L_j L_j^T$$

Each outer product $L_j L_j^T$ is an $n \times n$ symmetric matrix of exactly rank 1.

96 Criteria for Negative Definiteness

Theorem Problem 3.5.16: Diagonal Entries

Prove that if a non-positive diagonal entry appears anywhere in a symmetric matrix K, then K is **not** positive definite.

Proof

Suppose that the k-th diagonal entry of K is non-positive, i.e., $a_{kk} \leq 0$. Consider the standard basis vector e_k (which has a 1 in the k-th position and 0s elsewhere). Evaluate the quadratic form at $x = e_k$:

$$q(e_k) = e_k^T K e_k = a_{kk}$$

Since $a_{kk} \leq 0$, we have found a nonzero vector e_k for which $q(e_k)$ is not strictly positive. By definition, K fails to be positive definite.

Theorem Problem 3.5.17 & 3.5.18: Determinantal Criteria for Negative Definiteness

Formulate a determinantal criterion similar to Sylvester's Criterion to test if a symmetric matrix N is negative definite.

Derivation: By definition, N is negative definite if and only if $K = -N$ is positive definite. Sylvester's criterion states that K is positive definite $\iff$ every leading principal minor $D_k(K) > 0$. Since $K = -N$, the k-th leading principal submatrix of K is exactly -1 times the k-th leading principal submatrix of N. Using the property of determinants that $\det(cA) = c^k \det(A)$ for a $k \times k$ matrix, we have:

$$D_k(K) = D_k(-N) = (-1)^k D_k(N)$$

We require $D_k(K) > 0$ for all k. Therefore:

$$(-1)^k D_k(N) > 0$$

This establishes a strict pattern of alternating signs for the principal minors of N:

- For $k = 1$: $(-1)^1 D_1(N) > 0 \implies D_1(N) < 0$.
- For $k = 2$: $(-1)^2 D_2(N) > 0 \implies D_2(N) > 0$.
- For $k = 3$: $(-1)^3 D_3(N) > 0 \implies D_3(N) < 0$.

Conclusion: A symmetric matrix N is negative definite if and only if its leading principal minors alternate in sign, beginning with negative ($D_1 < 0, D_2 > 0, D_3 < 0, D_4 > 0, \dots$).
Problem 3.5.18 Application: True or False: A negative definite matrix must have a negative determinant. **Answer: False.** If N is an $n \times n$ negative definite matrix, its full determinant is $D_n(N)$. According to our derived criterion, the sign of $D_n(N)$ depends strictly on whether n is even or odd. For any even-dimensioned matrix $(2 \times 2, 4 \times 4)$, the full determinant of a negative definite matrix will be **positive**.

Example Example: Rank-1 Decomposition in $\mathbb{R}^3$

Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix}$. We first find its LDL^T factorization via Gaussian elimination.
Elimination Steps:

$$\begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1/2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & -1 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 5/2 \end{pmatrix} = U$$

Extracting L (using the inverse of the elimination multipliers) and the pivot matrix D:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & -1/2 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5/2 \end{pmatrix}, \quad L^T = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & -1/2 \\ 0 & 0 & 1 \end{pmatrix}$$

Rank-1 Expansion: Using the theorem formula $A = \sum_{j=1}^{3} d_j L_j L_j^T$, we multiply the columns of L by their transposes and scale by the pivots:

$$A = 1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 1 \\ -1/2 \end{pmatrix} \begin{pmatrix} 0 & 1 & -1/2 \end{pmatrix} + \frac{5}{2} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$$

Expanding these outer products yields the sum of three distinct rank-1 matrices:

$$= \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 1/2 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5/2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{pmatrix}$$

This perfectly reconstructs the original matrix A.

97 Alternate Criteria for Definiteness

Theorem Problem 3.5.15: Alternate Subdeterminant Criterion

There is an alternate criterion for positive definiteness based on subdeterminants of the matrix.

- **(a)** Prove that $K = \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$ is positive definite $\iff$ $a > 0, ad - b^2 > 0$, and $\det(K) > 0$. *Solution:* (Completed previously via explicitly deriving Sylvester's Criterion for 3×3 matrices).
- **(b)** Prove the general version: An $n \times n$ matrix K is positive definite $\iff$ its upper left square $k \times k$ submatrices have strictly positive determinants for all $k = 1, \ldots, n$. *Solution:* (To be formally proven at the conclusion of the chapter).

Theorem Problem 3.5.17: Determinantal Criteria for Negative Definiteness

Formulate a determinantal criterion similar to Exercise 3.5.15 for negative definite matrices. **Derivation:** By definition, a matrix N is negative definite $\iff$ $K = -N$ is positive definite. By Sylvester's criterion, K is positive definite $\iff$ the upper left square $k \times k$ submatrices $D_k(K)$ are strictly positive. Using the determinant property of scaling rows, $D_k(K) = D_k(-N) = (-1)^k D_k(N)$. Therefore, we require:

$$(-1)^k D_k(N) > 0 \quad \text{for all } k = 1, \ldots, n$$

This implies the leading principal minors of N must strictly alternate in sign, starting with a negative value:

- 1×1 submatrix is negative ($D_1 < 0$).
- 2×2 submatrix is positive ($D_2 > 0$).
- 3×3 submatrix is negative ($D_3 < 0$).

Explicit Cases:

- For $N = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ to be negative definite $\iff$ $a < 0$ and $ac - b^2 > 0$.
- For $N = \begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$ to be negative definite $\iff$ $a < 0$, $\det \begin{pmatrix} a & b \\ b & d \end{pmatrix} > 0$, and $\det(N) < 0$.

Example Problem 3.5.18: Determinant of Negative Definite Matrices

True or False: A negative definite matrix must have a negative determinant. **Answer: False.** According to the derived alternating criterion, the sign of the full determinant $\det(N)$ depends entirely on whether the matrix dimension n is odd or even. If n is even, the determinant is positive. *Example:* If $n = 2$, the matrix $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$ is negative definite $\implies$ $\det(N) = ac - b^2 > 0$.

Theorem Problem 3.5.16: Non-Positive Pivots

Let K be a symmetric matrix. Prove that if a non-positive diagonal entry appears anywhere during regular Gaussian elimination, then K is **not** positive definite.

Proof Intuitive Sketch

Suppose a non-positive entry occurs at the k-th diagonal position during Gaussian elimination. This means that multiplying the original matrix A by a unit lower diagonal matrix L_1 yields an upper triangular matrix $B = L_1 A$, where the k-th diagonal entry of B (b_{kk}) is non-positive (≤ 0). Let us test the quadratic form using the standard basis vector $v = e_k = (0, \dots, 0, 1, 0, \dots, 0)^T$.

$$e_k^T A e_k = e_k^T L_1 B e_k$$

Visually, multiplying e_k^T by L_1 acts upon the k-th row of L_1:

$$\begin{pmatrix} 0 & \dots & 1 & \dots & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ \vdots & \ddots & 0 \\ * & \dots & 1 \end{pmatrix} = \begin{pmatrix} * & \dots & * & 1 & 0 & \dots & 0 \end{pmatrix}$$

Multiplying this resultant row by the k-th column of B (extracted by Be_k):

$$\begin{pmatrix} * & \dots & * & 1 & 0 & \dots & 0 \end{pmatrix} \begin{pmatrix} * \\ \vdots \\ b_{kk} \\ 0 \\ \vdots \end{pmatrix} \approx b_{kk} \leq 0$$

Assuming the cross-terms vanish, $e_k^T A e_k \leq 0$. Since we found a non-zero vector e_k that yields a non-positive result, A cannot be positive definite.

Rigorous Correction for Pivot Isolation

While the visual sketch above elegantly captures the geometric intuition of isolating the pivot $b_{kk} \leq 0$, standard matrix arithmetic dictates that the dot product of the row of L_1 and the column of B leaves behind cross-terms ($\sum l_{ki} b_{ik} + b_{kk}$), which do not automatically cancel out to exactly b_{kk}.

The rigorous way to isolate the pivot: Instead of testing e_k, utilize the symmetric factorization $A = LDL^T$ (where D contains the pivots) and test the vector $x = (L^T)^{-1} e_k$. Since $(L^T)^{-1}$ is invertible, x is guaranteed to be a non-zero vector. Evaluate the quadratic form:

$$x^T A x = \left((L^T)^{-1} e_k\right)^T (LDL^T) \left((L^T)^{-1} e_k\right)$$

Applying transpose properties:

$$= e_k^T L^{-1} LDL^T (L^T)^{-1} e_k = e_k^T (I) D (I) e_k = e_k^T D e_k = d_{kk}$$

If the k-th pivot $d_{kk} \leq 0$, we have strictly proven that $x^T A x \leq 0$ for a non-zero vector x, conclusively proving A is not positive definite.

99 Orthogonal and Orthonormal Bases (Chapter 4)

Reference: Olver & Shakiban, Chapter 4.

Definition : Orthogonal and Orthonormal Basis

Let V be a real vector space with an inner product. Let $u_1, \ldots, u_n$ be a basis of V (so $\dim V = n$).

- This basis is called **orthogonal** if $\langle u_i, u_j \rangle = 0$ for all $i \neq j$.
- This basis is called **orthonormal** if it is orthogonal *and* $\langle u_i, u_i \rangle = 1$ for all i. (Equivalently, $\|u_i\| = 1$).

Example: In $V = \mathbb{R}^n$ with the standard dot product, the standard basis $e_1, \ldots, e_n$ (where $e_i = (0, \ldots, 1, \ldots, 0)^T$ with 1 in the i-th position) is an orthonormal basis.

Theorem Lemma: Normalizing a Basis

Suppose V is an inner product space and $v_1, \ldots, v_n$ is an orthogonal basis of V. If we define $u_i = \frac{v_i}{\|v_i\|}$ for $i = 1, \ldots, n$, then $u_1, \ldots, u_n$ is an orthonormal basis of V.

Proof

We check the conditions for an orthonormal basis:

- **Orthogonality:** For $i \neq j$, $\langle u_i, u_j \rangle = \left\langle \frac{v_i}{\|v_i\|}, \frac{v_j}{\|v_j\|} \right\rangle = \frac{1}{\|v_i\|\|v_j\|} \langle v_i, v_j \rangle = 0$.
- **Unit Length:** For all i, $\langle u_i, u_i \rangle = \left\langle \frac{v_i}{\|v_i\|}, \frac{v_i}{\|v_i\|} \right\rangle = \frac{1}{\|v_i\|^2} \langle v_i, v_i \rangle = \frac{\|v_i\|^2}{\|v_i\|^2} = 1$.

Example Example in $\mathbb{R}^3$

Let $V = \mathbb{R}^3$ with the standard dot product. Check that the following set is an orthogonal basis:

$$\left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 5 \\ -2 \\ 1 \end{pmatrix} \right\}$$

Check dot products:

- $v_1 \cdot v_2 = (1)(0) + (2)(1) + (-1)(2) = 0$
- $v_1 \cdot v_3 = (1)(5) + (2)(-2) + (-1)(1) = 5 - 4 - 1 = 0$
- $v_2 \cdot v_3 = (0)(5) + (1)(-2) + (2)(1) = 0 - 2 + 2 = 0$

Since they are mutually orthogonal and there are 3 of them in $\mathbb{R}^3$, they form a basis. To create an orthonormal basis, divide each by its norm ($\|v_1\| = \sqrt{6}, \|v_2\| = \sqrt{5}, \|v_3\| = \sqrt{30}$):

$$\left\{ \begin{pmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{pmatrix}, \begin{pmatrix} 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix}, \begin{pmatrix} 5/\sqrt{30} \\ -2/\sqrt{30} \\ 1/\sqrt{30} \end{pmatrix} \right\}$$

100 Properties of Orthogonal Sets

Theorem Proposition: Orthogonality implies Linear Independence

Let $v_1, \ldots, v_k \in V$ be non-zero mutually orthogonal elements (i.e., $v_i \neq 0$ for all i, and $\langle v_i, v_j \rangle = 0$ for all $i \neq j$). Then $v_1, \ldots, v_k$ are linearly independent.

Proof

Suppose a linear combination equals the zero vector:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0$$

where $c_i \in \mathbb{R}$. Take the inner product of both sides with a specific basis vector v_i:

$$\langle v_i, c_1 v_1 + \cdots + c_k v_k \rangle = \langle v_i, 0 \rangle = 0$$

Using bilinearity to expand the left side:

$$c_1 \langle v_i, v_1 \rangle + \cdots + c_i \langle v_i, v_i \rangle + \cdots + c_k \langle v_i, v_k \rangle = 0$$

Because the vectors are mutually orthogonal, every inner product $\langle v_i, v_j \rangle = 0$ except when $i = j$. The equation collapses to:

$$c_i \langle v_i, v_i \rangle = 0 \implies c_i \|v_i\|^2 = 0$$

Since $v_i \neq 0$, its norm $\|v_i\|^2 \neq 0$. Therefore, we must have $c_i = 0$. Since this holds for any i, all coefficients $c_i = 0$, which proves linear independence.

Theorem: If $v_1, \ldots, v_n$ are non-zero mutually orthogonal vectors, let $W = \operatorname{span}\{v_1, \ldots, v_n\} \subseteq V$. Then W is a subspace of dimension n with basis $v_1, \ldots, v_n$. In particular, if $\dim V = n$, then $v_1, \ldots, v_n$ is automatically a basis of V.

Theorem Theorem: Coordinates in an Orthonormal Basis

Let V be an inner product space, and let $u_1, \ldots, u_n$ be an **orthonormal** basis of V. For any $v \in V$, it can be written uniquely as a linear combination:

$$v = c_1 u_1 + c_2 u_2 + \cdots + c_n u_n$$

Then the coordinates c_i are given exactly by the inner product:

$$c_i = \langle v, u_i \rangle \quad \forall i = 1, \ldots, n$$

Furthermore, the squared norm of v satisfies Parseval's Identity:

$$\|v\|^2 = c_1^2 + \cdots + c_n^2 = \sum_{i=1}^{n} \langle v, u_i \rangle^2$$

Proof

Proof of coordinates:

$$\langle v, u_i \rangle = \left\langle \sum_{j=1}^{n} c_j u_j, u_i \right\rangle = \sum_{j=1}^{n} c_j \langle u_j, u_i \rangle$$

Since the basis is orthonormal, $\langle u_j, u_i \rangle = 1$ when $j = i$ and 0 otherwise. Thus the sum collapses to just c_i.

Proof of norm:

$$\|v\|^2 = \langle v, v\rangle = \left\langle \sum_{i=1}^{n} c_i u_i, \sum_{j=1}^{n} c_j u_j \right\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} c_i c_j \langle u_i, u_j\rangle$$

Because $\langle u_i, u_j\rangle = 0$ for $i \neq j$, the cross terms disappear. When $i = j$, $\langle u_i, u_i\rangle = 1$.

$$= \sum_{i=1}^{n} c_i c_i(1) = \sum_{i=1}^{n} c_i^2 = c_1^2 + \cdots + c_n^2$$

Example Finding Coordinates in an Orthonormal Basis

Consider $V = \mathbb{R}^3$ with the standard dot product, and the orthonormal basis $\{u_1, u_2, u_3\}$ we derived previously:

$$u_1 = \begin{pmatrix} 1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{pmatrix}, \quad u_2 = \begin{pmatrix} 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{pmatrix}, \quad u_3 = \begin{pmatrix} 5/\sqrt{30} \\ -2/\sqrt{30} \\ 1/\sqrt{30} \end{pmatrix}$$

Consider the vector $v = (1, 1, 1)^T \in \mathbb{R}^3$. We can write $v = c_1u_1 + c_2u_2 + c_3u_3$. Because the basis is orthonormal, we do not need to solve a system of equations; we simply compute the dot products $c_i = \langle v, u_i\rangle$:

- $c_1 = \langle v, u_1\rangle = 1\left(\frac{1}{\sqrt{6}}\right) + 1\left(\frac{2}{\sqrt{6}}\right) + 1\left(\frac{-1}{\sqrt{6}}\right) = \frac{2}{\sqrt{6}}$
- $c_2 = \langle v, u_2\rangle = 1(0) + 1\left(\frac{1}{\sqrt{5}}\right) + 1\left(\frac{2}{\sqrt{5}}\right) = \frac{3}{\sqrt{5}}$
- $c_3 = \langle v, u_3\rangle = 1\left(\frac{5}{\sqrt{30}}\right) + 1\left(\frac{-2}{\sqrt{30}}\right) + 1\left(\frac{1}{\sqrt{30}}\right) = \frac{4}{\sqrt{30}}$

Hence, $v = \frac{2}{\sqrt{6}}u_1 + \frac{3}{\sqrt{5}}u_2 + \frac{4}{\sqrt{30}}u_3$.

Theorem Theorem: Coordinates in a General Orthogonal Basis

Let V be an inner product space, and let $v_1, \ldots, v_n$ be an **orthogonal** basis of V (not necessarily unit length). Let $v \in V$, and express it as a linear combination:

$$v = a_1v_1 + a_2v_2 + \cdots + a_nv_n$$

where $a_i \in \mathbb{R}$ are the coordinates of v. Then:

$$a_i = \frac{\langle v, v_i\rangle}{\|v_i\|^2}$$

Furthermore, the squared norm of v is given by:

$$\|v\|^2 = \sum_{i=1}^{n} a_i^2 \|v_i\|^2 = \sum_{i=1}^{n} \frac{\langle v, v_i\rangle^2}{\|v_i\|^2}$$

Proof

To find a_i, take the inner product of v with v_i:

$$\langle v, v_i\rangle = \left\langle \sum_{j=1}^{n} a_j v_j, v_i \right\rangle = \sum_{j=1}^{n} a_j \langle v_j, v_i\rangle$$

Since the basis is orthogonal, $\langle v_j, v_i \rangle = 0$ for all $j \neq i$. The sum reduces to a single term:

$$\langle v, v_i \rangle = a_i \langle v_i, v_i \rangle = a_i \|v_i\|^2 \implies a_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2}$$

For the norm squared, we compute $\langle v, v \rangle$:

$$\langle v, v \rangle = \left\langle \sum_{i=1}^{n} a_i v_i, \sum_{j=1}^{n} a_j v_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \langle v_i, v_j \rangle$$

Again, all cross-terms where $i \neq j$ evaluate to 0 due to orthogonality. Thus:

$$\sum_{i=1}^{n} a_i a_i \langle v_i, v_i \rangle = \sum_{i=1}^{n} a_i^2 \|v_i\|^2$$

101 The Gram-Schmidt Process

Let W be a finite-dimensional real vector space with an inner product $\langle \cdot, \cdot \rangle$. Let $w_1, \ldots, w_n$ be a standard basis of W. We will construct an **orthogonal basis** $v_1, \ldots, v_n$ for W from this starting basis.

101.1 Deriving the Algorithm

Step 1: Define the first orthogonal vector simply as the first basis vector.

$$v_1 = w_1$$

Step 2: To find the second vector, we subtract a scalar multiple of v_1 from w_2:

$$v_2 = w_2 - cv_1$$

We want v_2 to be orthogonal to v_1, so we enforce the condition $\langle v_2, v_1 \rangle = 0$:

$$0 = \langle w_2 - cv_1, v_1 \rangle = \langle w_2, v_1 \rangle - c\langle v_1, v_1 \rangle$$

$$c = \frac{\langle w_2, v_1 \rangle}{\|v_1\|^2}$$

Thus, we define:

$$v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\|v_1\|^2} v_1$$

Since w_1, w_2 are linearly independent, w_2 cannot be a scalar multiple of w_1 (which is v_1). Therefore, $v_2 \neq 0$.

Step 3: Next, let $v_3 = w_3 - c_1 v_1 - c_2 v_2$. We require $\langle v_3, v_1 \rangle = 0$ and $\langle v_3, v_2 \rangle = 0$. Taking the inner product with v_1:

$$0 = \langle w_3 - c_1 v_1 - c_2 v_2, v_1 \rangle = \langle w_3, v_1 \rangle - c_1 \|v_1\|^2 - c_2(0) \implies c_1 = \frac{\langle w_3, v_1 \rangle}{\|v_1\|^2}$$

Taking the inner product with v_2:

$$0 = \langle w_3 - c_1 v_1 - c_2 v_2, v_2 \rangle = \langle w_3, v_2 \rangle - c_1(0) - c_2 \|v_2\|^2 \implies c_2 = \frac{\langle w_3, v_2 \rangle}{\|v_2\|^2}$$

Thus, we define:

$$v_3 = w_3 - \frac{\langle w_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle w_3, v_2 \rangle}{\|v_2\|^2} v_2$$

We have $v_3 \neq 0$ since w_1, w_2, w_3 are linearly independent.

General Step: Continuing in this way, suppose we have found mutually orthogonal, non-zero vectors $v_1, \ldots, v_{k-1}$ which are linear combinations of $w_1, \ldots, w_{k-1}$. We define the k-th orthogonal vector as:

$$v_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} v_i$$

This gives an inductive step to generate the full orthogonal basis $v_1, \ldots, v_n$. This entire sequence of operations is called the **Gram-Schmidt Process**.

To obtain an **orthonormal basis** $u_1, \ldots, u_n$, we simply normalize the results by defining $u_i = \frac{v_i}{\|v_i\|}$.

101.2 Gram-Schmidt Example in $\mathbb{R}^3$

Example Constructing an Orthonormal Basis

Let $V = \mathbb{R}^3$ equipped with the standard dot product. Given the basis:

$$w_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad w_3 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$$

Use the Gram-Schmidt process to find an orthogonal basis $\{v_1, v_2, v_3\}$, and then the orthonormal basis $\{u_1, u_2, u_3\}$.

Step 1: Find v_1

$$v_1 = w_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

(Note: $\|v_1\|^2 = 1^2 + (-1)^2 + 1^2 = 3$).

Step 2: Find v_2 We project w_2 onto v_1:

$$\langle w_2, v_1 \rangle = (1)(1) + (2)(-1) + (0)(1) = 1 - 2 = -1$$

$$v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} - \frac{-1}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \begin{pmatrix} 1/3 \\ -1/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 4/3 \\ 5/3 \\ 1/3 \end{pmatrix}$$

(Note: $\|v_2\|^2 = \frac{16}{9} + \frac{25}{9} + \frac{1}{9} = \frac{42}{9} = \frac{14}{3}$).

Step 3: Find v_3 We project w_3 onto both v_1 and v_2:

$$\langle w_3, v_1 \rangle = (0)(1) + (1)(-1) + (2)(1) = 1$$

$$\langle w_3, v_2 \rangle = (0)\left(\frac{4}{3}\right) + (1)\left(\frac{5}{3}\right) + (2)\left(\frac{1}{3}\right) = \frac{7}{3}$$

$$v_3 = w_3 - \frac{\langle w_3, v_1 \rangle}{\|v_1\|^2} v_1 - \frac{\langle w_3, v_2 \rangle}{\|v_2\|^2} v_2$$

$$v_3 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - \frac{1}{3} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} - \frac{7/3}{14/3} \begin{pmatrix} 4/3 \\ 5/3 \\ 1/3 \end{pmatrix}$$

Since $\frac{7/3}{14/3} = \frac{1}{2}$:

$$v_3 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1/3 \\ -1/3 \\ 1/3 \end{pmatrix} - \begin{pmatrix} 4/6 \\ 5/6 \\ 1/6 \end{pmatrix} = \begin{pmatrix} 0 - 2/6 - 4/6 \\ 6/6 + 2/6 - 5/6 \\ 12/6 - 2/6 - 1/6 \end{pmatrix} = \begin{pmatrix} -1 \\ 1/2 \\ 3/2 \end{pmatrix}$$

This yields the orthogonal basis $\{v_1, v_2, v_3\}$.

Step 4: Normalize to find u_1, u_2, u_3 Compute the lengths: $\|v_1\| = \sqrt{3}$ $\|v_2\| = \sqrt{42}/3$ $\|v_3\| = \sqrt{1 + 1/4 + 9/4} = \sqrt{14/4} = \frac{\sqrt{14}}{2}$

Dividing each v_i by its respective length:

$$u_1 = \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}, \quad u_2 = \begin{pmatrix} 4/\sqrt{42} \\ 5/\sqrt{42} \\ 1/\sqrt{42} \end{pmatrix}, \quad u_3 = \begin{pmatrix} -2/\sqrt{14} \\ 1/\sqrt{14} \\ 3/\sqrt{14} \end{pmatrix}$$

This is the required orthonormal basis.

102 The Alternative Gram-Schmidt Formulation (QR Decomposition)

Reference: Olver & Shakiban, Chapter 4.

The standard Gram-Schmidt process generates an orthogonal basis $v_1, \ldots, v_n$ from an original basis $w_1, \ldots, w_n$. We then normalize to get an orthonormal basis $u_1, \ldots, u_n$ where $u_i = \frac{v_i}{\|v_i\|}$.

102.1 Re-expressing the Basis

Let us look backward and express the original vectors w_i in terms of our new orthonormal basis vectors u_i. By observing the construction of the Gram-Schmidt process, we see a triangular relationship:

$$\begin{aligned} w_1 &= \|v_1\| u_1 \\ w_2 &= \|v_2\| u_2 + \frac{\langle w_2, v_1 \rangle}{\|v_1\|} u_1 \\ &\vdots \\ w_n &= \|v_n\| u_n + \sum_{i=1}^{n-1} \langle w_n, u_i \rangle u_i \end{aligned}$$

We can rewrite this system using a set of coefficients r_{ij}:

$$\begin{aligned} w_1 &= r_{11} u_1 \\ w_2 &= r_{12} u_1 + r_{22} u_2 \\ w_3 &= r_{13} u_1 + r_{23} u_2 + r_{33} u_3 \\ &\vdots \\ w_k &= r_{1k} u_1 + r_{2k} u_2 + \cdots + r_{kk} u_k \end{aligned}$$

Notice two important facts about the diagonal entries r_{ii}: 1. $r_{ii} = \|v_i\|$. 2. Since the vectors w_i are linearly independent, $\|v_i\| \neq 0$. Thus, $r_{ii} > 0$ for all i.

102.2 The New Formula

We can compute the coefficients r_{ij} much faster. Instead of remembering the messy fractional projections from the original Gram-Schmidt formula, we use the orthonormal properties of u. If we take the inner product of w_j with u_i (where $i < j$):

$$\langle w_j, u_i \rangle = \langle r_{1j} u_1 + \cdots + r_{jj} u_j, u_i \rangle$$

Because $\langle u_k, u_i \rangle = 0$ for all $k \neq i$, and $\langle u_i, u_i \rangle = 1$, the entire right side collapses to just r_{ij}:

$$r_{ij} = \langle w_j, u_i \rangle \quad \text{for } 1 \leq i < j$$

The Streamlined Algorithm: We want to extract the real numbers $r_{11}, r_{12}, \ldots$ and the orthonormal vectors $u_1, u_2, \ldots$ iteratively.

- **Step 1:** Define $r_{11} = \|w_1\|$, and set $u_1 = \frac{w_1}{r_{11}}$.

- **Step 2:** Compute $r_{12} = \langle w_2, u_1 \rangle$.
  Define $v_2 = w_2 - r_{12}u_1$. Compute $r_{22} = \|v_2\|$, and set $u_2 = \frac{v_2}{r_{22}}$.
- **Step 3:** Compute $r_{13} = \langle w_3, u_1 \rangle$ and $r_{23} = \langle w_3, u_2 \rangle$.
  Define $v_3 = w_3 - r_{13}u_1 - r_{23}u_2$. Compute $r_{33} = \|v_3\|$, and set $u_3 = \frac{v_3}{r_{33}}$.

General Step: Suppose we have computed $u_1, \ldots, u_{k-1}$. 1. Compute the coefficients: $r_{ik} = \langle w_k, u_i \rangle$ for $1 \le i \le k-1$. 2. Find the orthogonal vector: $v_k = w_k - \sum_{i=1}^{k-1} r_{ik}u_i$. 3. Normalize: $r_{kk} = \|v_k\|$, and $u_k = \frac{v_k}{r_{kk}}$.

102.3 Example Computation in $\mathbb{R}^3$

A Note on Arithmetic

The following is the corrected mathematical evaluation of the example begun in the handwritten notes. A slight arithmetic error in the notes regarding the dot products r_{13} and r_{23} led to a non-orthogonal v_3. This has been fixed below to provide a flawless reference.

Example Applying the Streamlined Process

Let us apply the new Gram-Schmidt process to the standard dot product basis:

$$w_1 = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad w_2 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad w_3 = \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix}$$

Step 1: Process w_1

$$r_{11} = \|w_1\| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

$$u_1 = \frac{w_1}{r_{11}} = \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}$$

Step 2: Process w_2 Compute the projection coefficient onto u_1:

$$r_{12} = \langle w_2, u_1 \rangle = (1)\left(\frac{1}{\sqrt{3}}\right) + (0) + (2)\left(\frac{1}{\sqrt{3}}\right) = \frac{3}{\sqrt{3}} = \sqrt{3}$$

Find the orthogonal vector v_2:

$$v_2 = w_2 - r_{12}u_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \sqrt{3}\begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

Normalize v_2:

$$r_{22} = \|v_2\| = \sqrt{0 + 1^2 + 1^2} = \sqrt{2} \implies u_2 = \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

Step 3: Process w_3 Compute the projection coefficients onto u_1 and u_2:

$$r_{13} = \langle w_3, u_1 \rangle = \frac{-2}{\sqrt{3}} + \frac{-2}{\sqrt{3}} + \frac{3}{\sqrt{3}} = \frac{-1}{\sqrt{3}}$$

$$r_{23} = \langle w_3, u_2 \rangle = 0 + \frac{2}{\sqrt{2}} + \frac{3}{\sqrt{2}} = \frac{5}{\sqrt{2}}$$

Find the orthogonal vector v_3:

$$\begin{aligned}
v_3 &= w_3 - r_{13}u_1 - r_{23}u_2 \\
&= \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix} - \left(\frac{-1}{\sqrt{3}}\right) \begin{pmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix} - \left(\frac{5}{\sqrt{2}}\right) \begin{pmatrix} 0 \\ 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \\
&= \begin{pmatrix} -2 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 1/3 \\ -1/3 \\ 1/3 \end{pmatrix} - \begin{pmatrix} 0 \\ 5/2 \\ 5/2 \end{pmatrix} \\
&= \begin{pmatrix} -6/3 + 1/3 \\ 12/6 - 2/6 - 15/6 \\ 18/6 + 2/6 - 15/6 \end{pmatrix} = \begin{pmatrix} -5/3 \\ -5/6 \\ 5/6 \end{pmatrix} = \frac{5}{6}\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}
\end{aligned}$$

Normalize v_3:

$$r_{33} = \|v_3\| = \frac{5}{6}\sqrt{(-2)^2 + (-1)^2 + 1^2} = \frac{5\sqrt{6}}{6}$$

$$u_3 = \frac{v_3}{r_{33}} = \frac{1}{\sqrt{6}}\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix}$$

Thus, the orthonormal basis is exactly u_1, u_2, u_3.

103 Orthogonal and Orthonormal Bases

Reference: Olver & Shakiban, Chapter 4, Exercises Pages 186-188.

Example Problem 4.1.1: Classifying Pairs in $\mathbb{R}^2$

Let $\mathbb{R}^2$ have the standard dot product. Classify the following pairs of vectors as (i) a basis, (ii) an orthogonal basis, or (iii) an orthonormal basis.

- **(a)** $v_1 = (1, 0)^T, \quad v_2 = (1, 1)^T$. *Answer:* They are linearly independent, so they form a **basis**. However, $v_1 \cdot v_2 = 1 \neq 0$, so they are **not orthogonal**.
- **(b)** $v_1 = (4/5, 3/5)^T, \quad v_2 = (-3/5, 4/5)^T$. *Answer:* $v_1 \cdot v_2 = -12/25 + 12/25 = 0$. Lengths: $\|v_1\| = \sqrt{16/25 + 9/25} = 1$, and $\|v_2\| = 1$. This is an **orthonormal basis**.
- **(c)** $v_1 = (-1/\sqrt{2}, 1/\sqrt{2})^T, \quad v_2 = (1/\sqrt{2}, 1/\sqrt{2})^T$. *Answer:* $v_1 \cdot v_2 = -1/2 + 1/2 = 0$. Lengths are exactly 1. This is an **orthonormal basis**.
- **(d)** $v_1 = (2, 1)^T, \quad v_2 = (-1, 2)^T$. *Answer:* $v_1 \cdot v_2 = -2 + 2 = 0$. However, $\|v_1\| = \sqrt{5} \neq 1$. This is an **orthogonal basis**, but not orthonormal.
- **(e)** $v_1 = (0, 1)^T, \quad v_2 = (0, -1)^T$. *Answer:* They are scalar multiples of each other ($v_2 = -v_1$), meaning they are linearly dependent. They do **not form a basis**.
- **(f)** $v_1 = (3/5, -4/5)^T, \quad v_2 = (4/5, 3/5)^T$. *Answer:* $v_1 \cdot v_2 = 12/25 - 12/25 = 0$. Lengths are 1. This is an **orthonormal basis**.

Example Problem 4.1.2 & 4.1.3: Classification in $\mathbb{R}^3$

4.1.2: Classify the following sets of vectors in $\mathbb{R}^3$ under the standard dot product:

- **(a)** $(1, 0, 0)^T, (0, -1, 0)^T, (0, 0, 1)^T$: All dot products are 0, all lengths are 1. **Orthonormal basis.**
- **(b)** $(4/13, 12/13, 3/13)^T, (-3/5, 0, 4/5)^T, (48/65, -25/65, 36/65)^T$: Computing dot product of 1 and 2: $-12/65 + 0 + 12/65 = 0$. All vectors have a norm of 1. It forms an **orthonormal**

basis.

- **(c)** $(0, -1/\sqrt{2}, 1/\sqrt{2})^T, (1/\sqrt{3}, 1/\sqrt{3}, 1/\sqrt{3})^T, (-1/\sqrt{2}, 0, 1/\sqrt{2})^T$: Dot product of 1 and 3: $0 + 0 + 1/2 = 1/2 \neq 0$. **Basis, but not orthogonal.**

4.1.3: Repeat 4.1.1 but use the weighted inner product $\langle v, w \rangle = v_1 w_1 + \frac{1}{2} v_2 w_2$. For pair (d), $v_1 = (2,1)^T, v_2 = (-1,2)^T$: $\langle v_1, v_2 \rangle = (2)(-1) + \frac{1}{2}(1)(2) = -2 + 1 = -1 \neq 0$. Under this new metric, they are **not orthogonal**.

Example Problem 4.1.4: Normalizing under a Custom Weight

Show that the standard basis vectors e_1, e_2, e_3 form an orthogonal basis with respect to the inner product $\langle v, w \rangle = v_1 w_1 + 2 v_2 w_2 + 3 v_3 w_3$. Then find an orthonormal basis for this inner product space.
Proof: Because the weights form a diagonal matrix, $\langle e_i, e_j \rangle = 0$ for all $i \neq j$. Thus, they are orthogonal. To find the orthonormal basis u_1, u_2, u_3, we must divide by their lengths measured in the *new* inner product:

- $\|e_1\|^2 = \langle e_1, e_1 \rangle = (1)^2 + 2(0) + 3(0) = 1 \implies u_1 = e_1$
- $\|e_2\|^2 = \langle e_2, e_2 \rangle = 0 + 2(1)^2 + 0 = 2 \implies u_2 = \frac{1}{\sqrt{2}} e_2 = (0, 1/\sqrt{2}, 0)^T$
- $\|e_3\|^2 = \langle e_3, e_3 \rangle = 0 + 0 + 3(1)^2 = 3 \implies u_3 = \frac{1}{\sqrt{3}} e_3 = (0, 0, 1/\sqrt{3})^T$

104 Designing Inner Products for Orthogonality

Example Problem 4.1.5: Enforcing Orthogonality

Find all values of a such that the vectors $v = (a, 1)^T$ and $w = (a, -1)^T$ form an orthogonal basis of $\mathbb{R}^2$ with respect to:
(a) The standard dot product:

$$v \cdot w = a^2 - 1 = 0 \implies a^2 = 1 \implies a = \pm 1$$

(b) The weighted inner product $\langle v, w \rangle = 3 v_1 w_1 + 2 v_2 w_2$**:**

$$3(a)(a) + 2(1)(-1) = 0 \implies 3a^2 - 2 = 0 \implies a = \pm\sqrt{2/3}$$

(c) The inner product given by the positive definite matrix $K = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$**:** We require $v^T K w = 0$.

$$\begin{pmatrix} a & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} a \\ -1 \end{pmatrix} = \begin{pmatrix} a & 1 \end{pmatrix} \begin{pmatrix} 2a + 1 \\ -a - 3 \end{pmatrix} = a(2a + 1) + 1(-a - 3) = 0$$

$$2a^2 + a - a - 3 = 0 \implies 2a^2 = 3 \implies a = \pm\sqrt{3/2}$$

Theorem Problem 4.1.9: The Limits of Diagonal Weights

True or False: If v_1, v_2, v_3 are a basis for $\mathbb{R}^3$, then they automatically form an orthogonal basis under some appropriately weighted inner product $\langle x, y \rangle = a x_1 y_1 + b x_2 y_2 + c x_3 y_3$ where $a, b, c > 0$.
Answer: False.
Proof by Counter-example: Consider the basis $v_1 = (1, 1, 0)^T$, $v_2 = (1, 0, 1)^T$, and $v_3 = (0, 1, 1)^T$. Assume there exists an inner product defined by the diagonal matrix $D = \text{diag}(a, b, c)$ that makes

them orthogonal. Evaluate $\langle v_1, v_2 \rangle = 0$:

$$v_1^T D v_2 = (1, 1, 0) \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = a(1)(1) + b(1)(0) + c(0)(1) = a$$

For them to be orthogonal, we strictly require $a = 0$. However, for D to define a valid inner product, it must be positive definite, which strictly requires $a > 0, b > 0, c > 0$. This is a contradiction. Therefore, simple scalar weighting is not sufficient to orthogonalize every arbitrary basis; off-diagonal elements (covariances) are required.

105 The Cross Product and Orthogonality

Theorem Problem 4.1.10: The Cross Product

The cross product between two vectors $v, w \in \mathbb{R}^3$ is defined as $u = v \times w$.

1. **(a) Show that u is orthogonal to both v and w under the standard dot product.** Using the scalar triple product property, $v \cdot (v \times w) = \det[v, v, w]$. Since a matrix with two identical rows has a determinant of 0, the dot product is 0. The same holds for w.

2. **(b) Show that $v \times w = 0 \iff v, w$ are parallel.** This follows from the norm identity: $\|v \times w\| = \|v\|\|w\| \sin\theta$. The cross product is the zero vector $\iff \sin\theta = 0$, meaning the angle is 0 or π (parallel).

3. **(c) Prove that if v, w are orthogonal non-zero vectors, then $v, w, v \times w$ forms an orthogonal basis of $\mathbb{R}^3$.** From (a), $v \times w$ is perpendicular to both. Since all three are mutually orthogonal and non-zero, they form a basis for 3D space.

4. **(d) True or False: If v, w are orthogonal unit vectors, then $u = v \times w$ completes an orthonormal basis. True.** Since they are unit vectors ($\|v\| = \|w\| = 1$) and orthogonal ($\theta = 90°, \sin 90° = 1$): $\|u\| = \|v\|\|w\| \sin(90°) = (1)(1)(1) = 1$. The third vector naturally has a unit length.

106 Matrices of Orthonormal Bases

Theorem Problem 4.1.13: Reverse-Engineering Inner Products

Let $v_1, \ldots, v_n$ be a basis of $\mathbb{R}^n$. Let $A = [v_1, v_2, \ldots, v_n]$ be the non-singular matrix whose columns are the basis vectors.

- **(a)** Show that $v_1, \ldots, v_n$ form an orthonormal basis for the inner product $\langle x, y \rangle = x^T K y$ if and only if $A^T K A = I$. *Proof:* The entry in the i-th row and j-th column of the product $A^T K A$ is exactly $v_i^T K v_j = \langle v_i, v_j \rangle$. For the basis to be orthonormal, this must equal 1 if $i = j$ and 0 otherwise. This is exactly the definition of the Identity matrix I.

- **(b)** Prove that every basis is orthonormal for some uniquely determined inner product. *Proof:* Since A forms a basis, it is invertible. We can solve $A^T K A = I$ for K:

  $$K = (A^T)^{-1} I A^{-1} = (AA^T)^{-1}$$

  Because A is invertible, AA^T is symmetric and strictly positive definite, meaning its inverse K is also symmetric and positive definite. Thus, it forms a valid inner product space.

Example Solving for specific inner products

Find the unique inner product matrix K that makes the following pairs an orthonormal basis.
(c) $v_1 = (1,1)^T$ and $v_2 = (2,3)^T$.

$$A = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \implies A^{-1} = \frac{1}{3-2}\begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}$$

$$K = (A^{-1})^T A^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}\begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 10 & -7 \\ -7 & 5 \end{pmatrix}$$

The inner product is $\langle x, y\rangle = 10x_1y_1 - 7x_1y_2 - 7x_2y_1 + 5x_2y_2$.
(d) $v_1 = (1,2)^T$ and $v_2 = (1,3)^T$.

$$A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} \implies A^{-1} = \begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix}$$

$$K = (A^{-1})^T A^{-1} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 3 & -1 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 13 & -5 \\ -5 & 2 \end{pmatrix}$$

Theorem Problem 4.1.14: Describing all Orthonormal Bases

Describe the set of *all* orthonormal bases of $\mathbb{R}^2$ under the inner product defined by $K = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$.

Proof

We seek all matrices A such that $A^TKA = I$. While expanding this into a system of equations yields a complex non-linear system representing an ellipse ($2a^2 - 2ac + 2c^2 = 1$, etc.), we can solve it instantly using matrix algebra. Since K is positive definite, it has a unique positive definite square root, $K^{1/2}$, such that $K = K^{1/2}K^{1/2}$. Substitute this into the required equation:

$$A^TK^{1/2}K^{1/2}A = I \implies (K^{1/2}A)^T(K^{1/2}A) = I$$

This tells us that the matrix $(K^{1/2}A)$ must be an **Orthogonal Matrix** Q (where $Q^TQ = I$). In $\mathbb{R}^2$, any orthogonal matrix Q is simply a rotation matrix $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ or a reflection. Therefore, $K^{1/2}A = Q \implies A = K^{-1/2}Q$. By varying the angle $\theta \in [0, 2\pi)$ in the rotation matrix Q, the formula $A = K^{-1/2}R(\theta)$ perfectly generates the coordinates of every possible orthonormal basis for this metric.

107 Exercise Solutions: Orthogonality in Abstract Spaces

Reference: Olver & Shakiban, Chapter 4, Exercises 4.1.15 - 4.1.20.

We now explore geometric theorems translated into abstract inner product spaces, focusing on orthogonal bases, polynomials, and solutions to differential equations.

107.1 Generalizing Euclidean Geometry

Example Problem 4.1.15: The Pythagorean Theorem

Let v, w be elements of an inner product space V. Prove that $\|v+w\|^2 = \|v\|^2 + \|w\|^2 \iff v$ and w are orthogonal. Explain why this formula can be viewed as a generalization of the Pythagorean Theorem.

Proof

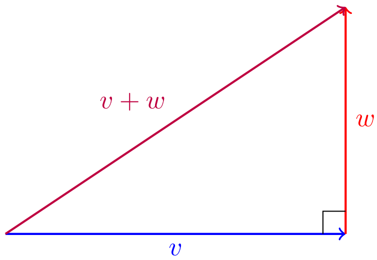
We expand the squared norm of the sum using the bilinearity and symmetry of the inner product:

$$\begin{aligned}\|v+w\|^2 &= \langle v+w, v+w\rangle \\ &= \langle v,v\rangle + \langle v,w\rangle + \langle w,v\rangle + \langle w,w\rangle \\ &= \|v\|^2 + 2\langle v,w\rangle + \|w\|^2\end{aligned}$$

For this to equal strictly $\|v\|^2 + \|w\|^2$, the middle term must vanish:

$$2\langle v,w\rangle = 0 \iff \langle v,w\rangle = 0$$

By definition, $\langle v,w\rangle = 0$ means v and w are orthogonal.
Geometric Explanation: In standard Euclidean space $\mathbb{R}^2$, two vectors being orthogonal means they meet at a right angle (90°). The vectors v and w form the perpendicular legs of a right triangle, and the vector sum $v + w$ forms the hypotenuse. The formula simplifies to $(\text{Leg}_1)^2 + (\text{Leg}_2)^2 = (\text{Hypotenuse})^2$, which is exactly the Pythagorean Theorem.



Example Problem 4.1.16: Diagonals of a Rhombus

Prove that if v_1, v_2 is a basis of an inner product space V, and their norms are equal ($\|v_1\| = \|v_2\|$), then $v_1 + v_2$ and $v_1 - v_2$ form an orthogonal basis of V.

Proof

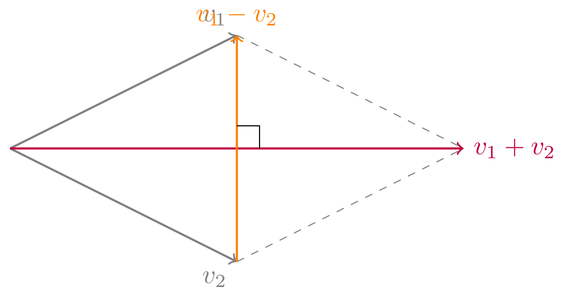
First, we prove they are orthogonal by evaluating their inner product:

$$\begin{aligned}\langle v_1+v_2, v_1-v_2\rangle &= \langle v_1,v_1\rangle - \langle v_1,v_2\rangle + \langle v_2,v_1\rangle - \langle v_2,v_2\rangle \\ &= \|v_1\|^2 - \langle v_1,v_2\rangle + \langle v_1,v_2\rangle - \|v_2\|^2 \\ &= \|v_1\|^2 - \|v_2\|^2\end{aligned}$$

Since we are given that $\|v_1\| = \|v_2\|$, their squares are equal, meaning:

$$\|v_1\|^2 - \|v_2\|^2 = 0$$

Thus, the vectors v_1+v_2 and v_1-v_2 are orthogonal. Because v_1, v_2 form a basis, they are linearly independent, ensuring $v_1 + v_2 \neq 0$ and $v_1 - v_2 \neq 0$. Any set of non-zero orthogonal vectors is linearly independent. Since there are 2 such vectors in a 2-dimensional space, they must form an orthogonal basis.
Geometric Note: Geometrically, v_1 and v_2 with equal lengths form a rhombus. The vectors v_1+v_2 and $v_1 - v_2$ represent the diagonals of this rhombus. This proves that the diagonals of a rhombus always intersect at right angles.



Example Problem 4.1.17: Gram Matrix of an Orthogonal Set

Suppose $v_1, \ldots, v_k$ are mutually orthogonal elements of an inner product space V. Write down their Gram matrix. Why is it non-singular?

Proof

The entries of the Gram matrix K are defined by $K_{ij} = \langle v_i, v_j \rangle$. Since the vectors are mutually orthogonal, $\langle v_i, v_j \rangle = 0$ for all $i \neq j$. The only non-zero entries exist on the main diagonal, where $i = j$, yielding $\langle v_i, v_i \rangle = \|v_i\|^2$. Thus, K is a purely diagonal matrix:

$$K = \begin{pmatrix} \|v_1\|^2 & 0 & \ldots & 0 \\ 0 & \|v_2\|^2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \|v_k\|^2 \end{pmatrix}$$

The determinant of a diagonal matrix is simply the product of its diagonal entries:

$$\det(K) = \|v_1\|^2 \|v_2\|^2 \ldots \|v_k\|^2$$

Since the vectors form a basis (or are at least mutually orthogonal non-zero elements), $\|v_i\|^2 > 0$ for all i. Therefore, the determinant is strictly positive ($\det(K) \neq 0$), which rigorously proves the matrix is non-singular (invertible).

108 Weighted Orthogonal Polynomials

Example Problem 4.1.18: Weighted Inner Product on $P^{(1)}$ and $P^{(2)}$

Let V be the vector space of polynomials equipped with the inner product $\langle p, q \rangle = \int_0^1 t\, p(t) q(t)\, dt$.

Proof (a) Prove this defines a valid inner product on $P^{(1)}$

Bilinearity and symmetry trivially hold due to the linearity of integrals. We must rigorously verify **Positivity**. For any linear polynomial $p(t) = at + b$:

$$\langle p, p \rangle = \int_0^1 t(p(t))^2\, dt = \int_0^1 t(at + b)^2\, dt$$

Since $t \geq 0$ on the interval $[0, 1]$ and $(at + b)^2 \geq 0$ for all real numbers, the integrand is always non-negative. Therefore, the integral ≥ 0. Suppose $\langle p, p \rangle = 0$. Since the integrand is continuous and non-negative, it must identically equal zero on the interval:

$$t(at + b)^2 = 0 \quad \text{for all } t \in [0, 1]$$

For $t \neq 0$, this requires $(at + b)^2 = 0 \implies at + b = 0$. Thus, $p(t)$ must be the zero polynomial $(a = 0, b = 0)$.

Proof (b) Find all polynomials orthogonal to p1(t)

Let $p(t) = at + b$. For $p(t)$ to be orthogonal to $p_1(t) = 1$, their inner product must be zero:

$$\langle 1, at + b \rangle = \int_0^1 t(1)(at + b)\, dt = \int_0^1 (at^2 + bt)\, dt = 0$$

Evaluate the integral:

$$\left[a\frac{t^3}{3} + b\frac{t^2}{2} \right]_0^1 = \frac{a}{3} + \frac{b}{2} = 0$$

Multiplying by 6 to clear the denominators yields:

$$2a + 3b = 0 \implies b = -\frac{2}{3}a$$

Thus, any polynomial of the form $p(t) = a\left(t - \frac{2}{3}\right)$ is orthogonal to $p_1(t) = 1$.

Proof (c) Construct an Orthonormal Basis for $P^{(1)}$

Step 1: Normalize $p_1(t) = 1$.

$$\|p_1\|^2 = \int_0^1 t(1)^2\, dt = \left[\frac{t^2}{2}\right]_0^1 = \frac{1}{2}$$

Thus, $\|p_1\| = \frac{1}{\sqrt{2}}$. The first unit basis vector is $u_1(t) = \frac{1}{1/\sqrt{2}} = \sqrt{2}$.
Step 2: Normalize the orthogonal polynomial. From part (b), let $v_2(t) = t - \frac{2}{3}$ (by choosing $a = 1, b = -2/3$). We compute its squared norm:

$$\begin{aligned}
\|v_2\|^2 &= \int_0^1 t\left(t - \frac{2}{3}\right)^2 dt \\
&= \int_0^1 t\left(t^2 - \frac{4}{3}t + \frac{4}{9}\right) dt \\
&= \int_0^1 \left(t^3 - \frac{4}{3}t^2 + \frac{4}{9}t\right) dt \\
&= \left[\frac{t^4}{4} - \frac{4t^3}{9} + \frac{4t^2}{18}\right]_0^1 = \frac{1}{4} - \frac{4}{9} + \frac{2}{9} = \frac{1}{4} - \frac{2}{9} = \frac{9-8}{36} = \frac{1}{36}
\end{aligned}$$

Thus, $\|v_2\| = \frac{1}{6}$. The second unit basis vector is $u_2(t) = \frac{t-2/3}{1/6} = 6t - 4$. The orthonormal basis for $P^{(1)}$ is $\left\{\sqrt{2},\ 6t - 4\right\}$.

Proof (d) Find an orthonormal basis for $P^{(2)}$

We must find a quadratic polynomial $p_3(t) = at^2 + bt + c$ orthogonal to both 1 and t (or equivalently, orthogonal to u_1 and u_2).
Condition 1: Orthogonal to 1

$$\langle 1, p_3 \rangle = \int_0^1 t(at^2 + bt + c)\, dt = \int_0^1 (at^3 + bt^2 + ct)\, dt = \frac{a}{4} + \frac{b}{3} + \frac{c}{2} = 0$$

Multiply by 12: $3a + 4b + 6c = 0$.
Condition 2: Orthogonal to t

$$\langle t, p_3 \rangle = \int_0^1 t^2(at^2 + bt + c)\, dt = \int_0^1 (at^4 + bt^3 + ct^2)\, dt = \frac{a}{5} + \frac{b}{4} + \frac{c}{3} = 0$$

Multiply by 60: $12a + 15b + 20c = 0$.

Solve the System: From Condition 1, $6c = -3a - 4b \implies 20c = -10a - \frac{40}{3}b$. Substitute into Condition 2:

$$12a + 15b - 10a - \frac{40}{3}b = 0 \implies 2a + \left(\frac{45-40}{3}\right)b = 0 \implies 2a + \frac{5}{3}b = 0 \implies 6a + 5b = 0$$

Choose $a = 10$, which gives $b = -12$. Substitute back to find c: $3(10) + 4(-12) + 6c = 0 \implies 30 - 48 + 6c = 0 \implies 6c = 18 \implies c = 3$. Thus, the orthogonal quadratic polynomial is $v_3(t) = 10t^2 - 12t + 3$.
Normalize $v_3(t)$:

$$\begin{aligned}
\|v_3\|^2 &= \int_0^1 t(10t^2 - 12t + 3)^2\,dt \\
&= \int_0^1 t(100t^4 - 240t^3 + 204t^2 - 72t + 9)\,dt \\
&= \int_0^1 (100t^5 - 240t^4 + 204t^3 - 72t^2 + 9t)\,dt \\
&= \frac{100}{6} - \frac{240}{5} + \frac{204}{4} - \frac{72}{3} + \frac{9}{2} \\
&= \frac{50}{3} - 48 + 51 - 24 + 4.5 = 16.666... - 21 + 4.5 = \frac{1}{6}
\end{aligned}$$

Since $\|v_3\| = \frac{1}{\sqrt{6}}$, the normalized vector is $u_3(t) = \sqrt{6}(10t^2 - 12t + 3)$. The orthonormal basis for $P^{(2)}$ is $\left\{\sqrt{2},\ 6t - 4,\ \sqrt{6}(10t^2 - 12t + 3)\right\}$.

109 Orthogonal Bases for Differential Equations

Example Problem 4.1.19: Trigonometric Solutions

Explain why the functions $\cos x$ and $\sin x$ form an orthogonal basis for the space of solutions to the differential equation $y'' + y = 0$ under the standard L^2 inner product on $[-\pi, \pi]$.

Proof

The general solution to the second-order linear ODE $y'' + y = 0$ is well known to be spanned by the basis $\{\cos x, \sin x\}$, making it a 2-dimensional vector space. We test for orthogonality under the given inner product:

$$\langle \cos x, \sin x \rangle = \int_{-\pi}^{\pi} \cos x \sin x\,dx$$

Using the double angle identity $\sin(2x) = 2\sin x \cos x$:

$$\int_{-\pi}^{\pi} \frac{1}{2}\sin(2x)\,dx = \left[-\frac{1}{4}\cos(2x)\right]_{-\pi}^{\pi} = -\frac{1}{4}(1) - \left(-\frac{1}{4}(1)\right) = 0$$

(Alternatively, notice that $\cos x \sin x$ is an odd function evaluated over a symmetric interval, so its integral is immediately 0). Since they are orthogonal and span the solution space, they form an orthogonal basis.

Example Problem 4.1.20: Exponential Solutions and Gram-Schmidt

Do the functions $e^{x/2}$ and $e^{-x/2}$ form an orthogonal basis for the space of solutions to the differential equation $4y'' - y = 0$ under the L^2 inner product on $[0, 1]$? If not, construct an orthogonal basis.

Proof

First, we check their orthogonality:

$$\langle e^{x/2}, e^{-x/2} \rangle = \int_0^1 e^{x/2} e^{-x/2}\, dx = \int_0^1 e^0\, dx = \int_0^1 1\, dx = 1$$

Since the inner product is exactly $1 \neq 0$, these fundamental solutions are **not** orthogonal.
Applying Gram-Schmidt: Let $w_1 = e^{x/2}$ and $w_2 = e^{-x/2}$. We will construct an orthogonal basis $\{v_1, v_2\}$. Set $v_1 = w_1 = e^{x/2}$. We need to project w_2 onto v_1. We need the norm squared of v_1:

$$\|v_1\|^2 = \int_0^1 (e^{x/2})^2\, dx = \int_0^1 e^x\, dx = [e^x]_0^1 = e - 1$$

Now, apply the Gram-Schmidt formula for the second vector:

$$\begin{aligned} v_2 &= w_2 - \frac{\langle w_2, v_1 \rangle}{\|v_1\|^2} v_1 \\ &= e^{-x/2} - \frac{1}{e-1} e^{x/2} \end{aligned}$$

Thus, the functions $v_1(x) = e^{x/2}$ and $v_2(x) = e^{-x/2} - \frac{1}{e-1} e^{x/2}$ form an orthogonal basis for the solution space of the differential equation.
Check:

$$\int_0^1 e^{x/2} \left(e^{-x/2} - \frac{e^{x/2}}{e-1} \right) dx = \int_0^1 1 dx - \frac{1}{e-1} \int_0^1 e^x dx = 1 - \frac{e-1}{e-1} = 0$$

(Validated).

110 Exercise Solutions: Orthogonal Bases and Coordinates

Reference: Olver & Shakiban, Chapter 4, Exercises 4.1.21 - 4.1.26.

We now practice finding coordinates of vectors relative to orthogonal and orthonormal bases across different spaces and inner products.

110.1 Standard Euclidean Space

Example Problem 4.1.21: Coordinates in $\mathbb{R}^3$

Consider the vectors $v_1 = (1, 1, 1)^T$, $v_2 = (1, 1, -2)^T$, and $v_3 = (-1, 1, 0)^T$.
(a) Prove that they form an orthogonal basis of $\mathbb{R}^3$ w.r.t the standard dot product. We evaluate their mutual inner products:

- $\langle v_1, v_2 \rangle = 1(1) + 1(1) + 1(-2) = 1 + 1 - 2 = 0$
- $\langle v_1, v_3 \rangle = 1(-1) + 1(1) + 1(0) = -1 + 1 + 0 = 0$
- $\langle v_2, v_3 \rangle = 1(-1) + 1(1) + (-2)(0) = -1 + 1 + 0 = 0$

They are mutually orthogonal. Because they are non-zero and there are 3 of them in $\mathbb{R}^3$, they are linearly independent and automatically form a basis. (Alternatively, the determinant of the matrix formed by them is $3 \neq 0$).
(b) Use orthogonality to write $v = (1, 2, 3)^T$ as a linear combination. Let $v = c_1 v_1 + c_2 v_2 + c_3 v_3$. The coordinates are given by $c_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2}$. First, calculate the squared norms: $\|v_1\|^2 = 3$, $\|v_2\|^2 = 6$, $\|v_3\|^2 = 2$.

- $c_1 = \frac{1(1)+2(1)+3(1)}{3} = \frac{6}{3} = 2$
- $c_2 = \frac{1(1)+2(1)+3(-2)}{6} = \frac{1+2-6}{6} = \frac{-3}{6} = -\frac{1}{2}$

- $c_3 = \frac{1(-1)+2(1)+3(0)}{2} = \frac{-1+2}{2} = \frac{1}{2}$

Thus, $v = 2v_1 - \frac{1}{2}v_2 + \frac{1}{2}v_3$.
(c) Verify the formula $\|v\|^2 = c_1^2\|v_1\|^2 + c_2^2\|v_2\|^2 + c_3^2\|v_3\|^2$. The actual squared norm is $\|v\|^2 = 1^2 + 2^2 + 3^2 = 14$. Using the coordinate formula:

$$\sum_{i=1}^{3} c_i^2\|v_i\|^2 = (2)^2(3) + \left(-\frac{1}{2}\right)^2(6) + \left(\frac{1}{2}\right)^2(2) = 12 + \frac{6}{4} + \frac{2}{4} = 12 + \frac{8}{4} = 12 + 2 = 14$$

The formula holds.
(d) Construct an orthonormal basis. Divide each vector by its norm ($u_i = v_i/\|v_i\|$):

$$u_1 = \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{pmatrix}, \quad u_2 = \begin{pmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ -2/\sqrt{6} \end{pmatrix}, \quad u_3 = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

(e) Write v in the orthonormal basis and verify $\|v\|^2 = d_1^2 + d_2^2 + d_3^2$. Let $v = d_1u_1 + d_2u_2 + d_3u_3$. Since the basis is orthonormal, $d_i = \langle v, u_i \rangle$.

- $d_1 = \frac{6}{\sqrt{3}} = 2\sqrt{3}$
- $d_2 = \frac{-3}{\sqrt{6}} = -\frac{\sqrt{3}}{\sqrt{2}}$
- $d_3 = \frac{1}{\sqrt{2}}$

Check the norm: $d_1^2 + d_2^2 + d_3^2 = 12 + \frac{3}{2} + \frac{1}{2} = 12 + \frac{4}{2} = 14$. This perfectly matches $\|v\|^2 = 14$.

Problem 4.1.22: Solving the Skipped Fractional Problem

Prove that $v_1 = \left(\frac{3}{5}, 0, \frac{4}{5}\right)^T$, $v_2 = \left(-\frac{4}{13}, \frac{12}{13}, \frac{3}{13}\right)^T$, $v_3 = \left(-\frac{48}{65}, -\frac{25}{65}, \frac{36}{65}\right)^T$ form an orthonormal basis, find the coordinates of $v = (1,1,1)^T$, and verify the length formula.
1. Verify Orthonormality:

- $\|v_1\|^2 = \frac{9}{25} + 0 + \frac{16}{25} = \frac{25}{25} = 1$.
- $\|v_2\|^2 = \frac{16+144+9}{169} = \frac{169}{169} = 1$.
- $\|v_3\|^2 = \frac{2304+625+1296}{4225} = \frac{4225}{4225} = 1$.
- $\langle v_1, v_2 \rangle = -\frac{12}{65} + 0 + \frac{12}{65} = 0$.
- $\langle v_1, v_3 \rangle = -\frac{144}{325} + 0 + \frac{144}{325} = 0$.
- $\langle v_2, v_3 \rangle = \frac{192-300+108}{845} = \frac{300-300}{845} = 0$.

Thus, they form an orthonormal basis.
2. Find Coordinates of $v = (1,1,1)^T$: Since the basis is orthonormal, $c_i = \langle v, v_i \rangle$ (which is just the sum of the components of v_i).

- $c_1 = \frac{3}{5} + \frac{4}{5} = \frac{7}{5}$.
- $c_2 = -\frac{4}{13} + \frac{12}{13} + \frac{3}{13} = \frac{11}{13}$.
- $c_3 = -\frac{48}{65} - \frac{25}{65} + \frac{36}{65} = -\frac{37}{65}$.

3. Verify Formula $\|v\|^2 = c_1^2 + c_2^2 + c_3^2$: The actual norm is $\|v\|^2 = 1^2 + 1^2 + 1^2 = 3$. Summing the squared coordinates (finding a common denominator of $65^2 = 4225$):

$$c_1^2 + c_2^2 + c_3^2 = \left(\frac{7}{5}\right)^2 + \left(\frac{11}{13}\right)^2 + \left(-\frac{37}{65}\right)^2 = \frac{49}{25} + \frac{121}{169} + \frac{1369}{4225}$$

$$= \frac{49(169) + 121(25) + 1369}{4225} = \frac{8281 + 3025 + 1369}{4225} = \frac{12675}{4225} = 3$$

The formula is perfectly verified.

110.2 Coordinates Under Custom Inner Products

Example Problem 4.1.23: Custom Positive Definite Metric

Let $\mathbb{R}^2$ have the inner product defined by the positive definite matrix $K = \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}$. The inner product formula is $\langle x, y \rangle = 2x_1y_1 - x_1y_2 - x_2y_1 + 3x_2y_2$.
(a) Show that $v_1 = (1,1)^T$ and $v_2 = (-2,1)^T$ form an orthogonal basis.

$$\langle v_1, v_2 \rangle = 2(1)(-2) - (1)(1) - (1)(-2) + 3(1)(1) = -4 - 1 + 2 + 3 = 0$$

Since they are orthogonal and non-zero, they form a basis.
(b) Write $v = (3,2)^T$ as a linear combination of v_1, v_2. We use $c_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2}$. Let's find the squared norms first:

- $\|v_1\|^2 = 2(1)(1) - 1(1) - 1(1) + 3(1)(1) = 2 - 1 - 1 + 3 = 3$
- $\|v_2\|^2 = 2(-2)(-2) - (-2)(1) - (1)(-2) + 3(1)(1) = 8 + 2 + 2 + 3 = 15$

Now, find the projections of v:

- $\langle v, v_1 \rangle = 2(3)(1) - 3(1) - 2(1) + 3(2)(1) = 6 - 3 - 2 + 6 = 7 \implies c_1 = 7/3$
- $\langle v, v_2 \rangle = 2(3)(-2) - 3(1) - 2(-2) + 3(2)(1) = -12 - 3 + 4 + 6 = -5 \implies c_2 = -5/15 = -1/3$

Thus, $v = \frac{7}{3}v_1 - \frac{1}{3}v_2$.
(c) Verify the formula $\|v\|^2 = c_1^2\|v_1\|^2 + c_2^2\|v_2\|^2$. Actual norm: $\|v\|^2 = 2(3)(3) - (3)(2) - (2)(3) + 3(2)(2) = 18 - 6 - 6 + 12 = 18$. Coordinate formula: $\left(\frac{7}{3}\right)^2(3) + \left(-\frac{1}{3}\right)^2(15) = \frac{49}{9}(3) + \frac{1}{9}(15) = \frac{49}{3} + \frac{5}{3} = \frac{54}{3} = 18$. Matches!
(d) Find an orthonormal basis u_1, u_2 for this inner product. Divide by their respective K-norms:

$$u_1 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{15}}\begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

(e) Write v in the ON basis and verify $\|v\|^2 = d_1^2 + d_2^2$. Let $v = d_1u_1 + d_2u_2$. $d_1 = \langle v, u_1 \rangle = \frac{1}{\sqrt{3}}\langle v, v_1 \rangle = \frac{7}{\sqrt{3}}$. $d_2 = \langle v, u_2 \rangle = \frac{1}{\sqrt{15}}\langle v, v_2 \rangle = -\frac{5}{\sqrt{15}} = -\frac{\sqrt{5}}{\sqrt{3}}$. Check norm: $d_1^2 + d_2^2 = \left(\frac{7}{\sqrt{3}}\right)^2 + \left(-\frac{\sqrt{5}}{\sqrt{3}}\right)^2 = \frac{49}{3} + \frac{5}{3} = \frac{54}{3} = 18$. Matches!

110.3 Abstract Theorems and Counter-Examples

Proof Problem 4.1.24: Inner Products of Coordinate Vectors

Let $u_1, \ldots, u_n$ be an **orthonormal** basis for V. Let $v = c_1u_1 + \cdots + c_nu_n$ and $w = d_1u_1 + \cdots + d_nu_n$.
(a) Prove that $\langle v, w \rangle = c_1d_1 + \cdots + c_nd_n$.
Using the bilinearity of the inner product, we can expand the expression into a double sum:

$$\langle v, w \rangle = \left\langle \sum_{i=1}^{n} c_iu_i, \sum_{j=1}^{n} d_ju_j \right\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} c_id_j\langle u_i, u_j \rangle$$

Because the basis is orthonormal, $\langle u_i, u_j \rangle = 0$ whenever $i \neq j$. All cross-terms vanish. When

$i = j$, $\langle u_i, u_i \rangle = 1$. The sum collapses to a single index:

$$\sum_{i=1}^{n} c_i d_i(1) = c_1 d_1 + \cdots + c_n d_n$$

(b) Write down the corresponding formula if the basis is only orthogonal. If the basis is orthogonal but not normalized, then $\langle u_i, u_i \rangle = \|u_i\|^2$. The formula becomes:

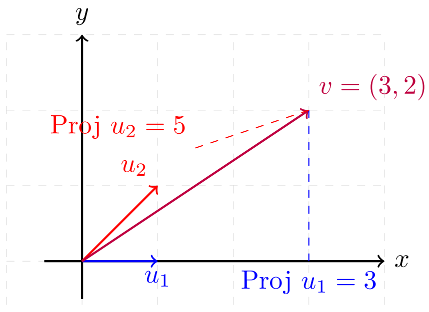
$$\langle v, w \rangle = \sum_{i=1}^{n} c_i d_i \|u_i\|^2$$

Example Problem 4.1.25: Why Orthogonality is Required

Demonstrate by a specific example why the formula $\|v\|^2 = \sum \langle v, u_i \rangle^2$ is **not** valid if the basis $u_1, \ldots, u_n$ is not an orthonormal basis.
Counter-Example: Let $V = \mathbb{R}^2$ with the standard dot product. Choose a non-orthogonal basis: $u_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $u_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. Let $v = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$. We know its true squared length is $\|v\|^2 = 3^2 + 2^2 = 13$. Now, let's try to apply the orthonormal coordinate formula blindly:

- $\langle v, u_1 \rangle = 3(1) + 2(0) = 3$.
- $\langle v, u_2 \rangle = 3(1) + 2(1) = 5$.

According to the formula, $\|v\|^2$ should be $3^2 + 5^2 = 9 + 25 = \mathbf{34}$. Since $13 \neq 34$, the formula utterly fails. The "cross-talk" (covariance) between the non-orthogonal basis vectors artificially inflates the coordinates.



Geometric Failure: When axes are not at 90°, projecting the vector perpendicularly onto the axes overestimates the lengths needed to reach the point, breaking the Pythagorean theorem.

110.4 Coordinates in Function Spaces

Example Problem 4.1.26: Projecting onto Orthogonal Polynomials

Let $V = P^{(2)}$ with the standard L^2 inner product on $[0, 1]$. In previous exercises, we established the following orthogonal basis:

$$p_1(x) = 1, \quad p_2(x) = x - \frac{1}{2}, \quad p_3(x) = x^2 - x + \frac{1}{6}$$

Their squared norms are $\|p_1\|^2 = 1$, $\|p_2\|^2 = \frac{1}{12}$, and $\|p_3\|^2 = \frac{1}{180}$. Write the standard monomials $1, x, x^2$ as linear combinations of this orthogonal basis.
1. For $f(x) = 1$: Trivially, $1 = 1p_1(x) + 0p_2(x) + 0p_3(x)$.
2. For $g(x) = x$: Let $x = c_1 p_1 + c_2 p_2 + c_3 p_3$. We use $c_i = \frac{\langle x, p_i \rangle}{\|p_i\|^2}$.

- $c_1 = \frac{1}{1}\int_0^1 x(1)\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$.
- $c_2 = 12\int_0^1 x\left(x-\frac{1}{2}\right)\,dx = 12\left[\frac{x^3}{3}-\frac{x^2}{4}\right]_0^1 = 12\left(\frac{1}{3}-\frac{1}{4}\right) = 12\left(\frac{1}{12}\right) = 1$.
- $c_3 = 180\int_0^1 x\left(x^2-x+\frac{1}{6}\right)\,dx = 180\left[\frac{x^4}{4}-\frac{x^3}{3}+\frac{x^2}{12}\right]_0^1 = 180\left(\frac{1}{4}-\frac{1}{3}+\frac{1}{12}\right) = 180(0) = 0$.

Thus, $x = \frac{1}{2}p_1(x) + p_2(x)$.

3. For $h(x) = x^2$**:** Let $x^2 = d_1p_1 + d_2p_2 + d_3p_3$. We use $d_i = \frac{\langle x^2, p_i\rangle}{\|p_i\|^2}$.

- $d_1 = \frac{1}{1}\int_0^1 x^2(1)\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$.
- $d_2 = 12\int_0^1 x^2\left(x-\frac{1}{2}\right)\,dx = 12\left[\frac{x^4}{4}-\frac{x^3}{6}\right]_0^1 = 12\left(\frac{1}{4}-\frac{1}{6}\right) = 12\left(\frac{1}{12}\right) = 1$.
- $d_3 = 180\int_0^1 x^2\left(x^2-x+\frac{1}{6}\right)\,dx = 180\left[\frac{x^5}{5}-\frac{x^4}{4}+\frac{x^3}{18}\right]_0^1 = 180\left(\frac{1}{5}-\frac{1}{4}+\frac{1}{18}\right)$. Finding a common denominator of 180: $\frac{36}{180}-\frac{45}{180}+\frac{10}{180} = \frac{1}{180}$. Thus, $d_3 = 180\left(\frac{1}{180}\right) = 1$.

Thus, $x^2 = \frac{1}{3}p_1(x) + p_2(x) + p_3(x)$. *(Self-Check:* $1/3 + (x-1/2) + (x^2-x+1/6) = x^2 + x - x + 1/3 - 1/2 + 1/6 = x^2$. *Verified!)*

111 The Gram-Schmidt Process (Exercises)

Reference: Olver & Shakiban, Chapter 4, Pages 196-197.

111.1 Standard Applications in $\mathbb{R}^3$ and $\mathbb{R}^4$

Example Problem 4.2.1: Orthonormal Bases in $\mathbb{R}^3$

Use the Gram-Schmidt process to determine an orthonormal basis for $\mathbb{R}^3$ starting with the following sets of vectors.

(a) $w_1 = (1,0,1)^T,\ w_2 = (1,1,0)^T,\ w_3 = (-1,1,1)^T$

- **Step 1:** $v_1 = w_1 = (1,0,1)^T$. (Note $\|v_1\|^2 = 2$).
- **Step 2:** $v_2 = w_2 - \frac{\langle w_2,v_1\rangle}{\|v_1\|^2}v_1 = (1,1,0)^T - \frac{1}{2}(1,0,1)^T = \left(\frac{1}{2},1,-\frac{1}{2}\right)^T$.
  (Note $\|v_2\|^2 = \frac{1}{4}+1+\frac{1}{4} = \frac{3}{2}$).
- **Step 3:** $v_3 = w_3 - \frac{\langle w_3,v_1\rangle}{\|v_1\|^2}v_1 - \frac{\langle w_3,v_2\rangle}{\|v_2\|^2}v_2$.
  $\langle w_3, v_1\rangle = -1+0+1 = 0$.
  $\langle w_3, v_2\rangle = (-1)\left(\frac{1}{2}\right) + (1)(1) + (1)\left(-\frac{1}{2}\right) = -\frac{1}{2}+1-\frac{1}{2} = 0$.
  Since w_3 is already orthogonal to both v_1 and v_2, $v_3 = w_3 = (-1,1,1)^T$. (Note $\|v_3\|^2 = 3$).

Normalize: Divide each v_i by its norm to get the orthonormal basis u_1, u_2, u_3:

$$u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\1\end{pmatrix},\quad u_2 = \sqrt{\frac{2}{3}}\begin{pmatrix}1/2\\1\\-1/2\end{pmatrix} = \frac{1}{\sqrt{6}}\begin{pmatrix}1\\2\\-1\end{pmatrix},\quad u_3 = \frac{1}{\sqrt{3}}\begin{pmatrix}-1\\1\\1\end{pmatrix}$$

(b) $w_1 = (1,1,-1)^T,\ w_2 = (0,1,-1)^T,\ w_3 = (1,0,-1)^T$

- **Step 1:** $v_1 = (1,1,-1)^T$. (Note $\|v_1\|^2 = 3$).
- **Step 2:** $v_2 = w_2 - \frac{2}{3}v_1 = (0,1,-1)^T - \left(\frac{2}{3},\frac{2}{3},-\frac{2}{3}\right)^T = \left(-\frac{2}{3},\frac{1}{3},-\frac{1}{3}\right)^T$.
  (Note $\|v_2\|^2 = \frac{4}{9}+\frac{1}{9}+\frac{1}{9} = \frac{2}{3}$).

- **Step 3:** $\langle w_3, v_1\rangle = 2$. And $\langle w_3, v_2\rangle = (1)\left(-\frac{2}{3}\right) + 0 + (-1)\left(-\frac{1}{3}\right) = -\frac{1}{3}$.
  $v_3 = (1,0,-1)^T - \frac{2}{3}(1,1,-1)^T - \frac{-1/3}{2/3}\left(-\frac{2}{3}, \frac{1}{3}, -\frac{1}{3}\right)^T$
  $v_3 = (1,0,-1)^T - \left(\frac{2}{3}, \frac{2}{3}, -\frac{2}{3}\right)^T + \left(-\frac{1}{3}, \frac{1}{6}, -\frac{1}{6}\right)^T = \left(0, -\frac{1}{2}, -\frac{1}{2}\right)^T$.
  (Note $\|v_3\|^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$).

Normalize:

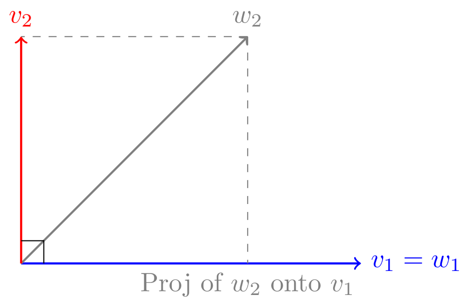
$$u_1 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\-1\end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}-2\\1\\-1\end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\-1\\-1\end{pmatrix}$$

(c) $w_1 = (1,2,3)^T,\ w_2 = (4,5,0)^T,\ w_3 = (2,3,-1)^T$

- **Step 1:** $v_1 = (1,2,3)^T$. ($\|v_1\|^2 = 14$).
- **Step 2:** $\langle w_2, v_1\rangle = 14$. $v_2 = w_2 - \frac{14}{14}v_1 = (4,5,0)^T - (1,2,3)^T = (3,3,-3)^T$. ($\|v_2\|^2 = 27$).
- **Step 3:** $\langle w_3, v_1\rangle = 5$. $\langle w_3, v_2\rangle = 18$.
  $v_3 = w_3 - \frac{5}{14}v_1 - \frac{18}{27}v_2 = (2,3,-1)^T - \left(\frac{5}{14}, \frac{10}{14}, \frac{15}{14}\right)^T - (2,2,-2)^T = \left(-\frac{5}{14}, \frac{4}{14}, -\frac{1}{14}\right)^T$.

Normalize:

$$u_1 = \frac{1}{\sqrt{14}}\begin{pmatrix}1\\2\\3\end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\-1\end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{42}}\begin{pmatrix}-5\\4\\-1\end{pmatrix}$$



Geometric Intuition: The Gram-Schmidt process extracts the purely orthogonal component v_2 by subtracting the parallel projection from the original vector w_2.

Example Problem 4.2.2: Gram-Schmidt in $\mathbb{R}^4$

(a) $w_1 = (1,0,0,1)^T,\ w_2 = (0,-1,0,1)^T,\ w_3 = (0,0,1,1)^T,\ w_4 = (1,1,1,1)^T$

- $v_1 = (1,0,0,1)^T$. ($\|v_1\|^2 = 2$).
- $v_2 = w_2 - \frac{1}{2}v_1 = \left(-\frac{1}{2}, -1, 0, \frac{1}{2}\right)^T$. ($\|v_2\|^2 = \frac{3}{2}$).
- $\langle w_3, v_1\rangle = 1$, $\langle w_3, v_2\rangle = 1/2$.
  $v_3 = w_3 - \frac{1}{2}v_1 - \frac{1/2}{3/2}v_2 = (0,0,1,1)^T - \left(\frac{1}{2}, 0, 0, \frac{1}{2}\right)^T - \left(-\frac{1}{6}, -\frac{1}{3}, 0, \frac{1}{6}\right)^T = \left(-\frac{1}{3}, \frac{1}{3}, 1, \frac{1}{3}\right)^T$. ($\|v_3\|^2 = \frac{4}{3}$).
- $\langle w_4, v_1\rangle = 2$, $\langle w_4, v_2\rangle = -1$, $\langle w_4, v_3\rangle = 4/3$.
  $v_4 = w_4 - \frac{2}{2}v_1 - \frac{-1}{3/2}v_2 - \frac{4/3}{4/3}v_3$
  $v_4 = (1,1,1,1)^T - (1,0,0,1)^T + \left(-\frac{1}{3}, -\frac{2}{3}, 0, \frac{1}{3}\right)^T - \left(-\frac{1}{3}, \frac{1}{3}, 1, \frac{1}{3}\right)^T = (0,0,0,0)^T$.

Conclusion: Because $v_4 = \mathbf{0}$, the original vector w_4 was a linear combination of the first three vectors ($w_4 = w_1 - w_2 + w_3$). Thus, they only span a 3-dimensional subspace of $\mathbb{R}^4$. Normalizing the first three yields the orthonormal basis:

$$u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}-1\\-2\\0\\1\end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{12}}\begin{pmatrix}-1\\1\\3\\1\end{pmatrix}$$

(b) $w_1 = (1,0,0,1)^T$, $w_2 = (4,1,0,0)^T$, $w_3 = (1,0,2,1)^T$, $w_4 = (0,2,0,1)^T$ Applying the standard process (see notes for arithmetic) yields the orthogonal basis:

$$v_1 = \begin{pmatrix}1\\0\\0\\1\end{pmatrix}, \quad v_2 = \begin{pmatrix}2\\1\\0\\-2\end{pmatrix}, \quad v_3 = \begin{pmatrix}0\\0\\2\\0\end{pmatrix}, \quad v_4 = \begin{pmatrix}-1/2\\2\\0\\1/2\end{pmatrix}$$

Normalized basis:

$$u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\0\\0\\1\end{pmatrix}, \quad u_2 = \frac{1}{3}\begin{pmatrix}2\\1\\0\\-2\end{pmatrix}, \quad u_3 = \begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad u_4 = \frac{\sqrt{2}}{6}\begin{pmatrix}-1\\4\\0\\1\end{pmatrix}$$

Theorem Problem 4.2.3: Linear Dependence Trap

Apply Gram-Schmidt to $w_1 = (1,-1,0)^T$, $w_2 = (0,1,2)^T$, $w_3 = (2,-1,2)^T$, $w_4 = (2,-2,1)^T$. What happens?

Proof of Failure: We begin the process normally:

- $v_1 = w_1 = (1,-1,0)^T$.
- $v_2 = w_2 - \frac{-1}{2}v_1 = (1/2, 1/2, 2)^T$.

However, upon evaluating the third vector, we find:

$$v_3 = w_3 - \frac{\langle w_3, v_1\rangle}{\|v_1\|^2}v_1 - \frac{\langle w_3, v_2\rangle}{\|v_2\|^2}v_2 = \cdots = \begin{pmatrix}0\\0\\0\end{pmatrix}$$

Because $v_3 = \mathbf{0}$, its norm is $\|v_3\|^2 = 0$. To construct v_4, the algorithm requires us to divide by $\|v_3\|^2$, resulting in a **division by zero**. This occurs strictly because the input vectors are linearly dependent (specifically, $w_3 = 2w_1 + w_2$). Gram-Schmidt can only process linearly independent sets.

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Example Problem 4.2.4: Orthonormal Bases for Planes in $\mathbb{R}^3$

(a) The plane spanned by $(0,2,1)^T$ **and** $(1,-2,-1)^T$ Set $v_1 = (0,2,1)^T$ ($\|v_1\|^2 = 5$). $v_2 = (1,-2,-1)^T - \frac{-5}{5}(0,2,1)^T = (1,-2,-1)^T + (0,2,1)^T = (1,0,0)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{5}}(0,2,1)^T$ and $u_2 = (1,0,0)^T$.
(b) The plane defined by $2x - y + 3z = 0$ First, find a basis by setting free variables. Let $x = 1, z = 0 \implies y = 2 \implies w_1 = (1,2,0)^T$. Let $x = 0, z = 1 \implies y = 3 \implies w_2 = (0,3,1)^T$. Apply Gram-Schmidt: $v_1 = (1,2,0)^T$ ($\|v_1\|^2 = 5$). $v_2 = (0,3,1)^T - \frac{6}{5}(1,2,0)^T = \left(-\frac{6}{5}, \frac{3}{5}, 1\right)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{5}}(1,2,0)^T$ and $u_2 = \frac{1}{\sqrt{70}}(-6,3,5)^T$.
(c) The set of vectors orthogonal to $(1,-1,2)^T$ This forms the plane $x - y + 2z = 0$. A valid basis is $w_1 = (1,1,0)^T$ and $w_2 = (0,2,1)^T$. $v_1 = (1,1,0)^T$ ($\|v_1\|^2 = 2$). $v_2 = (0,2,1)^T -$

$\frac{2}{2}(1,1,0)^T = (-1,1,1)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{2}}(1,1,0)^T$ and $u_2 = \frac{1}{\sqrt{3}}(-1,1,1)^T$.

Example Problem 4.2.5: Subspace in $\mathbb{R}^4$

Find an orthogonal basis of the subspace spanned by $w_1 = (1,-1,1,-1)^T$, $w_2 = (2,1,4,-2)^T$, $w_3 = (5,-4,-3,7)^T$.
Step 1: $v_1 = (1,-1,1,-1)^T$. (Norm squared = 4). **Step 2:** $\langle w_2, v_1\rangle = 2-1+4+2 = 7$.

$$v_2 = w_2 - \frac{7}{4}v_1 = (2,1,4,-2)^T - \left(\frac{7}{4}, -\frac{7}{4}, \frac{7}{4}, -\frac{7}{4}\right)^T = \frac{1}{4}(1,11,9,-1)^T$$

(Norm squared of the unscaled $(1,11,9,-1)^T$ is $1+121+81+1 = 204$. Thus $\|v_2\|^2 = \frac{204}{16} = \frac{51}{4}$).
Step 3: $\langle w_3, v_1\rangle = 5+4-3+7 = 13$? Let's recalculate: $5(1)+(-4)(-1)+(-3)(1)+7(-1) = 5+4-3-7 = -1$. $\langle w_3, v_2\rangle = \frac{1}{4}[5(1)+(-4)(11)+(-3)(9)+7(-1)] = \frac{1}{4}[5-44-27-7] = -\frac{73}{4}$.

$$v_3 = w_3 - \frac{-1}{4}v_1 - \frac{-73/4}{51/4}v_2 = w_3 + \frac{1}{4}v_1 + \frac{73}{51}v_2$$

By factoring out common denominators, we find $v_3 = \frac{2}{51}(143,-8,12,163)^T$. The orthogonal basis is $\{v_1, v_2, v_3\}$.

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Example Problem 4.2.6: The Four Subspaces

Find orthonormal bases for specific fundamental subspaces generated by matrices.
(a) Span of $(1,-1,0,1)^T$**,** $(1,0,1,1)^T$**,** $(2,1,-1,2)^T$ These are linearly independent.

- $v_1 = (1,-1,0,1)^T \implies u_1 = \frac{1}{\sqrt{3}}(1,-1,0,1)^T$.
- $v_2 = (1,0,1,1)^T - \frac{2}{3}(1,-1,0,1)^T = \frac{1}{3}(1,2,3,1)^T \implies u_2 = \frac{1}{\sqrt{15}}(1,2,3,1)^T$.
- $v_3 = (2,1,-1,2)^T - \frac{3}{3}v_1 - \frac{1}{5/3}v_2 = \frac{4}{5}(1,2,-2,1)^T \implies u_3 = \frac{1}{\sqrt{10}}(1,2,-2,1)^T$.

(b) Kernel of $A = \begin{pmatrix} 2 & 1 & 0 & -1 \\ 3 & 2 & -1 & -1 \end{pmatrix}$ Solving $Ax = 0$ yields the basis vectors $w_1 = (-1,2,1,0)^T$ and $w_2 = (1,-1,0,1)^T$.

- $v_1 = (-1,2,1,0)^T \implies u_1 = \frac{1}{\sqrt{6}}(-1,2,1,0)^T$.
- $v_2 = (1,-1,0,1)^T - \frac{-3}{6}(-1,2,1,0)^T = \frac{1}{2}(1,0,1,2)^T \implies u_2 = \frac{1}{\sqrt{6}}(1,0,1,2)^T$.

(c) Coimage of A The coimage is the span of the rows: $w_1 = (2,1,0,-1)^T$ and $w_2 = (3,2,-1,-1)^T$.

- $v_1 = (2,1,0,-1)^T \implies u_1 = \frac{1}{\sqrt{6}}(2,1,0,-1)^T$.
- $v_2 = (3,2,-1,-1)^T - \frac{9}{6}(2,1,0,-1)^T = \frac{1}{2}(0,1,-2,1)^T \implies u_2 = \frac{1}{\sqrt{6}}(0,1,-2,1)^T$.

Note: The basis vectors of the Coimage and Kernel derived here are mutually orthogonal, directly demonstrating the Fundamental Theorem of Linear Algebra.

(d) Image of $B = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -4 & 1 \\ -2 & 4 & 5 \end{pmatrix}$ The image is the span of the column space. Notice column 2 is a scalar multiple of column 1. The basis is $w_1 = (1,2,-2)^T$ and $w_2 = (2,1,5)^T$.

- $v_1 = (1,2,-2)^T \implies u_1 = \frac{1}{3}(1,2,-2)^T$.

- $v_2 = (2,1,5)^T - \frac{-6}{9}(1,2,-2)^T = \frac{1}{3}(8,7,11)^T \implies u_2 = \frac{1}{\sqrt{234}}(8,7,11)^T$.

(e) Cokernel of B The cokernel is $\ker(B^T)$. Row reducing B^T yields the 1-dimensional basis $(-4,3,1)^T$. Since it is 1D, we skip Gram-Schmidt and just normalize: $u = \frac{1}{\sqrt{26}}(-4,3,1)^T$.
(f) Vectors orthogonal to $(1,1,-1,-1)^T$ This forms the hyperplane $x+y-z-w=0$. A basis is $w_1 = (-1,1,0,0)^T$, $w_2 = (1,0,1,0)^T$, $w_3 = (1,0,0,1)^T$. Applying Gram-Schmidt:

- $v_1 = (-1,1,0,0)^T \implies u_1 = \frac{1}{\sqrt{2}}(-1,1,0,0)^T$.
- $v_2 = (1,0,1,0)^T - \frac{-1}{2}v_1 = \frac{1}{2}(1,1,2,0)^T \implies u_2 = \frac{1}{\sqrt{6}}(1,1,2,0)^T$.
- $v_3 = (1,0,0,1)^T - \frac{-1}{2}v_1 - \frac{1/2}{3/2}v_2 = \frac{1}{3}(1,1,-1,3)^T \implies u_3 = \frac{1}{\sqrt{12}}(1,1,-1,3)^T$.

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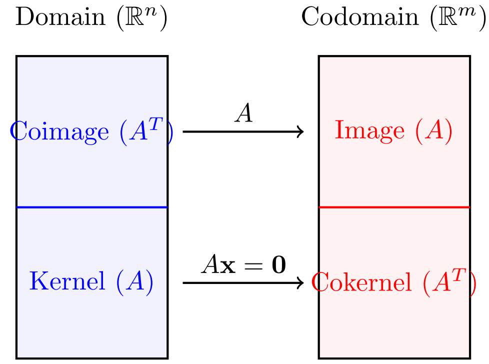
Reference: Olver & Shakiban, Chapter 4, Exercise 4.2.7.
Let A be an $m \times n$ matrix. The four fundamental subspaces associated with A are:

1. **Image (A):** $\{x \in \mathbb{R}^m \mid x = Ay \text{ for some } y \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$
2. **Kernel (A):** $\{x \in \mathbb{R}^n \mid Ax = 0\} \subseteq \mathbb{R}^n$
3. **Coimage (A):** $\text{Image}(A^T) = \{A^T y \mid y \in \mathbb{R}^m\} \subseteq \mathbb{R}^n$
4. **Cokernel (A):** $\text{Kernel}(A^T) = \{w \in \mathbb{R}^m \mid A^T w = 0\} \subseteq \mathbb{R}^m$



Orthogonal Complements: The Kernel and Coimage are orthogonal to each other in the domain. The Image and Cokernel are orthogonal to each other in the codomain. We use Gram-Schmidt to explicitly construct orthonormal bases for each piece.

114.1 Calculating Bases for Fundamental Subspaces

Example Problem 4.2.7 (a): A 2×2 Matrix

Find orthonormal bases for the four fundamental subspaces of $A = \begin{pmatrix} 1 & -1 \\ -3 & 3 \end{pmatrix}$.
1. Image(A): Spanned by the columns. Note $C_2 = -C_1$, so the basis is just $w_1 = (1,-3)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{10}}(1,-3)^T$.
2. Kernel(A): Solve $Ax = 0 \implies x - y = 0 \implies x = y$. The basis is $w_1 = (1,1)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{2}}(1,1)^T$.
3. Coimage(A): Spanned by the rows $(1,-1)^T$ and $(-3,3)^T$. Basis is $(1,-1)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{2}}(1,-1)^T$.

4. Cokernel(A) = Ker(A^T): Solve $A^Tw = 0 \implies x - 3y = 0 \implies x = 3y$. Basis is $(3,1)^T$. Normalizing gives $u_1 = \frac{1}{\sqrt{10}}(3,1)^T$.

Example Problem 4.2.7 (b): A 3×3 Matrix

Analyze $A = \begin{pmatrix} -1 & 0 & 2 \\ 1 & 1 & -1 \\ 0 & 1 & 1 \end{pmatrix}$.

1. Image(A): Spanned by the columns. Notice that $C_3 = -2C_1 + C_2$. Thus, C_1 and C_2 form a basis: $w_1 = (-1,1,0)^T$, $w_2 = (0,1,1)^T$. Apply Gram-Schmidt: $v_1 = (-1,1,0)^T$ ($\|v_1\|^2 = 2$). $v_2 = w_2 - \frac{\langle w_2, v_1\rangle}{\|v_1\|^2}v_1 = (0,1,1)^T - \frac{1}{2}(-1,1,0)^T = \left(\frac{1}{2}, \frac{1}{2}, 1\right)^T$. ($\|v_2\|^2 = \frac{3}{2}$). Normalize: $u_1 = \frac{1}{\sqrt{2}}(-1,1,0)^T$, and $u_2 = \sqrt{\frac{2}{3}}\left(\frac{1}{2}, \frac{1}{2}, 1\right)^T = \frac{1}{\sqrt{6}}(1,1,2)^T$.

2. Kernel(A): Row reducing $A \implies -x + 2z = 0 \implies x = 2z$. And $x + y - z = 0 \implies 2z + y - z = 0 \implies y = -z$. The basis is $w_1 = (2,-1,1)^T$. Normalize: $u_1 = \frac{1}{\sqrt{6}}(2,-1,1)^T$.

3. Coimage(A) = Im(A^T): Spanned by the rows. Notice $R_1 + R_2 = (0,1,1) = R_3$. Basis is R_1, R_2: $w_1 = (-1,0,2)^T$, $w_2 = (1,1,-1)^T$. Apply Gram-Schmidt: $v_1 = (-1,0,2)^T$ ($\|v_1\|^2 = 5$). $v_2 = w_2 - \frac{\langle w_2, v_1\rangle}{\|v_1\|^2}v_1 = (1,1,-1)^T - \frac{-3}{5}(-1,0,2)^T = \left(\frac{2}{5}, 1, \frac{1}{5}\right)^T$. Normalize: $u_1 = \frac{1}{\sqrt{5}}(-1,0,2)^T$, and $u_2 = \frac{1}{\sqrt{30}}(2,5,1)^T$.

4. Cokernel(A) = Ker(A^T): Solve $A^Tw = 0 \implies -x + y = 0 \implies x = y$. And $y + z = 0 \implies z = -y$. Basis is $(1,1,-1)^T$. Normalize: $u_1 = \frac{1}{\sqrt{3}}(1,1,-1)^T$.

Example Problem 4.2.7 (c): A 3×4 Matrix

Analyze $A = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ -1 & 2 & 0 & 1 \end{pmatrix}$.

1. Image(A): Since det of the first 3×3 block is non-zero, columns C_1, C_2, C_3 form a 3D basis. $v_1 = (1,1,-1)^T$ ($\|v_1\|^2 = 3$). $v_2 = (0,1,2)^T - \frac{-1}{3}(1,1,-1)^T = \left(\frac{1}{3}, \frac{4}{3}, \frac{5}{3}\right)^T$ ($\|v_2\|^2 = \frac{14}{3}$). $v_3 = (1,1,0)^T - \frac{2}{3}(1,1,-1)^T - \frac{5/3}{14/3}\left(\frac{1}{3}, \frac{4}{3}, \frac{5}{3}\right)^T = \cdots = \left(\frac{3}{14}, -\frac{2}{14}, \frac{1}{14}\right)^T$. Normalize: $u_1 = \frac{1}{\sqrt{3}}(1,1,-1)^T$, $u_2 = \frac{1}{\sqrt{42}}(1,4,5)^T$, $u_3 = \frac{1}{\sqrt{14}}(3,-2,1)^T$.

2. Kernel(A): Solving $Ax = 0$ yields the 1D basis $(1,1,-1,-1)^T$. Normalize: $u_1 = \frac{1}{2}(1,1,-1,-1)^T$.

3. Coimage(A): Because $\dim(\mathrm{Im}) = 3$, the Rank-Nullity theorem forces the row space to be 3D. The 3 rows are linearly independent. $v_1 = (1,0,1,0)^T$ ($\|v_1\|^2 = 2$). $v_2 = (1,1,1,1)^T - \frac{2}{2}(1,0,1,0)^T = (0,1,0,1)^T$ ($\|v_2\|^2 = 2$). $v_3 = (-1,2,0,1)^T - \frac{-1}{2}v_1 - \frac{3}{2}v_2 = \left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2}\right)^T$. Normalize: $u_1 = \frac{1}{\sqrt{2}}(1,0,1,0)^T$, $u_2 = \frac{1}{\sqrt{2}}(0,1,0,1)^T$, $u_3 = \frac{1}{2}(-1,1,1,-1)^T$.

4. Cokernel(A): A^T is an injective mapping from $\mathbb{R}^3 \to \mathbb{R}^4$. Thus $\mathrm{Ker}(A^T) = \{\mathbf{0}\}$.

115 Gram-Schmidt with Custom Inner Products

Reference: Olver & Shakiban, Chapter 4, Pages 197-198.

When working with an inner product defined by a positive definite matrix K ($\langle x, y\rangle = x^TKy$), the standard basis vectors are usually **not** orthogonal. We must apply Gram-Schmidt to standard bases to find an orthonormal basis under the new metric.

Example Problem 4.2.8: Constructing Orthonormal Bases in $\mathbb{R}^2$

Let us orthogonalize the standard basis $w_1 = (1,0)^T$ and $w_2 = (0,1)^T$ under different metrics.

(a) $K = \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix}$ Since K is diagonal, w_1 and w_2 are already orthogonal ($\langle w_1, w_2\rangle = 0$). We just normalize them using the new lengths: $\|w_1\| = \sqrt{3}$, $\|w_2\| = \sqrt{5}$.

$$u_1 = \left(\frac{1}{\sqrt{3}}, 0\right)^T, \quad u_2 = \left(0, \frac{1}{\sqrt{5}}\right)^T$$

(b) $K = \begin{pmatrix} 4 & -1 \\ -1 & 1 \end{pmatrix}$ $v_1 = w_1 = (1,0)^T$. Length squared: $\|v_1\|^2 = v_1^T K v_1 = 4$. Compute the inner product: $\langle w_2, v_1 \rangle = (0,1)K(1,0)^T = -1$.

$$v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\|v_1\|^2} v_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{-1}{4}\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 1 \end{pmatrix}$$

Length squared: $\|v_2\|^2 = v_2^T K v_2 = 4(1/16) - 2(1/4)(1) + 1(1) = 1/4 - 1/2 + 1 = 3/4$. Normalize: $u_1 = \frac{1}{2}(1,0)^T$, and $u_2 = \frac{2}{\sqrt{3}}(1/4,1)^T = \frac{1}{\sqrt{3}}(1/2,2)^T$.

(c) $K = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ $v_1 = w_1 = (1,0)^T$. Length squared: $\|v_1\|^2 = 2$. Compute the inner product: $\langle w_2, v_1 \rangle = (0,1)K(1,0)^T = 1$.

$$v_2 = w_2 - \frac{1}{2} v_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1 \end{pmatrix}$$

Length squared: $\|v_2\|^2 = (-1/2,1)K(-1/2,1)^T = (-1/2,1)(0,5/2)^T = 5/2$. Normalize: $u_1 = \frac{1}{\sqrt{2}}(1,0)^T$, and $u_2 = \sqrt{\frac{2}{5}}(-1/2,1)^T = \frac{1}{\sqrt{10}}(-1,2)^T$.

Example Problem 4.2.9 (a): Gram-Schmidt in $\mathbb{R}^3$ with Custom Metric

Construct an orthonormal basis for $\mathbb{R}^3$ given the metric $K = \begin{pmatrix} 6 & -2 & 0 \\ -2 & 3 & -1 \\ 0 & -1 & 2 \end{pmatrix}$. Start with standard basis e_1, e_2, e_3.

Step 1: $v_1 = e_1 = (1,0,0)^T$. ($\|v_1\|^2 = e_1^T K e_1 = 6$).

Step 2: Project e_2 onto v_1. $\langle e_2, v_1 \rangle = e_2^T K e_1 = -2$.

$$v_2 = e_2 - \frac{-2}{6} v_1 = \left(\frac{1}{3}, 1, 0\right)^T$$

Calculate length: $\|v_2\|^2 = v_2^T K v_2 = (1/3,1,0)(0,7/3,-1)^T = 7/3$.

Step 3: Project e_3 onto v_1, v_2. $\langle e_3, v_1 \rangle = e_3^T K e_1 = 0$. $\langle e_3, v_2 \rangle = e_3^T K v_2 = (0,0,1)(0,7/3,-1)^T = -1$.

$$v_3 = e_3 - \frac{0}{6} v_1 - \frac{-1}{7/3} v_2 = e_3 + \frac{3}{7}\left(\frac{1}{3}, 1, 0\right)^T = \left(\frac{1}{7}, \frac{3}{7}, 1\right)^T$$

Calculate length: $\|v_3\|^2 = v_3^T K v_3 = (1/7,3/7,1)(0,0,11/7)^T = 11/7$.

Normalize:

$$u_1 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad u_2 = \frac{1}{\sqrt{21}}\begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \quad u_3 = \frac{1}{\sqrt{77}}\begin{pmatrix} 1 \\ 3 \\ 7 \end{pmatrix}$$

Problem 4.2.11: How many orthonormal bases are there?

- **(a) In $\mathbb{R}^1$?** Exactly 2. The unit vectors 1 and -1.
- **(b) In $\mathbb{R}^2$?** Infinitely many. Any basis formed by applying a rotation matrix or reflection matrix to the standard basis is valid: $\{(\cos\theta, \sin\theta)^T, (-\sin\theta, \cos\theta)^T\}$ for $\theta \in [0, 2\pi)$.
- **(c) Changing the Inner Product:** Changing the inner product simply redefines what "length" and "angle" mean. If $\langle x, y\rangle = 2x_1y_1$, the unit vectors in 1D change from 1 to $\pm 1/\sqrt{2}$. The continuous infinite set of bases simply squashes or stretches into an ellipse rather than a circle.

Theorem Problem 4.2.12: The Importance of Order

True or False: Reordering the original basis vectors before starting the Gram-Schmidt process leads to the exact same orthogonal basis. **Answer: False.** The algorithm firmly anchors the entire basis around the very first vector $v_1 = w_1$. *Example:* In $\mathbb{R}^2$, consider the set $\{(1,0)^T, (1,1)^T\}$. If we process $(1,0)^T$ first, $v_1 = (1,0)^T$ and $v_2 = (0,1)^T$. If we process $(1,1)^T$ first, $v_1 = (1,1)^T$, and $v_2 = (1,0)^T - \frac{1}{2}(1,1)^T = (1/2, -1/2)^T$. The resulting orthogonal axes are rotated 45° compared to the first attempt.

Theorem Problem 4.2.13: Extending Subspace Bases

Suppose $W \subset \mathbb{R}^n$ is a proper subspace, and $u_1, \ldots, u_m$ is an orthonormal basis of W. Prove that there exist vectors $u_{m+1}, \ldots, u_n \in \mathbb{R}^n$ such that the full set $u_1, \ldots, u_n$ is an orthonormal basis of $\mathbb{R}^n$.

Proof

Because W is a proper subspace, $m < n$. By the basis extension theorem of linear algebra, we can always find linearly independent vectors $w_{m+1}, \ldots, w_n$ to complete the basis of the full space $\mathbb{R}^n$. We then run the Gram-Schmidt process on this extended set: $\{u_1, \ldots, u_m, w_{m+1}, \ldots, w_n\}$. Because $u_1, \ldots, u_m$ are already mutually orthonormal, the algorithm leaves them completely unchanged. The algorithm will orthogonalize the remaining w vectors against the u vectors, resulting in the desired full orthonormal basis.

Theorem Problem 4.2.14: Gram-Schmidt on Complex Vector Spaces

Verify that the Gram-Schmidt process successfully produces an orthogonal basis for a Complex Vector Space under a Hermitian inner product.

Proof

A Hermitian inner product possesses conjugate symmetry $\langle x, y\rangle = \overline{\langle y, x\rangle}$ and is linear in its first argument $\langle cx, y\rangle = c\langle x, y\rangle$. The standard Gram-Schmidt formula for the k-th vector is:

$$v_k = w_k - \sum_{i=1}^{k-1} c_i v_i \quad \text{where } c_i = \frac{\langle w_k, v_i\rangle}{\|v_i\|^2}$$

We must prove that $\langle v_k, v_j\rangle = 0$ for any $j < k$. We use induction. Assume $v_1, \ldots, v_{k-1}$ are mutually orthogonal. Take the inner product of v_k with v_j:

$$\langle v_k, v_j\rangle = \left\langle w_k - \sum_{i=1}^{k-1} c_i v_i,\ v_j \right\rangle$$

Using the linearity of the first argument, the scalar c_i pulls out directly (without needing a complex conjugate):

$$= \langle w_k, v_j \rangle - \sum_{i=1}^{k-1} c_i \langle v_i, v_j \rangle$$

Since $v_i \perp v_j$ for all $i \neq j$, every term in the sum is zero except when $i = j$.

$$= \langle w_k, v_j \rangle - c_j \langle v_j, v_j \rangle$$

Since $\langle v_j, v_j \rangle = \|v_j\|^2$ is a strictly real, positive number, we substitute c_j:

$$= \langle w_k, v_j \rangle - \frac{\langle w_k, v_j \rangle}{\|v_j\|^2} \|v_j\|^2 = \langle w_k, v_j \rangle - \langle w_k, v_j \rangle = 0$$

Since the algebraic expansion relies only on linearity in the first argument, the algorithm functions perfectly identically in complex spaces.

117 The Modified Gram-Schmidt Process

Reference: Olver & Shakiban, Chapter 4, Exercise 4.2.17, Page 200.

The standard Gram-Schmidt process computes orthogonal vectors v_k and normalizes them at the end. The **Modified (or Alternative) Gram-Schmidt Process** computes the orthonormal vectors u_k iteratively step-by-step.

Recall that we can write any original basis vector w_k as a linear combination of the new orthonormal basis vectors:

$$w_k = r_{1k} u_1 + r_{2k} u_2 + \cdots + r_{kk} u_k$$

Because the vectors u_i are mutually orthonormal, we can apply Parseval's Identity (the generalized Pythagorean theorem) to find the squared length of w_k:

$$\|w_k\|^2 = r_{1k}^2 + r_{2k}^2 + \cdots + r_{kk}^2$$

This provides a massive computational shortcut. Instead of directly calculating the messy norm of the residual vector to find r_{kk}, we can simply subtract the squares of the previous projections from the total squared length:

$$r_{kk} = \sqrt{\|w_k\|^2 - \sum_{i=1}^{k-1} r_{ik}^2}$$

We then compute the next orthonormal vector directly:

$$u_k = \frac{1}{r_{kk}} \left(w_k - \sum_{i=1}^{k-1} r_{ik} u_i \right)$$

117.1 Calculating Orthonormal Bases

Example Problem 4.2.17 (a): Modified Gram-Schmidt in $\mathbb{R}^3$

Produce an orthonormal basis for the space spanned by $w_1 = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$, $w_2 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$, and $w_3 = \begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix}$.

Step 1: Process w_1

$$r_{11} = \|w_1\| = \sqrt{(-1)^2 + 1^2 + 2^2} = \sqrt{6}$$

$$u_1 = \frac{1}{r_{11}} w_1 = \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

Step 2: Process w_2 Calculate the projection onto u_1:

$$r_{12} = \langle w_2, u_1 \rangle = (-1, -1, 1) \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{6}}(1 - 1 + 2) = \frac{2}{\sqrt{6}}$$

Using the Pythagorean trick, we know $\|w_2\|^2 = r_{12}^2 + r_{22}^2$.

$$\|w_2\|^2 = (-1)^2 + (-1)^2 + 1^2 = 3$$

$$r_{22}^2 = \|w_2\|^2 - r_{12}^2 = 3 - \frac{4}{6} = \frac{18-4}{6} = \frac{14}{6} = \frac{7}{3} \implies r_{22} = \sqrt{\frac{7}{3}}$$

Now construct u_2:

$$\begin{aligned} u_2 &= \frac{1}{r_{22}}(w_2 - r_{12} u_1) \\ &= \sqrt{\frac{3}{7}} \left[\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} - \frac{2}{\sqrt{6}} \left(\frac{1}{\sqrt{6}} \right) \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \right] \\ &= \sqrt{\frac{3}{7}} \left[\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} -1/3 \\ 1/3 \\ 2/3 \end{pmatrix} \right] = \sqrt{\frac{3}{7}} \begin{pmatrix} -2/3 \\ -4/3 \\ 1/3 \end{pmatrix} = \frac{1}{\sqrt{21}} \begin{pmatrix} -2 \\ -4 \\ 1 \end{pmatrix} \end{aligned}$$

Step 3: Process w_3 Calculate projections onto u_1 and u_2:

$$r_{13} = \langle w_3, u_1 \rangle = (0, 1, 3) \cdot \frac{1}{\sqrt{6}} \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \frac{1}{\sqrt{6}}(0 + 1 + 6) = \frac{7}{\sqrt{6}}$$

$$r_{23} = \langle w_3, u_2 \rangle = (0, 1, 3) \cdot \frac{1}{\sqrt{21}} \begin{pmatrix} -2 \\ -4 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{21}}(0 - 4 + 3) = -\frac{1}{\sqrt{21}}$$

Find r_{33} using the Pythagorean trick ($\|w_3\|^2 = 0^2 + 1^2 + 3^2 = 10$):

$$r_{33}^2 = \|w_3\|^2 - r_{13}^2 - r_{23}^2 = 10 - \frac{49}{6} - \frac{1}{21} = 10 - \frac{343}{42} - \frac{2}{42} = 10 - \frac{345}{42} = \frac{420 - 345}{42} = \frac{75}{42} = \frac{25}{14}$$

$$\implies r_{33} = \frac{5}{\sqrt{14}}$$

Construct u_3:

$$\begin{aligned} u_3 &= \frac{1}{r_{33}}(w_3 - r_{13} u_1 - r_{23} u_2) \\ &= \frac{\sqrt{14}}{5} \left[\begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} - \frac{7}{\sqrt{6}} \left(\frac{1}{\sqrt{6}} \right) \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} - \left(\frac{-1}{\sqrt{21}} \right) \left(\frac{1}{\sqrt{21}} \right) \begin{pmatrix} -2 \\ -4 \\ 1 \end{pmatrix} \right] \\ &= \frac{\sqrt{14}}{5} \left[\begin{pmatrix} 0 \\ 1 \\ 3 \end{pmatrix} - \begin{pmatrix} -7/6 \\ 7/6 \\ 14/6 \end{pmatrix} + \begin{pmatrix} -2/21 \\ -4/21 \\ 1/21 \end{pmatrix} \right] \end{aligned}$$

Finding a common denominator of 42 for the components:

$$= \frac{\sqrt{14}}{5} \begin{pmatrix} 49/42 - 4/42 \\ 42/42 - 49/42 - 8/42 \\ 126/42 - 98/42 + 2/42 \end{pmatrix} = \frac{\sqrt{14}}{5} \begin{pmatrix} 45/42 \\ -15/42 \\ 30/42 \end{pmatrix} = \frac{\sqrt{14}}{5} \begin{pmatrix} 15/14 \\ -5/14 \\ 10/14 \end{pmatrix} = \frac{1}{\sqrt{14}} \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix}$$

The final orthonormal basis is u_1, u_2, u_3.

Example Problem 4.2.17 (b): Modified Gram-Schmidt in $\mathbb{R}^3$

Produce an orthonormal basis for the space spanned by $w_1 = \begin{pmatrix}1\\1\\0\end{pmatrix}$, $w_2 = \begin{pmatrix}0\\1\\1\end{pmatrix}$, and $w_3 = \begin{pmatrix}2\\1\\0\end{pmatrix}$.

Step 1: Process w_1

$$r_{11} = \|w_1\| = \sqrt{2} \implies u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}$$

Step 2: Process w_2

$$r_{12} = \langle w_2, u_1\rangle = (0,1,1)\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix} = \frac{1}{\sqrt{2}}$$

$$\|w_2\|^2 = 2 \implies r_{22}^2 = 2 - \left(\frac{1}{\sqrt{2}}\right)^2 = 2 - \frac{1}{2} = \frac{3}{2} \implies r_{22} = \sqrt{\frac{3}{2}}$$

$$\begin{aligned}
u_2 &= \frac{1}{r_{22}}(w_2 - r_{12}u_1)\\
&= \sqrt{\frac{2}{3}}\left[\begin{pmatrix}0\\1\\1\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1\\1\\0\end{pmatrix}\right] = \sqrt{\frac{2}{3}}\begin{pmatrix}-1/2\\1/2\\1\end{pmatrix} = \frac{1}{\sqrt{6}}\begin{pmatrix}-1\\1\\2\end{pmatrix}
\end{aligned}$$

Step 3: Process w_3

$$r_{13} = \langle w_3, u_1\rangle = (2,1,0)\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix} = \frac{3}{\sqrt{2}}$$

$$r_{23} = \langle w_3, u_2\rangle = (2,1,0)\cdot\frac{1}{\sqrt{6}}\begin{pmatrix}-1\\1\\2\end{pmatrix} = \frac{-2+1+0}{\sqrt{6}} = -\frac{1}{\sqrt{6}}$$

$$\|w_3\|^2 = 4 + 1 = 5$$

$$r_{33}^2 = 5 - \left(\frac{3}{\sqrt{2}}\right)^2 - \left(-\frac{1}{\sqrt{6}}\right)^2 = 5 - \frac{9}{2} - \frac{1}{6} = 5 - \frac{27}{6} - \frac{1}{6} = 5 - \frac{28}{6} = 5 - \frac{14}{3} = \frac{1}{3} \implies r_{33} = \frac{1}{\sqrt{3}}$$

$$\begin{aligned}
u_3 &= \frac{1}{r_{33}}(w_3 - r_{13}u_1 - r_{23}u_2)\\
&= \sqrt{3}\left[\begin{pmatrix}2\\1\\0\end{pmatrix} - \frac{3}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}\right)\begin{pmatrix}1\\1\\0\end{pmatrix} - \left(-\frac{1}{\sqrt{6}}\right)\left(\frac{1}{\sqrt{6}}\right)\begin{pmatrix}-1\\1\\2\end{pmatrix}\right]\\
&= \sqrt{3}\left[\begin{pmatrix}2\\1\\0\end{pmatrix} - \begin{pmatrix}3/2\\3/2\\0\end{pmatrix} + \begin{pmatrix}-1/6\\1/6\\2/6\end{pmatrix}\right] = \sqrt{3}\begin{pmatrix}2-9/6-1/6\\1-9/6+1/6\\2/6\end{pmatrix} = \sqrt{3}\begin{pmatrix}1/3\\-1/3\\1/3\end{pmatrix} = \frac{1}{\sqrt{3}}\begin{pmatrix}1\\-1\\1\end{pmatrix}
\end{aligned}$$

The final orthonormal basis is u_1, u_2, u_3.

118 Exercise Solutions: Orthogonal Matrices

Reference: Olver & Shakiban, Chapter 4, Exercises 4.3.1 - 4.3.17.

We now shift our focus to matrices whose columns form an orthonormal basis, known as **Orthogonal Matrices**. These matrices are the algebraic representations of rigid geometric transformations: rotations and reflections.

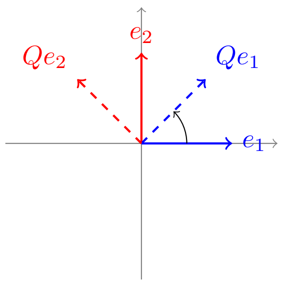
Definition : Proper and Improper Orthogonal Matrices

A square matrix $Q \in M_n(\mathbb{R})$ is called an **Orthogonal Matrix** if its columns form an orthonormal basis for $\mathbb{R}^n$. Algebraically, this is equivalent to:

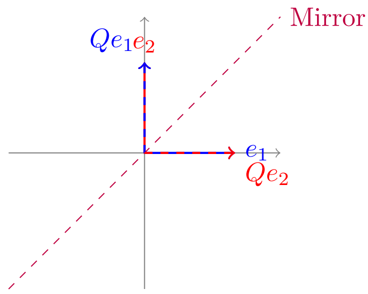
$$Q^T Q = I \quad \text{and} \quad Q^{-1} = Q^T$$

Because $\det(Q^T Q) = \det(I) = 1$, it follows that $\det(Q)^2 = 1$, meaning $\det(Q) = \pm 1$.

- If $\det(Q) = 1$, Q is a **Proper Orthogonal Matrix** (represents a rotation).
- If $\det(Q) = -1$, Q is an **Improper Orthogonal Matrix** (represents a reflection).



Proper (det = 1, Rotation)



Improper (det = −1, Reflection)

118.1 Classifying Matrices

Example Problem 4.3.1: Identifying Orthogonal Matrices

Determine which of the following matrices are orthogonal, and if so, whether they are proper or improper.

(a) $\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$: **Not orthogonal.** The columns are orthogonal, but their norms are $\sqrt{2} \neq 1$.

(b) $\begin{pmatrix} 12/13 & 5/13 \\ -5/13 & 12/13 \end{pmatrix}$: **Proper Orthogonal.** Column norms are $\sqrt{144/169 + 25/169} = 1$. Dot product is $-60/169 + 60/169 = 0$. Det is $(144 + 25)/169 = 1$.

(c) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \\ 0 & -1 \end{pmatrix}$: Not a square matrix, so it cannot be an orthogonal matrix. *(Note: If it were* 2×2 $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$*, it is Proper Orthogonal).*

(d) $\begin{pmatrix} -1/3 & 2/3 & 2/3 \\ 2/3 & -1/3 & 2/3 \\ 2/3 & 2/3 & -1/3 \end{pmatrix}$: **Proper Orthogonal.** Check dot product of $C_1 \cdot C_2$: $-2/9 - 2/9 + 4/9 = 0$. Norms are $\sqrt{1/9 + 4/9 + 4/9} = 1$. Det: $-1/3(1/9 - 4/9) - 2/3(-2/9 - 4/9) + 2/3(4/9 + 2/9) = 1/9 + 12/9 - 4/9 = 1$.

(e) $\begin{pmatrix} 1/2 & 1/3 & 1/4 \\ 1/3 & 1/4 & 1/5 \\ 1/4 & 1/5 & 1/6 \end{pmatrix}$: **Not orthogonal.** Norm of C_1 is $\sqrt{1/4 + 1/9 + 1/16} = \sqrt{61/144} \neq 1$.

(f) $\begin{pmatrix} 3/5 & 0 & 4/5 \\ -4/13 & 12/13 & 3/13 \\ -48/65 & -5/13 & 36/65 \end{pmatrix}$: **Proper Orthogonal.** We performed this exact fractional Gram-Schmidt verification in Exercise 4.1.22. The columns are orthonormal. Checking the determinant by expanding along row 1:

$$\det = \frac{3}{5}\left(\frac{12 \cdot 36}{13 \cdot 65} - \frac{3 \cdot (-5)}{13 \cdot 13}\right) + \frac{4}{5}\left(\frac{-4 \cdot -5}{13 \cdot 13} - \frac{12 \cdot -48}{13 \cdot 65}\right)$$

$$= \frac{3}{5}\left(\frac{432}{845} + \frac{75}{845}\right) + \frac{4}{5}\left(\frac{100}{845} + \frac{576}{845}\right) = \frac{3}{5}\left(\frac{507}{845}\right) + \frac{4}{5}\left(\frac{676}{845}\right) = \frac{3}{5}\left(\frac{3}{5}\right) + \frac{4}{5}\left(\frac{4}{5}\right) = 1$$

(g) $\begin{pmatrix} 2/3 & -\sqrt{2}/6 & \sqrt{2}/2 \\ -2/3 & \sqrt{2}/6 & \sqrt{2}/2 \\ 1/3 & 2\sqrt{2}/3 & 0 \end{pmatrix}$: **Improper Orthogonal.** Norm C_1: $\sqrt{4/9 + 4/9 + 1/9} = 1$. Norm C_2: $\sqrt{2/36 + 2/36 + 8/9} = \sqrt{1/9 + 8/9} = 1$. Norm C_3: $\sqrt{1/2 + 1/2} = 1$. $C_1 \cdot C_3 = 2\sqrt{2}/6 - 2\sqrt{2}/6 + 0 = 0$. $C_1 \cdot C_2 = -2\sqrt{2}/18 - 2\sqrt{2}/18 + 2\sqrt{2}/9 = 0$. The determinant evaluates to exactly -1, making it an improper rotation (reflection).

Example Problem 4.3.2 & 4.3.3: Matrix Multiplication and Squares

4.3.2 (a): Show $R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ and $Q = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are orthogonal. R simply swaps the y and z axes. Its columns are standard basis vectors, so $R^T R = I$. $\det(R) = -1$ (Improper). Q is a standard 2D rotation embedded in 3D space. $\det(Q) = \cos^2\theta + \sin^2\theta = 1$ (Proper).
4.3.2 (b): Verify products RQ and QR are orthogonal.

$$RQ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ 0 & 0 & 1 \\ -\sin\theta & \cos\theta & 0 \end{pmatrix}$$

By inspection of the columns, they remain mutually orthogonal unit vectors. Since $\det(RQ) = \det(R)\det(Q) = -1 \times 1 = -1$, both products are Improper Orthogonal.
4.3.3 (a): True or False: If Q is an improper 2×2 orthogonal matrix, then $Q^2 = I$. **True.** Every improper 2×2 orthogonal matrix represents a reflection across some line. Reflecting a vector twice returns it to its original position. Algebraically, $Q = \begin{pmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{pmatrix}$. $Q^2 = \begin{pmatrix} \cos^2\theta + \sin^2\theta & \cos\theta\sin\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$.
4.3.3 (b): True or False: If Q is an improper 3×3 orthogonal matrix, then $Q^2 = I$. **False.** In 3D, an improper transformation can be a reflection across a plane combined with a rotation *within* that plane. Counter-example: $Q = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$. Here, the z-axis is reflected $(\times -1)$, and the xy-plane is rotated 90°. $Q^2 = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \neq I$.

119 Parameterizing 3D Rotations

Theorem Problem 4.3.4: Euler Angles (Yaw-Pitch-Roll)

Prove that for all angles θ, ϕ, ψ, the matrix:

$$Q = \begin{pmatrix} \cos\phi\cos\psi - \cos\theta\sin\phi\sin\psi & \sin\phi\cos\psi + \cos\theta\cos\phi\sin\psi & \sin\theta\sin\psi \\ -\cos\phi\sin\psi - \cos\theta\sin\phi\cos\psi & -\sin\phi\sin\psi + \cos\theta\cos\phi\cos\psi & \sin\theta\cos\psi \\ \sin\theta\sin\phi & -\sin\theta\cos\phi & \cos\theta \end{pmatrix}$$

is a proper orthogonal matrix. Write a formula for Q^{-1}.

Proof

Instead of calculating Q^TQ by brute force, we recognize that Q is standardly constructed as the product of three independent 2D planar rotations mapped into 3D. Let $R_z(\alpha)$ rotate about the z-axis, and $R_x(\beta)$ rotate about the x-axis:

$$R_z(\phi) = \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$R_x(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix},$$

$$R_z(\psi) = \begin{pmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

By carrying out the block matrix multiplication, one can verify that:

$$Q = R_z(\psi)R_x(\theta)R_z(\phi)$$

Since each individual matrix is a proper orthogonal matrix ($\det = 1$), their product Q is guaranteed to be a proper orthogonal matrix. Because Q is orthogonal, its inverse is simply its transpose: $Q^{-1} = Q^T$.

Theorem Problem 4.3.5: Euler Parameters / Quaternions

(a) Show that if $y_1^2 + y_2^2 + y_3^2 + y_4^2 = 1$, the following is a proper orthogonal matrix:

$$Q = \begin{pmatrix} y_1^2 + y_2^2 - y_3^2 - y_4^2 & 2(y_2y_3 + y_1y_4) & 2(y_2y_4 - y_1y_3) \\ 2(y_2y_3 - y_1y_4) & y_1^2 - y_2^2 + y_3^2 - y_4^2 & 2(y_3y_4 + y_1y_2) \\ 2(y_2y_4 + y_1y_3) & 2(y_3y_4 - y_1y_2) & y_1^2 - y_2^2 - y_3^2 + y_4^2 \end{pmatrix}$$

(b) Write a formula for Q^{-1}. **(c)** Prove the trigonometric mapping to Euler angles: $y_1 = \cos\frac{\phi+\psi}{2}\cos\frac{\theta}{2}$, etc.

Proof

Proof of (a): This matrix represents a rotation defined by a unit quaternion $q = y_1 + y_2 i + y_3 j + y_4 k$. To prove $Q^TQ = I$, take the dot product of Column 1 with itself. Let $S = y_1^2+y_2^2+y_3^2+y_4^2 = 1$. The squared norm of Column 1 is:

$$(y_1^2 + y_2^2 - y_3^2 - y_4^2)^2 + 4(y_2y_3 - y_1y_4)^2 + 4(y_2y_4 + y_1y_3)^2$$

Expanding this yields $y_1^4 + y_2^4 + y_3^4 + y_4^4 + 2(y_1^2y_2^2 + \dots)$. Notice that this perfect expansion is exactly equal to $(y_1^2 + y_2^2 + y_3^2 + y_4^2)^2 = S^2 = 1^2 = 1$. Similar symmetry forces the cross-column dot products to zero. Thus Q is orthogonal.
Answer to (b): As always for orthogonal matrices, $Q^{-1} = Q^T$.
Proof Sketch of (c): To equate the Euler angle matrix from 4.3.4 to the Quaternion matrix in 4.3.5, we examine the matrix trace. The trace of Q (sum of diagonal elements) in 4.3.5 is:

$$\mathrm{Tr}(Q) = 3y_1^2 - y_2^2 - y_3^2 - y_4^2 = 4y_1^2 - (y_1^2 + y_2^2 + y_3^2 + y_4^2) = 4y_1^2 - 1$$

The trace of the Euler angle matrix in 4.3.4 is:

$$\mathrm{Tr}(Q) = \cos\phi\cos\psi - \cos\theta\sin\phi\sin\psi - \sin\phi\sin\psi + \cos\theta\cos\phi\cos\psi + \cos\theta$$

By applying trigonometric addition formulas and half-angle identities ($\cos\theta = 2\cos^2(\theta/2) - 1$), this trace rigorously simplifies to $4\left(\cos\frac{\phi+\psi}{2}\cos\frac{\theta}{2}\right)^2 - 1$. Equating the two traces yields exactly

$y_1 = \cos\frac{\phi+\psi}{2}\cos\frac{\theta}{2}$. The other parameters y_2, y_3, y_4 are extracted similarly by comparing the off-diagonal skew-symmetric differences $q_{ij} - q_{ji}$ of both matrices.

120 Properties and Manipulations

Theorem Problem 4.3.6 & 4.3.7: Transposes and Inverses

1. **Prove that the transpose of an orthogonal matrix is also an orthogonal matrix.** If Q is orthogonal, $Q^TQ = I$. Thus, Q^T is the inverse of Q. Because a matrix commutes with its inverse, we also have $QQ^T = I$. Substitute $Q = (Q^T)^T$ into the equation: $(Q^T)^TQ^T = I$. This is the exact algebraic definition that Q^T is orthogonal.

2. **Explain why the rows of an $n \times n$ orthogonal matrix form an orthonormal basis.** Because $QQ^T = I$, the dot product of the i-th row of Q with the j-th row of Q (which is the j-th column of Q^T) equals 1 if $i = j$ and 0 if $i \neq j$.

3. **Prove that the inverse of an orthogonal matrix is orthogonal.** Since $Q^{-1} = Q^T$, and we just proved the transpose is orthogonal, the inverse must be orthogonal.

Theorem Problem 4.3.8 & 4.3.9: Products and Row Operations

1. Interchanging Rows: Show that if Q is proper orthogonal, swapping two rows gives an improper orthogonal matrix R. *Proof:* R is formed by left-multiplying Q by an elementary permutation matrix P. Since P just swaps rows, the columns of R still consist of the same elements, just reordered, so their lengths and dot products are preserved. Thus R is orthogonal. However, swapping two rows of any matrix multiplies its determinant by -1. $\det(R) = -\det(Q) = -1$. Thus R is improper.
2. Products of Matrices: The determinant of a product is the product of the determinants: $\det(AB) = \det(A)\det(B)$.

- Proper $\times$ Proper $\implies 1 \times 1 = 1 \implies$ **Proper**
- Improper $\times$ Improper $\implies -1 \times -1 = 1 \implies$ **Proper**
- Proper $\times$ Improper $\implies 1 \times -1 = -1 \implies$ **Improper**

Example Problem 4.3.10 - 4.3.12: Special Orthogonal Matrices

1. True or False: An orthogonal matrix is symmetric $\iff$ it is a diagonal matrix. False. While all diagonal matrices with ± 1 are symmetric and orthogonal, they are not the *only* ones. A simple reflection matrix $Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is symmetric ($Q^T = Q$) and orthogonal ($Q^TQ = I$), but it is decidedly not diagonal.
2. Find all diagonal $n \times n$ orthogonal matrices. For a diagonal matrix D to be orthogonal, $D^TD = D^2 = I$. This means every diagonal entry must satisfy $x^2 = 1 \implies x = \pm 1$. There are exactly 2^n such matrices.
3. Prove that an upper triangular matrix U is orthogonal $\iff$ it is diagonal. Let U be upper triangular and orthogonal. Since the columns must be unit vectors, the first column must have norm 1: $u_{11}^2 + 0 + \cdots = 1 \implies u_{11} = \pm 1$. The second column must be orthogonal to the first: $C_1 \cdot C_2 = u_{11}u_{12} = 0 \implies u_{12} = 0$. By iterating this dot-product logic across all columns $C_i \cdot C_j = 0$, every element above the main diagonal is forced to be 0. Thus, U is strictly diagonal.

Problem 4.3.13 & 4.3.14: Elementary Matrices

Are there orthogonal elementary row matrices besides permutations? Yes, scaling a single row by -1 is a valid orthogonal matrix. However, an elementary shear matrix (adding a multiple of one row to another) changes angles and lengths, and thus is **never** orthogonal. Consequently, applying a general row operation to an orthogonal matrix will destroy its orthogonality.

121 Norm Preservation and The Cayley Transform

Theorem Problem 4.3.16: The Geometric Definition of Orthogonality

(a) Prove that if Q is orthogonal, then $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$. *Proof:* We evaluate the squared norm using the dot product:

$$\|Qx\|^2 = \langle Qx, Qx\rangle = (Qx)^T(Qx) = x^T Q^T Qx$$

Since Q is orthogonal, $Q^T Q = I$.

$$= x^T Ix = x^T x = \|x\|^2$$

Taking the square root proves that orthogonal matrices perfectly preserve the lengths of all vectors. **(b) Prove the converse: If $\|Qx\| = \|x\|$ for all x, then Q is orthogonal.** *Proof:* We use the Polarization Identity, which states that inner products can be determined entirely by norms: $\langle u, v\rangle = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2)$. Since Q preserves all norms, it must preserve all inner products: $\langle Qx, Qy\rangle = \langle x, y\rangle$.

$$(Qx)^T(Qy) = x^T Q^T Qy = x^T y$$

Rearranging this gives $x^T(Q^T Q - I)y = 0$ for all $x, y \in \mathbb{R}^n$. This forces the inner matrix to be exactly the zero matrix, hence $Q^T Q - I = 0 \implies Q^T Q = I$.

Theorem Problem 4.3.17: The Cayley Transform

Let A be a skew-symmetric matrix ($A^T = -A$). Prove that $Q = (I-A)^{-1}(I+A)$ is an orthogonal matrix.

Proof

First, we must ensure the inverse $(I - A)^{-1}$ actually exists. Suppose $(I - A)$ is singular. Then there exists a non-zero vector $v \neq 0$ such that $(I - A)v = 0 \implies v = Av$. Let us examine the squared norm of v:

$$\|v\|^2 = v^T v = v^T(Av) = (A^T v)^T v$$

Since A is skew-symmetric ($A^T = -A$):

$$= (-Av)^T v = -(Av)^T v = -v^T v = -\|v\|^2$$

The only way a number can equal its own negative is if it is zero: $\|v\|^2 = 0 \implies v = 0$. This contradicts our premise that $v \neq 0$. Thus, $(I - A)$ is strictly non-singular and invertible.
Now we prove $QQ^T = I$.

$$QQ^T = (I - A)^{-1}(I + A)\left[(I - A)^{-1}(I + A)\right]^T$$

Using the transpose property $(XY)^T = Y^T X^T$:

$$= (I - A)^{-1}(I + A)(I + A)^T\left((I - A)^{-1}\right)^T$$

Substitute $A^T = -A$:

$$(I + A)^T = I^T + A^T = I - A$$

$$\left((I-A)^{-1}\right)^T = ((I-A)^T)^{-1} = (I-A^T)^{-1} = (I+A)^{-1}$$

So the expression becomes:

$$QQ^T = (I-A)^{-1}(I+A)(I-A)(I+A)^{-1}$$

Because polynomials of the same matrix commute, $(I+A)(I-A) = I - A^2 = (I-A)(I+A)$. We can rearrange the middle terms to cancel with the inverses on the outside:

$$QQ^T = (I-A)^{-1}(I-A)(I+A)(I+A)^{-1} = (I)(I) = I$$

Thus, the Cayley transform produces a valid orthogonal matrix.

122 Generalized Inverses and Transformations

Reference: Olver & Shakiban, Chapter 4, Exercises 4.3.18 - 4.3.23.

Theorem Problem 4.3.18: Inverse of an Orthogonal (Not Orthonormal) Matrix

Suppose S is an $n \times n$ matrix whose columns form an orthogonal (but not orthonormal) basis of $\mathbb{R}^n$. Find a formula for S^{-1} mimicking the formula $Q^{-1} = Q^T$ for an orthogonal matrix.

Proof

Let $S = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$. Thus $S^T = \begin{pmatrix} - & v_1^T & - \\ & \vdots & \\ - & v_n^T & - \end{pmatrix}$. When we compute the product $S^T S$, the (i, j) entry is exactly the dot product $v_i \cdot v_j$. Because the columns are mutually orthogonal, $v_i \cdot v_j = 0$ for $i \neq j$. Thus, $S^T S$ is a purely diagonal matrix:

$$S^T S = \begin{pmatrix} \|v_1\|^2 & 0 & \dots & 0 \\ 0 & \|v_2\|^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \|v_n\|^2 \end{pmatrix}$$

To isolate the identity matrix, we multiply both sides by the inverse of this diagonal matrix (which is just the diagonal matrix of reciprocals):

$$\begin{pmatrix} \frac{1}{\|v_1\|^2} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\|v_n\|^2} \end{pmatrix} S^T S = I_n$$

By definition of the inverse matrix, the term on the left multiplying S must be S^{-1}. Thus:

$$S^{-1} = \begin{pmatrix} \frac{1}{\|v_1\|^2} & & 0 \\ & \ddots & \\ 0 & & \frac{1}{\|v_n\|^2} \end{pmatrix} S^T = \begin{pmatrix} - & \frac{v_1^T}{\|v_1\|^2} & - \\ & \vdots & \\ - & \frac{v_n^T}{\|v_n\|^2} & - \end{pmatrix}$$

Theorem Problem 4.3.19: Inner Product Preservation

Let $v_1, \dots, v_n$ and $w_1, \dots, w_n$ be two sets of linearly independent vectors in $\mathbb{R}^n$. Show that $v_i \cdot v_j = w_i \cdot w_j$ for all $1 \leq i, j \leq n$ if and only if there exists an orthogonal matrix Q such that $w_i = Qv_i$ for all i.

Proof

Let $V = [v_1, \dots, v_n]$ and $W = [w_1, \dots, w_n]$ be the $n \times n$ matrices formed by these vectors as columns. The condition $v_i \cdot v_j = w_i \cdot w_j$ for all i, j is algebraically equivalent to the matrix equation:

$$V^T V = W^T W$$

($\Leftarrow$) Suppose there exists an orthogonal matrix Q such that $w_i = Qv_i$. This means $W = QV$. Substitute this into $W^T W$:

$$W^T W = (QV)^T (QV) = V^T Q^T QV$$

Since Q is orthogonal, $Q^T Q = I$. Thus:

$$W^T W = V^T IV = V^T V \implies w_i \cdot w_j = v_i \cdot v_j$$

($\Rightarrow$) Suppose $V^T V = W^T W$. Because $v_1, \dots, v_n$ are linearly independent, they form a basis for $\mathbb{R}^n$, meaning the matrix V is invertible. Define the matrix Q exactly as $Q = WV^{-1}$. (This guarantees $W = QV$). We must prove Q is orthogonal.

$$Q^T Q = (WV^{-1})^T (WV^{-1}) = (V^{-1})^T W^T W V^{-1}$$

Substitute the given premise $W^T W = V^T V$:

$$Q^T Q = (V^T)^{-1} V^T V V^{-1} = ((V^T)^{-1} V^T)(VV^{-1}) = I \cdot I = I$$

Since $Q^T Q = I$, Q is indeed an orthogonal matrix.

123 Non-Square Matrices and Change of Basis

Example Problem 4.3.20: Tall Matrices with Orthonormal Columns

Suppose $u_1, \dots, u_k$ form an orthonormal set of vectors in $\mathbb{R}^n$ with $k < n$. Let $Q = [u_1, \dots, u_k]$ denote the $n \times k$ matrix whose columns are these vectors.

- **Prove that $Q^T Q = I_k$.** The entry in the i-th row and j-th column of $Q^T Q$ is exactly $u_i \cdot u_j$. Since the set is orthonormal, $u_i \cdot u_j = 0$ if $i \neq j$ and 1 if $i = j$. This perfectly describes the $k \times k$ identity matrix I_k.
- **Is $QQ^T = I_n$? No.** The matrix QQ^T is an $n \times n$ matrix. However, matrix rank obeys the inequality $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$. Therefore, $\text{rank}(QQ^T) \leq \text{rank}(Q) = k$. Since $k < n$, the rank of QQ^T is strictly less than n. The identity matrix I_n must have a full rank of n. Thus, QQ^T cannot possibly equal I_n. *(Geometric Note: QQ^T actually represents the projection matrix onto the k-dimensional subspace spanned by the vectors).*

Theorem Problem 4.3.21: Transition Matrix between Orthonormal Bases

Let $u_1, \dots, u_n$ and $\hat{u}_1, \dots, \hat{u}_n$ be two orthonormal bases of an inner product space V. Prove that the transition matrix $Q = (q_{ij})$ defined by $\hat{u}_i = \sum_{j=1}^n q_{ji} u_j$ is an orthogonal matrix.

Proof

We evaluate the inner product $\langle \hat{u}_i, \hat{u}_j \rangle$. Since $\hat{u}$ is an orthonormal basis, we know this evaluates to 1 if $i = j$ and 0 if $i \neq j$. Expand using the transition definition:

$$\langle \hat{u}_i, \hat{u}_j \rangle = \left\langle \sum_{k=1}^n q_{ki} u_k, \sum_{l=1}^n q_{lj} u_l \right\rangle$$

Using bilinearity:

$$= \sum_{k=1}^{n} \sum_{l=1}^{n} q_{ki} q_{lj} \langle u_k, u_l \rangle$$

Because u is also an orthonormal basis, $\langle u_k, u_l \rangle = 0$ for $k \neq l$. The sum collapses to where $k = l$:

$$= \sum_{k=1}^{n} q_{ki} q_{kj} \langle u_k, u_k \rangle = \sum_{k=1}^{n} q_{ki} q_{kj}$$

Notice that q_{ki} is the element in the i-th column of Q, which corresponds to the i-th row of Q^T. Thus, $\sum_{k=1}^{n} q_{ki} q_{kj}$ is exactly the row-by-column dot product forming the (i, j) entry of the matrix product $Q^T Q$. Since we established $\langle \hat{u}_i, \hat{u}_j \rangle = \delta_{ij}$ (the identity matrix), we conclude $(Q^T Q)_{ij} = (I)_{ij} \implies Q^T Q = I$. Therefore, Q is an orthogonal matrix.

Example Problem 4.3.22: Rectangular Matrices with Orthogonal Columns

Let A be an $m \times n$ matrix whose columns $v_1, \ldots, v_n$ are nonzero mutually orthogonal vectors in $\mathbb{R}^m$. **(a) Explain why $m \geq n$.** Because the n columns are nonzero and mutually orthogonal, they are strictly linearly independent. The maximum number of linearly independent vectors in an m-dimensional space ($\mathbb{R}^m$) is exactly m. Thus, $n \leq m$.
(b) Prove that $A^T A$ is a diagonal matrix. What are the diagonal entries? The (i, j) entry of $A^T A$ is $v_i \cdot v_j$. Due to mutual orthogonality, this is 0 whenever $i \neq j$, making the matrix purely diagonal. When $i = j$, the entry is $v_i \cdot v_i = \|v_i\|^2$. Thus, the diagonal entries are the squared Euclidean norms of the column vectors.
(c) Is AA^T diagonal? No, in general. We construct a counterexample. Let $v_1 = (1, 1, 0)^T$ and $v_2 = (1, -1, 1)^T$ in $\mathbb{R}^3$. Note $v_1 \cdot v_2 = 1 - 1 + 0 = 0$. Let $A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}$. Compute AA^T:

$$AA^T = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

This result has non-zero off-diagonal elements, proving it is not a diagonal matrix.

Theorem Problem 4.3.23: Custom Inner Products

Let $K > 0$ be a positive definite matrix. Prove that an $n \times n$ matrix S satisfies $S^T K S = I_n$ if and only if the columns of S form an orthonormal basis of $\mathbb{R}^n$ with respect to the inner product $\langle v, w \rangle = v^T K w$.

Proof

Let the columns of S be denoted $u_1, \ldots, u_n$. The matrix multiplication $S^T K S$ can be viewed element-by-element. The element in the i-th row and j-th column of $S^T K S$ is calculated by multiplying the i-th row of S^T by the matrix K, and then by the j-th column of S. Since the i-th row of S^T is u_i^T, this product is exactly:

$$(S^T K S)_{ij} = u_i^T K u_j$$

By the definition of our custom inner product, $u_i^T K u_j = \langle u_i, u_j \rangle$. Setting the matrix $S^T K S = I_n$ implies that:

$$\langle u_i, u_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

This is the precise mathematical definition of an orthonormal basis.

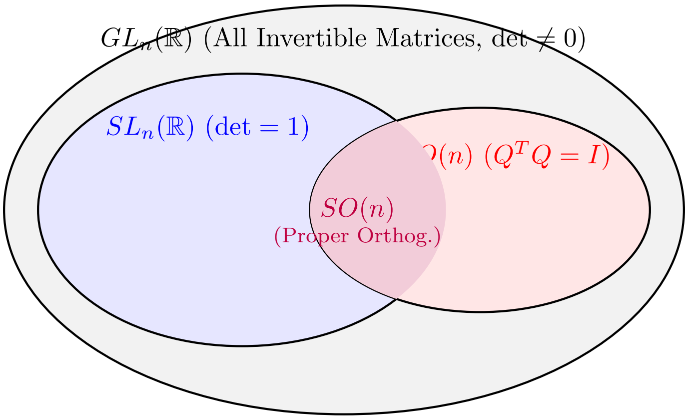
124 Matrix Groups in Abstract Algebra

Definition : Matrix Group

A set of $n \times n$ matrices $G \subseteq M_n(\mathbb{R})$ is called a **group** if it satisfies two conditions:

1. **Closure under multiplication:** Whenever $A, B \in G$, then $AB \in G$.
2. **Closure under inverses:** Whenever $A \in G$, then A is non-singular and $A^{-1} \in G$.

Note: As an immediate consequence of these two rules, the Identity matrix I_n must always belong to G. If $A \in G$, then $A^{-1} \in G$, and by the closure property, their product $AA^{-1} = I_n \in G$.



Theorem Problem 4.3.24: Examples of Valid Matrix Groups

Prove that the following sets of matrices form a group.
(i) General Linear Group $GL_n(\mathbb{R})$ (All nonsingular matrices): 1. If A, B are non-singular, $\det(AB) = \det(A)\det(B) \neq 0$, so AB is non-singular. 2. If $\det(A) \neq 0$, $\det(A^{-1}) = 1/\det(A) \neq 0$, so A^{-1} is non-singular.
(ii) Special Linear Group $SL_n(\mathbb{R})$ (All matrices with $\det = 1$): 1. If $\det(A) = 1$ and $\det(B) = 1$, then $\det(AB) = 1 \times 1 = 1$. 2. If $\det(A) = 1$, $\det(A^{-1}) = 1/1 = 1$.
(iii) Orthogonal Group $O(n)$ (All orthogonal matrices): 1. If A, B are orthogonal, $(AB)^T(AB) = B^TA^TAB = B^TIB = B^TB = I$. So AB is orthogonal. 2. If A is orthogonal ($A^TA = I$), then $(A^{-1})^TA^{-1} = (A^T)^TA^T = AA^T = I$. So A^{-1} is orthogonal.
(iv) Special Orthogonal Group $SO(n)$ (Proper orthogonal matrices): Requires orthogonality *and* $\det = 1$. Both properties are closed under multiplication and inverses as proven in (ii) and (iii).
(v) Symmetric Group S_n (All permutation matrices): 1. A permutation matrix is a bijection on the basis vectors. The composition of two bijections is a bijection. 2. The inverse of a bijection is a bijection. (Algebraically, $P^{-1} = P^T$, which remains a permutation matrix).
(vi) $SL_2(\mathbb{Z})$ (All 2×2 matrices with integer entries and $\det = 1$): 1. Integer arithmetic is closed under multiplication, and $\det(AB) = 1$. 2. The formula for the inverse is $A^{-1} = \frac{1}{\det A}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$. Since $\det A = 1$, no fractions are introduced. Thus A^{-1} contains only integers and $\det(A^{-1}) = 1$.

Theorem Problem 4.3.24: Sets that fail to be Groups

(c) Explain why the set of all nonsingular 2×2 matrices with integer entries does not form a group. While it is closed under multiplication, it is **not** closed under inverses. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. A has integer entries and $\det(A) = 5 \neq 0$. However, $A^{-1} = \frac{1}{5}\begin{pmatrix} 3 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 3/5 & -1/5 \\ -1/5 & 2/5 \end{pmatrix}$. The inverse contains fractions, so it leaves the set of integer matrices.

(d) Do the set of positive definite matrices form a group? No. The set of positive definite matrices is not closed under multiplication. Recall that a matrix MUST be symmetric to be considered positive definite. However, the product of two symmetric matrices is generally not symmetric unless they commute ($K_1K_2 = K_2K_1$). *Counter-example:* Let $K_1 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $K_2 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. Both are positive definite.

$$K_1K_2 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 2 & 4 \end{pmatrix}$$

Since the resulting matrix has $a_{12} = 1 \neq 2 = a_{21}$, it is not symmetric, and therefore cannot be positive definite.

125 QR Factorization Examples

Reference: Olver & Shakiban, Chapter 4, Exercises 211-212.

The Modified Gram-Schmidt process naturally yields the **QR Factorization** of a matrix. If $A = [w_1, \ldots, w_n]$ has linearly independent columns, we can factor it as $A = QR$, where:

- $Q = [u_1, \ldots, u_n]$ is an orthogonal matrix containing the orthonormal basis.
- R is an upper-triangular matrix containing the projection coefficients $r_{ij} = \langle w_j, u_i \rangle$, with the norms $r_{ii} = \|v_i\|$ strictly on the diagonal.

Example Problem 4.3.26: 3×3 **QR Factorization**

Write down the QR matrix factorization corresponding to the vectors $w_1 = (2, 2, -1)^T$, $w_2 = (0, 4, -1)^T$, $w_3 = (1, 2, -3)^T$.

Step 1: $r_{11} = \|w_1\| = \sqrt{9} = 3 \implies u_1 = \begin{pmatrix} 2/3 \\ 2/3 \\ -1/3 \end{pmatrix}$.

Step 2: $\|w_2\| = \sqrt{17}$. $r_{12} = \langle w_2, u_1 \rangle = 0 + 8/3 + 1/3 = 3$.

$$r_{22}^2 = \|w_2\|^2 - r_{12}^2 = 17 - 9 = 8 \implies r_{22} = 2\sqrt{2}$$

$$u_2 = \frac{1}{r_{22}}(w_2 - r_{12}u_1) = \frac{1}{2\sqrt{2}} \left[\begin{pmatrix} 0 \\ 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \right] = \frac{1}{2\sqrt{2}} \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix}$$

Step 3: $\|w_3\|^2 = 14$. $r_{13} = \langle w_3, u_1 \rangle = 2/3 + 4/3 + 3/3 = 3$. $r_{23} = \langle w_3, u_2 \rangle = -1/\sqrt{2} + 2/\sqrt{2} + 0 = 1/\sqrt{2}$.

$$r_{33}^2 = 14 - 3^2 - (1/\sqrt{2})^2 = 14 - 9 - 1/2 = 9/2 \implies r_{33} = \frac{3}{\sqrt{2}}$$

$$\begin{aligned} u_3 &= \frac{1}{r_{33}}(w_3 - r_{13}u_1 - r_{23}u_2) \\ &= \frac{\sqrt{2}}{3} \left[\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} - 3 \begin{pmatrix} 2/3 \\ 2/3 \\ -1/3 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{pmatrix} \right] \\ &= \frac{\sqrt{2}}{3} \left[\begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} - \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} + \begin{pmatrix} 1/2 \\ -1/2 \\ 0 \end{pmatrix} \right] = \frac{\sqrt{2}}{3} \begin{pmatrix} -1/2 \\ -1/2 \\ -2 \end{pmatrix} = \begin{pmatrix} -\frac{\sqrt{2}}{6} \\ -\frac{\sqrt{2}}{6} \\ -\frac{2\sqrt{2}}{3} \end{pmatrix} \end{aligned}$$

(Note: This can be equivalently written as $\frac{1}{3\sqrt{2}}(-1, -1, -4)^T$).

Resulting Matrices:

$$A = QR \implies \begin{pmatrix} 2 & 0 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & -3 \end{pmatrix} = \begin{pmatrix} 2/3 & -1/\sqrt{2} & -1/3\sqrt{2} \\ 2/3 & 1/\sqrt{2} & -1/3\sqrt{2} \\ -1/3 & 0 & -4/3\sqrt{2} \end{pmatrix} \begin{pmatrix} 3 & 3 & 3 \\ 0 & 2\sqrt{2} & 1/\sqrt{2} \\ 0 & 0 & 3/\sqrt{2} \end{pmatrix}$$

Example Problem 4.3.27: 2×2 **QR Factorizations**

Find the QR factorization for the following matrices:

(a) $A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$

- $w_1 = (1, 2)^T \implies r_{11} = \sqrt{5} \implies u_1 = (1/\sqrt{5}, 2/\sqrt{5})^T$.
- $w_2 = (-2, 1)^T \implies r_{12} = \langle w_2, u_1 \rangle = -2/\sqrt{5} + 2/\sqrt{5} = 0$.
- $r_{22} = \sqrt{\|w_2\|^2 - r_{12}^2} = \sqrt{5 - 0} = \sqrt{5}$.
- $u_2 = \frac{1}{\sqrt{5}}((-2, 1)^T - \mathbf{0}) = (-2/\sqrt{5}, 1/\sqrt{5})^T$.

$$A = QR = \begin{pmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{pmatrix} \begin{pmatrix} \sqrt{5} & 0 \\ 0 & \sqrt{5} \end{pmatrix}$$

(b) $A = \begin{pmatrix} 4 & 3 \\ 3 & 2 \end{pmatrix}$

- $w_1 = (4, 3)^T \implies r_{11} = 5 \implies u_1 = (4/5, 3/5)^T$.
- $w_2 = (3, 2)^T \implies r_{12} = \langle w_2, u_1 \rangle = 12/5 + 6/5 = 18/5$.
- $r_{22}^2 = \|w_2\|^2 - r_{12}^2 = 13 - \frac{324}{25} = \frac{325-324}{25} = \frac{1}{25} \implies r_{22} = \frac{1}{5}$.
- $u_2 = 5\left[\begin{pmatrix} 3 \\ 2 \end{pmatrix} - \frac{18}{5}\begin{pmatrix} 4/5 \\ 3/5 \end{pmatrix}\right] = 5\begin{pmatrix} 3 - 72/25 \\ 2 - 54/25 \end{pmatrix} = 5\begin{pmatrix} 3/25 \\ -4/25 \end{pmatrix} = \begin{pmatrix} 3/5 \\ -4/5 \end{pmatrix}$.

$$A = QR = \begin{pmatrix} 4/5 & 3/5 \\ 3/5 & -4/5 \end{pmatrix} \begin{pmatrix} 5 & 18/5 \\ 0 & 1/5 \end{pmatrix}$$

Solving General QR Problems

For parts (c) through (f) of Problem 4.3.27, the methodology remains exactly the same. Regardless of the size of the matrix, iteratively build the orthonormal columns of Q using Gram-Schmidt, and store the dot-product projections $\langle w_j, u_i \rangle$ in the upper triangle of R.

126 Solving Linear Systems with QR Factorization

Why compute the QR factorization? It drastically simplifies solving $Ax = b$. If $A = QR$, the system becomes $QRx = b$. Multiplying both sides by the transpose Q^T gives:

$$Q^T Q R x = Q^T b \implies Rx = Q^T b$$

Since R is an upper-triangular matrix, we can solve this new system using straightforward **back-substitution**, completely avoiding Gaussian elimination.

$$Rx = Q^T b$$

$$\begin{pmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \quad \begin{array}{l} \text{Substitute up} \\ \text{Solve first} \end{array}$$

Example Solving a System via QR

Find the QR factorization of the coefficient matrix to solve the system:

$$\begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

1. QR Factorization of A:

- $w_1 = (1, -1)^T \implies r_{11} = \sqrt{2} \implies u_1 = (1/\sqrt{2}, -1/\sqrt{2})^T$.
- $w_2 = (2, 3)^T \implies r_{12} = \langle w_2, u_1 \rangle = 2/\sqrt{2} - 3/\sqrt{2} = -1/\sqrt{2}$.
- $r_{22}^2 = \|w_2\|^2 - r_{12}^2 = 13 - 1/2 = 25/2 \implies r_{22} = 5/\sqrt{2}$.
- $u_2 = \frac{\sqrt{2}}{5} \left[\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \right] = \frac{\sqrt{2}}{5} \begin{pmatrix} 5/2 \\ 5/2 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$.

$$A = QR \implies \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & 5/\sqrt{2} \end{pmatrix}$$

2. Transform to $Rx = Q^T b$: Compute $c = Q^T b$:

$$c = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

Set up the upper triangular system:

$$\begin{pmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & 5/\sqrt{2} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -3/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$$

3. Back Substitution: From the bottom row: $\frac{5}{\sqrt{2}} y = \frac{1}{\sqrt{2}} \implies y = \frac{1}{5}$. Substitute y into the top row:

$$\sqrt{2}x - \frac{1}{\sqrt{2}} \left(\frac{1}{5} \right) = -\frac{3}{\sqrt{2}}$$

Divide everything by $\sqrt{2}$:

$$x - \frac{1}{10} = -\frac{3}{2} \implies x = -\frac{15}{10} + \frac{1}{10} = -\frac{14}{10} = -\frac{7}{5}$$

Solution: $x = -7/5, \quad y = 1/5$.

127 The Uniqueness of QR Factorization

Theorem Problem 4.3.30: Uniqueness Theorem

Prove that the QR factorization of a matrix A is strictly unique, assuming all diagonal entries of R are positive.

Proof

Suppose a matrix A has two valid factorizations: $A = Q_1R_1$ and $A = Q_2R_2$, where Q_1, Q_2 are orthogonal matrices and R_1, R_2 are upper triangular matrices with positive diagonal entries. We can equate them:

$$Q_1R_1 = Q_2R_2$$

Multiply on the left by Q_2^T and on the right by R_1^{-1}:

$$Q_2^TQ_1 = R_2R_1^{-1}$$

Let's analyze both sides of this new equation:

- **Left Side ($Q_2^TQ_1$):** The product of two orthogonal matrices is always an orthogonal matrix.
- **Right Side ($R_2R_1^{-1}$):** The inverse of an upper triangular matrix is upper triangular. The product of two upper triangular matrices is upper triangular. Furthermore, because both R_1 and R_2 have strictly positive diagonals, their product will have strictly positive diagonals.

Thus, the matrix $D = Q_2^TQ_1 = R_2R_1^{-1}$ is simultaneously an **orthogonal matrix** AND an **upper triangular matrix with positive diagonals**. From Exercise 4.3.12, the only way an upper triangular matrix can be orthogonal is if it is a purely diagonal matrix with entries of $+1$ or -1. Since we know the diagonal entries are positive, they must all be exactly $+1$. Therefore, $D = I$ (the Identity matrix).

$$Q_2^TQ_1 = I \implies Q_1 = Q_2$$
$$R_2R_1^{-1} = I \implies R_1 = R_2$$

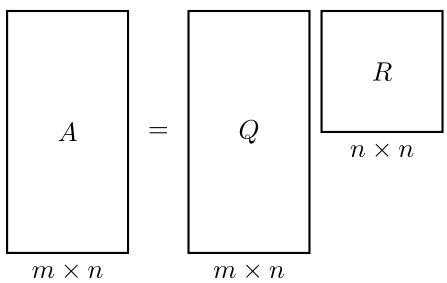
This proves the factorization is unique.

128 QR Factorization for Rectangular Matrices

The Gram-Schmidt algorithm does not require the input matrix to be square. It can be applied to any $m \times n$ rectangular matrix A (where $m \geq n$) as long as the n columns are linearly independent (i.e., $\text{rank}(A) = n$).

Theorem Problem 4.3.32 (a & b): Tall Matrix Factorization

Applying the Gram-Schmidt algorithm to an $m \times n$ matrix A of full column rank produces an $m \times n$ matrix Q with orthonormal columns and a non-singular $n \times n$ upper triangular matrix R, such that $A = QR$. The columns of Q form an orthonormal basis for Image(A).



Shape Match: Q takes the shape of A and satisfies $Q^TQ = I_n$ (though $QQ^T \neq I_m$). R is a square matrix that blends the orthogonal columns of Q back together.

Example Problem 4.3.32 (d): Factoring a 3×2 Matrix

Find the QR factorization of $A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 2 \end{pmatrix}$.

Step 1: $w_1 = (1,2,0)^T \implies r_{11} = \sqrt{1^2 + 2^2 + 0} = \sqrt{5}$.

$$u_1 = (1/\sqrt{5}, 2/\sqrt{5}, 0)^T$$

Step 2: $w_2 = (-1,3,2)^T$. $r_{12} = \langle w_2, u_1 \rangle = (-1)(1/\sqrt{5}) + (3)(2/\sqrt{5}) + 0 = 5/\sqrt{5} = \sqrt{5}$.

$$r_{22}^2 = \|w_2\|^2 - r_{12}^2 = (1+9+4) - (\sqrt{5})^2 = 14 - 5 = 9 \implies r_{22} = 3$$

$$u_2 = \frac{1}{3}\left[\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} - \sqrt{5} \begin{pmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{pmatrix} \right] = \frac{1}{3} \begin{pmatrix} -1-1 \\ 3-2 \\ 2-0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix}$$

Result:

$$A = QR = \begin{pmatrix} 1/\sqrt{5} & -2/3 \\ 2/\sqrt{5} & 1/3 \\ 0 & 2/3 \end{pmatrix} \begin{pmatrix} \sqrt{5} & \sqrt{5} \\ 0 & 3 \end{pmatrix}$$

Theorem Problem 4.3.32 (e): The Linear Dependence Trap

What happens to the algorithm if $\operatorname{rank}(A) < n$?
Answer: If the rank is strictly less than n, the columns are linearly dependent. At some stage k, the vector w_k will lie entirely within the subspace spanned by the previous vectors $w_1, \ldots, w_{k-1}$. When Gram-Schmidt subtracts the projections of w_k onto the previous vectors, the residual vector v_k will be exactly the zero vector $\mathbf{0}$. Consequently, $r_{kk} = \|v_k\| = 0$. The algorithm will attempt to divide by r_{kk} to normalize u_k, resulting in a fatal **division by zero** error, halting the program.

129 Orthogonal Projections onto Subspaces

Reference: Olver & Shakiban, Chapter 4.

We now extend the concept of projecting one vector onto another, to projecting a vector onto an entire subspace.

Definition : Orthogonal to a Subspace

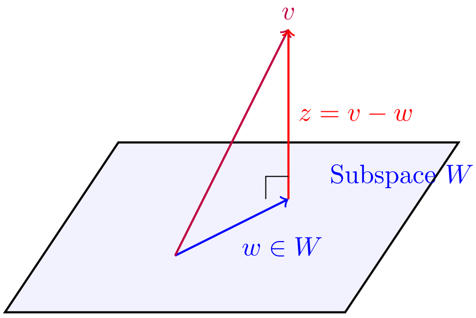
Let $W \subseteq V$ be a subspace of a real inner product space. A vector $z \in V$ is **orthogonal to** W (denoted $z \perp W$) if:

$$\langle z, w \rangle = 0 \quad \text{for all } w \in W$$

Note: If W is spanned by a basis $w_1, \ldots, w_n$, then verifying $z \perp W$ only requires checking that $\langle z, w_i \rangle = 0$ for all $i = 1, \ldots, n$.

Definition : Orthogonal Projection

Let $v \in V$. The **orthogonal projection** of v onto the subspace W is defined as the unique element $w \in W$ such that the difference vector $z = v - w$ is orthogonal to W ($z \perp W$).



Geometric Intuition: The projection w is the "shadow" of v on the plane W. The vector v can be uniquely decomposed as the sum of its projection w and the perpendicular "leftover" vector z.

Theorem Theorem: Projection via Orthonormal Basis

Let $u_1, \dots, u_n$ be an **orthonormal basis** of a finite-dimensional subspace $W \subseteq V$. Then the orthogonal projection of any vector $v \in V$ onto W is given by:

$$w = c_1 u_1 + \cdots + c_n u_n \quad \text{where} \quad c_i = \langle v, u_i \rangle \text{ for } 1 \le i \le n$$

Proof

First, we prove that the defined vector w actually satisfies the condition $z \perp W$. Let $z = v - w$. We take the inner product of z with any basis vector u_i:

$$\begin{aligned} \langle z, u_i \rangle &= \langle v - w, u_i \rangle = \langle v, u_i \rangle - \langle w, u_i \rangle \\ &= \langle v, u_i \rangle - \left\langle \sum_{j=1}^{n} c_j u_j, u_i \right\rangle \end{aligned}$$

Because the basis is orthonormal, $\langle u_j, u_i \rangle = 0$ for $i \neq j$ and 1 when $i = j$. Thus, the summation collapses to just c_i:

$$\langle z, u_i \rangle = \langle v, u_i \rangle - c_i = c_i - c_i = 0$$

Since z is orthogonal to every basis vector of W, we conclude $z \perp W$.
Uniqueness: Suppose there is another projection $w_1 = d_1 u_1 + \cdots + d_n u_n \in W$ such that $z_1 = v - w_1 \perp W$. Because $z_1 \perp W$, it must be orthogonal to u_i, meaning $\langle z_1, u_i \rangle = 0$.

$$\langle v - w_1, u_i \rangle = 0 \implies \langle v, u_i \rangle = \langle w_1, u_i \rangle$$

Since $\langle w_1, u_i \rangle = d_i$ (due to orthonormality) and $\langle v, u_i \rangle = c_i$, we have $d_i = c_i$ for all i. Thus $w_1 = w$ and $z_1 = z$, proving the projection is unique.

Theorem Theorem: Projection via Orthogonal Basis

If instead of an orthonormal basis, we only have an **orthogonal basis** $v_1, \dots, v_n$ of W, the orthogonal projection of v onto W requires scaling by the norms:

$$w = a_1 v_1 + \cdots + a_n v_n \quad \text{where} \quad a_i = \frac{\langle v, v_i \rangle}{\|v_i\|^2}$$

Consequently, the projection formula is:

$$w = \sum_{i=1}^{n} \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i$$

Example Example: Computing a Projection in $\mathbb{R}^3$

Let $W \subseteq \mathbb{R}^3$ under the standard dot product. Suppose W has an orthogonal basis $v_1 = \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. Let $v = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Find the orthogonal projection of v onto W.

Step 1: Compute necessary inner products and norms.

- $\|v_1\|^2 = (-1)^2 + 2^2 + 1^2 = 6$
- $\|v_2\|^2 = 1^2 + 1^2 + (-1)^2 = 3$
- $\langle v, v_1 \rangle = (1)(-1) + (0)(2) + (0)(1) = -1$
- $\langle v, v_2 \rangle = (1)(1) + (0)(1) + (0)(-1) = 1$

Step 2: Apply the projection formula.

$$w = \frac{\langle v, v_1 \rangle}{\|v_1\|^2} v_1 + \frac{\langle v, v_2 \rangle}{\|v_2\|^2} v_2$$

$$w = \frac{-1}{6} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \frac{1}{3} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/6 \\ -2/6 \\ -1/6 \end{pmatrix} + \begin{pmatrix} 2/6 \\ 2/6 \\ -2/6 \end{pmatrix} = \begin{pmatrix} 3/6 \\ 0 \\ -3/6 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 0 \\ -1/2 \end{pmatrix}$$

Alternative verification via Orthonormal Basis: If we first normalized the basis to $u_1 = \frac{1}{\sqrt{6}} v_1$ and $u_2 = \frac{1}{\sqrt{3}} v_2$, the projection $w = \langle v, u_1 \rangle u_1 + \langle v, u_2 \rangle u_2$ yields:

$$w = \left(\frac{-1}{\sqrt{6}} \right) \begin{pmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{pmatrix} + \left(\frac{1}{\sqrt{3}} \right) \begin{pmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ -1/\sqrt{3} \end{pmatrix} = \begin{pmatrix} 1/6 \\ -2/6 \\ -1/6 \end{pmatrix} + \begin{pmatrix} 1/3 \\ 1/3 \\ -1/3 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 0 \\ -1/2 \end{pmatrix}$$

Both methods produce the exact same shadow.

130 Orthogonal Subspaces and Complements

The Gram-Schmidt Process Re-interpreted

The orthogonal projection formula perfectly explains the mechanics of the Gram-Schmidt process. Recall we started with a basis $w_1, \dots, w_n$ and built v_k iteratively using:

$$v_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\|v_i\|^2} v_i$$

Notice that the summation term is exactly the orthogonal projection of w_k onto the subspace W_{k-1} spanned by the previously constructed orthogonal basis $v_1, \dots, v_{k-1}$. Therefore, $v_k = w_k - \text{Proj}_{W_{k-1}}(w_k)$. This defines v_k as the purely orthogonal error vector (z), guaranteeing that $v_k \perp W_{k-1}$.

Definition : Orthogonal Subspaces

Let V be an inner product space, and let $W, Z \subseteq V$ be two subspaces. W and Z are said to be **orthogonal to each other** (denoted $W \perp Z$) if:

$$\langle w, z \rangle = 0 \quad \text{for all } w \in W \text{ and all } z \in Z$$

Check condition: If W is spanned by $w_1, \dots, w_k$ and Z is spanned by $z_1, \dots, z_l$, then $W \perp Z \iff \langle w_i, z_j \rangle = 0$ for all $1 \le i \le k$ and $1 \le j \le l$.

Example Examples of Orthogonal Subspaces

Example 1: $\mathbb{R}^4$ under standard dot product Let $W = \text{span}(w_1)$ where $w_1 = (-1, 2, 0, 1)^T$. (A 1D line). Let $Z = \text{span}(z_1, z_2)$ where $z_1 = (3, 2, 0, -1)^T$ and $z_2 = (1, 0, 1, 1)^T$. (A 2D plane). Check the cross-basis dot products:

- $w_1 \cdot z_1 = -3 + 4 + 0 - 1 = 0$
- $w_1 \cdot z_2 = -1 + 0 + 0 + 1 = 0$

Since all basis combinations are orthogonal, $W \perp Z$.
Example 2: $\mathbb{R}^3$ planes and normals Let $W = \{(x, y, z) \in \mathbb{R}^3 \mid 2x - y + 3z = 0\}$. This is a 2D plane through the origin. Let $Z = \text{span}\left((2, -1, 3)^T\right)$. This is a 1D line defined by the normal vector of the plane. By the definition of a plane equation, the normal vector is orthogonal to every vector lying in the plane. Thus, $W \perp Z$.

Definition : The Orthogonal Complement

Let $W \subseteq V$ be a subspace of an inner product space V. The **orthogonal complement** of W, denoted as $W^\perp$ (read "W perp"), is the set of all vectors in V that are orthogonal to W:

$$W^\perp = \{v \in V \mid \langle v, w \rangle = 0 \quad \forall w \in W\}$$

131 Properties of the Orthogonal Complement

Theorem Proposition: Direct Sum Decomposition

Let V be an inner product space, and $W \subseteq V$ be a finite-dimensional subspace. Then $W^\perp$ is a subspace of V, $W \perp W^\perp$, and $W \cap W^\perp = \{0\}$. Furthermore, every vector $v \in V$ can be **uniquely** decomposed as:

$$v = w + z \quad \text{where } w \in W \text{ and } z \in W^\perp$$

(This is written algebraically as the direct sum $V = W \oplus W^\perp$).

Proof

Let $v \in V$. We define w to be the orthogonal projection of v onto W. Let $z = v - w$. By the properties of orthogonal projection previously proven, $z \perp W$, which by definition means $z \in W^\perp$. Thus $v = w + z$ with $w \in W$ and $z \in W^\perp$.
Uniqueness: Suppose there exists another decomposition $v = w_1 + z_1$ with $w_1 \in W$ and $z_1 \in W^\perp$. Subtracting the two equations yields:

$$w + z = w_1 + z_1 \implies z - z_1 = w_1 - w$$

Notice that the left side $(z - z_1)$ is a linear combination of vectors in $W^\perp$, so it belongs to $W^\perp$. The right side $(w_1 - w)$ is a linear combination of vectors in W, so it belongs to W. Since $W \cap W^\perp = \{0\}$ (because the only vector orthogonal to itself is the zero vector), both sides must equal the zero vector.

$$w_1 - w = 0 \implies w_1 = w \quad \text{and} \quad z - z_1 = 0 \implies z_1 = z$$

Thus, the decomposition is strictly unique.

Theorem Proposition: The Dimension Theorem

Let $W \subseteq V$ be a subspace. If V is a finite-dimensional space with $\dim V = m$, and $\dim W = n$, then:

$$\dim W^\perp = m - n$$

Proof

Let $v_1, \ldots, v_n$ be a basis for W. Suppose $\dim W^\perp = p$, and let $w_1, \ldots, w_p$ be a basis for $W^\perp$. By the previous proposition, any $v \in V$ can be written as $w + z$, meaning the union of the two bases $\{v_1, \ldots, v_n, w_1, \ldots, w_p\}$ spans V. We must show this union is linearly independent. Suppose a linear combination equals 0:

$$c_1 v_1 + \cdots + c_n v_n + d_1 w_1 + \cdots + d_p w_p = 0$$

Rearranging terms:

$$c_1 v_1 + \cdots + c_n v_n = -(d_1 w_1 + \cdots + d_p w_p)$$

Let this vector be u. The left side shows $u \in W$. The right side shows $u \in W^\perp$. Since $W \cap W^\perp = \{0\}$, we must have $u = 0$. Because v_i form a basis of W, $c_1 v_1 + \cdots + c_n v_n = 0 \implies c_i = 0$ for all i. Because w_j form a basis of $W^\perp$, $d_1 w_1 + \cdots + d_p w_p = 0 \implies d_j = 0$ for all j. Thus, the combined set of $n + p$ vectors is linearly independent and forms a full basis for V. Therefore, $n + p = \dim V = m \implies p = m - n$.

Theorem Proposition: The Double Complement

If $W \subseteq V$ is a finite-dimensional subspace of an inner product space V, then:

$$(W^\perp)^\perp = W$$

Proof

Let $w \in W$. By the definition of $W^\perp$, w is orthogonal to every vector in $W^\perp$. Because w is orthogonal to everything in $W^\perp$, it belongs to the orthogonal complement of $W^\perp$, which is $(W^\perp)^\perp$. This proves $W \subseteq (W^\perp)^\perp$. Using the dimension theorem, $\dim W^\perp = m - n$. Applying the theorem again to $W^\perp$, $\dim((W^\perp)^\perp) = m - (m - n) = n$. Since W is a subspace of $(W^\perp)^\perp$ and they have the exact same dimension n, they must be the identical set. Thus, $(W^\perp)^\perp = W$.

132 Orthogonality of Fundamental Subspaces

We now apply the theory of orthogonal complements to the four fundamental subspaces of an $m \times n$ real matrix $A \in M_{m \times n}(\mathbb{R})$. Recall definitions:

1. $\mathrm{Ker}(A) = \{x \in \mathbb{R}^n \mid Ax = 0\} \subseteq \mathbb{R}^n$ (Nullspace)
2. $\mathrm{Im}(A) = \{Ax \in \mathbb{R}^m \mid x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m$ (Column space)
3. $\mathrm{Coker}(A) = \{x \in \mathbb{R}^m \mid x^T A = 0\} \subseteq \mathbb{R}^m$ (Left nullspace)
4. $\mathrm{Coim}(A) = \{x^T A \in \mathbb{R}^n \mid x \in \mathbb{R}^m\} = \mathrm{Im}(A^T) \subseteq \mathbb{R}^n$ (Row space)

From earlier Rank-Nullity theory, we know:

$$\dim(\mathrm{Ker}(A)) + \dim(\mathrm{Coim}(A)) = n$$

$$\dim(\mathrm{Coker}(A)) + \dim(\mathrm{Im}(A)) = m$$

$$\dim(\mathrm{Coim}(A)) = \dim(\mathrm{Im}(A)) = r \text{ (The rank of A)}$$

Theorem Theorem: The Fundamental Theorem of Linear Algebra

Let $A \in M_{m\times n}(\mathbb{R})$. With respect to the standard Euclidean dot product:

1. $\mathrm{Ker}(A)$ and $\mathrm{Coim}(A)$ are orthogonal complements in $\mathbb{R}^n$.
$$\mathrm{Ker}(A) = \mathrm{Coim}(A)^\perp \subseteq \mathbb{R}^n$$
2. $\mathrm{Im}(A)$ and $\mathrm{Coker}(A)$ are orthogonal complements in $\mathbb{R}^m$.
$$\mathrm{Coker}(A) = \mathrm{Im}(A)^\perp \subseteq \mathbb{R}^m$$

Proof

Let $x \in \mathbb{R}^n$. By definition, $x \in \mathrm{Ker}(A) \iff Ax = 0$. Let r_i denote the i-th row of matrix A (for $1 \le i \le m$). The matrix multiplication $Ax = 0$ is structurally equivalent to stating that the dot product of every row of A with x equals zero:

$$r_i \cdot x = 0 \quad \text{for all } i = 1, \dots, m$$

This specifically means that x is orthogonal to the vectors $r_1, \dots, r_m$. Since $\mathrm{Coim}(A)$ (the row space) is exactly the span of $r_1, \dots, r_m$, x being orthogonal to the basis vectors means x is orthogonal to the entire subspace $\mathrm{Coim}(A)$. Therefore:

$$x \in \mathrm{Ker}(A) \iff x \perp \mathrm{Coim}(A) \iff x \in \mathrm{Coim}(A)^\perp$$

Thus, $\mathrm{Ker}(A) = \mathrm{Coim}(A)^\perp$.
By substituting A^T into this proven identity, the second statement $\mathrm{Coker}(A) = \mathrm{Im}(A)^\perp$ follows immediately by geometric symmetry.

133 Solving Linear Systems and the Cokernel

We utilize the orthogonality of the fundamental subspaces to determine when a linear system has a valid solution.

Theorem Theorem: Compatibility via the Cokernel

Let $A \in M_{m\times n}(\mathbb{R})$ and let $b \in \mathbb{R}^m$. The linear system $Ax = b$ has a solution if and only if $b \in \mathrm{Im}(A)$. Because $\mathrm{Im}(A)$ and $\mathrm{Coker}(A)$ are orthogonal complements in $\mathbb{R}^m$, this is logically equivalent to stating:
$$Ax = b \text{ has a solution} \iff b \in \mathrm{Coker}(A)^\perp$$

This means that b must be orthogonal to every vector in the cokernel of A. If $y_1, \dots, y_k$ form a basis for $\mathrm{Coker}(A)$, the system is compatible if and only if:

$$b \cdot y_i = 0 \quad \text{for all } i = 1, \dots, k$$

Example Example: A 3×3 System

Consider the matrix $A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ 1 & -2 & 3 \end{pmatrix}$. Notice that $\mathrm{rank}(A) = 2$, so $\dim(\mathrm{Im}(A)) = 2$, which implies $\dim(\mathrm{Coker}(A)) = 3 - 2 = 1$. By inspection, a linear combination of the rows yields zero: $(-1)R_1 + 2R_2 + 1R_3 = \mathbf{0}$. Thus, $y = (-1, 2, 1)^T$ is a basis for $\mathrm{Coker}(A)$.
For the system $Ax = b = (b_1, b_2, b_3)^T$ to have a solution, b must be orthogonal to y:

$$\langle b, y \rangle = -b_1 + 2b_2 + b_3 = 0$$

We can verify this by performing Gaussian elimination on the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 1 & -2 & b_2 \\ 1 & -2 & 3 & b_3 \end{array}\right) \xrightarrow{R_3 - R_1} \left(\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 1 & -2 & b_2 \\ 0 & -2 & 4 & b_3 - b_1 \end{array}\right)$$

$$\xrightarrow{R_3 + 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & -1 & b_1 \\ 0 & 1 & -2 & b_2 \\ 0 & 0 & 0 & b_3 - b_1 + 2b_2 \end{array}\right)$$

The system is consistent if and only if the zero row corresponds to a zero value: $b_3 - b_1 + 2b_2 = 0$. This perfectly matches our Cokernel orthogonality condition.

Example Example: A 4×3 System

Consider the linear system $Ax = b$ given by:

$$\begin{pmatrix} 1 & -1 & 3 \\ -1 & 2 & -4 \\ 2 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$$

The system has a solution $\iff$ $b \in \text{Coker}(A)^\perp$. We find the basis for $\text{Coker}(A)$ by solving $A^T y = 0$.

$$\begin{pmatrix} 1 & -1 & 2 & 1 \\ -1 & 2 & 3 & 0 \\ 3 & -4 & 1 & 2 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 7 & 2 \\ 0 & 1 & 5 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This yields the equations $y_1 + 7y_3 + 2y_4 = 0$ and $y_2 + 5y_3 + y_4 = 0$. Setting the free variables (y_3, y_4) to $(1, 0)$ and $(0, 1)$ respectively gives the basis:

$$y_1 = \begin{pmatrix} -7 \\ -5 \\ 1 \\ 0 \end{pmatrix}, \quad y_2 = \begin{pmatrix} -2 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

Thus, the vector b must satisfy:

$$-7b_1 - 5b_2 + b_3 = 0$$
$$-2b_1 - b_2 + b_4 = 0$$

134 Orthogonal Decomposition and Isomorphisms

Let $A \in M_{m \times n}(\mathbb{R})$. Because $\text{Ker}(A)$ and $\text{Coim}(A)$ are orthogonal complements in $\mathbb{R}^n$, any vector $x \in \mathbb{R}^n$ can be written **uniquely** as:

$$x = w + z \quad \text{where } w \in \text{Coim}(A) \text{ and } z \in \text{Ker}(A)$$

When we multiply this vector by A, we get:

$$Ax = A(w + z) = Aw + Az$$

Since $z \in \text{Ker}(A)$, $Az = 0$, leaving $Ax = Aw$. Thus, multiplication by A acts purely on the Coimage component. This establishes a powerful theorem:

Theorem Theorem: Coimage-Image Isomorphism

Multiplication by $A \in M_{m \times n}(\mathbb{R})$ defines an **isomorphism** between $\text{Coim}(A)$ and $\text{Im}(A)$. Both spaces have exactly the same dimension $r = \text{rank}(A)$. Furthermore, if $v_1, \ldots, v_r$ is a basis of

Coim(A), then $Av_1, \dots, Av_r$ is a basis of Im(A).

134.1 The Minimum Norm Solution

If $b \in \text{Im}(A)$, the system $Ax = b$ has a solution. If Ker(A) is non-trivial, there are infinitely many solutions forming a flat geometric space (a line, plane, etc.). The general solution takes the form $x = w + z$, where w is a particular solution and $z \in \text{Ker}(A)$.

Theorem Theorem: The Minimum Norm Solution

For any compatible linear system $Ax = b$, there exists a **unique** solution $w \in \text{Coim}(A)$. This specific solution w is distinguished by the fact that it has the strictly smallest length (minimum norm) of all possible solutions to the system:

$$\|w\| \leq \|x\| \quad \text{for all } x \text{ satisfying } Ax = b$$

Proof

Let x be any solution to $Ax = b$. By orthogonal decomposition, $x = w + z$ where $w \in \text{Coim}(A)$ and $z \in \text{Ker}(A)$. Because $w \perp z$, we apply the Pythagorean Theorem:

$$\|x\|^2 = \|w + z\|^2 = \|w\|^2 + \|z\|^2$$

Since norms are non-negative, $\|w\|^2 + \|z\|^2 \geq \|w\|^2$. Equality holds if and only if $\|z\|^2 = 0 \implies z = 0 \implies x = w$.

Algorithm to find the Minimum Norm Solution

To isolate the unique vector $w \in \text{Coim}(A)$, we must force the generic solution to be completely orthogonal to the Kernel. 1. Find a basis $z_1, \dots, z_{n-r}$ for Ker(A). 2. The minimum norm solution w must simultaneously satisfy the original system $Ax = b$ AND be orthogonal to every kernel basis vector. 3. Solve the augmented system of equations:

$$Ax = b$$

$$z_i \cdot x = 0 \quad \text{for } i = 1, \dots, n - r$$

This combined system has full rank n and will yield exactly one unique solution.

135 Comprehensive Example: Finding the Minimum Norm Solution

Example Solving the full 4×4 System

Consider the linear system $Ax = b$:

$$\begin{pmatrix} 1 & -1 & 2 & -2 \\ 0 & 1 & -2 & 1 \\ 1 & 3 & -5 & 2 \\ 5 & -1 & 9 & -6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \\ w \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 4 \\ 6 \end{pmatrix}$$

Step 1: Find a basis for Ker(A). Row reduce A:

$$\xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Equations: $x - w = 0 \implies x = w$, $y + w = 0 \implies y = -w$, and $z = 0$. The Kernel is 1-dimensional, spanned by the basis vector $\mathbf{z_1} = (1, -1, 0, 1)^T$.
Step 2: Find a basis for Coker(A) to verify b. Row reduce A^T:

$$\xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & -24 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Equations: $x - 2w = 0 \implies x = 2w$, $y - 24w = 0 \implies y = 24w$, $z + 7w = 0 \implies z = -7w$. The Cokernel is spanned by $\mathbf{y_1} = (2, 24, -7, 1)^T$. Verify compatibility: $b \cdot y_1 = (-1)(2) + (1)(24) + (4)(-7) + (6)(1) = -2 + 24 - 28 + 6 = 0$. Since $b \in \text{Coker}(A)^\perp$, the system has solutions.
Step 3: Find the General Solution. Using the RREF of the augmented matrix $[A|b]$, we get:

$$\begin{pmatrix} 1 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 & 3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$z = 1$, $y + w = 3 \implies y = 3 - w$, $x - w = 0 \implies x = w$. Letting $w = t$, the general solution is parameterized as $\mathbf{x} = (t, 3 - t, 1, t)^T$.
Step 4: Find the Minimum Norm Solution. We force the general solution to be orthogonal to the Kernel basis vector z_1:

$$\langle \mathbf{x}, \mathbf{z_1} \rangle = 0 \implies (t, 3 - t, 1, t) \cdot (1, -1, 0, 1)^T = 0$$

$$t - (3 - t) + 0 + t = 0 \implies 3t - 3 = 0 \implies t = 1$$

Substitute $t = 1$ into the general solution to get the unique minimum norm vector $\mathbf{w} \in \text{Coim}(A)$:

$$\mathbf{w} = (1, 2, 1, 1)^T$$

Step 5: Verifications. 1. *Minimization verification:* The squared length of the general solution is $\|x\|^2 = t^2 + (3 - t)^2 + 1^2 + t^2 = 3t^2 - 6t + 10 = 3(t - 1)^2 + 7$. The absolute minimum occurs at $t = 1$, where $\|w\| = \sqrt{7}$. 2. *Coimage verification:* Since $\mathbf{w} \in \text{Coim}(A)$, it must be a linear combination of the rows of A. Indeed:

$$\begin{pmatrix} 1 \\ 2 \\ 1 \\ 1 \end{pmatrix} = -4 \begin{pmatrix} 1 \\ -1 \\ 2 \\ -2 \end{pmatrix} - 17 \begin{pmatrix} 0 \\ 1 \\ -2 \\ 1 \end{pmatrix} + 5 \begin{pmatrix} 1 \\ 3 \\ -5 \\ 2 \end{pmatrix}$$

This is the exclusive and unique solution satisfying this property.

136 Calculating Orthogonal Projections

Reference: Olver & Shakiban, Chapter 4, Exercises 4.4.1 - 4.4.7.

Recall that the orthogonal projection of a vector v onto a subspace W with an **orthogonal basis** $v_1, \ldots, v_n$ is given by:

$$w = \text{Proj}_W(v) = \sum_{i=1}^{n} \frac{\langle v, v_i \rangle}{\|v_i\|^2} v_i$$

136.1 Projections in $\mathbb{R}^n$

Example Problem 4.4.2: Projections with Given Bases

Find the orthogonal projection of $v = (1, 2, -1)^T$ onto the following subspaces of $\mathbb{R}^3$.
(a) The line in the direction $u_1 = \frac{1}{\sqrt{3}}(-1, 1, 1)^T$ Since u_1 is a unit vector, we use the simplified formula $w = \langle v, u_1 \rangle u_1$:

$$\langle v, u_1 \rangle = \frac{1}{\sqrt{3}}((1)(-1) + (2)(1) + (-1)(1)) = \frac{-1 + 2 - 1}{\sqrt{3}} = 0$$

Since the inner product is 0, v is already orthogonal to the line. The projection is $w = \mathbf{0}$.
(b) The line spanned by $v_1 = (2, -1, 3)^T$

$$\|v_1\|^2 = 4 + 1 + 9 = 14$$

$$\langle v, v_1 \rangle = (1)(2) + (2)(-1) + (-1)(3) = 2 - 2 - 3 = -3$$

$$w = \frac{-3}{14}\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -3/7 \\ 3/14 \\ -9/14 \end{pmatrix}$$

(c) The plane spanned by $v_1 = (1, 0, 1)^T$ **and** $v_2 = (-1, 2, 1)^T$ First, verify orthogonality of the basis: $v_1 \cdot v_2 = -1 + 0 + 1 = 0$. (They are orthogonal).

$$\|v_1\|^2 = 2, \quad \|v_2\|^2 = 6$$

$$\langle v, v_1 \rangle = 1 + 0 - 1 = 0 \implies \text{No projection along } v_1$$

$$\langle v, v_2 \rangle = -1 + 4 - 1 = 2$$

$$w = 0v_1 + \frac{2}{6}v_2 = \frac{1}{3}\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/3 \\ 2/3 \\ 1/3 \end{pmatrix}$$

Example Problem 4.4.4: Projecting onto an Image

Find the orthogonal projection of $v = (1, 2, 3)^T$ onto the image of the matrix $A = \begin{pmatrix} 3 & 2 \\ 2 & -2 \\ 1 & -2 \end{pmatrix}$.

The image of A is spanned by its column vectors: $v_1 = (3, 2, 1)^T$ and $v_2 = (2, -2, -2)^T$. First, we must check if they are orthogonal:

$$v_1 \cdot v_2 = (3)(2) + (2)(-2) + (1)(-2) = 6 - 4 - 2 = 0$$

Since they are orthogonal, we can use them directly in the projection formula without needing Gram-Schmidt!

$$\|v_1\|^2 = 9 + 4 + 1 = 14, \quad \|v_2\|^2 = 4 + 4 + 4 = 12$$

$$\langle v, v_1 \rangle = (1)(3) + (2)(2) + (3)(1) = 3 + 4 + 3 = 10$$

$$\langle v, v_2 \rangle = (1)(2) + (2)(-2) + (3)(-2) = 2 - 4 - 6 = -8$$

The projection w is:

$$w = \frac{10}{14}\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} + \frac{-8}{12}\begin{pmatrix} 2 \\ -2 \\ -2 \end{pmatrix} = \frac{5}{7}\begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix} - \frac{2}{3}\begin{pmatrix} 2 \\ -2 \\ -2 \end{pmatrix} = \begin{pmatrix} 15/7 - 4/3 \\ 10/7 + 4/3 \\ 5/7 + 4/3 \end{pmatrix} = \begin{pmatrix} 17/21 \\ 58/21 \\ 43/21 \end{pmatrix}$$

Example Problem 4.4.5: When Gram-Schmidt is Mandatory

Find the orthogonal projection of $v = (1, 3, -1)^T$ onto the plane spanned by $w_1 = (-1, 2, 1)^T$ and $w_2 = (2, 1, -3)^T$.
Check Orthogonality: $w_1 \cdot w_2 = -2 + 2 - 3 = -3 \neq 0$. Because the basis vectors are not orthogonal, we **cannot** use the projection formula directly. We must first construct an orthogonal basis $\{z_1, z_2\}$ using Gram-Schmidt.
Step 1: Gram-Schmidt Let $z_1 = w_1 = (-1, 2, 1)^T$. ($\|z_1\|^2 = 6$).

$$z_2 = w_2 - \frac{\langle w_2, z_1 \rangle}{\|z_1\|^2} z_1 = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} - \frac{-3}{6} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} + \begin{pmatrix} -1/2 \\ 1 \\ 1/2 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 2 \\ -5/2 \end{pmatrix}$$

To avoid fractions, we can scale z_2 by 2 (scaling doesn't affect orthogonality). Let $z_2 = (3, 4, -5)^T$. ($\|z_2\|^2 = 9 + 16 + 25 = 50$).
Step 2: Projection Formula Now project v onto the orthogonal basis $\{z_1, z_2\}$:

$$\langle v, z_1 \rangle = -1 + 6 - 1 = 4$$

$$\langle v, z_2 \rangle = 3 + 12 + 5 = 20$$

$$w = \frac{4}{6} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \frac{20}{50} \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} = \frac{2}{3} \begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} + \frac{2}{5} \begin{pmatrix} 3 \\ 4 \\ -5 \end{pmatrix} = \begin{pmatrix} -2/3 + 6/5 \\ 4/3 + 8/5 \\ 2/3 - 10/5 \end{pmatrix} = \begin{pmatrix} 8/15 \\ 44/15 \\ -20/15 \end{pmatrix}$$

136.2 Custom Inner Products

Example Problem 4.4.6: Weighted Projection in $\mathbb{R}^4$

Find the orthogonal projection of $v = (1, 2, -1, 2)^T$ onto the span of $w_1 = (1, -1, 2, 5)^T$ and $w_2 = (2, 1, 0, -1)^T$ using the weighted inner product:

$$\langle x, y \rangle = 4x_1y_1 + 3x_2y_2 + 2x_3y_3 + x_4y_4$$

1. Check Orthogonality under the new metric:

$$\langle w_1, w_2 \rangle = 4(1)(2) + 3(-1)(1) + 2(2)(0) + 1(5)(-1) = 8 - 3 + 0 - 5 = 0$$

They are orthogonal! We can use the projection formula directly.
2. Calculate Norms and Inner Products:

- $\|w_1\|^2 = 4(1)^2 + 3(-1)^2 + 2(2)^2 + 1(5)^2 = 4 + 3 + 8 + 25 = 40$
- $\|w_2\|^2 = 4(2)^2 + 3(1)^2 + 2(0)^2 + 1(-1)^2 = 16 + 3 + 0 + 1 = 20$
- $\langle v, w_1 \rangle = 4(1)(1) + 3(2)(-1) + 2(-1)(2) + 1(2)(5) = 4 - 6 - 4 + 10 = 4$
- $\langle v, w_2 \rangle = 4(1)(2) + 3(2)(1) + 2(-1)(0) + 1(2)(-1) = 8 + 6 + 0 - 2 = 12$

3. Projection Formula:

$$w = \frac{4}{40} w_1 + \frac{12}{20} w_2 = \frac{1}{10} \begin{pmatrix} 1 \\ -1 \\ 2 \\ 5 \end{pmatrix} + \frac{3}{5} \begin{pmatrix} 2 \\ 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1/10 + 12/10 \\ -1/10 + 6/10 \\ 2/10 + 0 \\ 5/10 - 6/10 \end{pmatrix} = \begin{pmatrix} 13/10 \\ 1/2 \\ 1/5 \\ -1/10 \end{pmatrix}$$

137 Subspaces and Orthogonal Complements

Theorem Problem 4.4.8 (a): The Orthogonal Complement is a Subspace

Prove that the set of all vectors orthogonal to a given subspace $W \subseteq \mathbb{R}^n$ forms a valid vector subspace. This set is the orthogonal complement $W^\perp$.

Proof

By definition, $W^\perp = \{v \in \mathbb{R}^n \mid \langle v, w \rangle = 0 \text{ for all } w \in W\}$. To prove $W^\perp$ is a subspace, we must verify closure under addition and scalar multiplication.

1. **Closure under Addition:** Let $v_1, v_2 \in W^\perp$. By definition, for any $w \in W$, $\langle v_1, w \rangle = 0$ and $\langle v_2, w \rangle = 0$. Evaluate the sum:

$$\langle v_1 + v_2, w \rangle = \langle v_1, w \rangle + \langle v_2, w \rangle = 0 + 0 = 0$$

   Since the sum is orthogonal to every $w \in W$, $v_1 + v_2 \in W^\perp$.

2. **Closure under Scalar Multiplication:** Let $v \in W^\perp$ and $c \in \mathbb{R}$. Evaluate the scaled vector:

$$\langle cv, w \rangle = c\langle v, w \rangle = c(0) = 0$$

   Since it remains orthogonal to every $w \in W$, $cv \in W^\perp$.

Since it is closed under both operations (and trivially contains the zero vector), $W^\perp$ is a subspace.

Example Problem 4.4.8 (b): Finding a Basis for $W^\perp$

Find a basis for the set of all vectors in $\mathbb{R}^4$ that are orthogonal to the subspace spanned by $w_1 = (1, 2, 0, -1)^T$ and $w_2 = (2, 0, 3, 1)^T$.
Solution: The set of all orthogonal vectors is exactly the **Kernel** of the matrix formed by putting w_1 and w_2 into the rows. We solve $Ax = 0$:

$$\begin{pmatrix} 1 & 2 & 0 & -1 \\ 2 & 0 & 3 & 1 \end{pmatrix} \xrightarrow{R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -4 & 3 & 3 \end{pmatrix}$$

Let the free variables be x_3 and x_4. From Row 2: $-4x_2 + 3x_3 + 3x_4 = 0 \implies x_2 = \frac{3}{4}x_3 + \frac{3}{4}x_4$. From Row 1: $x_1 + 2\left(\frac{3}{4}x_3 + \frac{3}{4}x_4\right) - x_4 = 0 \implies x_1 + \frac{3}{2}x_3 + \frac{1}{2}x_4 = 0 \implies x_1 = -\frac{3}{2}x_3 - \frac{1}{2}x_4$. Setting $(x_3, x_4) = (2, 0) \implies v_1 = (-3, 3/2, 2, 0)^T \implies$ Scale by 2 $\implies (-6, 3, 4, 0)^T$. Setting $(x_3, x_4) = (0, 4) \implies v_2 = (-2, 3, 0, 4)^T$. The basis for $W^\perp$ is $\{(-6, 3, 4, 0)^T, (-2, 3, 0, 4)^T\}$.

138 The Projection Matrix

Reference: Olver & Shakiban, Chapter 4, Exercises 4.4.9 - 4.4.11.

Rather than calculating dot products every time we want to project a vector v onto a subspace W, we can construct a single **Projection Matrix** P. Multiplying any vector by P will instantly return its orthogonal projection w.

138.1 Projection Matrix for Orthonormal Bases

Theorem Problem 4.4.9: P

Let $u_1, \ldots, u_k$ be an **orthonormal basis** for a subspace $W \subseteq \mathbb{R}^m$. Let $A = [u_1, u_2, \ldots, u_k]$ be the $m \times k$ matrix containing these column vectors. Define the matrix $P = AA^T$.

1. **(a) Prove that $w = Pv$ is the orthogonal projection of v onto W.** We must prove that the residual $(v - w) \perp W$, meaning it is orthogonal to every column of A. We evaluate

$A^T(v-w)$:
$$A^T(v-Pv) = A^Tv - A^T(AA^T)v = A^Tv - (A^TA)A^Tv$$
Because the columns of A are orthonormal, $A^TA = I_k$.
$$= A^Tv - (I_k)A^Tv = A^Tv - A^Tv = 0$$
Since the dot products with all basis vectors equal zero, $v - w \perp W$.

2. **(b) Prove that P is symmetric.**
$$P^T = (AA^T)^T = (A^T)^TA^T = AA^T = P$$

3. **(c) Prove that P is idempotent ($P^2 = P$).**
$$P^2 = (AA^T)(AA^T) = A(A^TA)A^T = A(I_k)A^T = AA^T = P$$
Geometric meaning: Projecting a vector onto a surface, and then projecting that shadow onto the same surface again, does nothing to the shadow.

$$v = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \xrightarrow{\text{Multiply by } P} \quad w = Pv = \begin{pmatrix} x_p \\ y_p \\ z_p \end{pmatrix}$$

Matrix Projection: Instead of summing individual vector projections, the single matrix P computes the final coordinates instantly.

138.2 Projection Matrix for Any Basis

Theorem Problem 4.4.10: The Master Projection Formula

Let $w_1, \ldots, w_k$ be an **arbitrary basis** (not necessarily orthogonal) for a subspace $W \subseteq \mathbb{R}^m$. Let $A = [w_1, \ldots, w_k]$. The projection matrix is given by the general formula:
$$P = A(A^TA)^{-1}A^T$$

Proof

We rigorously verify the projection properties for this more complex formula. **1. Idempotence ($P^2 = P$):**
$$\begin{aligned} P^2 &= [A(A^TA)^{-1}A^T][A(A^TA)^{-1}A^T] \\ &= A(A^TA)^{-1}[A^TA](A^TA)^{-1}A^T \\ &= A(A^TA)^{-1}[I_k]A^T = A(A^TA)^{-1}A^T = P \end{aligned}$$

2. Symmetry ($P^T = P$): Using the properties $(XY)^T = Y^TX^T$ and $((M)^{-1})^T = (M^T)^{-1}$:
$$\begin{aligned} P^T &= \left[A(A^TA)^{-1}A^T\right]^T \\ &= (A^T)^T\left((A^TA)^{-1}\right)^T A^T \\ &= A\left((A^TA)^T\right)^{-1} A^T \\ &= A(A^TA)^{-1}A^T = P \end{aligned}$$

3. Orthogonality of Residual ($v - Pv \perp W$): We check $A^T(v - Pv) = 0$:
$$A^Tv - A^T[A(A^TA)^{-1}A^T]v = A^Tv - [A^TA](A^TA)^{-1}A^Tv = A^Tv - I_kA^Tv = 0$$

Example Problem 4.4.11: Applying the Master Formula

Find the orthogonal projection of $v = (1, 0, 0)^T$ onto the image of the matrix $A = \begin{pmatrix} 5 \\ -5 \\ 7 \end{pmatrix}$.

Since A is a single column, we can use the master projection matrix formula. First, compute the inner scalar $(A^T A)$:

$$A^T A = (5)^2 + (-5)^2 + (7)^2 = 25 + 25 + 49 = 99$$

Since it's a 1×1 matrix, its inverse is simply the scalar $\frac{1}{99}$. Now build the projection matrix $P = A(A^T A)^{-1} A^T$:

$$P = \frac{1}{99} \begin{pmatrix} 5 \\ -5 \\ 7 \end{pmatrix} \begin{pmatrix} 5 & -5 & 7 \end{pmatrix} = \frac{1}{99} \begin{pmatrix} 25 & -25 & 35 \\ -25 & 25 & -35 \\ 35 & -35 & 49 \end{pmatrix}$$

To find the projection w, multiply v by P:

$$w = Pv = \frac{1}{99} \begin{pmatrix} 25 & -25 & 35 \\ -25 & 25 & -35 \\ 35 & -35 & 49 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 25/99 \\ -25/99 \\ 35/99 \end{pmatrix}$$

(Note: Since $v = e_1$, multiplying by P simply extracted the first column of the projection matrix!)

139 Calculating Orthogonal Complements

Reference: Olver & Shakiban, Chapter 4, Exercises 4.4.12 & 4.4.13.

Example Problem 4.4.12: Finding $W^\perp$ in $\mathbb{R}^3$

Find the orthogonal complement $W^\perp$ of the subspaces $W \subseteq \mathbb{R}^3$ spanned by the indicated vectors. What is the dimension of $W^\perp$?

(a) $W = \textbf{span}\left((3, -1, 1)^T\right)$ The space W is a 1D line. By the dimension theorem, $\dim W^\perp = 3 - 1 = 2$. $W^\perp = \{(x, y, z) \mid 3x - y + z = 0\}$. This is a plane. We find a basis by setting free variables x, z: Let $x = 1, z = 0 \implies y = 3 \implies (1, 3, 0)^T$. Let $x = 0, z = 3 \implies y = 3 \implies (0, 3, -3)^T \implies (0, 1, -1)^T$. Basis: $\{(1, 3, 0)^T, (0, 1, -1)^T\}$.

(b) $W = \textbf{span}\left((1, 2, 3)^T, (2, 0, 1)^T\right)$ The space W is a 2D plane (vectors are linearly independent). Thus, $\dim W^\perp = 3 - 2 = 1$. $W^\perp = \{(x, y, z) \mid x + 2y + 3z = 0 \text{ and } 2x + z = 0\}$. From the second equation, $z = -2x$. Substitute into the first: $x + 2y + 3(-2x) = 0 \implies -5x + 2y = 0 \implies 2y = 5x$. Let $x = 2 \implies y = 5, z = -4$. Basis for $W^\perp$ is $\{(2, 5, -4)^T\}$.

(c) $W = \textbf{span}\left((1, 2, 3)^T, (2, 4, 6)^T\right)$ Notice the second vector is exactly $2\times$ the first vector. Thus, they are linearly dependent, and W is just a 1D line! $\dim W^\perp = 3 - 1 = 2$. $W^\perp = \{(x, y, z) \mid x + 2y + 3z = 0\}$.

(d) $W = \textbf{span}\left((1, 0, 1)^T, (-2, 3, 0)^T, (-1, 2, 0)^T\right)$ Checking the determinant of the matrix formed by these three vectors: $\det = 1(0 - 6) - 0 + 1(-4 - (-3)) = -6 - 1 = -7 \neq 0$. Since they are linearly independent, they span all of $\mathbb{R}^3$. $W = \mathbb{R}^3$. Thus, $W^\perp = \{\mathbf{0}\}$ and $\dim W^\perp = 0$.

Correction to Handwritten Notes on 4.4.13 (a)

In the handwritten notes for Problem 4.4.13(a), the orthogonal complement of the plane $3x + 4y - 5z = 0$ was incorrectly listed as having a dimension of 2 with basis $\{(5, 0, 3), (4, -3, 0)\}$. Those vectors lie *inside* the plane itself! The orthogonal complement of a 2D plane in 3D space is a 1D line defined by the plane's normal vector. Correct Answer: $\dim W^\perp = 1$, Basis $= \{(3, 4, -5)^T\}$.

Example Problem 4.4.13: Complements of Fundamental Subspaces

Find a basis for the orthogonal complement of the following subspaces of $\mathbb{R}^3$:
(b) The line in the direction $(-2,1,3)^T$ $W^\perp = \{(x,y,z) \mid -2x+y+3z=0\}$. $\dim W^\perp = 2$. Basis: $\{(1,2,0)^T,(3,0,2)^T\}$.
(c) The image of the matrix $A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ -2 & 0 & 2 & 1 \\ -1 & 2 & 1 & 4 \end{pmatrix}$ The image is spanned by the column vectors. We find the linearly independent columns. $C_3 = -C_1$. The first two columns $C_1 = (1,-2,-1)^T$ and $C_2 = (2,0,2)^T$ are linearly independent. Checking C_4, the rank of A is 2. We seek $W^\perp = \operatorname{Im}(A)^\perp = \operatorname{Coker}(A) = \operatorname{Ker}(A^T)$. Solve $A^T\mathbf{y} = 0$: $x - 2y - z = 0$ $2x + 2z = 0 \implies z = -x$. Substitute into first: $x - 2y - (-x) = 0 \implies 2x - 2y = 0 \implies y = x$. Let $x = 1 \implies y = 1, z = -1$. Basis for $W^\perp$ is $\{(1,1,-1)^T\}$.
(d) The cokernel of $A = \begin{pmatrix} 1 & 2 & 1 \\ -2 & 0 & 2 \\ -1 & 2 & 1 \end{pmatrix}$ By the Fundamental Theorem of Linear Algebra, $\operatorname{Coker}(A)^\perp = (\operatorname{Im}(A)^\perp)^\perp = \operatorname{Im}(A)$. Therefore, we simply need a basis for the column space of A. The first two columns are linearly independent: $\{(1,-2,-1)^T,(2,0,2)^T\}$. This forms the basis of $W^\perp$ immediately.

140 Orthogonal Decomposition of Vectors ($v = w + z$)

Example Problem 4.4.15: Decomposing Vectors

Decompose each vector with respect to the indicated subspace as $v = w + z$, where $w \in W$ and $z \in W^\perp$.
(a) $v = (1,2)^T$, $\quad W = \textbf{span}((-1,3)^T)$**.** The basis for W is a single vector, so it is automatically orthogonal.

$$w = \operatorname{Proj}_W(v) = \frac{\langle v, w_1\rangle}{\|w_1\|^2} w_1 = \frac{(1)(-1)+(2)(3)}{1+9}\begin{pmatrix} -1 \\ 3 \end{pmatrix} = \frac{5}{10}\begin{pmatrix} -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 3/2 \end{pmatrix}$$

$$z = v - w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} -1/2 \\ 3/2 \end{pmatrix} = \begin{pmatrix} 3/2 \\ 1/2 \end{pmatrix}$$

Check: $z \cdot w_1 = (3/2)(-1) + (1/2)(3) = -3/2 + 3/2 = 0$. Valid.
(b) $v = (2,-1,1)^T$, $\quad W = \textbf{span}((-3,2,1)^T,(-1,0,5)^T)$**. Step 1: Gram-Schmidt on** W**.** The given vectors are not orthogonal. $v_1 = (-3,2,1)^T$. ($\|v_1\|^2 = 14$).

$$v_2 = (-1,0,5)^T - \frac{3+0+5}{14}\begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix} - \frac{4}{7}\begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} = \frac{1}{7}\begin{pmatrix} 5 \\ -8 \\ 31 \end{pmatrix}$$

To avoid fractions in the basis, we scale $v_2 \to (5,-8,31)^T$. ($\|v_2\|^2 = 25 + 64 + 961 = 1050$).
Step 2: Project v **onto the orthogonal basis.** $\langle v, v_1\rangle = 2(-3) + (-1)(2) + 1(1) = -7$. $\langle v, v_2\rangle = 2(5) + (-1)(-8) + 1(31) = 10 + 8 + 31 = 49$.

$$w = \frac{-7}{14}\begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} + \frac{49}{1050}\begin{pmatrix} 5 \\ -8 \\ 31 \end{pmatrix} = -\frac{1}{2}\begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} + \frac{7}{150}\begin{pmatrix} 5 \\ -8 \\ 31 \end{pmatrix}$$

Finding a common denominator of 150:

$$w = \begin{pmatrix} 225/150 \\ -150/150 \\ -75/150 \end{pmatrix} + \begin{pmatrix} 35/150 \\ -56/150 \\ 217/150 \end{pmatrix} = \begin{pmatrix} 260/150 \\ -206/150 \\ 142/150 \end{pmatrix} = \begin{pmatrix} 26/15 \\ -103/75 \\ 71/75 \end{pmatrix}$$

Step 3: Find z**.**

$$z = v - w = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} - \begin{pmatrix} 26/15 \\ -103/75 \\ 71/75 \end{pmatrix} = \begin{pmatrix} 30/15 - 26/15 \\ -75/75 + 103/75 \\ 75/75 - 71/75 \end{pmatrix} = \begin{pmatrix} 4/15 \\ 28/75 \\ 4/75 \end{pmatrix}$$

(c) The Decomposition Shortcut (Projecting onto $W^\perp$): Let $v = (1,0,0)^T$ and $W = \text{Ker}\begin{pmatrix} 1 & 2 & -1 \\ 2 & 0 & 2 \end{pmatrix}$. Finding a basis for W and orthogonalizing it takes time. However, $W^\perp = \text{Coim}(A)$, which is spanned exactly by the rows of A: $y_1 = (1,2,-1)^T$ and $y_2 = (2,0,2)^T$. If we project v onto $W^\perp$ first, we instantly find z, and then $w = v - z$. Orthogonalize $W^\perp$: $z_1 = (1,2,-1)^T$ ($\|z_1\|^2 = 6$). $z_2 = (2,0,2)^T - \frac{0}{6}z_1 = (2,0,2)^T$. ($\|z_2\|^2 = 8$). Project v:

$$z = \text{Proj}_{W^\perp}(v) = \frac{1}{6}\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \frac{2}{8}\begin{pmatrix} 2 \\ 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/6 \\ 2/6 \\ -1/6 \end{pmatrix} + \begin{pmatrix} 3/6 \\ 0 \\ 3/6 \end{pmatrix} = \begin{pmatrix} 4/6 \\ 2/6 \\ 2/6 \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/3 \\ 1/3 \end{pmatrix}$$

Now find w:

$$w = v - z = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2/3 \\ 1/3 \\ 1/3 \end{pmatrix} = \begin{pmatrix} 1/3 \\ -1/3 \\ -1/3 \end{pmatrix}$$

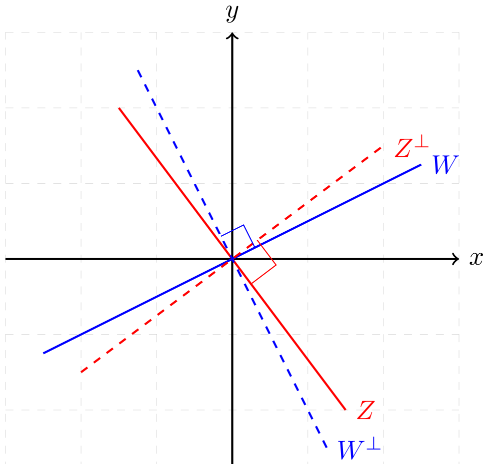
141 Theoretical Properties of Orthogonal Complements

Theorem Problem 4.4.20: Disjoint Subspaces

Let $W \subseteq V$ be a subspace. Prove that: **(a)** $W \cap W^\perp = \{0\}$
Proof: Let w be a vector such that $w \in W$ and $w \in W^\perp$. By the definition of the orthogonal complement, w must be orthogonal to every vector in W. Since w itself is in W, it must be orthogonal to itself: $\langle w, w \rangle = 0$. By the positivity axiom of inner products, this implies $w = 0$.
(b) $W \subseteq (W^\perp)^\perp$
Proof: Let $w \in W$. For every vector $z \in W^\perp$, we know by definition that $\langle w, z \rangle = 0$. This means w is orthogonal to every vector in $W^\perp$. Therefore, w fulfills the exact requirement to be an element of the orthogonal complement of $W^\perp$, which is $(W^\perp)^\perp$.

Theorem Problem 4.4.21 & 4.4.22: Trivial Complements and Subset Reversal

1. Prove $V^\perp = \{0\}$ and $\{0\}^\perp = V$: Let $v \in V^\perp$. Then $\langle v, w \rangle = 0$ for all $w \in V$. Choose $w = v$, yielding $\langle v, v \rangle = 0 \implies v = 0$. Let $v \in \{0\}^\perp$. Then $\langle v, 0 \rangle = 0$, which is trivially true for every vector in the universe V.
2. Prove that if $W_1 \subset W_2$, then $W_1^\perp \supset W_2^\perp$: Let $v \in W_2^\perp$. This means $\langle v, w \rangle = 0$ for every $w \in W_2$. Since every element of W_1 is also an element of W_2 (because $W_1 \subset W_2$), it must be true that $\langle v, w \rangle = 0$ for every $w \in W_1$. Thus, v satisfies the condition to belong to $W_1^\perp$. Since every element of $W_2^\perp$ is in $W_1^\perp$, we have $W_2^\perp \subset W_1^\perp$.



Complementary Subspaces: Two lines W and Z are complementary because they span $\mathbb{R}^2$ and intersect only at $\{0\}$. Their orthogonal complements (perpendicular lines) $W^\perp$ and $Z^\perp$ are also distinct lines that span $\mathbb{R}^2$, maintaining the complementary relationship.

Theorem Problem 4.4.23: Orthogonal Complements of Complementary Subspaces

Two subspaces $W, Z \subseteq \mathbb{R}^n$ are **complementary** if $W \cap Z = \{0\}$ and $W + Z = \mathbb{R}^n$. Prove that if W and Z are complementary, then $W^\perp$ and $Z^\perp$ are also complementary subspaces.

Proof

We must prove two conditions: $W^\perp \cap Z^\perp = \{0\}$ and $W^\perp + Z^\perp = \mathbb{R}^n$. **1. Intersection:** Let $v \in W^\perp \cap Z^\perp$. This means $v \perp W$ and $v \perp Z$. Because v is orthogonal to every vector in both basis sets, it is orthogonal to any linear combination of them. Since W and Z span the entire space $\mathbb{R}^n$, v is orthogonal to every vector in $\mathbb{R}^n$. The only vector orthogonal to the entire space is $v = 0$. Thus, $W^\perp \cap Z^\perp = \{0\}$.
2. Span: From the dimension theorem, if $\dim V = n$ and W, Z are complementary, then $\dim W + \dim Z = n$. Let $\dim W = k$, so $\dim Z = n - k$. By the orthogonal complement dimension theorem:

$$\dim W^\perp = n - k \quad \text{and} \quad \dim Z^\perp = n - (n - k) = k$$

Since $W^\perp$ and $Z^\perp$ only intersect at $\{0\}$, the dimension of their sum is the sum of their dimensions:

$$\dim(W^\perp + Z^\perp) = (n - k) + k = n$$

Since their sum is an n-dimensional subspace of $\mathbb{R}^n$, it must be the entire space $W^\perp + Z^\perp = \mathbb{R}^n$.

142 Orthogonal Polynomial Complements

Example Problem 4.4.19: Orthogonal Complements in Polynomial Spaces

Let $V = \mathcal{P}^{(4)}$ be the space of quartic polynomials with the standard L^2 inner product $\langle p, q \rangle = \int_{-1}^{1} p(x)q(x)\,dx$. Let $W = \mathcal{P}^{(2)}$ be the subspace of quadratic polynomials. Find an explicit basis for $W^\perp$.
Solution: A vector $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4$ belongs to $W^\perp$ if it is orthogonal to the standard basis of W: $\{1, x, x^2\}$. We compute the three necessary integrals:

1. $\int_{-1}^{1} p(x) \cdot 1\,dx = 2a_0 + \frac{2}{3}a_2 + \frac{2}{5}a_4 = 0$
2. $\int_{-1}^{1} p(x) \cdot x\,dx = \frac{2}{3}a_1 + \frac{2}{5}a_3 = 0 \implies a_3 = -\frac{5}{3}a_1$
3. $\int_{-1}^{1} p(x) \cdot x^2\,dx = \frac{2}{3}a_0 + \frac{2}{5}a_2 + \frac{2}{7}a_4 = 0$

Because V is 5-dimensional and W is 3-dimensional, $\dim W^\perp = 2$. We have two free variables to choose to generate the two basis polynomials.
Basis Polynomial 1 (Odd powers): Let $a_0 = a_2 = a_4 = 0$. Choose $a_1 = -3 \implies a_3 = 5$. This yields the polynomial $p_1(x) = 5x^3 - 3x$. *(This is proportional to the 3rd Legendre Polynomial).*
Basis Polynomial 2 (Even powers): Let $a_1 = a_3 = 0$. We must solve the system:

$$a_0 + \frac{1}{3}a_2 + \frac{1}{5}a_4 = 0 \implies a_0 = -\frac{1}{3}a_2 - \frac{1}{5}a_4$$

Substitute into the third condition:

$$\frac{1}{3}\left(-\frac{1}{3}a_2 - \frac{1}{5}a_4\right) + \frac{1}{5}a_2 + \frac{1}{7}a_4 = 0$$

$$\left(\frac{1}{5} - \frac{1}{9}\right)a_2 + \left(\frac{1}{7} - \frac{1}{15}\right)a_4 = 0 \implies \frac{4}{45}a_2 + \frac{8}{105}a_4 = 0 \implies \frac{1}{45}a_2 + \frac{2}{105}a_4 = 0$$

Multiply by 315: $7a_2 + 6a_4 = 0$. Choose $a_4 = 35 \implies a_2 = -30$. Then $a_0 = -\frac{1}{3}(-30) - \frac{1}{5}(35) = 10 - 7 = 3$. This yields the polynomial $p_2(x) = 35x^4 - 30x^2 + 3$. *(This is proportional to the 4th Legendre Polynomial).*
The orthogonal complement basis is exactly $\{5x^3 - 3x,\ 35x^4 - 30x^2 + 3\}$.

143 Geometry of Complementary Subspaces

Reference: Olver & Shakiban, Chapter 4, Exercises 4.4.23 - 4.4.27.

Example Problem 4.4.23(b): Visualizing Complementary Complements

We previously proved that if $W, Z \subseteq \mathbb{R}^n$ are complementary subspaces ($W \cap Z = \{0\}$ and $W + Z = \mathbb{R}^n$), then their orthogonal complements $W^\perp$ and $Z^\perp$ are also complementary. Sketch a picture illustrating this result when W and Z are lines in $\mathbb{R}^2$.
Solution: Let $V = \mathbb{R}^2$. Let W be the x-axis ($y = 0$) and Z be the line $y = x$. Any point $(a, b) \in \mathbb{R}^2$ can be uniquely written as $w + z$:

$$(a, b) = (a - b, 0) + (b, b) \quad \text{where } (a - b, 0) \in W,\ (b, b) \in Z$$

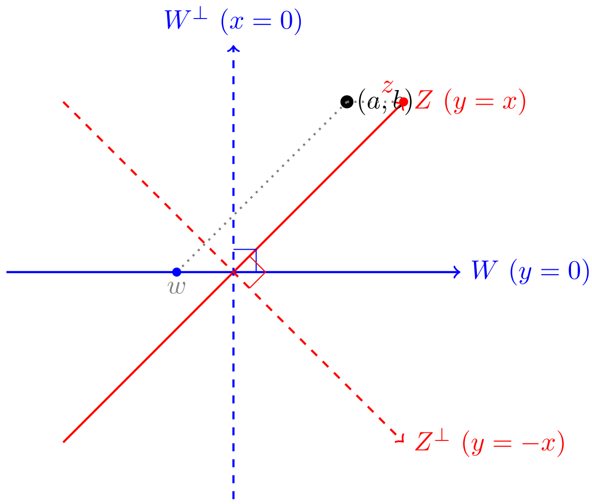
The orthogonal complements are the lines perpendicular to W and Z:

- $W^\perp = \{(0, y) \mid y \in \mathbb{R}\}$ (The y-axis, $x = 0$)
- $Z^\perp = \{(x, -x) \mid x \in \mathbb{R}\}$ (The line $y = -x$)

We can verify that $W^\perp$ and $Z^\perp$ are complementary by decomposing (a, b):

$$(a, b) = (0, a + b) + (a, -a) \quad \text{where } (0, a + b) \in W^\perp,\ (a, -a) \in Z^\perp$$

Since this decomposition is unique and they intersect only at the origin, $W^\perp \oplus Z^\perp = \mathbb{R}^2$.



Complementarity: Just as any point can be built by moving along W then Z, any point can also be built by moving along $W^\perp$ then $Z^\perp$.

Theorem Problem 4.4.24: Orthogonal Complement of a Sum

Prove that if W, Z are subspaces of an inner product space V, then:

$$(W + Z)^\perp = W^\perp \cap Z^\perp$$

Proof

We prove this by showing mutual containment (if $A \subseteq B$ and $B \subseteq A$, then $A = B$).
Part 1: $(W+Z)^\perp \subseteq W^\perp \cap Z^\perp$ Let $v \in (W+Z)^\perp$. By definition, $\langle v, u\rangle = 0$ for every $u \in W+Z$. Since $u = w+z$, we know $\langle v, w+z\rangle = 0$ for all $w \in W, z \in Z$. If we choose $z = 0$, we get $\langle v, w\rangle = 0$ for all $w \in W \implies v \in W^\perp$. If we choose $w = 0$, we get $\langle v, z\rangle = 0$ for all $z \in Z \implies v \in Z^\perp$. Since v is in both, $v \in W^\perp \cap Z^\perp$.
Part 2: $W^\perp \cap Z^\perp \subseteq (W+Z)^\perp$ Let $v \in W^\perp \cap Z^\perp$. Then $\langle v, w\rangle = 0$ for all $w \in W$, and $\langle v, z\rangle = 0$ for all $z \in Z$. Let $u \in W + Z$. We can write $u = w + z$. Evaluate the inner product:

$$\langle v, u\rangle = \langle v, w + z\rangle = \langle v, w\rangle + \langle v, z\rangle = 0 + 0 = 0$$

Since v is orthogonal to every element in $W+Z$, $v \in (W+Z)^\perp$. Thus, the sets are exactly equal.

Theorem Problem 4.4.26: Orthogonality via Spanning Sets

Prove Lemma 4.35: If $w_1, \ldots, w_k$ span W and $z_1, \ldots, z_l$ span Z, then W and Z are orthogonal subspaces if and only if $\langle w_i, z_j\rangle = 0$ for all $1 \le i \le k$ and $1 \le j \le l$.

Proof

($\Rightarrow$) If $W \perp Z$, then by definition every vector in W is orthogonal to every vector in Z. Since $w_i \in W$ and $z_j \in Z$, it is trivially true that $\langle w_i, z_j\rangle = 0$.
($\Leftarrow$) Suppose $\langle w_i, z_j\rangle = 0$ for all basis vectors. Let $w \in W$ and $z \in Z$. Because the vectors form spanning sets, we can write $w = \sum_{i=1}^k c_i w_i$ and $z = \sum_{j=1}^l d_j z_j$. Evaluate the inner product using bilinearity:

$$\langle w, z\rangle = \left\langle \sum_{i=1}^k c_i w_i, \sum_{j=1}^l d_j z_j \right\rangle = \sum_{i=1}^k \sum_{j=1}^l c_i d_j \langle w_i, z_j\rangle$$

Since every cross-term $\langle w_i, z_j\rangle = 0$ by hypothesis, the entire double sum is 0. Thus $w \perp z$ for all vectors, meaning $W \perp Z$.

144 The Breakdown of Orthogonal Complements

Reference: Olver & Shakiban, Chapter 4, Exercise 4.4.28.

In finite-dimensional spaces, if $W \subseteq V$, then $W \oplus W^\perp = V$. We always have a perfect orthogonal complement. We now prove that this foundational theorem **fails** in infinite-dimensional spaces.

Theorem Problem 4.4.28: The Infinite-Dimensional Anomaly

Let $V = C^0[a,b]$ be the space of continuous functions with the L^2 inner product $\langle f, g\rangle = \int_a^b f(x)g(x)dx$. Let $W = \{f \in C^0[a,b] \mid f(a) = 0 \text{ and } f(b) = 0\}$ be the subspace of functions tied down at the endpoints.

1. **(a)** Show that W has a complementary subspace Z of dimension 2.
2. **(b)** Prove that an orthogonal complement to W **does not exist**.

Proof Proof of (a): Constructing a Complement

We define a subspace Z spanned by two linear "tent" functions $g_1(x)$ and $g_2(x)$:

- Let $g_1(x)$ be a line such that $g_1(a) = 1$ and $g_1(b) = 0$. (e.g., $g_1(x) = \frac{b-x}{b-a}$)
- Let $g_2(x)$ be a line such that $g_2(a) = 0$ and $g_2(b) = 1$. (e.g., $g_2(x) = \frac{x-a}{b-a}$)

Let $Z = \text{span}(g_1, g_2)$. Since g_1, g_2 are linearly independent, $\dim Z = 2$. To prove Z is complementary to W, we must show $W \cap Z = \{0\}$ and $W + Z = V$.

1. Intersection: Suppose $f \in W \cap Z$. Since $f \in Z$, $f(x) = \lambda g_1(x) + \mu g_2(x)$. Since $f \in W$, it must satisfy the boundary conditions:

$$f(a) = \lambda(1) + \mu(0) = \lambda \implies \text{But } f \in W \text{ requires } f(a) = 0 \implies \lambda = 0$$

$$f(b) = \lambda(0) + \mu(1) = \mu \implies \text{But } f \in W \text{ requires } f(b) = 0 \implies \mu = 0$$

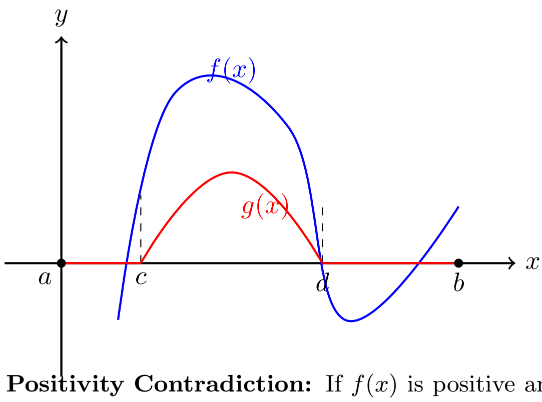
Thus $f(x) = 0g_1 + 0g_2 = 0$. $W \cap Z = \{0\}$.

2. Span: Let $f \in V$ be any continuous function. We construct a projection onto Z: Let $h(x) = f(a)g_1(x) + f(b)g_2(x)$. Clearly $h \in Z$. Now consider the residual function $w(x) = f(x) - h(x)$. Evaluate at the boundaries:

$$w(a) = f(a) - (f(a)(1) + f(b)(0)) = f(a) - f(a) = 0$$

$$w(b) = f(b) - (f(a)(0) + f(b)(1)) = f(b) - f(b) = 0$$

Since $w(a) = w(b) = 0$, the residual $w \in W$. Therefore, $f = w + h$ with $w \in W$ and $h \in Z$. Thus $V = W \oplus Z$.



The Positivity Contradiction: If $f(x)$ is positive anywhere, we can build a bump function $g(x)$ entirely within that region. Since $g(a) = g(b) = 0$, $g \in W$. But their product is strictly positive, violating $\langle f, g\rangle = 0$.

Proof Proof of (b): The Non-Existence of $W^\perp$

Suppose for the sake of contradiction that an orthogonal complement $Z = W^\perp$ *does* exist such that $W \oplus Z = V$. Let $f \in W^\perp$. By definition, for every function $g \in W$, we must have:

$$\langle f, g\rangle = \int_a^b f(x)g(x)\,dx = 0$$

Suppose $f(x)$ is not the zero function. Then $f(x) \neq 0$ at some point. Without loss of generality, assume $f(x_0) > 0$ for some $x_0 \in (a, b)$. Because f is continuous, there must exist a small sub-interval $[c, d] \subset (a, b)$ strictly inside the domain where $f(x) > 0$ everywhere on $[c, d]$.
We can construct a continuous "bump" function $g(x)$ such that:

- $g(x) > 0$ for $x \in (c, d)$
- $g(x) = 0$ everywhere else in $[a, b]$.

(For example, a triangle/tent function localized entirely between c and d). Notice that $g(a) = 0$ and $g(b) = 0$. Therefore, $g \in W$. Now evaluate the inner product:

$$\langle f, g\rangle = \int_a^b f(x)g(x)\,dx = \int_c^d f(x)g(x)\,dx$$

Because both $f(x) > 0$ and $g(x) > 0$ strictly on the interval (c, d), their product is strictly positive, meaning the integral must be strictly greater than zero:

$$\int_c^d f(x)g(x)\,dx > 0$$

This is a direct contradiction of the requirement that $\langle f, g\rangle = 0$. (A symmetric argument applies if we assume $f(x_0) < 0$).
Therefore, our assumption that $f(x)$ is non-zero must be false. The only function that is orthogonal to every function in W is the identically zero function $f(x) = 0$. Thus, $W^\perp = \{0\}$. However, we proved in part (a) that any complementary subspace to W must have dimension 2. Since $\dim(\{0\}) = 0 \neq 2$, the set $W^\perp$ cannot possibly serve as the complementary subspace $(W \oplus \{0\} = W \neq V)$. Thus, W does not admit an orthogonal complement.

145 Eigenvalues and Eigenvectors (Chapter 8)

Reference: Lecture 7 Notes.

We now transition from solving static linear systems to analyzing the dynamic properties of linear transformations. We look for special vectors that do not change direction when multiplied by a matrix.

Definition : Eigenvalues and Eigenvectors

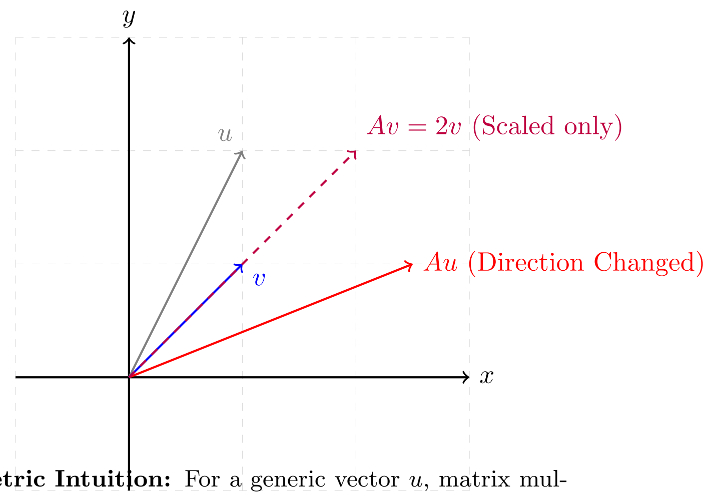
Let $A \in M_n(\mathbb{R})$ be a square matrix. A scalar $\lambda \in \mathbb{C}$ is called an **eigenvalue** of A if there exists a *nonzero* vector $v \neq 0$ such that:

$$Av = \lambda v$$

The vector v is called an **eigenvector** of A associated to the eigenvalue λ.

The Zero Vector Exclusion

Why do we insist $v \neq 0$? Because if $v = 0$, then $A(0) = 0 = \lambda(0)$ is trivially true for *every* scalar $\lambda \in \mathbb{R}$. Excluding the zero vector makes eigenvalues mathematically meaningful.



Geometric Intuition: For a generic vector u, matrix multiplication rotates and scales it. For an eigenvector v, matrix multiplication acts purely as a scalar stretch (by λ).

Theorem Theorem: The Characteristic Equation

A scalar λ is an eigenvalue of $A \in M_n(\mathbb{R})$ if and only if the matrix $(A - \lambda I_n)$ is singular (i.e., its rank is less than n). **Corollary:** λ is an eigenvalue of A if and only if it is a solution to the **characteristic equation**:

$$\det(A - \lambda I) = 0$$

Example Example: 2×2 Matrix

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$.
1. Find Eigenvalues: Set $\det(A - \lambda I) = 0$.

$$\det \begin{pmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{pmatrix} = (3-\lambda)^2 - 1 = \lambda^2 - 6\lambda + 9 - 1 = \lambda^2 - 6\lambda + 8 = 0$$

Factoring the polynomial: $(\lambda - 4)(\lambda - 2) = 0$. So A has two eigenvalues: $\lambda_1 = 4$ and $\lambda_2 = 2$.
2. Find Eigenvectors: For $\lambda_1 = 4$, solve $(A - 4I)v = 0$:

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies -x + y = 0 \implies x = y$$

Thus, $v_1 = (1, 1)^T$ is an eigenvector. (Any multiple $c(1, 1)^T$ is also an eigenvector).
For $\lambda_2 = 2$, solve $(A - 2I)v = 0$:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies x + y = 0 \implies x = -y$$

Thus, $v_2 = (-1, 1)^T$ is an eigenvector.

146 Eigenspaces and Multiplicity

Note that if v is an eigenvector, then any scalar multiple cv is also an eigenvector:

$$A(cv) = c(Av) = c(\lambda v) = \lambda(cv)$$

Definition : Eigenspace

Let $A \in M_n(\mathbb{R})$ and let λ be an eigenvalue of A. The **eigenspace** $V_\lambda \subseteq \mathbb{C}^n$ is the subspace of all eigenvectors associated to λ, together with the zero vector:

$$V_\lambda = \{v \in \mathbb{C}^n \mid Av = \lambda v\} = \{v \in \mathbb{C}^n \mid (A - \lambda I)v = 0\} = \mathrm{Ker}(A - \lambda I)$$

Note: A scalar λ is an eigenvalue $\iff$ $V_\lambda \neq \{0\}$ $\iff$ $A - \lambda I$ is a singular matrix. In particular, 0 is an eigenvalue of A $\iff$ $\mathrm{Ker}(A) \neq \{0\}$ $\iff$ A is a singular matrix.

Example Example: A 3×3 Matrix with a Double Root

Consider $A = \begin{pmatrix} 0 & -1 & -1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$. Find its eigenvalues and eigenspaces.

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & -1 & -1 \\ 1 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{pmatrix} = -(\lambda - 1)^2(\lambda - 2) = 0$$

A has two eigenvalues: $\lambda_1 = 1$ (a **double eigenvalue** since it occurs with algebraic multiplicity 2) and $\lambda_2 = 2$ (a **simple eigenvalue** with algebraic multiplicity 1).
Eigenspace for $\lambda_1 = 1$: Solve $(A - I)v = 0$.

$$\begin{pmatrix} -1 & -1 & -1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies x + y + z = 0 \implies x = -y - z$$

We have two free variables (y, z). Set $y = 1, z = 0 \implies v_1 = (-1, 1, 0)^T$. Set $y = 0, z = 1 \implies v_2 = (-1, 0, 1)^T$. $V_1 = \mathrm{span}((-1, 1, 0)^T, (-1, 0, 1)^T)$. It is a 2-dimensional eigenspace.
Eigenspace for $\lambda_2 = 2$: Solve $(A - 2I)v = 0$.

$$\begin{pmatrix} -2 & -1 & -1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \xrightarrow{RREF} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \implies x = -z, \ y = z$$

Set $z = 1 \implies v_3 = (-1, 1, 1)^T$. $V_2 = \mathrm{span}((-1, 1, 1)^T)$. It is a 1-dimensional eigenspace.

147 Complex Eigenvalues and Conjugate Pairs

Example Example: Solving for Complex Eigenvectors

Let $A = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & -2 \\ 2 & 2 & -1 \end{pmatrix}$. The characteristic equation is:

$$\det(A - \lambda I) = -\lambda^3 + \lambda^2 - 3\lambda - 5 = -(\lambda + 1)(\lambda^2 - 2\lambda + 5) = -(\lambda + 1)(\lambda - (1 + 2i))(\lambda - (1 - 2i))$$

The roots are $\lambda_1 = -1$, $\lambda_2 = 1 + 2i$, and $\lambda_3 = 1 - 2i$. Solving for $\lambda_1 = -1$ yields the real eigenspace $V_{-1} = \mathrm{span}((-1, 1, 1)^T)$.
Solve for $\lambda_2 = 1 + 2i$:

$$A - (1 + 2i)I = \begin{pmatrix} -2i & 2 & 0 \\ 0 & -2i & -2 \\ 2 & 2 & -2-2i \end{pmatrix}$$

Set $(A - (1 + 2i)I)v = 0$. Using the first two rows: 1. $-2ix + 2y = 0 \implies 2y = 2ix \implies y = ix$ 2. $-2iy - 2z = 0 \implies 2z = -2iy \implies z = -iy$ Substitute $y = ix$ into the second equation: $z = -i(ix) = -i^2x = x$. Let $x = 1$. Then $y = i$ and $z = 1$. Thus, $V_{1+2i} = \mathrm{span}_{\mathbb{C}}\left((1, i, 1)^T\right)$.

Solve for $\lambda_3 = 1 - 2i$: By doing the same algebra, we find $y = -ix$ and $z = x$. Thus, $V_{1-2i} = \text{span}_{\mathbb{C}}\left((1, -i, 1)^T\right)$.

Notice that $\lambda_3 = \overline{\lambda_2}$, and its associated eigenvector $v_3 = \overline{v_2}$. This is not a coincidence.

Theorem Proposition: Conjugate Eigenpairs

Suppose $A \in M_n(\mathbb{R})$ is a real matrix (a matrix with strictly real entries). If $\lambda = \mu + i\nu$ is a complex eigenvalue of A with corresponding complex eigenvector $v = x + iy$, then its complex conjugate $\overline{\lambda} = \mu - i\nu$ is also an eigenvalue of A, and its corresponding eigenvector is $\overline{v} = x - iy$.

Proof

By definition, we are given $Av = \lambda v$. Take the complex conjugate of both sides of the equation:

$$\overline{Av} = \overline{\lambda v} \implies \overline{A}\,\overline{v} = \overline{\lambda}\,\overline{v}$$

Because A has purely real entries, conjugating the matrix does nothing to it: $\overline{A} = A$. Substitute this back into the equation:

$$A\overline{v} = \overline{\lambda}\,\overline{v}$$

This is the exact algebraic definition stating that $\overline{\lambda}$ is an eigenvalue of A with eigenvector $\overline{v}$.

148 Properties of the Characteristic Polynomial

Definition : The Characteristic Polynomial

For $A \in M_n(\mathbb{R})$, the characteristic polynomial is defined as $P_A(\lambda) = \det(A - \lambda I)$. It takes the form:

$$P_A(\lambda) = c_n\lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0$$

where the coefficients are given by:

- $c_n = (-1)^n$
- $c_{n-1} = (-1)^{n-1}\text{tr}(A)$ (where trace is the sum of diagonal entries)
- $c_0 = P_A(0) = \det(A)$

By the Fundamental Theorem of Algebra, $P_A(\lambda)$ factors perfectly over the complex numbers into n roots (not necessarily distinct):

$$P_A(\lambda) = (-1)^n(\lambda - \lambda_1)(\lambda - \lambda_2)\ldots(\lambda - \lambda_n)$$

Thus, an $n \times n$ matrix has exactly n complex eigenvalues (counted with multiplicity). Comparing polynomial expansions yields two fundamental identities:

$$\det(A) = \lambda_1\lambda_2\ldots\lambda_n \quad \text{(Product of eigenvalues)}$$

$$\text{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n \quad \text{(Sum of eigenvalues)}$$

Definition : Algebraic Multiplicity

An eigenvalue λ_i is said to have **algebraic multiplicity** k if the linear factor $(\lambda - \lambda_i)$ appears exactly k times in the factorization of $P_A(\lambda)$. An eigenvalue with algebraic multiplicity 1 is called a **simple eigenvalue**.

Theorem Proposition: Transpose Eigenvalues

For any $A \in M_n(\mathbb{R})$, A and its transpose A^T have the exact same characteristic polynomial, and therefore the exact same eigenvalues with the same algebraic multiplicities.

Proof

We use the fundamental property that the determinant of a matrix equals the determinant of its transpose ($\det(M) = \det(M^T)$).

$$P_A(\lambda) = \det(A - \lambda I) = \det((A - \lambda I)^T)$$

Since the transpose is a linear operator and the identity matrix is symmetric ($I^T = I$):

$$= \det(A^T - (\lambda I)^T) = \det(A^T - \lambda I) = P_{A^T}(\lambda)$$

Since their polynomials are identical, their roots are identical.

149 Linear Independence and Completeness

Theorem Lemma: Distinct Eigenvalues imply Linear Independence

Let $A \in M_n(\mathbb{R})$. If $\lambda_1, \dots, \lambda_k$ are strictly **distinct** eigenvalues of A ($\lambda_i \neq \lambda_j$ for $i \neq j$), then their corresponding eigenvectors $v_1, \dots, v_k$ are linearly independent.

Proof

We prove this by induction on k. **Base Case ($k = 1$):** Since v_1 is an eigenvector, $v_1 \neq 0$ by definition. A set of one non-zero vector is linearly independent. **Inductive Step:** Assume the theorem holds for $k - 1$ distinct eigenvectors. We must show it holds for k. Suppose a linear combination equals zero:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \quad \text{— } \textbf{(Equation 1)}$$

Multiply Equation 1 by the matrix A. Since $Av_i = \lambda_i v_i$:

$$c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \cdots + c_k \lambda_k v_k = 0 \quad \text{— } \textbf{(Equation 2)}$$

Now, multiply Equation 1 by the scalar λ_k:

$$c_1 \lambda_k v_1 + c_2 \lambda_k v_2 + \cdots + c_k \lambda_k v_k = 0 \quad \text{— } \textbf{(Equation 3)}$$

Subtract Equation 3 from Equation 2. The k-th term perfectly cancels out:

$$c_1(\lambda_1 - \lambda_k)v_1 + c_2(\lambda_2 - \lambda_k)v_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} = 0$$

We now have a linear combination of $k-1$ eigenvectors equaling zero. By our inductive hypothesis, $v_1, \dots, v_{k-1}$ are linearly independent, so every coefficient in this new equation must be exactly zero:

$$c_i(\lambda_i - \lambda_k) = 0 \quad \text{for all } 1 \leq i \leq k - 1$$

Because the eigenvalues are distinct, $(\lambda_i - \lambda_k) \neq 0$. Therefore, we must have $c_i = 0$ for all $1 \leq i \leq k - 1$. Substitute $c_1 = c_2 = \cdots = c_{k-1} = 0$ back into Equation 1:

$$0 + 0 + \cdots + c_k v_k = 0 \implies c_k v_k = 0$$

Since $v_k \neq 0$, $c_k = 0$. Since all $c_i = 0$, the full set of k eigenvectors is linearly independent.

Corollary: If an $n \times n$ matrix A has n distinct real eigenvalues, then their corresponding eigenvectors

$v_1, \ldots, v_n$ form a basis of $\mathbb{R}^n$.

Definition : Complete Matrix

An eigenvalue λ is called **complete** if the dimension of its eigenspace (its geometric multiplicity) equals its algebraic multiplicity in $P_A(\lambda)$. An $n \times n$ matrix A is called **complete** if all of its eigenvalues are complete.

Theorem Theorem: Complete Matrices Span the Space

A matrix $A \in M_n(\mathbb{R})$ is complete if and only if its eigenvectors span $\mathbb{C}^n$ (i.e., there exists a basis of the entire space consisting exclusively of eigenvectors of A). In particular, any $n \times n$ matrix with n distinct eigenvalues is automatically complete.

Example Example: A Non-Complete (Defective) Matrix

Consider the upper-triangular shift matrix $A = \begin{pmatrix} a & 1 & 0 \\ 0 & a & 1 \\ 0 & 0 & a \end{pmatrix}$. The characteristic polynomial is $(a-\lambda)^3 = 0$. Thus $\lambda = a$ is an eigenvalue with an **algebraic multiplicity of 3**. However, let's find the dimension of its eigenspace V_a by solving $(A - aI)v = 0$:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \implies y = 0, \; z = 0$$

The only free variable is x. Thus, $V_a = \text{span}((1,0,0)^T)$. Because the dimension of the eigenspace is 1, which is strictly less than its algebraic multiplicity of 3, the matrix A is **not complete** (also called "defective"). It is impossible to form a basis of $\mathbb{R}^3$ using only the eigenvectors of A.

150 Complex Vector Space (POST MIDSEM)

Reference: Olver & Shakiban, Chapter 3, Page 172 onwards.

We now extend our study of linear algebra from the field of real numbers ($\mathbb{R}$) to the field of complex numbers ($\mathbb{C}$).

150.1 Review of Complex Numbers

A complex number belongs to the set $\mathbb{C} = \{x + iy \mid x, y \in \mathbb{R} \text{ and } i = \sqrt{-1}\}$.

- **Addition:** $(x + iy) + (u + iv) = (x + u) + i(y + v)$
- **Multiplication:** $(x + iy)(u + iv) = (xu - yv) + i(xv + yu)$
- **Complex Conjugate:** The conjugate of $z = x + iy$ is $\bar{z} = x - iy$.

Given $z = x + iy$, we define its real and imaginary parts:

$$\text{Re}(z) = x = \frac{z + \bar{z}}{2}, \quad \text{Im}(z) = y = \frac{z - \bar{z}}{2i}$$

Properties of conjugates: $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{zw} = \bar{z}\bar{w}$.

The squared magnitude of a complex number is a positive real number:

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2 \quad \text{where } |z| = \sqrt{x^2 + y^2}$$

This allows us to perform division easily. The inverse of z is:

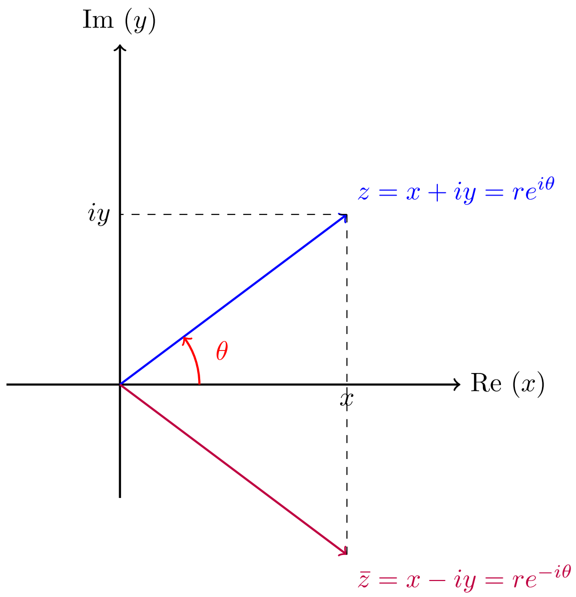
$$\frac{1}{z} = \frac{\bar{z}}{|z|^2} \quad (\text{if } z \neq 0) \implies \frac{w}{z} = \frac{w\bar{z}}{|z|^2} = \frac{(u + iv)(x - iy)}{x^2 + y^2}$$

Definition : Polar Form and Euler's Formula

A complex number can be expressed in polar coordinates. Let $r = |z| = \sqrt{x^2 + y^2}$, and let the angle θ be the argument $\arg(z) = \mathrm{ph}(z)$ (phase).

$$x = r\cos\theta, \quad y = r\sin\theta \implies z = r(\cos\theta + i\sin\theta)$$

By **Euler's Formula**, $e^{i\theta} = \cos\theta + i\sin\theta$. Thus, we can write $z = re^{i\theta}$.



The Complex Plane (Argand Diagram):
The conjugate $\bar{z}$ is a perfect reflection across
the Real axis, meaning its phase is exactly $-\theta$.

Multiplication in polar form is highly intuitive: $|zw| = |z||w|$ and $\mathrm{ph}(zw) = \mathrm{ph}(z) + \mathrm{ph}(w)$. Note that the phase is defined up to an integer multiple of 2π.

150.2 Complex Vector Spaces

Definition : The Space $\mathbb{C}^n$

The set $\mathbb{C}^n$ is the collection of all n-dimensional column vectors with complex entries:

$$\mathbb{C}^n = \left\{(z_1, \ldots, z_n)^T \mid z_j \in \mathbb{C} \text{ for } 1 \le j \le n\right\}$$

We can split any complex vector into a real vector and an imaginary vector: $z = x + iy$ where $x = (x_1, \ldots, x_n)^T \in \mathbb{R}^n$ and $y = (y_1, \ldots, y_n)^T \in \mathbb{R}^n$.

Example: Let $z = \begin{pmatrix} 1+2i \\ -3 \\ 5i \end{pmatrix}$. We can write this as $\begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix} + i\begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$. The conjugate of a vector is applied component-wise: $\bar{z} = x - iy = \begin{pmatrix} 1 \\ -3 \\ 0 \end{pmatrix} - i\begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix} = \begin{pmatrix} 1-2i \\ -3 \\ -5i \end{pmatrix}$.

A **complex vector space** is a set V satisfying the exact same vector space axioms of addition and scalar multiplication as real vector spaces, except the scalars are allowed to belong to $\mathbb{C}$ instead of just $\mathbb{R}$. All concepts derived in Math I (span, linear independence, basis, dimension) carry over directly, using terms like *complex span*, *linearly independent over* $\mathbb{C}$, and *complex dimension*.

151 The Hermitian Dot Product

The notion of an inner product on a complex vector space requires a slight modification from the real case. If we used the standard Euclidean dot product $z^T z$, the length of a vector might not be a real positive number! For example, if $z = (i)$, then $z^T z = i^2 = -1$. Lengths cannot be negative.

Definition : The Hermitian Dot Product

The correct generalization of the Euclidean dot product for $\mathbb{C}^n$ is the **Hermitian Dot Product**. Let $z, w \in \mathbb{C}^n$. We define:

$$z \cdot w = z^T \bar{w} = z_1\bar{w}_1 + z_2\bar{w}_2 + \cdots + z_n\bar{w}_n \in \mathbb{C}$$

Example Example: Computing Hermitian Dot Products

Let $z = \begin{pmatrix} 1+i \\ 3+2i \end{pmatrix}$ and $w = \begin{pmatrix} 1+2i \\ i \end{pmatrix}$. Compute $z \cdot w$:

$$\begin{aligned} z \cdot w = z^T \bar{w} &= (1+i)\overline{(1+2i)} + (3+2i)\overline{(i)} \\ &= (1+i)(1-2i) + (3+2i)(-i) \\ &= (1-2i+i+2) + (-3i+2) = (3-i) + (2-3i) = 5-4i \end{aligned}$$

Now compute $w \cdot z$:

$$\begin{aligned} w \cdot z = w^T \bar{z} &= (1+2i)\overline{(1+i)} + (i)\overline{(3+2i)} \\ &= (1+2i)(1-i) + (i)(3-2i) \\ &= (1-i+2i+2) + (3i+2) = (3+i) + (2+3i) = 5+4i \end{aligned}$$

Notice that $w \cdot z = \overline{z \cdot w}$.

151.1 Properties of the Hermitian Inner Product

Because of the conjugate, $z \cdot z$ is always a positive real number:

$$z \cdot z = z_1\bar{z}_1 + \cdots + z_n\bar{z}_n = |z_1|^2 + \cdots + |z_n|^2 \in \mathbb{R}_{\geq 0}$$

Thus, we can define the norm $\|z\| = \sqrt{z \cdot z} \geq 0$, with $\|z\| = 0 \iff z = 0$. For $z = (1+i, 3+2i)^T$, its length is $\|z\| = \sqrt{1^2 + 1^2 + 3^2 + 2^2} = \sqrt{2+13} = \sqrt{15}$.

Theorem Axioms of a Complex Inner Product

An inner product on a complex vector space V is a pairing $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{C}$ that satisfies:

1. **Sesquilinearity (1.5 Linearity):** It is linear in the first variable, and *conjugate linear* in the second variable.
$$\langle cu + dv, w \rangle = c\langle u, w \rangle + d\langle v, w \rangle$$
$$\langle u, cv + dw \rangle = \bar{c}\langle u, v \rangle + \bar{d}\langle u, w \rangle$$
2. **Conjugate Symmetry:** $\langle v, w \rangle = \overline{\langle w, v \rangle}$.
3. **Positivity:** $\|v\|^2 = \langle v, v \rangle \geq 0$, and $\langle v, v \rangle = 0 \iff v = 0$.

Correction and Verification of Complex Cauchy-Schwarz

Let $v = (1+i, 2i, -3)^T$ and $w = (2-i, 2+2i, i)^T \in \mathbb{C}^3$. We evaluate the norms and inner product to verify the complex Cauchy-Schwarz inequality ($|\langle v, w \rangle| \leq \|v\|\|w\|$).
1. Norms: $\|v\|^2 = (1^2+1^2) + (2^2) + (-3)^2 = 2+4+9 = 15 \implies \|v\| = \sqrt{15}$. $\|w\|^2 = (2^2 + (-1)^2) + (2^2 + 2^2) + (1^2) = 5+8+1 = 14 \implies \|w\| = \sqrt{14}$.

2. Inner Product $\langle v, w\rangle = v^T\bar{w}$:

$$\begin{aligned}\langle v, w\rangle &= (1+i)\overline{(2-i)} + (2i)\overline{(2+2i)} + (-3)\overline{(i)}\\ &= (1+i)(2+i) + (2i)(2-2i) + (-3)(-i)\\ &= (2+i+2i-1) + (4i+4) + 3i\\ &= (1+3i) + (4+4i) + 3i = 5+10i\end{aligned}$$

(Note: A minor arithmetic slip in the handwritten notes resulted in $-5+11i$; the correct evaluation is $5+10i$).

3. Cauchy-Schwarz Check: $|\langle v, w\rangle| = |5+10i| = \sqrt{25+100} = \sqrt{125} \approx 11.18$. $\|v\|\|w\| = \sqrt{15}\sqrt{14} = \sqrt{210} \approx 14.49$. Since $\sqrt{125} \leq \sqrt{210}$, the inequality firmly holds.

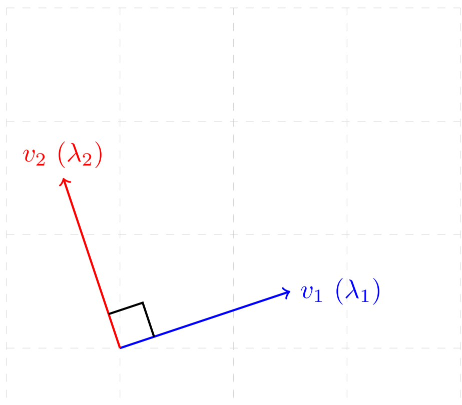
152 The Spectral Theorem for Real Symmetric Matrices

We now present one of the most powerful theorems in linear algebra, describing the perfect behavior of symmetric matrices.

Theorem Theorem: Real Symmetric Matrices

Let $A \in M_n(\mathbb{R})$ be a real symmetric matrix ($A = A^T$). Then:

1. All eigenvalues of A are strictly **real** numbers.
2. Eigenvectors corresponding to distinct eigenvalues are strictly **orthogonal** to each other.
3. There exists an **orthonormal basis** of $\mathbb{R}^n$ consisting entirely of eigenvectors of A.



The Spectral Theorem: For a symmetric matrix, you can always find a set of eigenvectors that act just like the standard x and y axes—they are mutually perpendicular and span the entire space.

Proof Proof of (a): All Eigenvalues are Real

Consider the Hermitian dot product on $\mathbb{C}^n$ given by $\langle z, w\rangle = z^T\bar{w}$. Consider the linear transformation $T : \mathbb{C}^n \to \mathbb{C}^n$ given by $z \mapsto Az$. Let $\lambda \in \mathbb{C}$ be a complex eigenvalue of A, and let $z \in \mathbb{C}^n$ be its associated eigenvector ($z \neq 0$).
We evaluate the inner product $\langle Az, w\rangle$:

$$\langle Az, w\rangle = (Az)^T\bar{w} = z^TA^T\bar{w}$$

Since A is a symmetric matrix, $A^T = A$:

$$= z^TA\bar{w}$$

Since A is a real matrix, it is equal to its own complex conjugate ($A = \bar{A}$). Therefore, $A\bar{w} =$

$\bar{A}\bar{w} = \overline{Aw}$:

$$= z^T \overline{Aw} = \langle z, Aw \rangle$$

Now, we apply this identity to the eigenvector z:

$$\langle Az, z \rangle = \langle \lambda z, z \rangle = \lambda \langle z, z \rangle = \lambda \|z\|^2$$

Conversely, moving A to the other side:

$$\langle z, Az \rangle = \langle z, \lambda z \rangle = \bar{\lambda} \langle z, z \rangle = \bar{\lambda} \|z\|^2$$

Because $\langle Az, z \rangle = \langle z, Az \rangle$, we must have:

$$\lambda \|z\|^2 = \bar{\lambda} \|z\|^2$$

Since z is an eigenvector, $z \neq 0 \implies \|z\|^2 \neq 0$. We can divide both sides by $\|z\|^2$, leaving:

$$\lambda = \bar{\lambda}$$

The only complex numbers that equal their own conjugates are real numbers. Thus, $\lambda \in \mathbb{R}$.

Proof Proof of (b): Distinct Eigenvectors are Orthogonal

Now that we know all eigenvalues of A are real, we can conclude that the associated eigenvectors have strictly real coordinates. We restrict our view back to $\mathbb{R}^n$ with the standard Euclidean dot product $\langle x, y \rangle = x^T y$.
Let $\lambda \neq \mu$ be two distinct real eigenvalues. Let $x \in \mathbb{R}^n$ be an eigenvector associated to λ ($Ax = \lambda x$) and $y \in \mathbb{R}^n$ be an eigenvector associated to μ ($Ay = \mu y$).
Evaluate the inner product $\langle Ax, y \rangle$:

$$\langle Ax, y \rangle = (Ax)^T y = x^T A^T y = x^T A y = \langle x, Ay \rangle$$

Substitute the eigenvalue relations:

$$\langle \lambda x, y \rangle = \langle x, \mu y \rangle \implies \lambda \langle x, y \rangle = \mu \langle x, y \rangle$$

Bringing terms to one side gives:

$$(\lambda - \mu) \langle x, y \rangle = 0$$

Since the eigenvalues are distinct, $(\lambda - \mu) \neq 0$. Therefore, we must strictly have:

$$\langle x, y \rangle = 0$$

Thus, the eigenvectors are orthogonal.

153 Linear Transformations (Chapter 7)

Reference: Olver & Shakiban, Chapter 7.

We formalize the concept of mappings between vector spaces.

Definition : Linear Transformation

A **linear map** (or linear transformation) from a real vector space V to another real vector space W is a function $T : V \to W$ such that:

1. **Additivity:** $T(v + w) = T(v) + T(w)$ for all $v, w \in V$.
2. **Homogeneity:** $T(\lambda v) = \lambda T(v)$ for all $v \in V$ and $\forall \lambda \in \mathbb{R}$.

Note: If V and W are complex vector spaces, the scalar λ belongs to $\mathbb{C}$.

Linear transformations preserve the fundamental structure of the vector space. Because of this, the action of a linear transformation on the entire infinite space can be perfectly captured just by knowing what it does to a finite set of basis vectors.

Theorem Theorem: Matrix Representation of a Linear Transformation

Let $T : V \to W$ be a linear transformation. Suppose V has a basis $v_1, \dots, v_n$ and W has a basis $w_1, \dots, w_m$. Let $v \in V$. It can be written uniquely as $v = x_1 v_1 + \cdots + x_n v_n$ for coordinates $x_i \in \mathbb{R}$. Let $w = T(v) \in W$. It can be written uniquely as $w = y_1 w_1 + \cdots + y_m w_m$ for coordinates $y_i \in \mathbb{R}$.

Then the output coordinates y are calculated from the input coordinates x via standard matrix multiplication:

$$\begin{pmatrix} y_1 \\ \vdots \\ y_m \end{pmatrix} = B \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \qquad \Longrightarrow \qquad y = Bx$$

where $B = (b_{ij})$ is an $m \times n$ matrix. The columns of B are entirely determined by where T sends the basis vectors of V:

$$T(v_j) = b_{1j} w_1 + \cdots + b_{mj} w_m \quad \text{for } 1 \le j \le n$$

Proof

By definition, $T(v_j) = \sum_{i=1}^m b_{ij} w_i$. We evaluate the transformation on a general vector $v = \sum_{j=1}^n x_j v_j$. By the linearity of T, we can pull the summation and the scalars x_j outside the function:

$$T(v) = T\left(\sum_{j=1}^n x_j v_j\right) = \sum_{j=1}^n x_j T(v_j)$$

Substitute the basis expansion for $T(v_j)$:

$$= \sum_{j=1}^n x_j \left(\sum_{i=1}^m b_{ij} w_i\right)$$

Rearrange the double summation to group terms by the output basis vectors w_i:

$$= \sum_{i=1}^m \left(\sum_{j=1}^n b_{ij} x_j\right) w_i$$

We defined the output vector coordinates as $T(v) = \sum_{i=1}^m y_i w_i$. Because the coordinate representation in a basis is unique, the coefficients on w_i must match exactly:

$$y_i = \sum_{j=1}^n b_{ij} x_j \quad \text{for } 1 \le i \le m$$

This is the precise algebraic definition of the matrix-vector multiplication $y = Bx$.

154 Matrix Representations of Linear Transformations

Reference: Olver & Shakiban, Chapter 7.

We have seen that a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by matrix multiplication. However, the exact numbers inside that matrix depend entirely on the **basis** we choose to view the space with.

Example Example: Two Different Matrix Representations

Consider the linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ given by:

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x - y \\ 2x + 4y \end{pmatrix}$$

1. Representation in the Standard Basis: Let $e_1 = (1, 0)^T$ and $e_2 = (0, 1)^T$. We evaluate T on these basis vectors:

$$T(e_1) = T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad T(e_2) = T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}$$

Placing these output vectors as the columns of a matrix gives us the standard representation A:

$$A = \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix} \quad \Longrightarrow \quad T(v) = Av$$

2. Representation in a New Basis: Now, suppose we view the exact same space using a different basis: $v_1 = (1, -1)^T$ and $v_2 = (1, -2)^T$. Let's see what the transformation T does to these specific vectors:

$$T(v_1) = T\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 - (-1) \\ 2(1) + 4(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \end{pmatrix} = 2\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 2v_1$$

$$T(v_2) = T\begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 - (-2) \\ 2(1) + 4(-2) \end{pmatrix} = \begin{pmatrix} 3 \\ -6 \end{pmatrix} = 3\begin{pmatrix} 1 \\ -2 \end{pmatrix} = 3v_2$$

Since $T(v_1) = 2v_1 + 0v_2$ and $T(v_2) = 0v_1 + 3v_2$, the matrix representing T with respect to this new basis is purely diagonal:

$$B = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

If a vector is given in coordinates of the new basis, $v = c_1v_1 + c_2v_2$, the transformation is simply $T(v) = 2c_1v_1 + 3c_2v_2$. *(Note: We were able to diagonalize the transformation because we cleverly chose v_1 and v_2 to be the eigenvectors of A).*

155 Change of Basis and Coordinate Vectors

How do we translate a vector from standard coordinates to coordinates in a new basis?

Theorem Theorem: The Transition Matrix S

Suppose $v_1, \ldots, v_n$ is a basis of $\mathbb{R}^n$. Consider a vector $x \in \mathbb{R}^n$. In the standard basis $e_1, \ldots, e_n$, we write $x = (x_1, \ldots, x_n)^T$. Let $y_1, \ldots, y_n \in \mathbb{R}$ be the coordinates of x with respect to the new basis $v_1, \ldots, v_n$:

$$x = y_1v_1 + y_2v_2 + \cdots + y_nv_n$$

Let S be the $n \times n$ matrix whose j-th column is the vector v_j (written in standard coordinates): $S = [v_1, v_2, \ldots, v_n]$. Then the old and new coordinates are related exactly by:

$$x = Sy$$

Proof

By definition, the j-th basis vector is a column vector in $\mathbb{R}^n$: $v_j = (s_{1j}, s_{2j}, \ldots, s_{nj})^T$. We expand the linear combination defining x:

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = y_1 \begin{pmatrix} s_{11} \\ \vdots \\ s_{n1} \end{pmatrix} + y_2 \begin{pmatrix} s_{12} \\ \vdots \\ s_{n2} \end{pmatrix} + \cdots + y_n \begin{pmatrix} s_{1n} \\ \vdots \\ s_{nn} \end{pmatrix}$$

Evaluating the i-th row of this sum yields:

$$x_i = y_1 s_{i1} + y_2 s_{i2} + \cdots + y_n s_{in} = \sum_{j=1}^{n} s_{ij} y_j$$

This is the precise definition of multiplying the matrix S by the column vector y. Thus, $x = Sy$. Because $v_1, \ldots, v_n$ form a valid basis, they are linearly independent, ensuring the matrix S is non-singular. Thus, we can find the new coordinates from the old coordinates by multiplying by the inverse: $y = S^{-1}x$.

156 Similar Matrices and Change of Basis

We now merge the previous two concepts. If A describes a transformation in the standard basis, and B describes the same transformation in a new basis, how are A and B related algebraically?

Theorem Theorem: Change of Basis for Transformations

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. Let $A \in M_n(\mathbb{R})$ be the matrix representation of T with respect to the standard basis. Let $v_1, \ldots, v_n$ be a new basis of $\mathbb{R}^n$, and let $B \in M_n(\mathbb{R})$ be the representation of T with respect to this new basis. If S is the transition matrix whose columns are $v_1, \ldots, v_n$, then:

$$B = S^{-1}AS$$

Two matrices A and B related by this equation are called **Similar Matrices**.

$$\begin{tikzcd}
\mathbb{R}^n \text{ (Std Coords)} \arrow[r, "A"] & \mathbb{R}^n \text{ (Std Coords)} \arrow[d, "S^{-1}"] \\
\mathbb{R}^n \text{ (New Coords)} \arrow[u, "S"] \arrow[r, "B = S^{-1}AS"'] & \mathbb{R}^n \text{ (New Coords)}
\end{tikzcd}$$

Proof

We construct the columns of the new matrix B. By definition, the j-th column of B is the vector $T(v_j)$ written in the coordinates of the new basis.

1. Start with the j-th new basis vector. Its standard coordinates are exactly the j-th column of S.

2. Apply the transformation in standard coordinates: The j-th column of the matrix product AS is $A \times$ (j-th col of S) $= Av_j = T(v_j)$.

3. Convert these output standard coordinates back into the new basis coordinates by multiplying by S^{-1}.

Thus, the j-th column of B is the j-th column of the combined matrix $S^{-1}AS$. Since this holds for all columns, $B = S^{-1}AS$.

Example Example: Verifying the Change of Basis Formula

Let's return to our example transformation where $A = \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix}$ and our new basis vectors are $v_1 = (1, -1)^T$ and $v_2 = (1, -2)^T$.

1. Construct the Transition Matrix S:

$$S = \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix}$$

2. Find the inverse S^{-1}: The determinant is $(1)(-2) - (1)(-1) = -2 + 1 = -1$.

$$S^{-1} = \frac{1}{-1} \begin{pmatrix} -2 & -1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}$$

3. Compute $S^{-1}AS$: First, evaluate $S^{-1}A$:

$$S^{-1}A = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2(1) + 1(2) & 2(-1) + 1(4) \\ -1(1) - 1(2) & -1(-1) - 1(4) \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ -3 & -3 \end{pmatrix}$$

Next, multiply by S:

$$(S^{-1}A)S = \begin{pmatrix} 4 & 2 \\ -3 & -3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} 4(1) + 2(-1) & 4(1) + 2(-2) \\ -3(1) - 3(-1) & -3(1) - 3(-2) \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

This perfectly matches our diagonal matrix B, mathematically verifying the formula $B = S^{-1}AS$.

157 Formalizing Complex Spaces and the Spectral Theorem

Reference: Digital Whiteboard Lecture Notes (April 21, 2026).

While we previously introduced complex eigenvalues and the Spectral Theorem, we will now construct the rigorous algebraic proofs that govern these structures.

157.1 Complex Isomorphisms and Eigenvalues

Example Example: Complex Eigenvalues of a Real Rotation

Consider the 2×2 real matrix representing a $90°$ rotation:

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

To find the eigenvalues, we solve the characteristic polynomial:

$$\det(A - \lambda I) = \det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 + 1 = 0$$

The eigenvalues are strictly complex: $\lambda_1 = i$ and $\lambda_2 = -i$.

To find the eigenvector for $\lambda_1 = i$, we solve $(A - iI)z = 0$:

$$\begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies x - iy = 0 \implies x = iy$$

Let $y = -i$. Then $x = i(-i) = -i^2 = 1$. Thus, $z_1 = \begin{pmatrix} 1 \\ -i \end{pmatrix} \in \mathbb{C}^2$ is an eigenvector associated with i. (By the conjugate pair theorem, $z_2 = \begin{pmatrix} 1 \\ i \end{pmatrix}$ is the eigenvector for $-i$).

Isomorphism between $\mathbb{C}^n$ and $\mathbb{R}^{2n}$

The complex space $\mathbb{C}^n$ has a standard $\mathbb{C}$-basis $e_1 = (1, 0, \dots, 0)^T, \dots, e_n = (0, \dots, 1)^T$. Its dimension over the complex field is n. However, if we restrict our scalar field to $\mathbb{R}$, we can view $\mathbb{C}^n$ as a real vector space. Because every complex component $z_j = x_j + iy_j$ requires two real numbers to define it, there is a natural isomorphism to a real space of twice the dimension:

$$\mathbb{C}^n \cong \mathbb{R}^{2n}$$

$$(z_1, \dots, z_n) \longleftrightarrow (x_1, y_1, x_2, y_2, \dots, x_n, y_n)$$

157.2 Sesquilinearity of the Hermitian Inner Product

The Euclidean dot product $X^T X = \sum x_i^2 \geq 0$ works perfectly in $\mathbb{R}^n$. However, extending this naively to $\mathbb{C}^n$ breaks the positivity axiom. For instance, if $z = (i, 0)^T$, then $z^T z = i^2 = -1 < 0$.

To ensure lengths are real and positive, we define the **Hermitian Dot Product** mapping $\mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$:

$$\langle z, w \rangle = z^T \bar{w} = z_1 \bar{w}_1 + z_2 \bar{w}_2 + \cdots + z_n \bar{w}_n$$

This guarantees that $\|z\|^2 = z \cdot z = |z_1|^2 + \cdots + |z_n|^2 \geq 0$.

Theorem Proposition: Sesquilinearity

The Hermitian inner product is **sesquilinear** ("one-and-a-half linear"). It is strictly complex linear in the first variable, and conjugate-linear (antilinear) in the second variable.

Proof

1. Linearity in the First Variable: Let $z, z', w \in \mathbb{C}^n$ and $\lambda \in \mathbb{C}$.

$$\langle z + z', w \rangle = (z + z')^T \bar{w} = z^T \bar{w} + (z')^T \bar{w} = \langle z, w \rangle + \langle z', w \rangle$$

$$\langle \lambda z, w \rangle = (\lambda z)^T \bar{w} = \lambda (z^T \bar{w}) = \lambda \langle z, w \rangle$$

2. Conjugate-Linearity in the Second Variable:

$$\langle z, w + w' \rangle = z^T (\overline{w + w'}) = z^T (\bar{w} + \bar{w}') = z^T \bar{w} + z^T \bar{w}' = \langle z, w \rangle + \langle z, w' \rangle$$

However, pulling a scalar out of the second variable conjugates the scalar:

$$\langle z, \lambda w \rangle = z^T (\overline{\lambda w}) = z^T (\bar{\lambda} \bar{w}) = \bar{\lambda} (z^T \bar{w}) = \bar{\lambda} \langle z, w \rangle$$

3. Conjugate Symmetry (Anti-symmetry):

$$\overline{\langle z, w \rangle} = \overline{z_1 \bar{w}_1 + \dots} = \bar{z}_1 w_1 + \cdots = w_1 \bar{z}_1 + \cdots = \langle w, z \rangle$$

157.3 Proving the Spectral Theorem

Theorem The Spectral Theorem for Symmetric Matrices

Let $A \in M_n(\mathbb{R})$ be a real symmetric matrix ($A = A^T$). Then:

1. **(a)** All eigenvalues of A are strictly real.
2. **(b)** Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.
3. **(c)** There exists an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A. Hence, there exists an orthogonal matrix $S \in O(n)$ such that $S^{-1}AS = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Proof Proof of (a): Real Eigenvalues

Let $\lambda \in \mathbb{C}$ be an eigenvalue of A, and let $z \in \mathbb{C}^n$ be its corresponding non-zero eigenvector ($Az = \lambda z$). We evaluate the Hermitian inner product $\langle Az, z\rangle$:

$$\langle Az, z\rangle = (Az)^T\bar{z} = z^T A^T \bar{z}$$

Because A is a symmetric matrix, $A^T = A$:

$$= z^T A\bar{z}$$

Because A is a matrix with strictly real entries, taking its complex conjugate does nothing: $\bar{A} = A$. Therefore, $A\bar{z} = \bar{A}\bar{z} = \overline{Az}$. Substitute this in:

$$= z^T(\overline{Az}) = \langle z, Az\rangle$$

Now, substitute the eigenvector relationship $Az = \lambda z$ into both sides of the identity $\langle Az, z\rangle = \langle z, Az\rangle$:

- **Left Side:** $\langle Az, z\rangle = \langle \lambda z, z\rangle = \lambda\langle z, z\rangle$.
- **Right Side:** $\langle z, Az\rangle = \langle z, \lambda z\rangle = \bar{\lambda}\langle z, z\rangle$ *(by sesquilinearity)*.

Equating them yields:

$$\lambda\langle z, z\rangle = \bar{\lambda}\langle z, z\rangle$$

Because $z \neq 0$, its squared norm $\langle z, z\rangle \neq 0$. We divide both sides by $\langle z, z\rangle$:

$$\lambda = \bar{\lambda}$$

The only complex numbers equal to their own conjugates are real numbers. Thus, $\lambda \in \mathbb{R}$.

$$A\mathbf{v} = \lambda\mathbf{v} \xrightarrow{[v_1 \dots v_n]} AS = SB \xrightarrow{S^{-1}} S^{-1}AS = \begin{pmatrix}\lambda_1 & & 0\\ & \ddots & \\ 0 & & \lambda_n\end{pmatrix}$$

Proof Proof of (b): Orthogonal Eigenvectors

Because the eigenvalues are real, the corresponding eigenvectors lie in $\mathbb{R}^n$. We revert to the standard Euclidean dot product $\langle v, w\rangle = v^T w$.
Let $\lambda \neq \mu$ be two distinct eigenvalues of A. Let v be an eigenvector for λ ($Av = \lambda v$), and let w be an eigenvector for μ ($Aw = \mu w$). Evaluate the inner product $\langle Av, w\rangle$:

$$\langle Av, w\rangle = (Av)^T w = v^T A^T w = v^T Aw = \langle v, Aw\rangle$$

Substitute the eigenvalue definitions:

$$\langle \lambda v, w\rangle = \langle v, \mu w\rangle$$

$$\lambda\langle v, w\rangle = \mu\langle v, w\rangle \implies (\lambda - \mu)\langle v, w\rangle = 0$$

Since the eigenvalues are strictly distinct, $(\lambda - \mu) \neq 0$. Therefore, we are mathematically forced to conclude:

$$\langle v, w\rangle = 0$$

Thus, eigenvectors corresponding to distinct eigenvalues are perfectly orthogonal.